Biomath Booklet

Chapters 1 – 2

Contents

		Lecture Problems	Textbook Problems	Past Exam Questions	Test Your Understanding	Page
1.1	What Is a Function?	9	26		3	**3**
1.2	Linear Functions	11	34	3	3	**29**
1.3	Average Rate of Change and Relative Change	10	50	2	6	**59**
1.5	Exponential Functions	6	31	1	5	**97**
1.6	The Natural Logarithm	6	41		1	**122**
1.7	Exponential Growth and Decay	8	35	3	6	**136**
1.8	New Functions From Old	10	43	1	1	**160**
1.9	Proportionality and Power Functions	5	30	1	2	**196**
1.10	Periodic Functions	7	32	3	1	**210**
2.1	Instantaneous Rate of Change	7	24	2	1	**233**
2.2	The Derivative Function	7	31	3		**254**
2.3	Interpretations of the Derivative	8	46	8	1	**282**
2.4	The Second Derivative	3	28	5	1	**319**

1.1 What Is a Function?

In this section

1. Functions, Domain & Range 3 lecture · 4 textbook
2. Function Notation & Interpretation 3 lecture · 6 textbook
3. Horizontal & Vertical Intercepts 3 lecture · 6 textbook
4. Behavior & Graphs of Functions 10 textbook
5. Test Your Understanding 3 questions

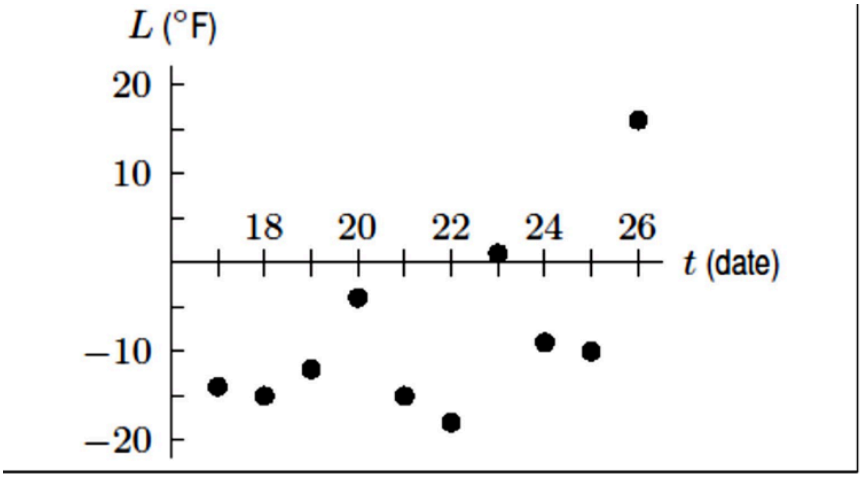
1.1.1 Functions, Domain & Range

Lecture Problems

Ex 1 $f(x) = x + 8$, Domain $= \{1, 2, 3, 4\}$.

Find the range.

Ex 2 The function giving the daily low temperatures in International Falls is represented by the graph and the table. Find the domain and range of L = f(t).



Date (t)	17	18	19	20	21
Low temp (°F)	−14	−15	−12	−4	−15

Date (t)	22	23	24	25	26
Low temp (°F)	−18	1	−9	−10	16

Domain What is the domain?

Range What is the range?

Ex 3 Other functions arise naturally as graphs. The figure contains electrocardiogram (EKG) pictures showing the heartbeat patterns of two patients, one normal and one not.

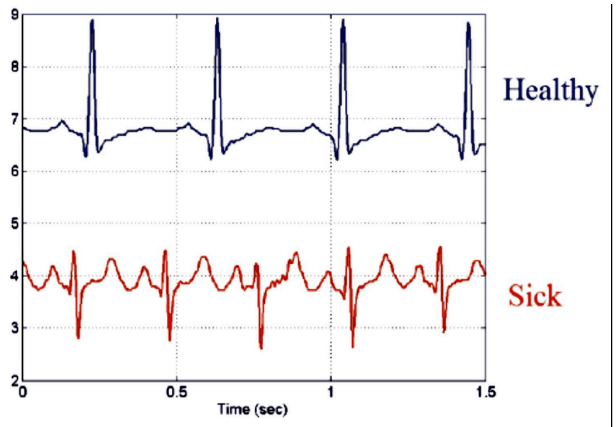


Figure: EKG readings on two patients

Why are such functions usually shown as graphs instead of formulas?

Textbook Problems

Problem 7 For the functions in Problems 7–11, find f(5).

$f(x) = 2x + 3$

Problem 8 Find f(5):

$f(x) = 10x - x^2$

Problem 11 Find f(5) from the table:

x	1	2	3	4	5	6	7	8
f(x)	2.3	2.8	3.2	3.7	4.1	5.0	5.6	6.2

f(5) = ?

Problem 12 Let $y = f(x) = x^2 + 2$.

a. Find the value of y when x is zero.

b. What is f(3)?

c. What values of x give y a value of 11?

d. Are there any values of x that give y a value of 1?

1.1.2 Function Notation & Interpretation

Lecture Problems

Ex 6 The value of a car, V, is a function of the age of the car, a, so V = g(a).

i. Interpret the statement g(5) = 9 in terms of the value of a car if V is in thousands of dollars and a is in years.

ii. In the same units, the value of a Honda is approximated by g(a) = 13.78 − 0.8a. Find and interpret the vertical and horizontal intercepts of the graph of this depreciation function g.

Ex 7

Same as Textbook Problem 4

The concentration of carbon dioxide, C = f(t), in the atmosphere, in parts per million (ppm), is a function of years, t, since 1960.

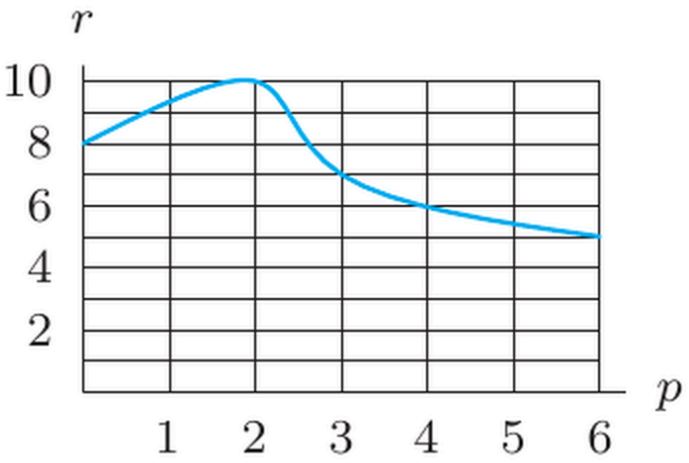
i. Interpret f(40) = 370 in terms of carbon dioxide.

ii. What is the meaning of f(50)?

Ex 8

Same as Textbook Problem 6

The graph of r = f(p) is given in the figure.



i. What is the value of r when p = 0?

ii. What is the value of r when p = 3?

iii. What is f(2)?

Textbook Problems

Problem 2 The population of a city, P, in millions, is a function of t, the number of years since 1970, so P = f(t).

Explain the meaning of the statement f(35) = 12 in terms of the population of this city.

Problem 3 Let W = f(t) represent wheat production in Argentina, in millions of metric tons, where t is years since 1990.

Interpret the statement f(12) = 9 in terms of wheat production.

Problem 4

Same as Lecture Example 7

The concentration of carbon dioxide, C = f(t), in the atmosphere, in parts per million (ppm), is a function of years, t, since 1960.

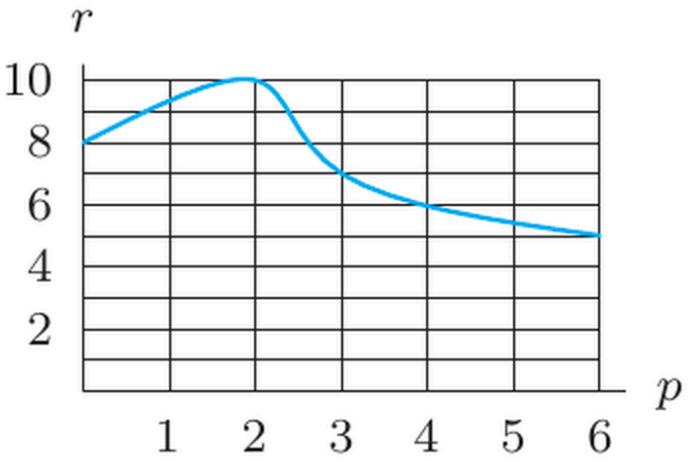
a. Interpret f(40) = 370 in terms of carbon dioxide.

b. What is the meaning of f(50)?

Problem 6

Same as Lecture Example 8

The graph of r = f(p) is in Figure 1.6.

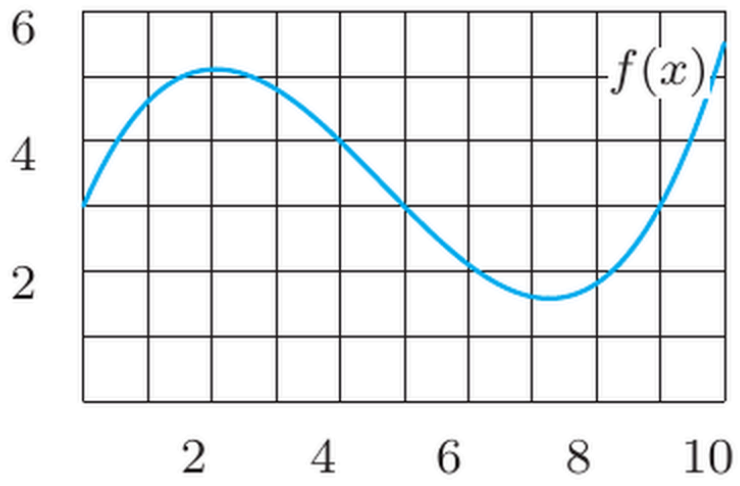


(a) What is the value of r when p is 0?

(a) When p is 3?

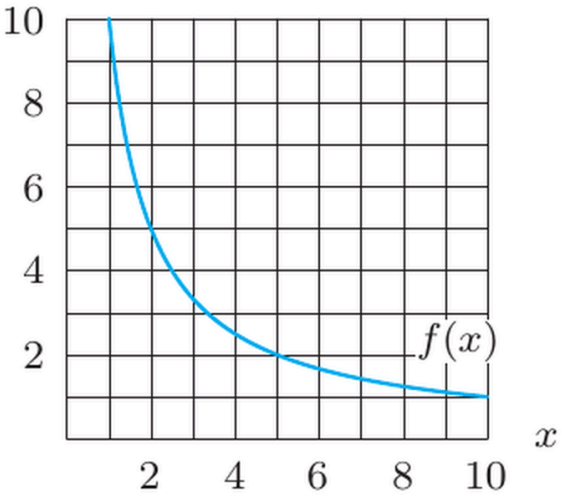
(b) What is f(2)?

Problem 9 Find f(5) from the graph:



f(5) ≈ ?

Problem 10 Find f(5) from the graph:



f(5) = ?

1.1.3 Horizontal & Vertical Intercepts

Lecture Problems

Ex 4 All snow tree crickets chirp at essentially the same rate at the same temperature. The chirp rate, C, in chirps per minute, can be computed using the formula $C = f(T) = 4T - 160$, where T is the temperature in °F.

a. Solve $f(T) = 0$, and interpret the result.

b. Plot the function. Which description matches the graph?

Ex 5 Find the horizontal intercept (x-intercept) and the vertical intercept (y-intercept) of each of the following functions:

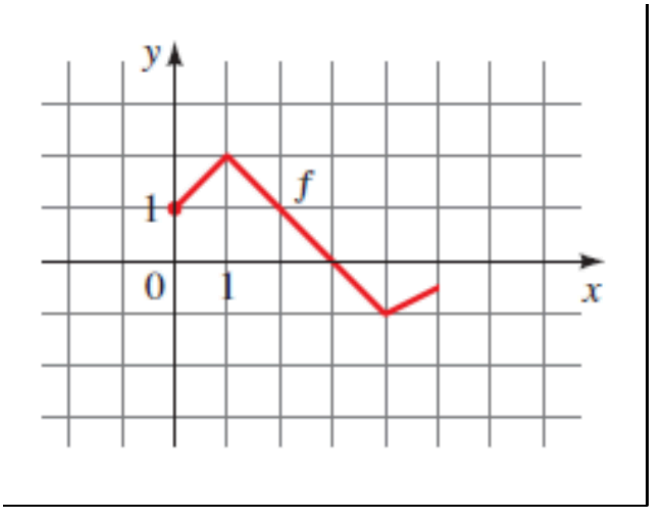
i. Table

Date	5	6	7	8
Snowfall	6.8	12.2	0	9.3

Date	9	10	11	12
Snowfall	14.5	1.9	0	0.1

---

ii. Graph



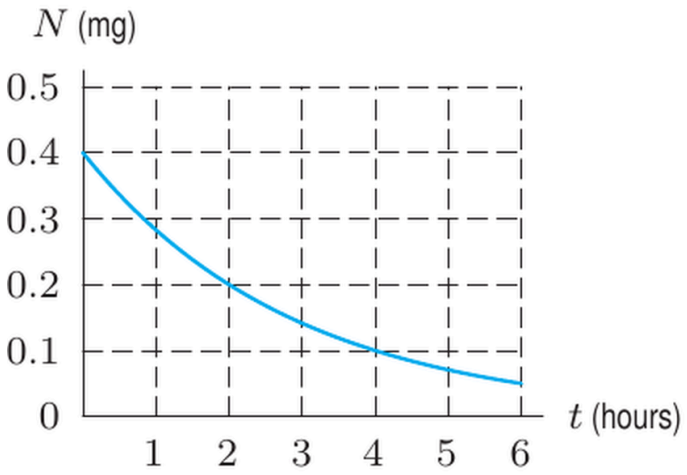
---

iii. Formula: $y = f(x) = 4x - 12$.

Ex 9

Same as Textbook Problem 14

The figure shows the amount of nicotine, N = f(t), in mg, in a person's bloodstream as a function of the time, t in hours, since the person finished smoking a cigarette.



i. Estimate f(3) and interpret it in terms of nicotine.

ii. About how many hours have passed before the nicotine level is down to 0.1 mg?

iii. What is the vertical intercept? What does it represent in terms of nicotine?

iv. If this function had a horizontal intercept, what would it represent?

Textbook Problems

Problem 13 The use of CFCs has declined since the 1987 Montreal Protocol came into force to reduce the use of substances that deplete the ozone layer. World annual CFC consumption, C = f(t), in million tons, is a function of time, t, in years since 1987.

a. Interpret f(10) = 0.2 in terms of CFCs.

---

b. Interpret the vertical intercept of the graph of this function in terms of CFCs.

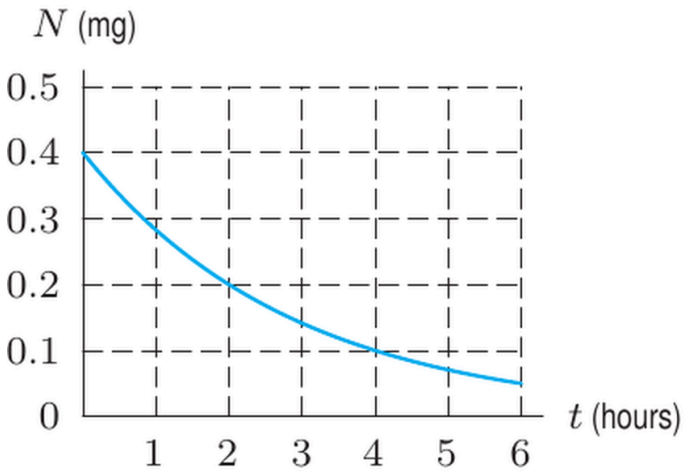
---

c. Interpret the horizontal intercept of the graph of this function in terms of CFCs.

Problem 14

Same as Lecture Example 9

Figure 1.7 shows the amount of nicotine, N = f(t), in mg, in a person's bloodstream as a function of the time, t, in hours, since the person finished smoking a cigarette.



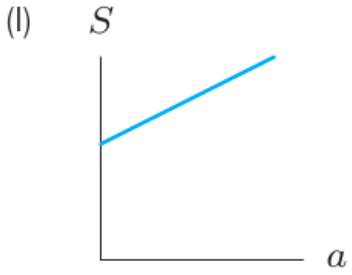
a. Estimate f(3) and interpret it in terms of nicotine.

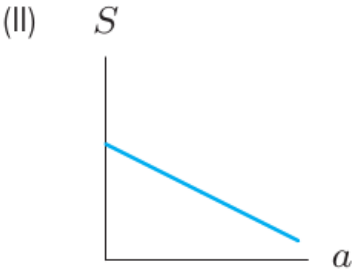
b. About how many hours have passed before the nicotine level is down to 0.1 mg?

c. What is the vertical intercept? What does it represent in terms of nicotine?

d. If this function had a horizontal intercept, what would it represent?

Problem 15 The number of sales per month, S, is a function of the amount, a (in dollars), spent on advertising that month, so S = f(a).



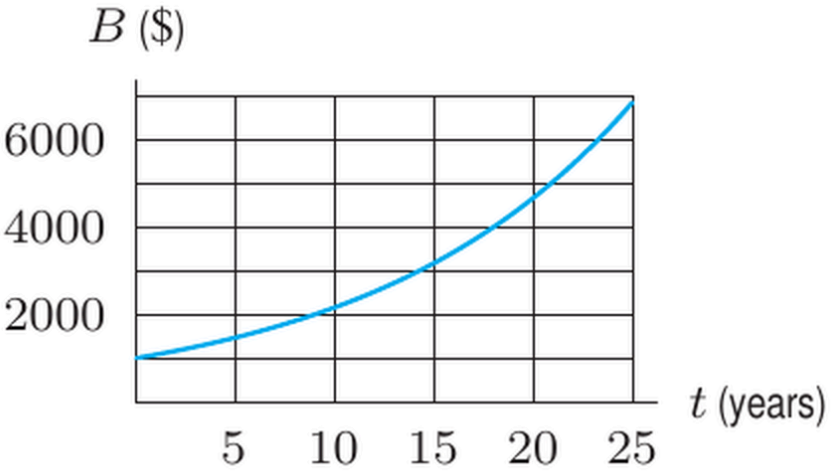


a. Interpret the statement f(1000) = 3500.

b. Which of the graphs in Figure 1.8 is more likely to represent this function?

c. What does the vertical intercept of the graph of this function represent, in terms of sales and advertising?

Problem 16 A deposit is made into an interest-bearing account. Figure 1.9 shows the balance, B, in the account t years later.

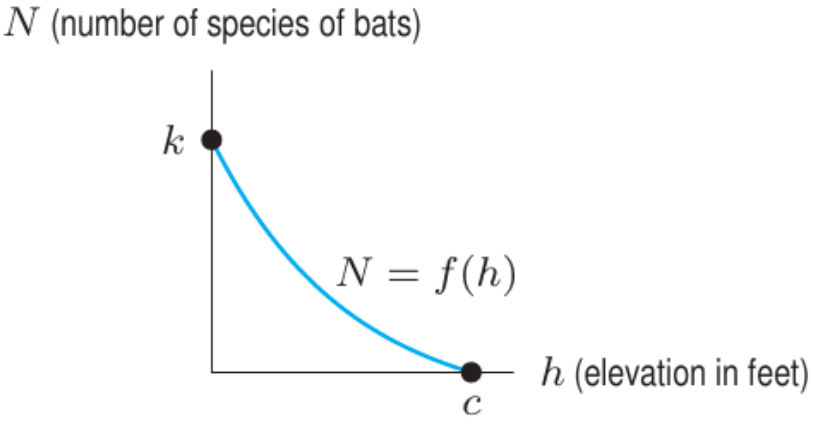


a. What was the original deposit?

b. Estimate f(10) and interpret it.

c. When does the balance reach $5000?

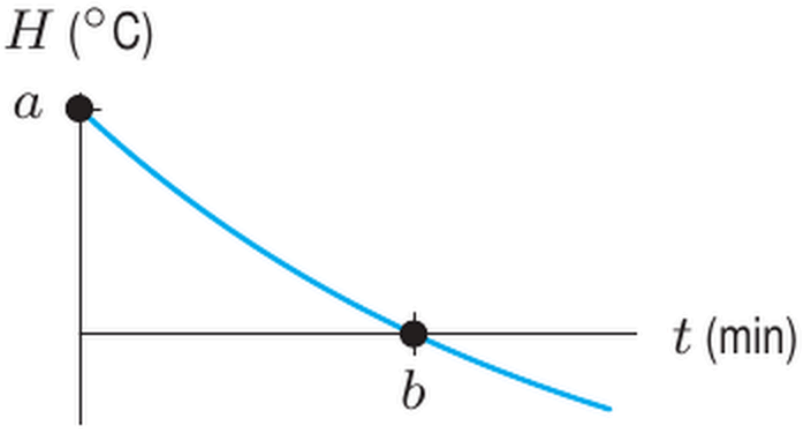
Problem 18 In the Andes mountains in Peru, the number, N, of species of bats is a function of the elevation, h, in feet above sea level, so N = f(h).



a. Interpret the statement f(500) = 100 in terms of bat species.

b. What are the meanings of the vertical intercept, k, and horizontal intercept, c, in Figure 1.10?

Problem 20 An object is put outside on a cold day at time t = 0. Its temperature, H = f(t), in °C, is graphed in Figure 1.11 (t in minutes).



a. What does the statement f(30) = 10 mean in terms of temperature? Include units for 30 and for 10 in your answer.

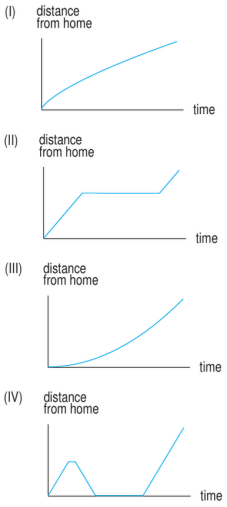
---

b. Explain what the vertical intercept, a, and the horizontal intercept, b, represent in terms of temperature of the object and time outside.

1.1.4 Behavior & Graphs of Functions

Textbook Problems

Problem 1 Which graph in Figure 1.5 best matches each of the following stories? Write a story for the remaining graph.



a. I had just left home when I realized I had forgotten my books, and so I went back to pick them up.

b. Things went fine until I had a flat tire.

c. I started out calmly but sped up when I realized I was going to be late.

Remaining graph Which story fits the remaining graph, (I)?

Problem 5 The population of Washington DC grew from 1900 to 1950, stayed approximately constant during the 1950s, and decreased from about 1960 to 2005.

Graph the population as a function of years since 1900. Which description matches the graph?

Problem 17 When a patient with a rapid heart rate takes a drug, the heart rate plunges dramatically and then slowly rises again as the drug wears off.

Sketch the heart rate against time from the moment the drug is administered. Which description matches the sketch?

Problem 19 After an injection, the concentration of a drug in a patient's body increases rapidly to a peak and then slowly decreases. Graph the concentration of the drug in the body as a function of the time since the injection was given. Assume that the patient has none of the drug in the body before the injection. Label the peak concentration and the time it takes to reach that concentration.

Graph Which description matches the graph?

Peak What does the peak on the graph represent?

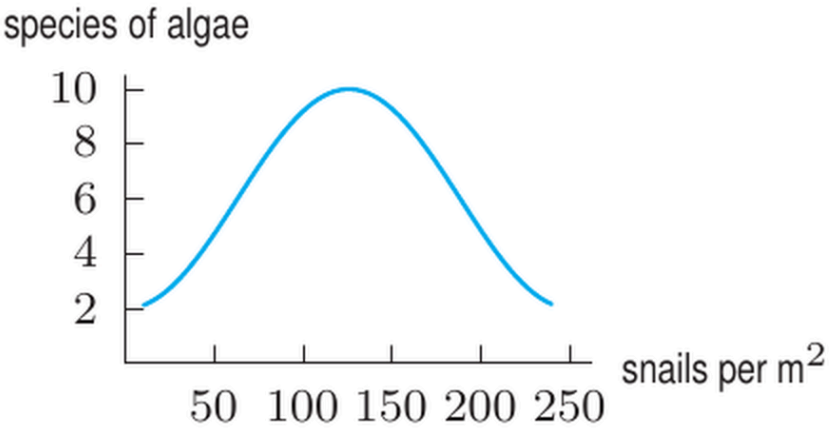
Problem 21 Financial investors know that, in general, the higher the expected rate of return on an investment, the higher the corresponding risk.

a. Graph this relationship, showing expected return as a function of risk. Expected return as a function of risk is:

---

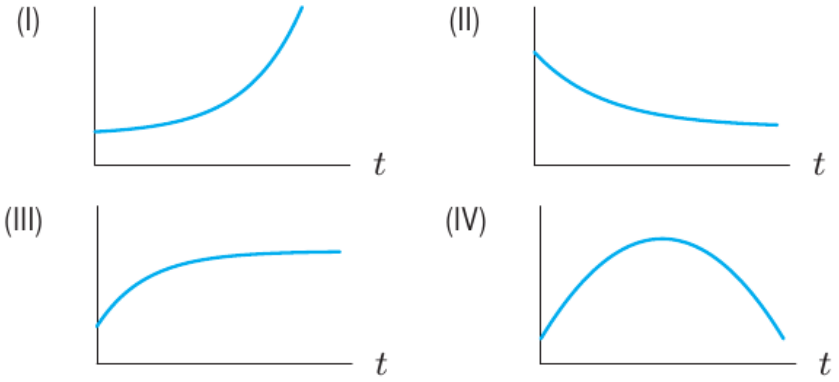
b. On the figure from part (a), mark a point with high expected return and low risk. Where is it?

Problem 22 In tide pools on the New England coast, snails eat algae.



Describe what Figure 1.12 tells you about the effect of snails on the diversity of algae. Does the graph support the statement that diversity peaks at intermediate predation levels?

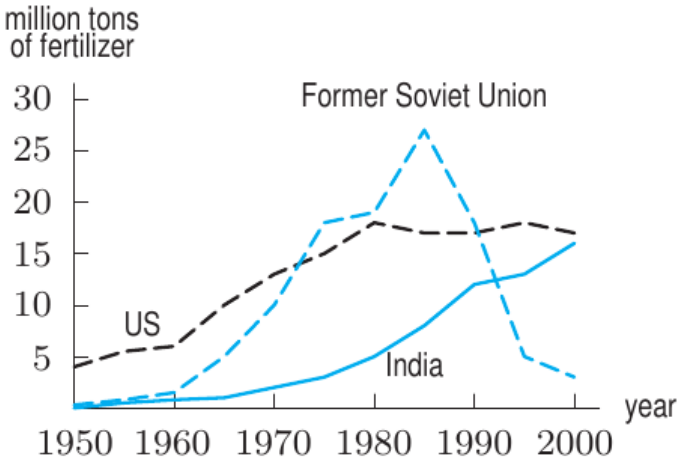
Problem 23 A potato is put in an oven to bake at time t = 0.



(a) Which of the graphs in Figure 1.13 could represent the potato's temperature as a function of time?

(b) What does the vertical intercept represent in terms of the potato's temperature?

Problem 24 Figure 1.14 shows fifty years of fertilizer use in the US, India, and the former Soviet Union.



a. Estimate fertilizer use in 1970 in the US, India, and the former Soviet Union.

b. India Describe how fertilizer use has changed in India over this 50-year period.

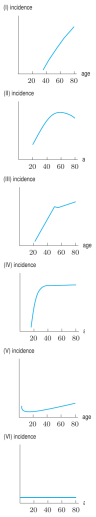
b. US Describe how fertilizer use has changed in the US over this 50-year period.

b. Former Soviet Union Describe how fertilizer use has changed in the former Soviet Union over this 50-year period.

Problem 25 The gas mileage of a car (in miles per gallon) is highest when the car is going about 45 miles per hour and is lower when the car is going faster or slower than 45 mph.

Graph gas mileage as a function of speed of the car. Which description matches the graph?

Problem 26 The six graphs in Figure 1.15 show frequently observed patterns of age-specific cancer incidence rates, in number of cases per 1000 people, as a function of age. The scales on the vertical axes are equal.



a. Which graph shows an incidence rate that rises quickly with age and then stays constant?

---

a. Which graph shows an incidence rate that increases with age and then decreases slightly after about 60–70?

---

a. Which graph shows an incidence rate that does not depend on age?

---

b. Which graph shows a relatively high incidence rate for children?

---

c. Which graph shows a brief decrease in the incidence rate at around age 50?

---

d. Which graph or graphs might represent a cancer that is caused by toxins which build up in the body over time? (For example, lung cancer.)

1.1.5 Test Your Understanding

Test Your Understanding

TYU 1 · P3 The amount of chlorophyll in a plant leaf (in mg) as a function of time (in weeks) is described by

$$A(t) = 10 + \frac{t}{t^2 + 1}, \quad \text{for } t \in [0, \infty).$$

True or False: The initial amount of this drug is 0 mg.

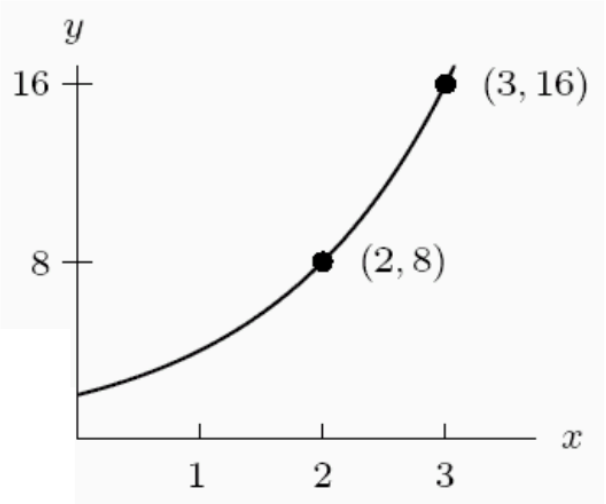
TYU 1 · P4 The number of Drosophila, N(t), is given by N(t) = 500t / (1 + 49t).

True or False: The number of Drosophilas at t = 1 is 10.

TYU 1 · P5 The curve below describes the population size, y, after x years.

The population size …… over time.

A. always decays
B. always grows
C. can both grow and decay
D. grows to a saturation level
E. decays to zero



Answers · Section 1.1

1.1.1 Functions, Domain & Range

Ex 1 **{9, 10, 11, 12}** — f(1) = 1 + 8 = 9, f(2) = 2 + 8 = 10, f(3) = 3 + 8 = 11, f(4) = 4 + 8 = 12. Range = {9, 10, 11, 12}.

Ex 2 **Domain {17, 18, 19, 20, 21, 22, 23, 24, 25, 26}** — The inputs are the dates t: Domain = {17, 18, 19, 20, 21, 22, 23, 24, 25, 26}.

Range {−18, −15, −14, −12, −10, −9, −4, 1, 16} — The outputs are the temperatures. Listed once each (−15 appears twice): Range = {−18, −15, −14, −12, −10, −9, −4, 1, 16}.

Ex 3 **The pattern of repetitions, which the doctor needs, is easier to see on a graph** — Although it is possible to construct a formula to approximate an EKG function, this is seldom done. The pattern of repetitions is what a doctor needs to know, and it is more easily seen from a graph than from a formula.

Problem 7 **13** — f(5) = 2(5) + 3 = 10 + 3 = 13.

Problem 8 **25** — f(5) = 10(5) − 5^2 = 50 − 25 = 25.

Problem 11 **4.1** — In the table, the column x = 5 gives f(5) = 4.1.

Problem 12

a. **2** — y = f(0) = 0^2 + 2 = 2.

b. **11** — f(3) = 3^2 + 2 = 9 + 2 = 11.

c. **x = 3 and x = −3** — x^2 + 2 = 11 → x^2 = 9 → x = ±3.

d. **No, because x^2 would have to equal −1** — x^2 + 2 = 1 → x^2 = −1, which has no real solution. So no value of x gives y = 1 (the smallest value of y is 2).

1.1.2 Function Notation & Interpretation

Ex 6 **i. A 5-year-old car is worth $9,000** — The input 5 is the age (years) and the output 9 is the value (thousands of dollars): when the car is 5 years old, its value is 9 thousand dollars.

ii. Vertical 13.78: a new car is worth $13,780; horizontal a ≈ 17.2: the car is worth $0 after about 17.2 years — Vertical intercept: g(0) = 13.78 − 0.8(0) = 13.78 → a new Honda (age 0) is worth $13,780. Horizontal intercept: 13.78 − 0.8a = 0 → a = 13.78 ÷ 0.8 ≈ 17.2 → the car is worth $0 after about 17.2 years.

Ex 7 **i. In 2000, the CO_2 concentration was 370 ppm** — t = 40 means 40 years after 1960, i.e. the year 2000. So in the year 2000, the concentration of carbon dioxide in the atmosphere was 370 parts per million.

ii. The CO_2 concentration in the year 2010 — f(50) represents the concentration of carbon dioxide in the atmosphere in the year 1960 + 50 = 2010.

Ex 8 **i. 8** — At p = 0 the graph starts at r = 8.

ii. 7 — At p = 3 the graph is at r = 7.

iii. 10 — f(2) is the value of r when p = 2: the highest point of the graph, r = 10.

Problem 2 **In 2005, the population of the city was 12 million** — t = 35 → 1970 + 35 = 2005. In 2005, the population of the city was 12 million.

Problem 3 **In 2002, wheat production was 9 million metric tons** — t = 12 → 1990 + 12 = 2002. In 2002, wheat production in Argentina was 9 million metric tons.

Problem 4 **a. In 2000, the CO_2 concentration was 370 ppm** — t = 40 means 40 years after 1960, i.e. the year 2000. So in the year 2000, the concentration of carbon dioxide in the atmosphere was 370 parts per million.

b. The CO_2 concentration in the year 2010 — f(50) represents the concentration of carbon dioxide in the atmosphere in the year 1960 + 50 = 2010.

Problem 6 **(a) 8** — At p = 0 the graph starts at r = 8.

(a) 7 — At p = 3 the graph is at r = 7.

(b) 10 — f(2) is the value of r when p = 2: the highest point of the graph, r = 10.

Problem 9 **About 3.3** — At x = 5 the curve is going down between its high point (near x = 2) and its low point (near x = 7.5); its height there is about 3.3.

Problem 10 **2** — At x = 5 the curve is at height 2, so f(5) = 2.

1.1.3 Horizontal & Vertical Intercepts

Ex 4 **a. T = 40°F: at 40°F the crickets stop chirping** — 4T − 160 = 0 → 4T = 160 → T = 40. At 40°F the chirp rate is 0: the crickets do not chirp.

b. An increasing straight line through (40, 0) and (100, 240) — C = 4T − 160 is linear with a positive coefficient (4), so it is an increasing straight line. It crosses the T-axis at T = 40 (C = 0), and at T = 100: C = 4(100) − 160 = 240. Plot (40, 0) and (100, 240) and join them.

Ex 5 **i. Horizontal intercepts: Date = 7 and Date = 11; vertical intercept: none** — Horizontal intercepts (output = 0): snowfall is 0 on Date 7 and Date 11 → x = 7 and x = 11. Vertical intercept (input = 0): the table has no Date = 0, so there is no vertical intercept.

ii. Horizontal intercept: x = 3; vertical intercept: y = 1 — The graph crosses the horizontal axis at x = 3 (y = 0) and the vertical axis at y = 1 (x = 0).

iii. Horizontal: x = 3; vertical: y = −12 — Horizontal intercept: set y = 0 → 4x − 12 = 0 → x = 3. Vertical intercept: set x = 0 → y = 4(0) − 12 = −12.

Ex 9

i. **f(3) ≈ 0.15 mg: 3 hours after smoking, about 0.15 mg of nicotine is in the blood** — At t = 3 the curve is at about 0.15 mg (the doctor's answer; the booklet key reads it as 0.14). So f(3) ≈ 0.15 mg: three hours after the person finished smoking, about 0.15 mg of nicotine is in the bloodstream.

ii. **About 4 hours** — The curve reaches N = 0.1 mg at about t = 4 hours.

iii. **0.4 mg: the amount of nicotine when the person finished smoking** — At t = 0 the graph is at N = 0.4 mg: the amount of nicotine in the bloodstream right when the person finished smoking.

iv. **The time when all the nicotine is gone from the blood** — A horizontal intercept is where N = 0: the time when all the nicotine has left the bloodstream.

Problem 13

a. **In 1997, world CFC consumption was 0.2 million tons** — t = 10 → 1987 + 10 = 1997. In 1997, world CFC consumption was 0.2 million tons.

b. **The CFC consumption in 1987** — The vertical intercept is f(0), the value at t = 0, i.e. CFC consumption in 1987.

c. **The year when CFC consumption becomes zero** — The horizontal intercept is where C = 0: the year when world CFC consumption drops to zero.

Problem 14

a. **f(3) ≈ 0.15 mg: 3 hours after smoking, about 0.15 mg of nicotine is in the blood** — At t = 3 the curve is at about 0.15 mg (the doctor's answer; the booklet key reads it as 0.14). So f(3) ≈ 0.15 mg: three hours after the person finished smoking, about 0.15 mg of nicotine is in the bloodstream.

b. **About 4 hours** — The curve reaches N = 0.1 mg at about t = 4 hours.

c. **0.4 mg: the amount of nicotine when the person finished smoking** — At t = 0 the graph is at N = 0.4 mg: the amount of nicotine in the bloodstream right when the person finished smoking.

d. **The time when all the nicotine is gone from the blood** — A horizontal intercept is where N = 0: the time when all the nicotine has left the bloodstream.

Problem 15

a. **Spending $1000 on advertising in a month gives 3500 sales that month** — Input a = 1000 dollars of advertising; output S = 3500 sales. Spending $1000 on advertising in a month results in 3500 sales that month.

b. **(I), because sales increase as advertising increases** — More advertising should give more sales, so the function is increasing: graph (I).

c. **The number of sales in a month with no advertising** — The vertical intercept is f(0): the number of sales in a month when nothing (a = 0) is spent on advertising.

Problem 16

a. **About $1000** — The original deposit is the balance at t = 0: about $1000.

b. **f(10) ≈ $2200: the balance after 10 years** — At t = 10 the curve is at about B = 2200: after 10 years the balance is about $2200.

c. **After about 21 years** — B = 5000 is reached at about t = 21 years.

Problem 18

a. At 500 feet above sea level there are 100 species of bats — Input h = 500 feet, output N = 100 species: at an elevation of 500 feet there are 100 species of bats.

b. k: number of bat species at sea level; c: elevation above which there are no bats — k = f(0): the number of bat species at sea level (h = 0). c is where N = 0: the elevation above which there are no bat species.

Problem 20

a. After 30 minutes outside, the object's temperature is 10°C — Input t = 30 minutes; output H = 10°C. Thirty minutes after being put outside, the object's temperature is 10°C.

b. a: the initial temperature at t = 0; b: the minutes until the temperature reaches 0°C — a = f(0): the temperature of the object when it was put outside (t = 0). b is where H = 0: the number of minutes until the object's temperature reaches 0°C.

1.1.4 Behavior & Graphs of Functions

Problem 1

a. (IV) — Distance from home goes up, then back down to zero (returned home), then up again: graph (IV).

b. (II) — Distance increases, then stays constant (stopped with the flat tire): graph (II).

c. (III) — Distance increases slowly at first, then faster and faster (the curve gets steeper): graph (III).

Remaining graph "I started walking fast, then slowed down." — Graph (I) rises steeply at first and then less steeply: fast at the start, then slower.

Problem 5

Increasing from 0 to 50, constant from 50 to 60, then decreasing to 105 — 1900–1950 → t = 0 to 50 (increasing); the 1950s → t = 50 to 60 (constant); 1960–2005 → t = 60 to 105 (decreasing).

Problem 17

Drops steeply at first, then rises slowly — Heart rate starts high, falls sharply (plunges), then increases gradually as the drug wears off.

Problem 19

Graph At 0, rises steeply to a peak, then decreases slowly — The concentration starts at 0 (no drug before the injection), increases quickly to the peak concentration at time t(peak), then decreases slowly.

Peak The highest concentration and the time t(peak) it takes to reach it — The peak is the highest point of the graph: the peak concentration, reached at time t(peak) after the injection.

Problem 21

a. An increasing function — Higher risk goes with higher expected return, so return increases as risk increases. (An opportunity with high return and low risk would lie above the graph, at the upper left.)

b. Above the curve, at the upper left — Low risk = left side; high return = top. Such a point lies above the curve, at the upper left. (Investors hope to find such opportunities.)

Problem 22

It increases, peaks near 125 snails/m², then decreases — The number of algae species rises to a peak at about 125 snails per m² and then falls, so diversity is highest at intermediate predation levels. Yes, the graph supports the statement.

Problem 23

(a) (III) — The potato's temperature rises quickly at first, then levels off as it approaches the oven temperature: graph (III).

(b) The initial (room) temperature of the potato — The vertical intercept is the temperature at t = 0: the potato's initial (room) temperature when it is put in the oven.

Problem 24

a. US ≈ 13, India ≈ 2, former Soviet Union ≈ 10 — Above 1970: the US curve is at about 13, India at about 2 and the former Soviet Union at about 10 million tons.

b. India It increased steadily — India: increased steadily. US: increased until about 1980, then leveled off. Former Soviet Union: rose to a peak around 1985, then dropped sharply.

b. US It increased until about 1980, then leveled off — US (black dashed curve): increased until about 1980, then leveled off.

b. Former Soviet Union It rose to a peak around 1985, then dropped sharply — Former Soviet Union (blue dashed curve): rose to a peak around 1985, then dropped sharply.

Problem 25

Rises to a maximum at 45 mph, then falls — Mileage increases up to the highest point at 45 mph, then decreases for higher speeds: a curve with a peak at 45.

Problem 26

a. (IV) — Graph (IV): the rate increases quickly, then stays constant.

a. (II) — Graph (II): the rate increases, then decreases slightly after age 60–70.

a. (VI) — Graph (VI) is a horizontal line: the incidence rate is constant at all ages.

b. (V) — Graph (V) starts high at young ages, drops, then increases slowly: a cancer such as leukemia.

c. (III) — Graph (III) increases, dips briefly at about age 50, then increases again: a cancer such as breast cancer.

d. (I) — If toxins build up over time, the risk keeps growing with age: graph (I), which increases with age (for example, lung cancer).

1.1.5 Test Your Understanding

TYU 1 · P3

The initial amount is A(0):

$$A(0) = 10 + \frac{0}{0+1} = 10 \text{ mg}$$

The initial amount is 10 mg, not 0 mg.

Answer: False

TYU 1 · P4

$$N(1) = \frac{500(1)}{1+49(1)} = \frac{500}{50} = 10$$

Answer: True

TYU 1 · P5

The curve is increasing everywhere (from (2, 8) up to (3, 16) and still rising) and does not level off.

Answer: B — always grows

1.2 Linear Functions

In this section

1. Slope of a Straight Line 2 lecture · 4 textbook
2. Slope-Intercept Form & Interpretation 2 lecture · 11 textbook · 1 past exam
3. Point-Slope Form & Finding Equations 3 lecture · 9 textbook · 2 past exam
4. Recognizing Linear Data from Tables 2 lecture · 8 textbook
5. Families of Linear Functions 2 lecture · 2 textbook
6. Test Your Understanding 3 questions

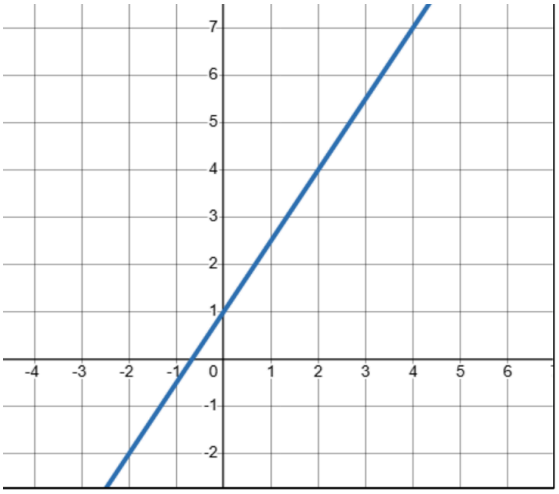
1.2.1 Slope of a Straight Line

Lecture Problems

Ex 1 Find the slope of the line passes through the points (2, 3) and (4, 9).

m = ?

Ex 2 Find the slope of the line given in the figure.



m = ?

Textbook Problems

Problem 1 For Problems 1–4, determine the slope and the y-intercept of the line whose equation is given.

$7y + 12x - 2 = 0$

Problem 2 Determine the slope and the y-intercept of the line:

$3x + 2y = 8$

Problem 3 Determine the slope and the y-intercept of the line:

$12x = 6y + 4$

Problem 4 Determine the slope and the y-intercept of the line:

$-4y + 2x + 8 = 0$

1.2.2 Slope-Intercept Form & Interpretation

Lecture Problems

Ex 3 If y is the world record time to run the mile, in seconds, and t is the number of years since 1900, then records show that, approximately, $y = g(t) = 260 - 0.4t$. Explain the meaning of the intercept, 260, and the slope, −0.4, in terms of the world record time to run the mile and sketch the graph.

Intercept What is the meaning of the intercept, 260?

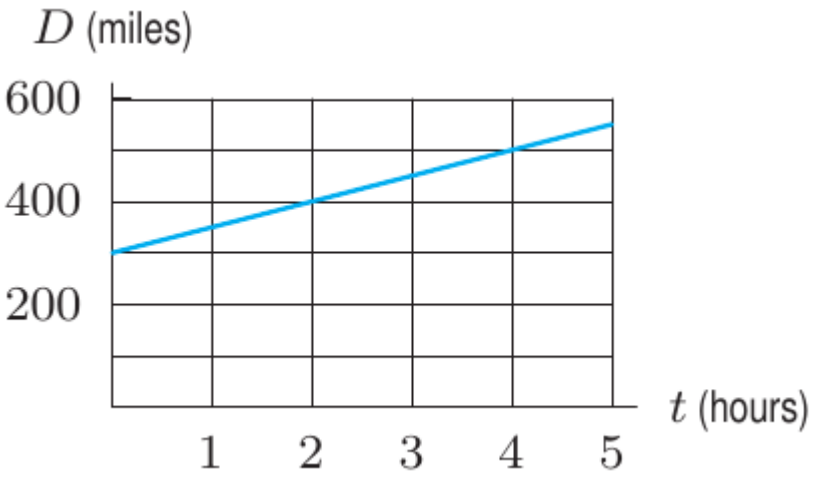
Slope What is the meaning of the slope, −0.4?

Sketch To sketch the graph, which two points lie on it?

Ex 4

Same as Textbook Problem 20

The figure shows the distance from home, in miles, of a person on a 5-hour trip.



i. Estimate the vertical intercept. Give units and interpret it in terms of distance from home.

ii. Estimate the slope of this linear function. Give units and interpret it in terms of distance from home.

iii. Give a formula for distance, D, from home as a function of time, t in hours.

Textbook Problems

Problem 11 A city's population was 30,700 in the year 2000 and is growing by 850 people a year.

a. Give a formula for the city's population, P, as a function of the number of years, t, since 2000.

b. What is the population predicted to be in 2010?

c. When is the population expected to reach 45,000?

Problem 12 A cell phone company charges a monthly fee of $25 plus $0.05 per minute.

Find a formula for the monthly charge, C, in dollars, as a function of the number of minutes, m, the phone is used during the month.

Problem 13 A company rents cars at $40 a day and 15 cents a mile. Its competitor's cars are $50 a day and 10 cents a mile.

a. For each company, give a formula for the cost of renting a car for a day as a function of the distance traveled.

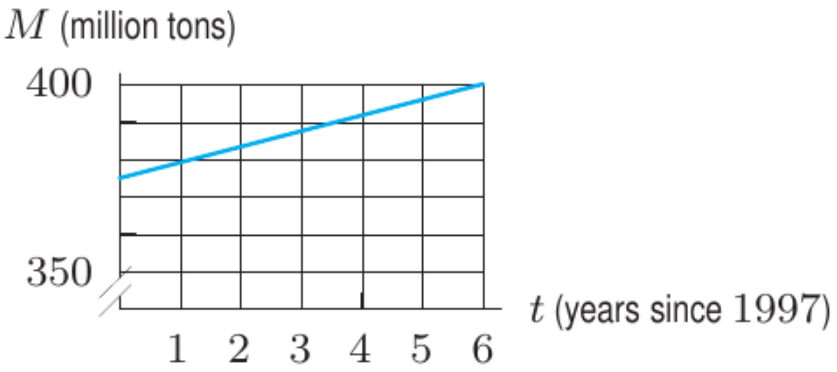
---

b. On the same axes, graph both functions. Where do the two graphs cross?

---

c. How should you decide which company is cheaper?

Problem 14 World milk production rose at an approximately constant rate between 1997 and 2003. See Figure 1.23.



a. Estimate the vertical intercept and interpret it in terms of milk production.

---

b. Estimate the slope and interpret it in terms of milk production.

---

c. Give an approximate formula for milk production, M, as a function of t.

Problem 15 Annual revenue R from McDonald's restaurants worldwide can be estimated by $R = 19.1 + 1.8t$, where R is in billion dollars and t is in years since January 1, 2005.

a. What is the slope of this function? Include units. Interpret the slope in terms of McDonald's revenue.

b. What is the vertical intercept of this function? Include units. Interpret the vertical intercept in terms of McDonald's revenue.

c. What annual revenue does the function predict for 2010?

d. When is annual revenue predicted to hit 30 billion dollars?

Problem 16 Let y be the percent increase in annual US national production during a year when the unemployment rate changes by u percent. (For example, $u = 2$ if unemployment increases from 4% to 6%.) Okun's law states that $y = 3.5 - 2u$.

a. What is the meaning of the number 3.5 in Okun's law?

b. What is the effect on national production of a year when unemployment rises from 5% to 8%?

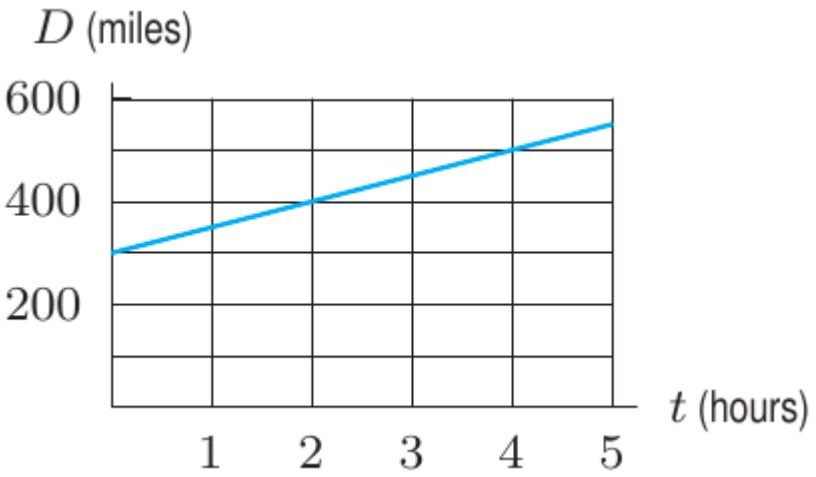
c. What change in the unemployment rate corresponds to a year when production is the same as the year before?

d. What is the meaning of the coefficient -2 in Okun's law?

Problem 20

Same as Lecture Example 4

Figure 1.24 shows the distance from home, in miles, of a person on a 5-hour trip.



a. Estimate the vertical intercept. Give units and interpret it in terms of distance from home.

b. Estimate the slope of this linear function. Give units, and interpret it in terms of distance from home.

c. Give a formula for distance, D, from home as a function of time, t in hours.

Problem 28 Problems 28–33 concern the maximum heart rate (MHR), which is the maximum number of times a person's heart can safely beat in one minute.

Maximum heart rate (MHR), in beats per minute, a = age in years: **Females: MHR = 226 − a**, **Males: MHR = 220 − a**.

Which of the following is the correct statement?

Problem 29

Maximum heart rate (MHR), in beats per minute, a = age in years: **Females: MHR = 226 − a**, **Males: MHR = 220 − a**.

Which of the following is the correct statement for a male and female of the same age?

Problem 30

Maximum heart rate (MHR), in beats per minute, a = age in years: **Females: MHR = 226 − a**, **Males: MHR = 220 − a**.

What can be said about the ages of a male and a female with the same maximum heart rate?

Problem 31

Maximum heart rate (MHR), in beats per minute, a = age in years: **Females: MHR = 226 − a**, **Males: MHR = 220 − a**.

Recently it has been suggested that a more accurate predictor of MHR for both males and females is given by MHR = 208 − 0.7a.

a. At what age do the old and new formulas give the same MHR for females? For males?

b. Which of the following is true?

c. When testing for heart disease, doctors ask patients to walk on a treadmill until their heart rates reach 85 percent of the MHR. For a 65-year-old male, what is the difference in beats per minute between the heart rate reached if the old formula is used and the heart rate reached if the new formula is used?

Past Exam Questions

2024 Q2 A population of cells contains 2400 cells initially.

a. If the population grows by 70 cells per hour, find a formula for the number of cells, N, as a function of the number of hours, t.

b. Predict the number of cells at t = 10 hours.

c. When will the number of cells reach 86400 cells?

1.2.3 Point-Slope Form & Finding Equations

Lecture Problems

Ex 5 Find an equation of the line passes through the point (2, 5) and has slope m = 7.

Equation = ?

Ex 6 Find an equation of the line passes through the points (1, 2) and (4, −7).

Equation = ?

Ex 7 The solid waste generated each year in the cities of the US is increasing. The solid waste generated, in millions of tons, was 238.3 in 2000 and 251.3 in 2006.

i. Assuming that the amount of solid waste (W) generated by US cities is a linear function of time (t), find a formula for this function by finding the equation of the line through these two points.

---

ii. Use this formula to predict the amount of solid waste generated in the year 2020.

Textbook Problems

Problem 5 For Problems 5–8, find an equation for the line that passes through the given points.

(0, 2) and (2, 3)

Problem 6 Find an equation for the line that passes through the given points.

(0, 0) and (1, 1)

Problem 7 Find an equation for the line that passes through the given points.

(−2, 1) and (2, 3)

Problem 8 Find an equation for the line that passes through the given points.

(4, 5) and (2, −1)

Problem 22 World grain production was 1241 million tons in 1975 and 2048 million tons in 2005, and has been increasing at an approximately constant rate.

a. Find a linear function for world grain production, P, in million tons, as a function of t, the number of years since 1975.

b. Using units, interpret the slope in terms of grain production.

c. Using units, interpret the vertical intercept in terms of grain production.

d. According to the linear model, what is the predicted world grain production in 2015?

e. According to the linear model, when is grain production predicted to reach 2500 million tons?

Problem 24 Annual sales of music compact discs (CDs) have declined since 2000. Sales were 942.5 million in 2000 and 384.7 million in 2008.

a. Find a formula for annual sales, S, in millions of music CDs, as a linear function of the number of years, t, since 2000.

b. Give units for and interpret the slope and the vertical intercept of this function.

c. Use the formula to predict music CD sales in 2012.

Problem 25 In a California town, the monthly charge for waste collection is $8 for 32 gallons of waste and $12.32 for 68 gallons of waste.

a. Find a linear formula for the cost, C, of waste collection as a function of the number of gallons of waste, w.

b. What is the slope of the line found in part (a)? Give units and interpret your answer in terms of the cost of waste collection.

c. What is the vertical intercept of the line found in part (a)? Give units and interpret your answer in terms of the cost of waste collection.

Problem 26 The number of species of coastal dune plants in Australia decreases as the latitude, in °S, increases. There are 34 species at 11°S and 26 species at 44°S.

a. Find a formula for the number, N, of species of coastal dune plants in Australia as a linear function of the latitude, l, in °S.

b. Give units for and interpret the slope and the vertical intercept of this function.

c. Graph this function between l = 11°S and l = 44°S. Which points are the two ends of the graph?

Problem 34 An Australian study found that, if other factors are constant (education, experience, etc.), taller people receive higher wages for the same work. The study reported a "height premium" for men of 3% of the hourly wage for a 10 cm increase in height; for women the height premium reported was 2%. We assume that hourly wages are a linear function of height, with slope given by the height premium at the average hourly wage for that gender.

a. The average hourly wage for a 178 cm Australian man is AU$29.40. Express the average hourly wage of an Australian man as a function of his height, x cm.

b. The average hourly wage for a 164 cm Australian woman is AU$24.78. Express the average hourly wage of an Australian woman as a function of her height, y cm.

c. What is the difference in average hourly wages between men and women of height 178 cm?

d. Is there a height for which men and women are predicted to have the same wage? If so, what is it?

Past Exam Questions

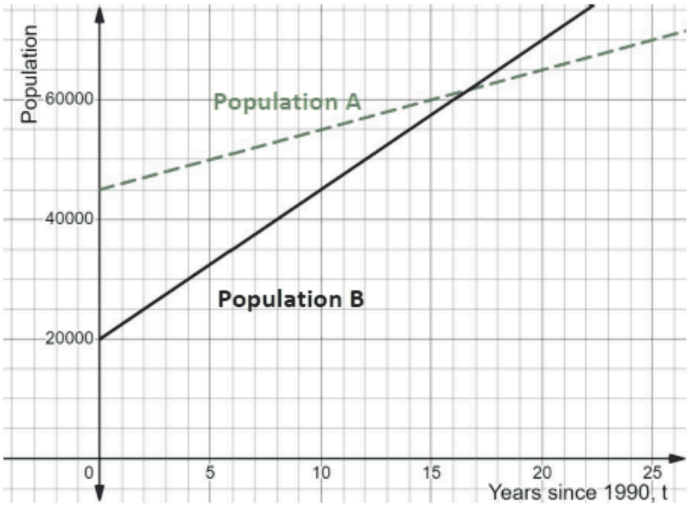
2024 Q3 The concentration of carbon dioxide in the atmosphere, C (in ppm), has been rising steadily with time, t, in years since 1950.

a. If the carbon dioxide concentration was 325 ppm in 1970 and 385 ppm in 2000, find a formula for C as a linear function of t.

b. Provide units and interpretation of the slope of this function.

c. In what year do you expect the concentration of carbon dioxide to reach 500 ppm?

2022 Q3 The figure below shows the size of two animal populations as a function of time.



a. Find a formula for the size of each animal population, P, as a function of the number of years since 1990, t.

b. Which population is growing faster than the other? Explain why.

1.2.4 Recognizing Linear Data from Tables

Lecture Problems

Ex 8 Which of the following tables of values could represent a linear function?

a)

x	0	1	2	3
f(x)	25	30	35	40

b)

x	0	2	4	6
g(x)	10	16	26	40

c)

t	20	30	40	50
h(t)	2.4	2.2	2.0	1.8

Which tables could be linear?

Ex 9 The data in the following table lie on a line. Find formulas for each of the following functions, and give units for the slope in each case:

p (dollars)	5	10	15	20
q (tons)	100	90	80	70

i. q as a function of p.

ii. p as a function of q.

Textbook Problems

Problem 17 Which of the following tables could represent linear functions?

(a)

x	0	1	2	3
y	27	25	23	21

(b)

t	15	20	25	30
s	62	72	82	92

(c)

u	1	2	3	4
w	5	10	18	28

Which tables could be linear?

Problem 18 For each table in Problem 17 that could represent a linear function, find a formula for that function.

(a)

x	0	1	2	3
y	27	25	23	21

(b)

t	15	20	25	30
s	62	72	82	92

(c)

u	1	2	3	4
w	5	10	18	28

Formulas = ?

Problem 19 A company's pricing schedule in Table 1.3 is designed to encourage large orders. (A gross is 12 dozen.) Find a formula for:

q (order size, gross)	3	4	5	6
p (price/dozen)	15	12	9	6

a. q as a linear function of p.

---

b. p as a linear function of q.

Problem 21 The percentage of people, P, below the poverty level in the US is given in Table 1.4.

Year (since 2000)	0	1	2	3
P (percentage)	11.3	11.7	12.1	12.5

a. Find a formula for the percentage in poverty as a linear function of time in years since 2000.

b. Use the formula to predict the percentage in poverty in 2006.

c. What is the difference between the prediction and the actual percentage, 12.3%?

Problem 23 Search and rescue teams work to find lost hikers. Members of the search team separate and walk parallel to one another through the area to be searched. Table 1.5 shows the percent, P, of lost individuals found for various separation distances, d, of the searchers.

Separation distance d (ft)	20	40	60	80	100
Approximate percent found, P	90	80	70	60	50

a. Explain how you know that the percent found, P, could be a linear function of separation distance, d.

b. Find P as a linear function of d.

c. What is the slope of the function? Give units and interpret the answer.

d. What are the vertical and horizontal intercepts of the function? Give units and interpret the answers.

Problem 27 Table 1.6 gives the average weight, w, in pounds, of American men in their sixties for height, h, in inches.

h (inches)	68	69	70	71
w (pounds)	166	171	176	181

h (inches)	72	73	74	75
w (pounds)	186	191	196	201

a. How do you know that the data in this table could represent a linear function?

b. Find weight, w, as a linear function of height, h. What is the slope of the line? What are the units for the slope?

c. Find height, h, as a linear function of weight, w. What is the slope of the line? What are the units for the slope?

Problem 32

Maximum heart rate (MHR), in beats per minute, a = age in years: **Females: MHR = 226 − a**, **Males: MHR = 220 − a**.

Experiments suggest that the female MHR decreases by 12 beats per minute by age 21, and by 19 beats per minute by age 33. Is this consistent with MHR being approximately linear with age?

Problem 33

Maximum heart rate (MHR), in beats per minute, a = age in years: **Females: MHR = 226 − a**, **Males: MHR = 220 − a**.

Experiments suggest that the male MHR decreases by 9 beats per minute by age 21, and by 26 beats per minute by age 33. Is this consistent with MHR being approximately linear with age?

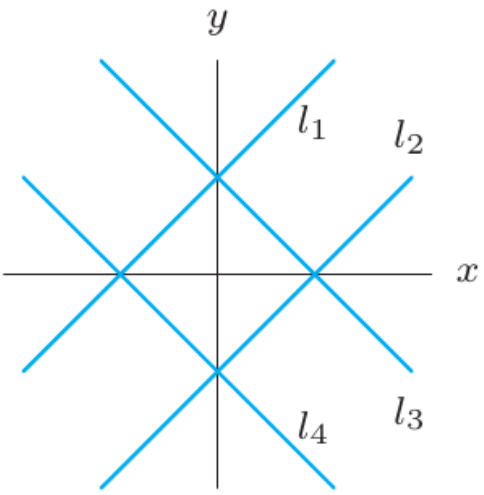
1.2.5 Families of Linear Functions

Lecture Problems

Ex 10

Same as Textbook Problem 9

The figure below shows four lines given by y = b + mx. Match the lines to the conditions on the parameters m and b.



a) m > 0, b > 0

b) m < 0, b > 0

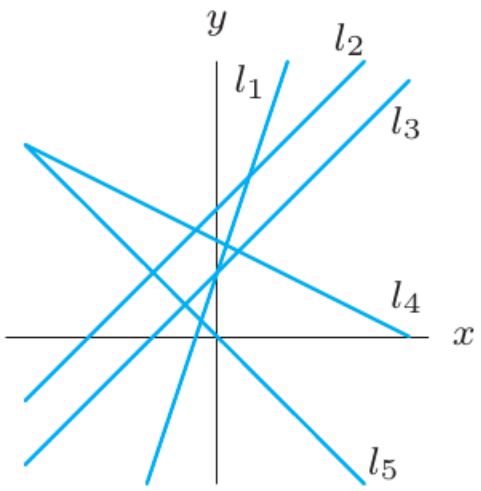
c) m > 0, b < 0

d) m < 0, b < 0

Ex 11

Same as Textbook Problem 10

Use the figure below.



i. Which two lines in the figure below have the same slope?

---

i. Of these two lines, which has the larger y-intercept?

---

ii. Which two lines have the same y-intercept?

---

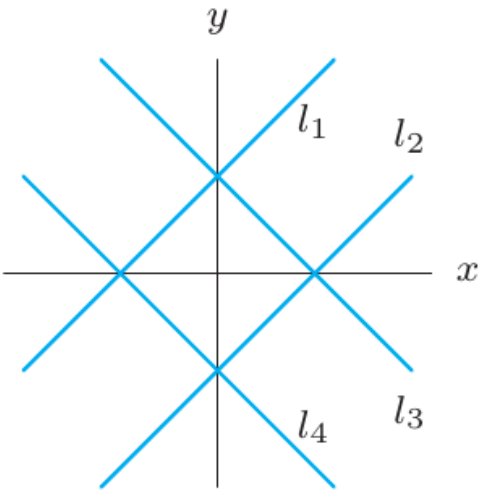
ii. Of these two lines, which has the larger slope?

Textbook Problems

Problem 9

Same as Lecture Example 10

Figure 1.21 shows four lines given by equation y = b + mx. Match the lines to the conditions on the parameters m and b.



(a) m > 0, b > 0

(b) m < 0, b > 0

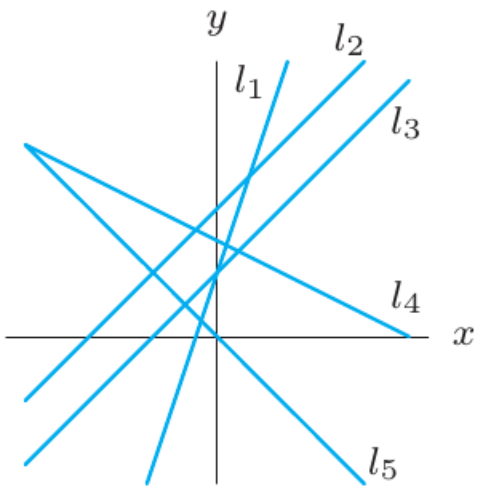
(c) m > 0, b < 0

(d) m < 0, b < 0

Problem 10

Same as Lecture Example 11

Use Figure 1.22.



(a) Which two lines in Figure 1.22 have the same slope?

(a) Of these two lines, which has the larger y-intercept?

(b) Which two lines have the same y-intercept?

(b) Of these two lines, which has the larger slope?

1.2.6 Test Your Understanding

Test Your Understanding

TYU 2 · P3 A population of 100 million pathogens loses 2 million pathogens every 2 hours. Let P be the size of this population at t hours.

a. The mathematical function that best describes this behavior is

A. Hyperbolic
B. Logistic
C. Linear
D. Exponential
E. Power

b. What is the best formula to describe this model?

A. $P = -2t + 100$
B. $P = 100(0.8)^{t/50}$
C. $P = 2t + 100$
D. $P = -t + 100$
E. $P = t + 100$

c. How many pathogens (in millions) are present after 50 hours?

A. 0
B. 80
C. 200
D. 50
E. 150

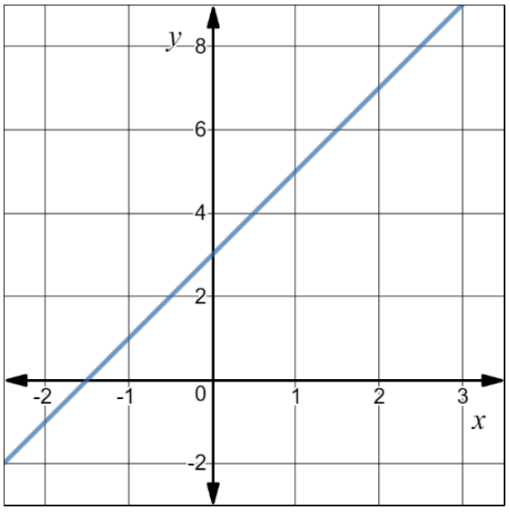
d. This pathogenic population will die out after …… hours.

A. 50
B. −50
C. 100
D. −100
E. 0

TYU 2 · P6 The graph in the figure below describes a function.

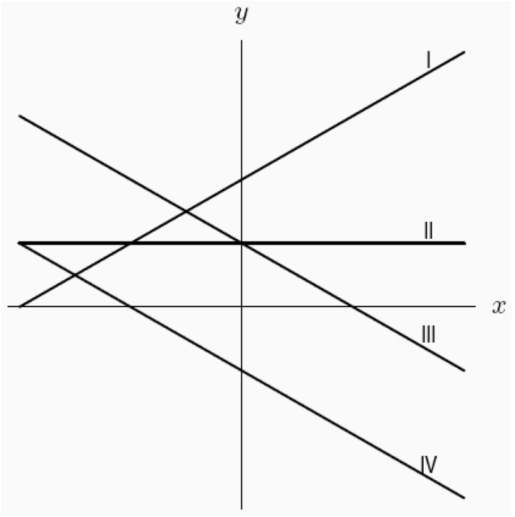
Which of these formulas best describes the graph of the given function?

A. $y = 3x - 2$
B. $y = 2x - 1.5$
C. $y = 2x + 3$
D. $y = -2x + 3$
E. $y = 3x + 2$



TYU 2 · P7 Consider the graph of the following functions. Decide whether each statement is True or False.

1. Both lines III and IV have the smallest rate of change.
2. Line I has the greatest y-intercept.
3. Line I has the smallest x-intercept.



Answers · Section 1.2

1.2.1 Slope of a Straight Line

Ex 1 **m = 3** — $m = (y_2 - y_1)/(x_2 - x_1) = (9 - 3)/(4 - 2) = 6/2 = 3$
Or: $m = (3 - 9)/(2 - 4) = -6/-2 = 3$ (same answer)
Positive slope ⇒ y = f(x) is increasing.

Ex 2 **m = 3/2** — Points from the line: (0, 1) and (2, 4)
$m = (y_2 - y_1)/(x_2 - x_1) = (4 - 1)/(2 - 0) = 3/2$

Problem 1 **m = −12/7, b = 2/7** — $7y = 2 - 12x \rightarrow y = 2/7 - (12/7)x$. So m = −12/7 and b = 2/7.

Problem 2 **m = −3/2, b = 4** — $2y = 8 - 3x \rightarrow y = 4 - (3/2)x$. So m = −3/2 and b = 4.

Problem 3 **m = 2, b = −2/3** — $6y = 12x - 4 \rightarrow y = 2x - 4/6 = 2x - 2/3$. So m = 2 and b = −2/3.

Problem 4 **m = 1/2, b = 2** — $4y = 2x + 8 \rightarrow y = (1/2)x + 2$. So m = 1/2 and b = 2.

1.2.2 Slope-Intercept Form & Interpretation

Ex 3 **Intercept In 1900, the world record time to run the mile was 260 seconds** — i) y = 260 when t = 0
The world record time to run the mile in year 1900 was 260 seconds.
Slope Since 1900, the world record time has been decreasing by 0.4 seconds every year — ii) m = −0.4
Since 1900, the world record time to run the mile is decreasing by 0.4 sec every year.
Sketch (0, 260) and (100, 220) — y = 260 − 0.4t
t = 0 ⇒ y = 260
t = 10 ⇒ y = 260 − 0.4(10) = 256
t = 100 ⇒ y = 260 − 0.4(100) = 220
Draw a decreasing line starting at 260 on the y-axis.

Ex 4 **i. 300 miles: at the beginning of the trip, the person was 300 miles from home** — y-intercept: y = 300 miles when t = 0
It means at the beginning of the person's trip, he was 300 miles away from home.
ii. 50 miles/hour: the distance from home increases by 50 miles each hour — Points (0, 300) and (2, 400)
$\text{slope} = (y_2 - y_1)/(x_2 - x_1) = (400 - 300)/(2 - 0) = 100/2 = 50$ miles/hour
Or: slope = Rise/Run = 200/4 = 50
Since the beginning of the trip up to 5 hours, the distance from home is increasing by 50 miles per hour.
iii. D = 50t + 300 — y = mx + b
m = 50, b = 300
D = 50t + 300

Problem 11

a. P = 30,700 + 850t — Starting value (t = 0) b = 30,700; rate of change m = 850 people per year. P = 30,700 + 850t.

b. 39,200 — 2010 → t = 10. P = 30,700 + 850(10) = 30,700 + 8,500 = 39,200.

c. t ≈ 16.8 years (in about 2017) — 45,000 = 30,700 + 850t → 850t = 14,300 → t = 14,300 / 850 ≈ 16.8 years after 2000, in about 2017.

Problem 12

C = 25 + 0.05m — Fixed fee = 25 (the intercept); cost per minute = 0.05 (the slope). C = 25 + 0.05m.

Problem 13

a. $C_1 = 40 + 0.15d$ and $C_2 = 50 + 0.10d$ — First company: \$40 fixed + \$0.15 per mile → $C_1 = 40 + 0.15d$. Competitor: \$50 fixed + \$0.10 per mile → $C_2 = 50 + 0.10d$.

b. At (200, 70): 200 miles, \$70 — 40 + 0.15d = 50 + 0.10d → 0.05d = 10 → d = 200 miles. Cost: 40 + 0.15(200) = 40 + 30 = \$70. The lines cross at (200, 70); C_1 starts lower but is steeper.

c. Under 200 miles the first company is cheaper; over 200 miles the competitor is cheaper — The graphs cross where 40 + 0.15d = 50 + 0.10d → 0.05d = 10 → d = 200 miles (cost \$70). For less than 200 miles the first company is cheaper; for more than 200 miles the competitor is cheaper.

Problem 14

a. ≈ 375 million tons: the milk production in 1997 — At t = 0 the line is at about 375 million tons, so world milk production in 1997 was about 375 million tons.

b. ≈ 4.2 million tons/year: production increases by about 4.2 million tons per year — From (0, 375) to (6, 400): slope ≈ (400 − 375) / 6 = 25/6 ≈ 4.2 million tons per year.

c. M ≈ 375 + 4.2t — Intercept ≈ 375, slope ≈ 4.2: M ≈ 375 + 4.2t.

Problem 15

a. 1.8 billion dollars/year: revenue increases by 1.8 billion dollars each year — The slope is 1.8 billion dollars per year: McDonald's revenue increases by 1.8 billion dollars every year.

b. 19.1 billion dollars: the revenue in 2005 — At t = 0 (2005), R = 19.1 billion dollars: McDonald's annual revenue in 2005.

c. 28.1 billion dollars — 2010 → t = 5. R = 19.1 + 1.8(5) = 19.1 + 9 = 28.1 billion dollars.

d. t ≈ 6.06 (early 2011) — 30 = 19.1 + 1.8t → 1.8t = 10.9 → t ≈ 6.06 years after 2005, early 2011.

Problem 16

a. Production increases by 3.5% in a year when unemployment does not change — When u = 0 (unemployment does not change), y = 3.5: national production increases by 3.5% that year.

b. Production decreases by 2.5% — u = 8 − 5 = 3. y = 3.5 − 2(3) = 3.5 − 6 = −2.5: production decreases by 2.5%.

c. Unemployment increases by 1.75 percentage points — Same production → y = 0. 0 = 3.5 − 2u → u = 1.75: unemployment rises by 1.75 percentage points.

d. Each 1-point increase in unemployment reduces production growth by 2 points — −2 is the slope: for each 1 percentage point increase in unemployment, the growth in production goes down by 2 percentage points.

Problem 20

a. 300 miles: at the beginning of the trip, the person was 300 miles from home — The y-intercept is D = 300 miles when t = 0: at the beginning of the trip, the person was 300 miles away from home.

b. 50 miles/hour: the distance from home increases by 50 miles each hour — Points (0, 300) and (2, 400): slope = (400 − 300) / (2 − 0) = 100/2 = 50 miles per hour. From the beginning of the trip up to 5 hours, the distance from home is increasing by 50 miles per hour.

c. D = 300 + 50t — D = b + mt with b = 300 and m = 50: D = 50t + 300.

Problem 28

As you age, your maximum heart rate decreases by one beat per minute per year — The slope of MHR = 226 − a (or 220 − a) is −1. Its units are (beats per minute) per year: MHR decreases by one beat per minute for each year of age.

Problem 29

The female's maximum heart rate exceeds the male's — Same age a: female 226 − a, male 220 − a. The female's MHR is always 6 beats per minute higher.

Problem 30

The female is 6 years older — 226 − a(female) = 220 − a(male) → a(female) − a(male) = 6. The female is 6 years older.

Problem 31

a. Females: 60 years; males: 40 years — Females: 226 − a = 208 − 0.7a → 18 = 0.3a → a = 60. Males: 220 − a = 208 − 0.7a → 12 = 0.3a → a = 40.

b. The new formula predicts a lower MHR for young people and a higher MHR for older people — At a = 0 the new formula gives 208, less than 226 or 220 (lower for the young). The new formula decreases more slowly (slope −0.7 instead of −1), so it becomes higher for older people. Statement (ii).

c. ≈ 6.4 beats per minute — Old: 220 − 65 = 155 → 0.85 × 155 = 131.75. New: 208 − 0.7(65) = 162.5 → 0.85 × 162.5 = 138.125. Difference ≈ 6.4 beats per minute.

2024 Q2

a. N = 2400 + 70t — Initial value b = 2400 cells; rate m = 70 cells per hour. N = 2400 + 70t.

b. 3,100 cells — N = 2400 + 70(10) = 2400 + 700 = 3100 cells.

c. t = 1200 hours — 86,400 = 2400 + 70t → 70t = 84,000 → t = 1200 hours.

1.2.3 Point-Slope Form & Finding Equations

Ex 5

7x − y = 9 — $y - y_0 = m(x - x_0)$

$y - 5 = 7(x - 2)$

$y - 5 = 7x - 14$

$7x - y = -5 + 14$

$7x - y = 9$ (or $y - 7x = -9$)

Ex 6 **$y + 3x = 5$** — 1) Find slope: $m = (y_2 - y_1)/(x_2 - x_1) = (-7 - 2)/(4 - 1) = -9/3 = -3$
2) Eq. of the line with $m = -3$, P(1, 2) (or you choose P(4, −7))
$y - y_1 = m(x - x_1)$
$y - 2 = -3(x - 1)$
$y - 2 = -3x + 3$
$y + 3x = 3 + 2$
$y + 3x = 5$ (or $y = -3x + 5$)

Ex 7 **i. $W = 2.167t - 4095.7$** — (t, W) ⇒ (2000, 238.3) and (2006, 251.3)
$m = (W_2 - W_1)/(t_2 - t_1) = (251.3 - 238.3)/(2006 - 2000) = 2.167$ millions of tons/year
$W - W_1 = m(t - t_1)$
$W - 238.3 = 2.167(t - 2000)$
$W - 238.3 = 2.167t - 4334$
$W = 2.167t - 4095.7$
ii. 281.64 millions of tons — t = 2020
$W = 2.167(2020) - 4095.7 = 281.64$ millions of tons

Problem 5 **$y = 2 + 0.5x$** — $m = (3 - 2) / (2 - 0) = 1/2$. The line passes through (0, 2), so b = 2: $y = 2 + 0.5x$.

Problem 6 **$y = x$** — $m = (1 - 0) / (1 - 0) = 1$ and b = 0: $y = x$.

Problem 7 **$y = 0.5x + 2$** — $m = (3 - 1) / (2 - (-2)) = 2/4 = 0.5$. $y - 1 = 0.5(x + 2) \rightarrow y = 0.5x + 2$.

Problem 8 **$y = 3x - 7$** — $m = (-1 - 5) / (2 - 4) = -6 / -2 = 3$. $y - 5 = 3(x - 4) \rightarrow y = 3x - 7$.

Problem 22 **a. $P = 1241 + 26.9t$** — Points (0, 1241) and (30, 2048): $m = (2048 - 1241) / 30 = 807/30 = 26.9$. $P = 1241 + 26.9t$.

b. Production increases by 26.9 million tons per year — Slope = 26.9 million tons per year: world grain production increases by 26.9 million tons every year.

c. Production in 1975 was 1241 million tons — At t = 0 (1975), P = 1241: world grain production in 1975 was 1241 million tons.

d. 2317 million tons — 2015 → t = 40. $P = 1241 + 26.9(40) = 1241 + 1076 = 2317$ million tons.

e. $t \approx 46.8$, in 2022 — $2500 = 1241 + 26.9t \rightarrow 26.9t = 1259 \rightarrow t \approx 46.8$ years after 1975, in 2022.

Problem 24 **a. $S = 942.5 - 69.725t$** — Points (0, 942.5) and (8, 384.7): $m = (384.7 - 942.5) / 8 = -557.8 / 8 = -69.725$. $S = 942.5 - 69.725t$.

b. Sales decrease by 69.725 million CDs per year; 942.5 million CDs were sold in 2000 — Slope = −69.725 million CDs per year: sales decrease by 69.725 million CDs each year. Intercept = 942.5 million CDs: sales in 2000.

c. ≈ 105.8 million CDs — 2012 → t = 12. $S = 942.5 - 69.725(12) = 942.5 - 836.7 = 105.8$ million CDs.

Problem 25

a. C = 4.16 + 0.12w — m = (12.32 − 8) / (68 − 32) = 4.32 / 36 = 0.12. C − 8 = 0.12(w − 32) → C = 0.12w − 3.84 + 8 → C = 4.16 + 0.12w.

b. $0.12 per gallon: each extra gallon of waste costs 12 cents — Slope = 0.12 dollars per gallon: each additional gallon of waste costs 12 cents more.

c. $4.16: the fixed charge, even with no waste — Intercept = $4.16 (when w = 0): the fixed charge for waste collection.

Problem 26

a. N ≈ 36.7 − 0.24l — m = (26 − 34) / (44 − 11) = −8/33 ≈ −0.24. N − 34 = −0.24(l − 11) → N ≈ 36.7 − 0.24l.

b. About 0.24 fewer species per degree south; 36.7 species at 0° (outside Australia, no real meaning) — Slope ≈ −0.24 species per degree: about 0.24 fewer species for each degree further south. Intercept 36.7 species at latitude 0°, which is outside Australia, so it has no real meaning here.

c. (11, 34) and (44, 26) — At l = 11, N = 34 and at l = 44, N = 26. The graph is a decreasing line segment from (11, 34) to (44, 26).

Problem 34

a. W = 13.70 + 0.0882x — Slope = 3% of 29.40 per 10 cm = 0.882 / 10 = 0.0882 dollars per cm. W − 29.40 = 0.0882(x − 178) → W = 29.40 − 15.70 + 0.0882x = 13.70 + 0.0882x.

b. W = 16.65 + 0.04956y — Slope = 2% of 24.78 per 10 cm = 0.4956 / 10 = 0.04956 dollars per cm. W − 24.78 = 0.04956(y − 164) → W = 24.78 − 8.13 + 0.04956y ≈ 16.65 + 0.04956y.

c. ≈ AU$3.93 per hour — Men at 178 cm: 29.40. Women at 178 cm: 16.65 + 0.04956(178) ≈ 25.47. Difference ≈ 29.40 − 25.47 ≈ AU$3.93 per hour.

d. Yes, h ≈ 76.4 cm — 13.70 + 0.0882h = 16.65 + 0.04956h → 0.03864h ≈ 2.95 → h ≈ 76.4 cm. This height (76.4 cm) is not realistic for adults.

2024 Q3

a. C = 285 + 2t — 1970 → t = 20; 2000 → t = 50. m = (385 − 325) / (50 − 20) = 60/30 = 2. C − 325 = 2(t − 20) → C = 2t + 285.

b. 2 ppm/year: CO_2 concentration increases by 2 ppm each year — Slope = 2 ppm per year: the CO_2 concentration increases by 2 ppm every year.

c. 2057 — 500 = 285 + 2t → 2t = 215 → t = 107.5. Year = 1950 + 107.5 = 2057.5, so in 2057.

2022 Q3

a. A: P = 45,000 + 1000t; B: P = 20,000 + 2500t — A: starts at 45,000 and reaches 60,000 at t = 15 → m = 15,000/15 = 1000 → P = 45,000 + 1000t. B: starts at 20,000 and reaches 60,000 at t = 16 → m = 40,000/16 = 2500 → P = 20,000 + 2500t.

b. B, because it has the larger slope — B grows by 2500 animals per year and A by 1000 per year. B has the larger slope, so it is growing faster.

1.2.4 Recognizing Linear Data from Tables

Ex 8 **a) and c) are linear; b) is not** — a) x: +1, +1, +1 and f(x): +5, +5, +5
The differences in y-values are constant for equal differences in x-values ⇒ it represents a linear function.
$m_1 = (30 - 25)/(1 - 0) = 5$, $m_2 = (35 - 30)/(2 - 1) = 5$, $m_3 = (40 - 35)/(3 - 2) = 5$
Constant slope ⇒ linear function.

b) $y_2 - y_1 = 6 \neq y_3 - y_2 = 10$, but changes in x-values are constant ⇒ not linear function.
$m_1 = (16 - 10)/(2 - 0) = 6/2 = 3$, $m_2 = (26 - 16)/(4 - 2) = 10/2 = 5$
$m_1 \neq m_2$ ⇒ not linear function.

c) Constant change in y-values for constant change in x-values ⇒ linear function.
$m_1 = (2.2 - 2.4)/(30 - 20) = -0.2/10 = -0.02$
$m_2 = (2 - 2.2)/(40 - 30) = -0.2/10 = -0.02$
$m_3 = (1.8 - 2)/(50 - 40) = -0.2/10 = -0.02$
Constant slope ⇒ linear function.

Ex 9 **i. $q = -2p + 110$; slope −2 tons/dollars** — (p, q) ⇒ (5, 100), (10, 90)
$m = (90 - 100)/(10 - 5) = -10/5 = -2$ tons/dollars
$q - q_1 = m(p - p_1)$ ⇒ $q - 100 = -2(p - 5)$
$q - 100 = -2p + 10$
$q = -2p + 110$ ($q = 110 - 2p$)

ii. $p = -0.5q + 55$; slope −0.5 dollars/tons — (q, p) ⇒ (100, 5), (90, 10)
$m = (p_2 - p_1)/(q_2 - q_1) = (10 - 5)/(90 - 100) = 5/(-10) = -0.5$ dollars/tons
$p - p_1 = m(q - q_1)$ ⇒ $p - 5 = -0.5(q - 100)$
$p - 5 = -0.5q + 50$
$p = -0.5q + 55$ ($p = 55 - 0.5q$)

Problem 17 **(a) and (b) only** — (a) y decreases by 2 each time → linear. (b) s increases by 10 for each +5 in t → linear. (c) w changes by 5, 8, 10 → not linear.

Problem 18 **$y = 27 - 2x$ and $s = 32 + 2t$** — (a) $m = -2$, $b = 27$ (at $x = 0$): $y = 27 - 2x$. (b) $m = 10/5 = 2$; $s - 62 = 2(t - 15)$ → $s = 2t + 32$.

Problem 19 **a. $q = 8 - p/3$** — Points (p, q) = (15, 3) and (12, 4): $m = (4 - 3) / (12 - 15) = -1/3$. $q - 3 = -(1/3)(p - 15)$ → $q = -p/3 + 5 + 3 = 8 - p/3$.

b. $p = 24 - 3q$ — p decreases by 3 for each +1 in q: $m = -3$. $p - 15 = -3(q - 3)$ → $p = -3q + 9 + 15 = 24 - 3q$.

Problem 21 **a. $P = 11.3 + 0.4t$** — P increases by 0.4 each year (constant) and P = 11.3 at t = 0: $P = 11.3 + 0.4t$.

b. 13.7% — 2006 → $t = 6$. $P = 11.3 + 0.4(6) = 11.3 + 2.4 = 13.7\%$.

c. 1.4 percentage points — $13.7 - 12.3 = 1.4$ percentage points (the model overestimates).

Problem 23

a. For every 20 ft increase in d, P decreases by 10 (constant difference) — For every 20 ft increase in d, P decreases by 10: the differences are constant, so P could be linear in d.

b. P = 100 − 0.5d — m = −10/20 = −0.5. P − 90 = −0.5(d − 20) → P = −0.5d + 10 + 90 = 100 − 0.5d.

c. −0.5 percent per foot: each extra foot of separation lowers the percent found by 0.5 — Slope = −0.5 percent per foot: each extra foot of separation lowers the percent of lost people found by 0.5.

d. Vertical: 100% (searchers side by side); horizontal: d = 200 ft (nobody is found) — d = 0 → P = 100%: when the searchers walk side by side, everyone is found. P = 0 → 0.5d = 100 → d = 200 ft: at 200 ft apart, nobody is found.

Problem 27

a. w increases by 5 for every 1 inch increase in h — For every 1 inch increase in h, w increases by 5 pounds: constant differences, so the data could be linear.

b. w = 5h − 174; slope 5 pounds per inch — m = 5 lb/inch. w − 166 = 5(h − 68) → w = 5h − 340 + 166 = 5h − 174.

c. h = 0.2w + 34.8; slope 0.2 inches per pound — m = 1/5 = 0.2 inch/lb. h − 68 = 0.2(w − 166) → h = 0.2w − 33.2 + 68 = 0.2w + 34.8.

Problem 32

Yes, the rates are almost equal (12/21 ≈ 0.57 and 19/33 ≈ 0.58) — 12/21 ≈ 0.57 and 19/33 ≈ 0.58 beats per minute per year. The rates are almost the same, so it is consistent with an approximately linear function.

Problem 33

No, the rates are very different (9/21 ≈ 0.43 and 26/33 ≈ 0.79) — 9/21 ≈ 0.43 and 26/33 ≈ 0.79. The rates are very different, so it is not consistent with MHR being approximately linear.

1.2.5 Families of Linear Functions

Ex 10

a) l_1 — m > 0, b > 0 → l_1 (b is on the y-axis)

b) l_3 — m < 0, b > 0 → l_3

c) l_2 — m > 0, b < 0 → l_2

d) l_4 — m < 0, b < 0 → l_4

Ex 11

i. l_2 and l_3 — Same slope = parallel lines.

l_2 and l_3 have same slopes.

i. l_2 — Larger y-intercept = b value is larger.

l_2 has larger y-intercept.

ii. l_1 and l_3 — Same y-intercept = they intersect the y-axis at the same point.

l_1 and l_3 have same y-intercept.

ii. l_1 — Larger slope = line moving up faster.

l_1 has larger slope.

Problem 9

(a) $\mathbf{l_1}$ — m > 0: the line rises; b > 0: it crosses the y-axis above the origin → l_1.

(b) $\mathbf{l_3}$ — m < 0: the line falls; b > 0: it crosses the y-axis above the origin → l_3.

(c) $\mathbf{l_2}$ — m > 0: the line rises; b < 0: it crosses the y-axis below the origin → l_2.

(d) $\mathbf{l_4}$ — m < 0: the line falls; b < 0: it crosses the y-axis below the origin → l_4.

Problem 10

(a) $\mathbf{l_2}$ **and** $\mathbf{l_3}$ — l_2 and l_3 are parallel, so they have the same slope.

(a) $\mathbf{l_2}$ — l_2 crosses the y-axis higher than l_3, so l_2 has the larger y-intercept (larger b value).

(b) $\mathbf{l_1}$ **and** $\mathbf{l_3}$ — l_1 and l_3 intersect the y-axis at the same point, so they have the same y-intercept.

(b) $\mathbf{l_1}$ — l_1 is steeper (moving up faster), so l_1 has the larger slope.

1.2.6 Test Your Understanding

TYU 2 · P3

(a) The population loses a fixed amount (2 million) every 2 hours → constant rate of change.

Answer: C — Linear

(b) Slope = −2 million ÷ 2 hours = −1 million/hour; initial value = 100 million.

$$P = -t + 100$$

Answer: D

(c)

$$P(50) = -50 + 100 = 50 \text{ million pathogens}$$

Answer: D — 50

(d) Dies out when P = 0:

$$-t + 100 = 0 \Rightarrow t = 100 \text{ hours}$$

Answer: C — 100

TYU 2 · P6

From the graph: vertical intercept b = 3, and the line passes through (0, 3) and (1, 5).

$$m = \frac{5 - 3}{1 - 0} = 2$$

$$y = 2x + 3$$

Answer: C

TYU 2 · P7

1. True — III and IV are parallel with the same negative slope, which is smaller than the zero slope of II and the positive slope of I.

2. True — line I crosses the y-axis highest.

3. True — line I crosses the x-axis farthest to the left.

1.3 Average Rate of Change and Relative Change

In this section

1. Average Rate of Change from Tables & Data 2 lecture · 14 textbook
2. Average Rate of Change from Graphs & Formulas 2 lecture · 7 textbook
3. Relative Change & Relative Rate of Change 2 lecture · 13 textbook · 2 past exam
4. Increasing, Decreasing & Concavity 3 lecture · 11 textbook
5. Average Velocity 1 lecture · 5 textbook
6. Test Your Understanding 6 questions

1.3.1 Average Rate of Change from Tables & Data

Lecture Problems

Ex 1 The table shows the height of the winning pole vault at the Olympics during the 1960s and 1990s. Winning height in men's Olympic pole vault (approximate):

Year	1960	1964	1968	…	1992	1996	2000
Height (inches)	185	201	213	…	228	233	232

i. Find the rate of change of the winning height between 1960 and 1968, and between 1992 and 2000.

ii. In which of these two periods did the height increase faster?

Ex 2

Similar problem (different numbers): Textbook Problem 24

The table below shows the production of tobacco in the US (millions of pounds).

Year	2003	2004	2005	2006
Production	802	882	645	728

Year	2007	2008	2009	2010
Production	788	801	823	719

i. What is the average rate of change (ARC) in tobacco production between 2003 and 2010? Give units and interpret your answer in terms of tobacco production.

---

ii. During this seven-year period, is there any interval during which the average rate of change was positive? If so, when?

Textbook Problems

Problem 15 Table 1.11 shows world bicycle production, in millions.

Year	1950	1960	1970	1980	1990	2000
Bicycles	11	20	36	62	92	101

a. Find the change in bicycle production between 1950 and 2000. Give units.

---

b. Find the average rate of change in bicycle production between 1950 and 2000. Give units and interpret your answer in terms of bicycle production.

Problem 16 Table 1.12 gives the net sales of The Gap, Inc (millions of dollars).

Year	2003	2004	2005
Sales	15,854	16,267	16,019

Year	2006	2007	2008
Sales	15,923	15,763	14,526

a. Find the change in net sales between 2005 and 2008.

b. Find the average rate of change in net sales between 2005 and 2008. Give units and interpret your answer.

c. From 2003 to 2008, were there any one-year intervals during which the average rate of change was positive? If so, when?

Problem 17 Table 1.13 shows attendance at NFL football games, in millions of fans.

Year	2003	2004	2005	2006	2007
Attendance	21.64	21.71	21.79	22.20	22.26

a. Find the average rate of change in the attendance from 2003 to 2007. Give units.

b. Find the annual increase in the attendance for each year from 2003 to 2007. (Your answer should be four numbers.)

c. Is the average rate of change found in part (a) the average of the four yearly changes found in part (b)?

Problem 19 Table 1.14 gives sales of Pepsico (millions of dollars).

Year	2003	2004	2005
Sales	26,971	29,261	32,562

Year	2006	2007	2008
Sales	35,137	39,474	45,251

a. Find the change in sales between 2003 and 2008.

b. Find the average rate of change in sales between 2003 and 2008. Give units and interpret your answer.

Problem 20 Table 1.15 shows world population, P (billions of people), world passenger automobile production, A (millions of cars), and world cell phone subscribers, C (millions of subscribers).

Year	1995	2000	2005
P (billions)	5.68	6.07	6.45
A (millions)	36.1	41.3	45.9
C (millions)	91	740	2168

a. P Find the average rate of change of P between 1995 and 2005, with units.

a. A Find the average rate of change of A between 1995 and 2005, with units.

a. C Find the average rate of change of C between 1995 and 2005, with units.

b. (i) Between 1995 and 2005, which increased faster: population or the number of automobiles?

b. (ii) Between 1995 and 2005, which increased faster: population or the number of cell phone subscribers?

Problem 24

Similar problem (different numbers): Lecture Example 2

Table 1.16 shows the production of tobacco in the US (millions of pounds).

Year	1996	1997	1998	1999
Production	1517	1787	1480	1293

Year	2000	2001	2002	2003
Production	1053	991	879	831

a. What is the average rate of change in tobacco production between 1996 and 2003? Give units and interpret your answer in terms of tobacco production.

b. During this seven-year period, is there any interval during which the average rate of change was positive? If so, when?

Problem 25 Do you expect the average rate of change (in units per year) of each of the following to be positive or negative? Explain your reasoning.

a. Number of acres of rain forest in the world.

b. Population of the world.

c. Number of polio cases each year in the US, since 1950.

d. Height of a sand dune that is being eroded.

e. Cost of living in the US.

Problem 27 Table 1.17 shows the total US labor force, L, in thousands of workers. Find the average rate of change between 1940 and 2000; between 1940 and 1960; between 1980 and 2000. Give units and interpret your answers in terms of the labor force.

Year	1940	1960	1980	2000
L	47,520	65,778	99,303	136,891

1940–2000 ARC between 1940 and 2000?

1940–1960 ARC between 1940 and 1960?

1980–2000 ARC between 1980 and 2000?

Problem 28 The total world marine catch of fish, in metric tons, was 17 million in 1950 and 99 million in 2001.

What was the average rate of change in the marine catch during this period? Give units and interpret your answer.

Problem 29 Table 1.18 gives the revenues, R, of General Motors (billions of dollars).

Year	2003	2004	2005
R	184.0	192.9	193.1

Year	2006	2007	2008
R	205.6	181.1	149.0

a. Find the change in revenues between 2003 and 2008.

b. Find the average rate of change in revenues between 2003 and 2008. Give units and interpret your answer.

c. From 2003 to 2008, were there any one-year intervals during which the average rate of change was negative? If so, which?

Problem 30 The number of US households with cable television was 12,168,450 in 1977 and 73,365,880 in 2003.

Estimate the average rate of change in the number of US households with cable television during this 26-year period. Give units and interpret your answer.

Problem 32 Table 1.19 shows the concentration, c, of creatinine in the bloodstream of a dog.

t (minutes)	2	4	6	8	10
c (mg/ml)	0.439	0.383	0.336	0.298	0.266

a. (i) Including units, find the average rate at which the concentration is changing between the 6th and 8th minutes.

a. (ii) Including units, find the average rate between the 8th and 10th minutes.

b. Explain the sign and relative magnitudes of your results in terms of creatinine.

Problem 33 The population of the world reached 1 billion in 1804, 2 billion in 1927, 3 billion in 1960, 4 billion in 1974, 5 billion in 1987 and 6 billion in 1999. Find the average rate of change of the population of the world, in people per minute, during each of these intervals. (1 year = 525,600 minutes.)

1804–1927 People per minute?

1927–1960 People per minute?

1960–1974 People per minute?

1974–1987 People per minute?

1987–1999 People per minute?

Problem 36 Table 1.20 gives the sales, S, of Intel Corporation (millions of dollars).

Year	2003	2004	2005
S	30,100	34,200	38,800

Year	2006	2007	2008
S	35,400	38,300	37,600

a. Find the change in sales between 2003 and 2008.

b. Find the average rate of change in sales between 2003 and 2008. Give units and interpret your answer.

1.3.2 Average Rate of Change from Graphs & Formulas

Lecture Problems

Ex 3 Use the given figure to estimate the average rate of change (ARC) of the number of farms in the US between 1950 and 1970.

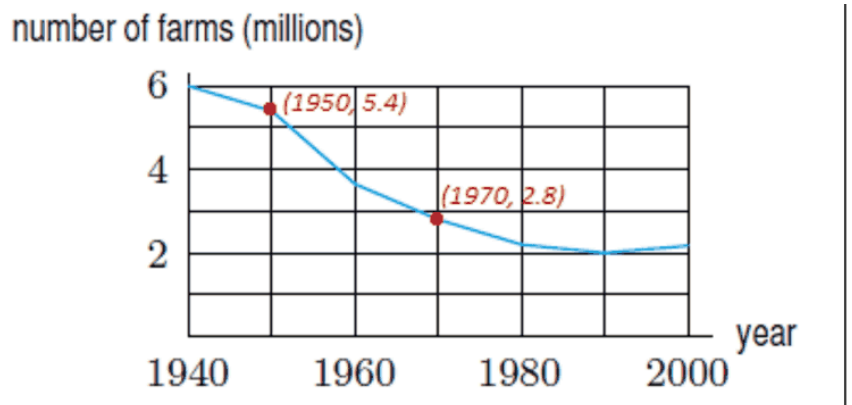


Figure : Number of farms in the US (in millions)

ARC = ?

Ex 4 $f(x) = \sqrt{x}$

(a) Find the average rate of change (ARC) of $y = f(x) = \sqrt{x}$ between $x = 1$ and $x = 4$.

(b) Graph $f(x)$ and represent this average rate of change as the slope of a line. Which line is it?

Textbook Problems

Problem 12 $f(x) = 2x^2$

Find the average rate of change of $f(x) = 2x^2$ between $x = 1$ and $x = 3$.

Problem 13 $f(x) = 3x^2 + 4$

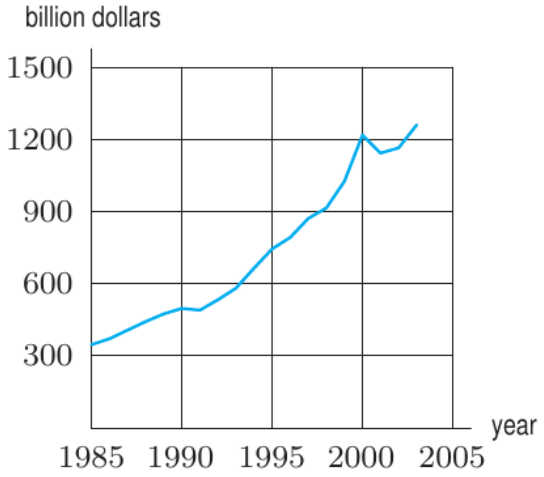
ARC Find the average rate of change of $f(x) = 3x^2 + 4$ between $x = -2$ and $x = 1$.

Graph Illustrate your answer graphically. The ARC is the slope of which line?

Problem 14 When a deposit of \$1000 is made into an account paying 8% interest, compounded annually, the balance, \$B, in the account after t years is given by $B = 1000(1.08)^t$.

Find the average rate of change in the balance over the interval $t = 0$ to $t = 5$. Give units and interpret your answer in terms of the balance in the account.

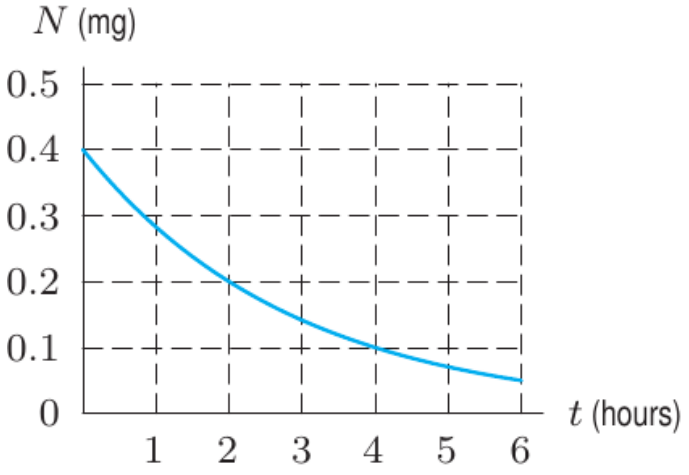
Problem 18 Figure 1.34 shows the total value of US imports, in billions of dollars.



a. Was the value of the imports higher in 1985 or in 2003? Approximately how much higher?

b. Estimate the average rate of change of US imports between 1985 and 2003. Give units and interpret your answer in terms of imports.

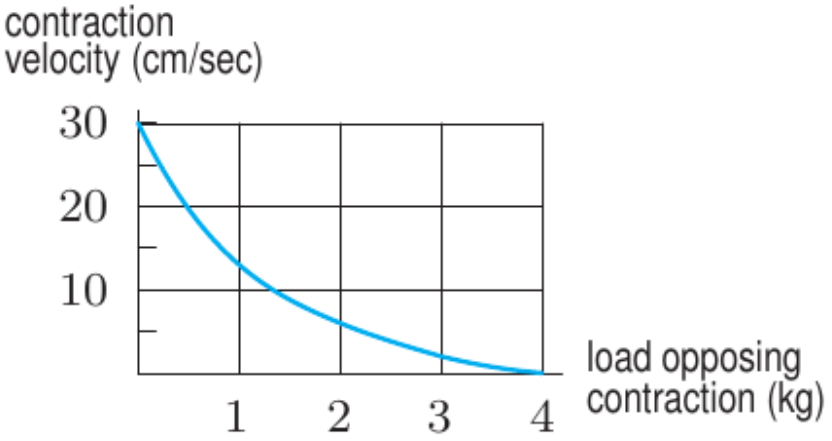
Problem 31 Figure 1.7 in Problem 14 of Section 1.1 shows the amount of nicotine N = f(t), in mg, in a person's bloodstream as a function of the time, t, in hours, since the last cigarette.



a. Is the average rate of change in nicotine level positive or negative? Explain.

b. Find the average rate of change in the nicotine level between t = 0 and t = 3. Give units and interpret your answer in terms of nicotine.

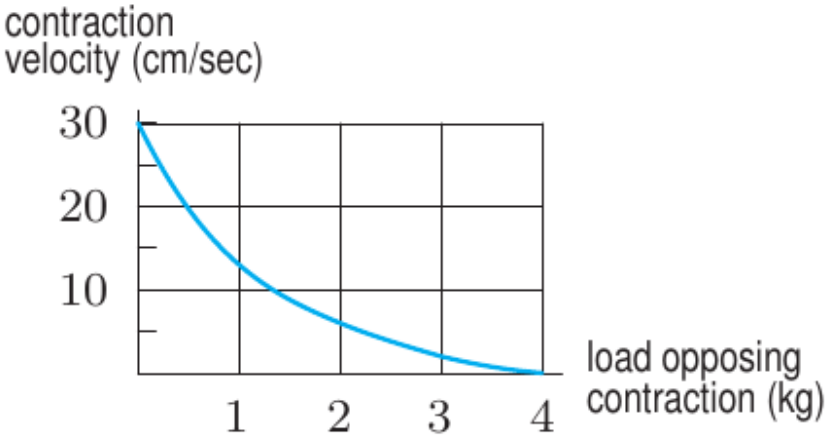
Problem 34 Problems 34–35 refer to Figure 1.38, which shows the contraction velocity of a muscle as a function of the load it pulls against.



a. In terms of the muscle, interpret the vertical intercept.

b. In terms of the muscle, interpret the horizontal intercept.

Problem 35 Use Figure 1.38 (contraction velocity of a muscle against load).



(a) Find the change in muscle contraction velocity when the load changes from 1 kg to 3 kg. Give units.

(b) Find the average rate of change in the contraction velocity between 1 kg and 3 kg. Give units.

1.3.3 Relative Change & Relative Rate of Change

Lecture Problems

Ex 5 Find the relative change (RC) in the price of a $75.99 pair of jeans if the sale price is $52.99.

RC = ?

Ex 6 Use the given figure to estimate the relative rate of change (RRC) of the number of farms in the US between 1950 and 1970.

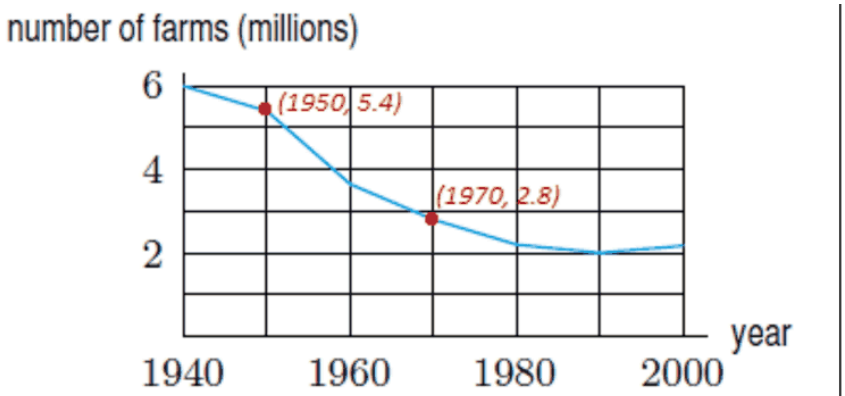


Figure : Number of farms in the US (in millions)

RRC = ?

Textbook Problems

Problem 5 In Problems 5–8, find the relative, or percent, change.

S changes from 400 to 450

Problem 6 Find the relative, or percent, change.

B changes from 12,000 to 15,000

Problem 7 Find the relative, or percent, change.

R changes from 50 to 47

Problem 8 Find the relative, or percent, change.

W changes from 0.3 to 0.05

Problem 42 In Problems 42–45, which relative change is bigger in magnitude? Justify your answer.

The change in the Dow Jones average from 164.6 to 77.9 in 1931; the change from 13261.8 to 8776.4 in 2008.

Problem 43 Which relative change is bigger in magnitude?

The change in the US population from 5.2 million to 7.2 million from 1800 to 1810; the change from 151.3 to 179.3 (million) from 1950 to 1960.

Problem 44 Which relative change is bigger in magnitude?

An increase in class size from 5 to 10; an increase in class size from 30 to 50.

Problem 45 Which relative change is bigger in magnitude?

An increase in sales from \$100,000 to \$500,000; an increase in sales from \$20,000,000 to \$20,500,000.

Problem 46 Find the relative change of a population if it changes

a. From 1000 to 2000

b. From 2000 to 1000

c. From 1,000,000 to 1,001,000

Problem 47 On Black Monday, October 28, 1929, the stock market on Wall Street crashed. The Dow Jones average dropped from 298.94 to 260.64 in one day.

What was the relative change in the index?

Problem 48 On May 11, 2009, the cost to mail a letter in the US was raised from 42 cents to 44 cents.

Find the relative change in the cost.

Problem 49 The US Consumer Price Index (CPI) is a measure of the cost of living. The inflation rate is the annual relative rate of change of the CPI. Use the January data in Table 1.22 to estimate the inflation rate for each of years 2005–2008.

Year	2005	2006	2007	2008	2009
CPI	190.7	198.3	202.416	211.08	211.143

2005 Inflation rate in 2005?

2006 Inflation rate in 2006?

2007 Inflation rate in 2007?

2008 Inflation rate in 2008?

Problem 50 During 2008 the US economy stopped growing and began to shrink. Table 1.23 gives quarterly data on the US Gross Domestic Product (GDP).

t (years since 2008)	0	0.25	0.5	0.75	1.0
GDP (trillion dollars)	14.15	14.29	14.41	14.2	14.09

a. t = 0 Estimate the relative growth rate (percent per year) at t = 0.

---

a. t = 0.25 Estimate the relative growth rate at t = 0.25.

---

a. t = 0.5 Estimate the relative growth rate at t = 0.5.

---

a. t = 0.75 Estimate the relative growth rate at t = 0.75.

---

b. Economists say an economy is in recession if the GDP decreases for two quarters in a row. Was the US in recession in 2008?

Past Exam Questions

2024 Q1 To assess how quickly a wound heals, its area was measured every five days.

Time, t (days)	0	5	10	15	20	25	30
Area, A (mm^2)	200	165	140	120	105	95	85

a. Find the average rate of change of the wound's area between day 10 and day 25. Provide units and interpret this value.

---

b. Find the relative rate of change of the wound's area between day 10 and day 25. Provide units and interpret this value.

2022 Q1 The table below gives information about the number of patients diagnosed with pancreatic cancer in the United States.

Year	1997	2002	2004
Number of cases	27000	30300	31860

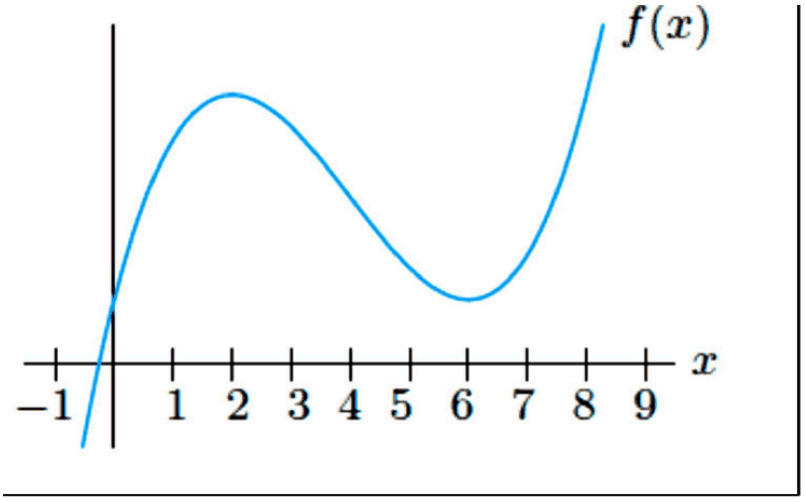
a. Find the average rate of change of the number of cases from 1997 to 2004. Give units and interpret your answer in terms of the number of diagnosed pancreatic cases.

b. Find the relative rate of change of the number of cases from 1997 to 2004. Give units and interpret your answer in terms of the number of diagnosed pancreatic cases.

1.3.4 Increasing, Decreasing & Concavity

Lecture Problems

Ex 7 Using the given figure, estimate the intervals over which:



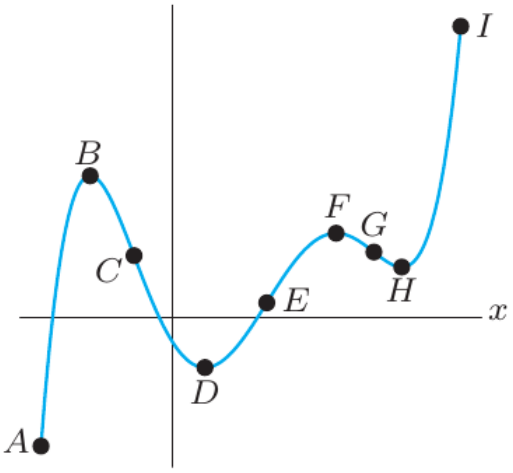
(a) The function is increasing; decreasing.

(b) The graph is concave up; concave down.

Ex 8

Same as Textbook Problem 10

For which pairs of consecutive points in the given figure is the function:



i. Increasing and concave up?

ii. Increasing and concave down?

iii. Decreasing and concave up?

iv. Decreasing and concave down?

Ex 9 Use the following values of f(t).

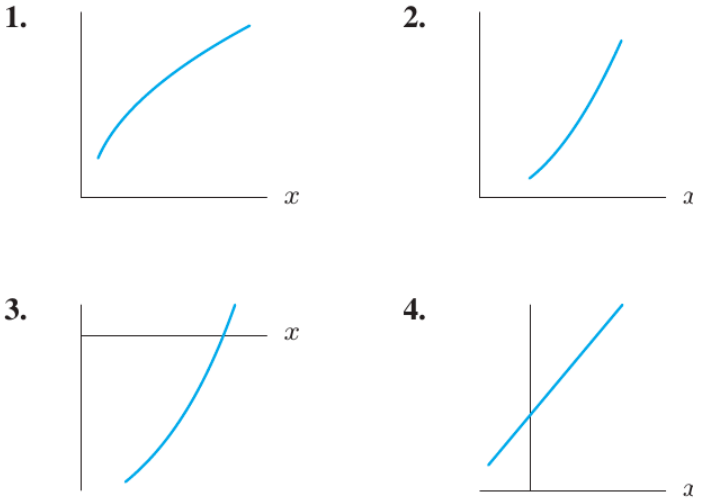
t	0	5	10	15	20	25	30
f(t)	12.6	13.1	14.1	16.2	20.0	29.6	42.7

(i) Does f appear to be increasing or decreasing?

(ii) Do you think its graph is concave up or concave down?

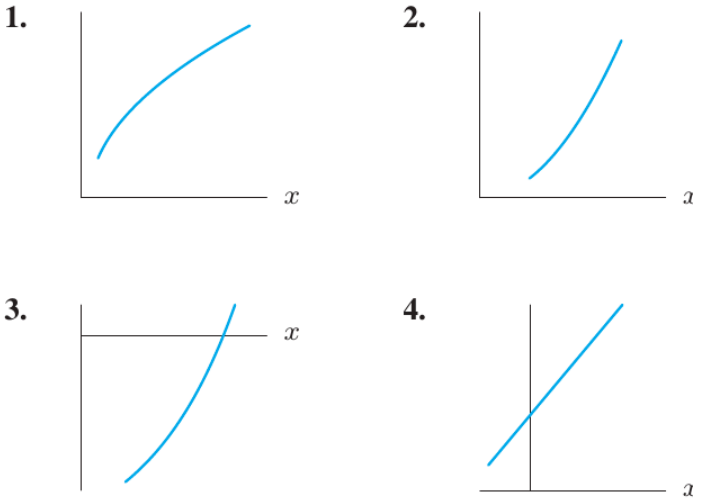
Textbook Problems

Problem 1 In Problems 1–4, decide whether the graph is concave up, concave down, or neither.



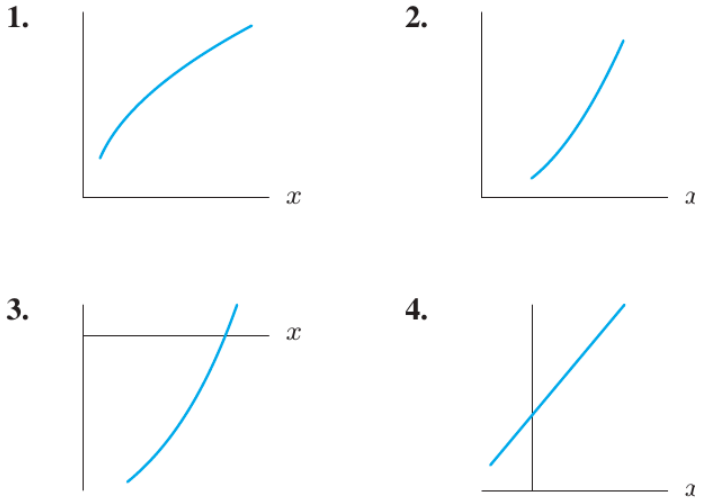
Graph 1

Problem 2 Decide whether the graph is concave up, concave down, or neither.



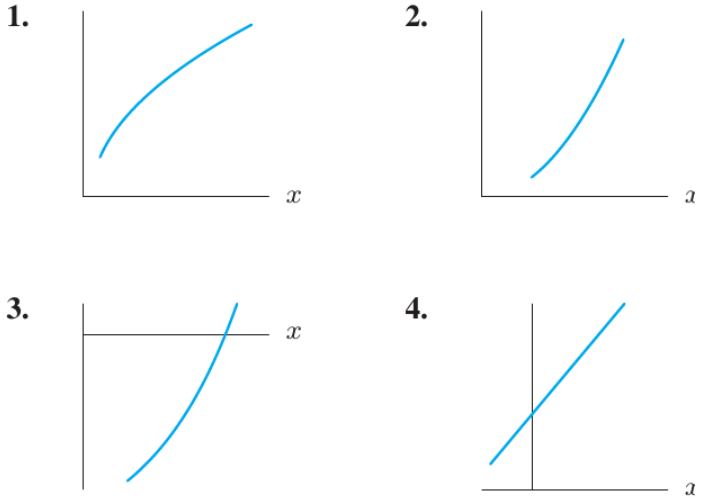
Graph 2

Problem 3 Decide whether the graph is concave up, concave down, or neither.



Graph 3

Problem 4 Decide whether the graph is concave up, concave down, or neither.



Graph 4

Problem 9 Table 1.10 gives values of a function w = f(t).

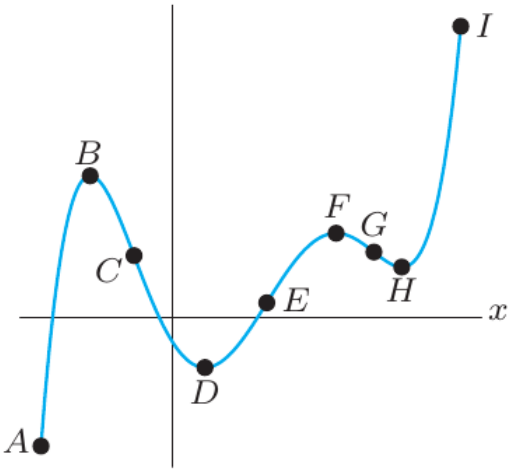
t	0	4	8	12	16	20	24
w	100	58	32	24	20	18	17

Is this function increasing or decreasing? Is the graph of this function concave up or concave down?

Problem 10

Same as Lecture Example 8

For which pairs of consecutive points in Figure 1.33 is the function graphed:



a. Increasing and concave up

b. Increasing and concave down

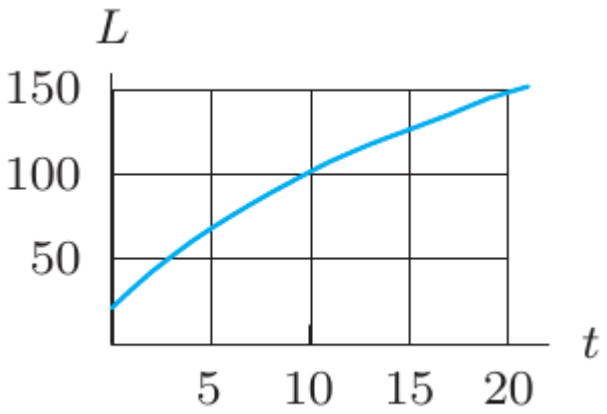
c. Decreasing and concave up

d. Decreasing and concave down

Problem 11 Graph a function f(x) which is increasing everywhere and concave up for negative x and concave down for positive x.

Which description matches such a graph?

Problem 26 Figure 1.37 shows the length, L, in cm, of a sturgeon (a type of fish) as a function of the time, t, in years.



a. Is the function increasing or decreasing? Is the graph concave up or concave down?

b. Estimate the average rate of growth of the sturgeon between t = 5 and t = 15. Give units and interpret your answer in terms of the sturgeon.

Problem 38 Values of F(t), G(t), and H(t) are in Table 1.21.

t	10	20	30	40	50	60
F(t)	15	22	28	33	37	40
G(t)	15	18	21	24	27	30
H(t)	15	17	20	24	29	35

Which graph is concave up and which is concave down? Which function is linear?

Problem 39 Experiments suggest that the male maximum heart rate decreases by 9 beats per minute during the first 21 years of his life, and by 26 beats per minute during the first 33 years.

If you model the maximum heart rate as a function of age, should you use a function that is increasing or decreasing, concave up or concave down?

Problem 40 A car starts slowly and then speeds up. Eventually the car slows down and stops.

Graph the distance that the car has traveled against time. Which description matches the graph?

1.3.5 Average Velocity

Lecture Problems

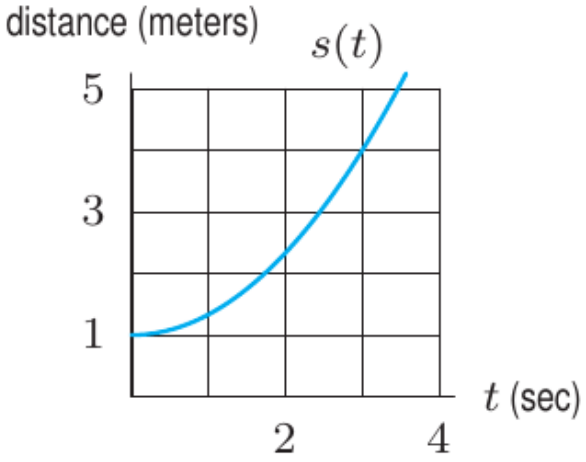
Ex 10 A grapefruit is thrown up in the air. The height of the grapefruit above the ground first increases and then decreases. Height, y, of the grapefruit above the ground t seconds after it is thrown:

t (sec)	0	1	2	3	4	5	6
y (feet)	6	90	142	162	150	106	30

Find the average velocity of the grapefruit over the interval t = 4 to t = 6. Explain the sign of your answer.

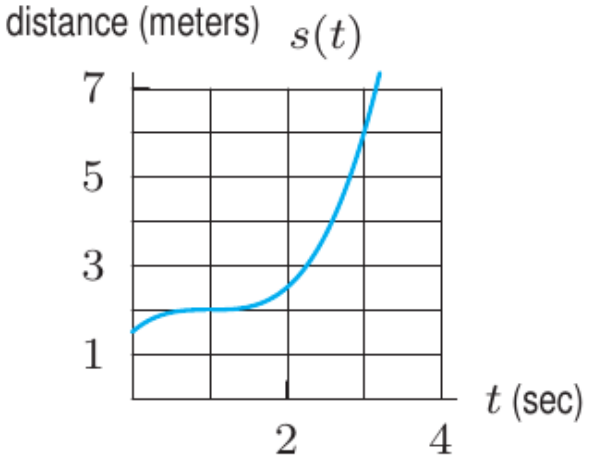
Textbook Problems

Problem 21 Figure 1.35 shows a particle's distance from a point.



What is the particle's average velocity from t = 0 to t = 3?

Problem 22 Figure 1.36 shows a particle's distance from a point.



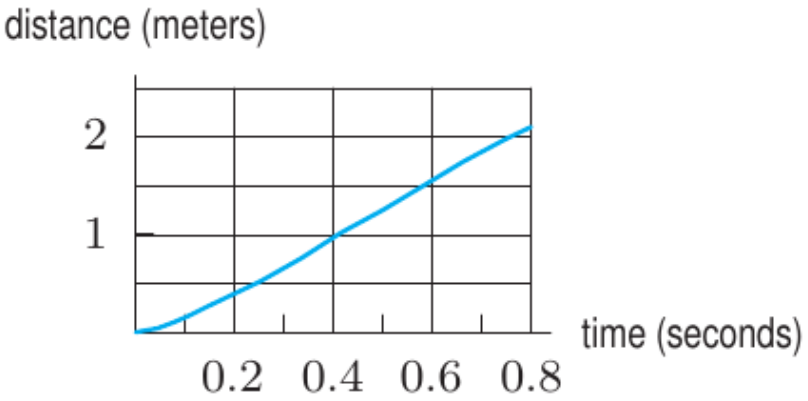
What is the particle's average velocity from t = 1 to t = 3?

Problem 23 At time t in seconds, a particle's distance s(t), in cm, from a point is given in the table.

t	0	3	6	10	13
s(t)	0	72	92	144	180

What is the average velocity of the particle from t = 3 to t = 10?

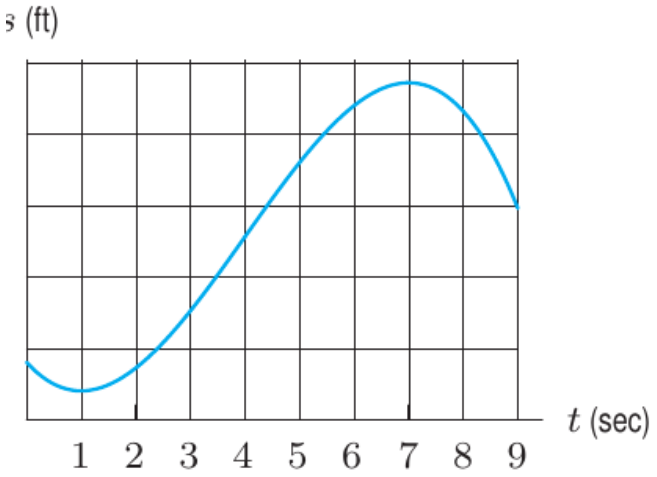
Problem 37 In an experiment, a lizard is encouraged to run as fast as possible. Figure 1.39 shows the distance run in meters as a function of the time in seconds.



a. If the lizard were running faster and faster, what would be the concavity of the graph? Does this match what you see?

b. Estimate the average velocity of the lizard during this 0.8 second experiment.

Problem 41 Figure 1.40 shows the position of an object at time t.



a. Draw a line on the graph whose slope represents the average velocity between t = 2 and t = 8. Which line is it?

b. Is average velocity greater between t = 0 and t = 3 or between t = 3 and t = 6?

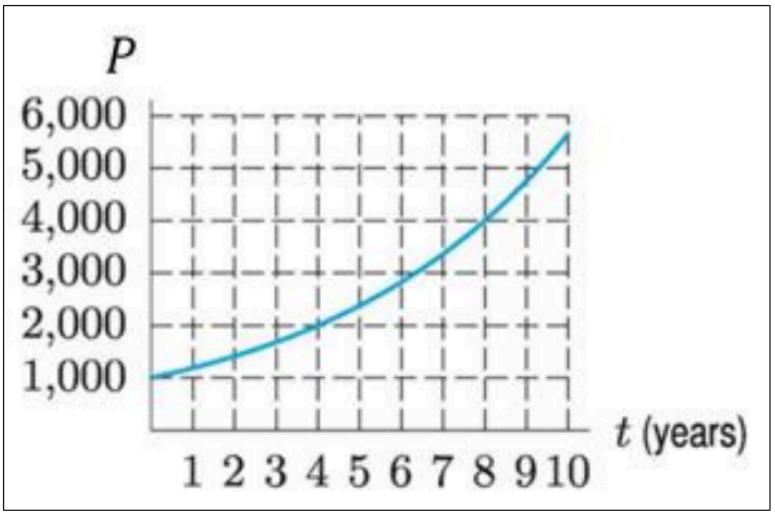
c. Is average velocity positive or negative between t = 6 and t = 9?

1.3.6 Test Your Understanding

Test Your Understanding

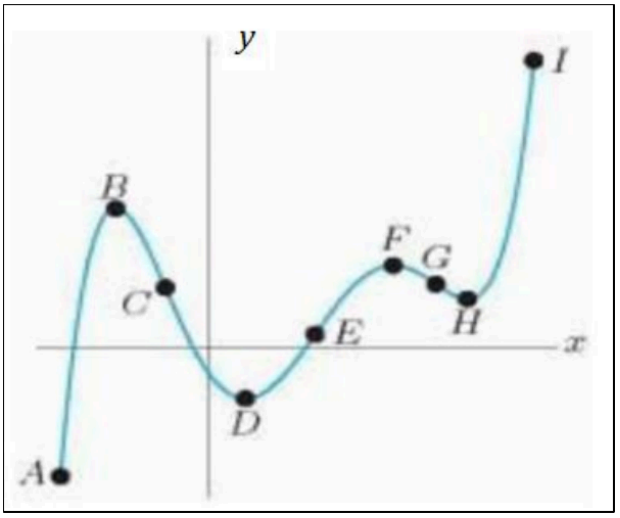
TYU 1 · P1 Consider the figure below where P is the population size at time t, in years. Decide whether each statement is True or False.

1. The population size is a function of time.
2. The population size reaches 2,000 in 4 years.
3. The mathematical statement P(10) = 5,600 means that in 10 years the population size will reach 5,600.
4. The population size increases with time.
5. The average rate of change in population size during the first 4 years is 250 people per year.
6. The relative change in population size during the first 4 years is 100%.
7. The vertical intercept represents the initial population size.
8. The observation that the graph has no horizontal intercept indicates that this population never dies out.
9. The rate of change in population size is always increasing.
10. The graph is concave down.
11. The slope is increasing.



TYU 1 · P2 Let the following be a graph of the function y = f(x). Decide whether each statement is True or False.

1. The quantity y is increasing and its rate of change is increasing between points E and F.
2. The quantity y is decreasing and its rate of change is increasing between points C and D.
3. The average rate of change between points A and B is greater than the average rate of change between points D and F.
4. The rate of change is negative between points B and D.



TYU 1 · P6 The time, t (in hours), that a drug stays in a person's system is a function of the quantity administered, q (in mg). What are the units of the rate of change of t with respect to q?

A. hours
B. mg
C. mg per hour
D. hour per mg
E. mg^2 per hour

TYU · Tobacco 1 Table 1.18 shows the production of tobacco in the US.

Year	**1996**	**1997**	**1998**	**1999**	**2000**	**2001**	**2002**	**2003**
Production (millions of pounds)	1517	1787	1480	1293	1053	991	879	831

a. What is the average rate of change in tobacco production between 1996 and 2003? Give units and interpret your answer in terms of tobacco production.
b. During this seven-year period, is there any interval during which the average rate of change was positive? If so, when?

TYU · Tobacco 2 Table 1.18 shows the production of tobacco in the US.

Year	**1996**	**1997**	**1998**	**1999**	**2000**	**2001**	**2002**	**2003**
Production (millions of pounds)	1517	1787	1480	1293	1053	991	879	831

a. What is the relative rate of change in tobacco production between 1996 and 2003? Give units and interpret your answer in terms of tobacco production.
b. During this seven-year period, is there any interval during which the relative rate of change was positive? If so, when?

TYU · Quantity Given the table below:

Year	**1979**	**1984**	**1989**	**1994**	**1999**	**2004**
Quantity	10	3	0.5	15	12	0.2

a. Find the rate of changes (ARC) between all intervals.
b. On what intervals is the graph increasing and/or decreasing?
c. Determine the intervals where the graph is concave up or concave down by studying the behavior of ARC.

Answers · Section 1.3

1.3.1 Average Rate of Change from Tables & Data

Ex 1

i. 1960–1968: 3.5 in/year; 1992–2000: 0.5 in/year — ARC = (213 − 185)/(1968 − 1960) = 28/8 = 3.5 inch/year. ARC = (232 − 228)/(2000 − 1992) = 4/8 = 0.5 inch/year.

ii. 1960–1968, because its ARC is larger — Since ARC_1 (3.5) is larger than ARC_2 (0.5), the height increased faster in the first period (the 1960s).

Ex 2

i. ≈ −11.86 million pounds/year: production decreased by about 11.86 million pounds per year — ARC = (719 − 802)/(2010 − 2003) = −83/7 ≈ −11.86 million pounds/year. Tobacco production decreased by about 11.86 million pounds per year between 2003 and 2010. (Note: the lecture notes write −11.84; −83/7 = −11.857 ≈ −11.86.)

ii. Yes: 2003–2004 and 2005–2009 — The ARC was positive during 2003–2004 (802 → 882) and 2005–2009 (645 → 728 → 788 → 801 → 823). E.g. ARC = (882 − 802)/(2004 − 2003) = 80 million pounds/year.

Problem 15

a. 90 million bicycles — Change = 101 − 11 = 90 million bicycles.

b. 1.8 million bicycles/year: production increased by 1.8 million bicycles per year — ARC = 90/(2000 − 1950) = 90/50 = 1.8 million bicycles per year.

Problem 16

a. −1,493 million dollars — Change = 14,526 − 16,019 = −1,493 million dollars.

b. ≈ −497.7 million dollars/year: sales decreased by about 497.7 million dollars per year — ARC = −1,493/(2008 − 2005) = −1,493/3 ≈ −497.7 million dollars/year.

c. Yes: 2003–2004 only — Only 2003–2004 (15,854 → 16,267) increased; every later year decreased.

Problem 17

a. 0.155 million fans/year — ARC = (22.26 − 21.64)/(2007 − 2003) = 0.62/4 = 0.155 million fans/year.

b. 0.07, 0.08, 0.41, 0.06 — 21.71 − 21.64 = 0.07; 21.79 − 21.71 = 0.08; 22.20 − 21.79 = 0.41; 22.26 − 22.20 = 0.06 (million fans).

c. Yes: (0.07 + 0.08 + 0.41 + 0.06)/4 = 0.155 — (0.07 + 0.08 + 0.41 + 0.06)/4 = 0.62/4 = 0.155 = the ARC from part (a). ✓

Problem 19

a. 18,280 million dollars — Change = 45,251 − 26,971 = 18,280 million dollars.

b. 3,656 million dollars/year: sales increased by 3,656 million dollars per year — ARC = 18,280/(2008 − 2003) = 18,280/5 = 3,656 million dollars/year.

Problem 20

a. P 0.077 billion people/year — ARC = (6.45 − 5.68)/10 = 0.77/10 = 0.077 billion people/year.

a. A 0.98 million cars/year — ARC = (45.9 − 36.1)/10 = 9.8/10 = 0.98 million cars/year.

a. C 207.7 million subscribers/year — ARC = (2168 − 91)/10 = 2077/10 = 207.7 million subscribers/year.

b. (i) Automobiles — P and A have different units, so compare how fast each grew relative to its 1995 value: automobiles 9.8/36.1 ≈ 27% versus population 0.77/5.68 ≈ 14% over the 10 years. Automobiles increased faster.

b. (ii) Cell phone subscribers — Cell phone subscribers grew from 91 to 2168 million (more than 20 times), while population grew only about 14%. Cell phone subscribers increased faster.

Problem 24

a. −98 million pounds/year: production decreased by 98 million pounds per year — ARC = (831 − 1517)/(2003 − 1996) = −686/7 = −98 million pounds/year.

b. Yes: 1996–1997 only — Only 1996–1997 (1517 → 1787) increased.

Problem 25

a. Negative — Negative: rain forests are being cut down, so the number of acres decreases.

b. Positive — Positive: the world population is increasing.

c. Negative — Negative: vaccination made polio cases decrease.

d. Negative — Negative: erosion makes the dune lower.

e. Positive — Positive: prices (cost of living) increase over time.

Problem 27

1940–2000 ≈ 1,489.5 thousand workers/year — ARC = (136,891 − 47,520)/60 = 89,371/60 ≈ 1,489.5 thousand workers/year: the labor force grew by about 1,489,500 workers per year.

1940–1960 912.9 thousand workers/year — ARC = (65,778 − 47,520)/20 = 18,258/20 = 912.9 thousand workers/year.

1980–2000 1,879.4 thousand workers/year — ARC = (136,891 − 99,303)/20 = 37,588/20 = 1,879.4 thousand workers/year.

Problem 28

≈ 1.61 million tons/year: the catch increased by about 1.61 million tons per year — ARC = (99 − 17)/(2001 − 1950) = 82/51 ≈ 1.61 million tons/year.

Problem 29

a. −35 billion dollars — Change = 149.0 − 184.0 = −35 billion dollars.

b. −7 billion dollars/year: revenue decreased by 7 billion dollars per year — ARC = −35/5 = −7 billion dollars/year.

c. Yes: 2006–2007 and 2007–2008 — 205.6 → 181.1 (2006–2007) and 181.1 → 149.0 (2007–2008) decreased.

Problem 30

≈ 2,353,747 households/year: cable households increased by about 2.35 million per year — ARC = (73,365,880 − 12,168,450)/26 = 61,197,430/26 ≈ 2,353,747 households/year.

Problem 32 **a. (i) −0.019 (mg/ml)/min** — ARC = (0.298 − 0.336)/(8 − 6) = −0.038/2 = −0.019 (mg/ml)/min.

a. (ii) −0.016 (mg/ml)/min — ARC = (0.266 − 0.298)/(10 − 8) = −0.032/2 = −0.016 (mg/ml)/min.

b. Negative: the concentration is decreasing, and it decreases more slowly over time — Both rates are negative, so creatinine concentration is decreasing. |−0.016| < |−0.019|, so it decreases more slowly over time.

Problem 33 **1804–1927 ≈ 15.5** — 1 billion people in 123 years: 1,000,000,000 ÷ (123 × 525,600) ≈ 15.5 people/min.

1927–1960 ≈ 57.7 — 1 billion in 33 years: 1,000,000,000 ÷ (33 × 525,600) ≈ 57.7 people/min.

1960–1974 ≈ 135.9 — 1 billion in 14 years: 1,000,000,000 ÷ (14 × 525,600) ≈ 135.9 people/min.

1974–1987 ≈ 146.4 — 1 billion in 13 years: 1,000,000,000 ÷ (13 × 525,600) ≈ 146.4 people/min.

1987–1999 ≈ 158.5 — 1 billion in 12 years: 1,000,000,000 ÷ (12 × 525,600) ≈ 158.5 people/min.

Problem 36 **a. 7,500 million dollars** — Change = 37,600 − 30,100 = 7,500 million dollars.

b. 1,500 million dollars/year: sales increased by 1,500 million dollars per year — ARC = 7,500/5 = 1,500 million dollars/year.

1.3.2 Average Rate of Change from Graphs & Formulas

Ex 3 **−0.13 million farms/year** — ARC = (2.8 − 5.4)/(1970 − 1950) = −2.6/20 = −0.13 million farms/year. The number of farms in the US decreased by 0.13 million (130,000) farms per year.

Ex 4 **(a) 1/3 ≈ 0.33** — ARC = (f(4) − f(1))/(4 − 1) = (2 − 1)/3 = 1/3 ≈ 0.33.

(b) The secant line through (1, 1) and (4, 2) — The ARC is the slope of the secant line joining the points (1, f(1)) = (1, 1) and (4, f(4)) = (4, 2).

Problem 12 **8** — f(1) = 2, f(3) = 18. ARC = (18 − 2)/(3 − 1) = 16/2 = 8.

Problem 13 **ARC −3** — f(−2) = 3(4) + 4 = 16, f(1) = 3 + 4 = 7. ARC = (7 − 16)/(1 − (−2)) = −9/3 = −3.

Graph The secant line through (−2, 16) and (1, 7) — The ARC = −3 is the slope of the secant line joining (−2, 16) and (1, 7) on the graph of $f(x) = 3x^2 + 4$.

Problem 14 **≈ 93.87 dollars/year: the balance grows by about $93.87 per year** — $B(5) = 1000(1.08)^5 \approx 1469.33$; B(0) = 1000. ARC = (1469.33 − 1000)/5 ≈ 93.87 dollars/year.

Problem 18 **a. 2003, higher by ≈ 920 billion dollars** — Imports were about 340 billion dollars in 1985 and about 1,260 billion dollars in 2003: 2003 was higher by ≈ 920 billion dollars.

b. ≈ 51 billion dollars/year: imports increased by about 51 billion dollars per year — ARC ≈ 920/(2003 − 1985) = 920/18 ≈ 51 billion dollars/year.

Problem 31 **a. Negative, because nicotine is decreasing** — The nicotine level decreases over time, so the average rate of change is negative.

b. ≈ −0.087 mg/hour: nicotine decreased by about 0.087 mg per hour — ARC = (f(3) − f(0))/3 ≈ (0.14 − 0.4)/3 = −0.26/3 ≈ −0.087 mg/hour. (If f(3) is read as 0.15, ARC ≈ −0.083 mg/hour; either way about −0.08 to −0.09.)

Problem 34 **a. 30 cm/sec: the contraction velocity with no load** — Vertical intercept = 30 cm/sec: the contraction velocity when there is no load.

b. 4 kg: the load at which the muscle cannot contract — Horizontal intercept = 4 kg: the load at which the contraction velocity is zero (the muscle cannot contract).

Problem 35 **(a) ≈ −11 cm/sec** — Velocity ≈ 14 cm/sec at 1 kg and ≈ 3 cm/sec at 3 kg. Change ≈ 3 − 14 = −11 cm/sec.

(b) ≈ −5.5 (cm/sec)/kg — ARC ≈ −11/(3 − 1) = −5.5 (cm/sec) per kg.

1.3.3 Relative Change & Relative Rate of Change

Ex 5 **≈ −0.303 = −30.3%** — RC = (52.99 − 75.99)/75.99 = −23/75.99 ≈ −0.303 = −30.3%. The discount in the price is about 30%.

Ex 6 **≈ −2.4% per year** — From Ex 3, ARC = −0.13 million/year. RRC = ARC/f(1950) = −0.13/5.4 ≈ −0.0241 = −2.41% per year. The number of farms decreased by about 2.41% per year.

Problem 5 **12.5%** — RC = (450 − 400)/400 = 50/400 = 0.125 = 12.5%.

Problem 6 **25%** — RC = (15,000 − 12,000)/12,000 = 3,000/12,000 = 0.25 = 25%.

Problem 7 **−6%** — RC = (47 − 50)/50 = −3/50 = −0.06 = −6%.

Problem 8 **≈ −83.3%** — RC = (0.05 − 0.3)/0.3 = −0.25/0.3 ≈ −0.833 = −83.3%.

Problem 42 **1931 (≈ −52.7%) is bigger than 2008 (≈ −33.8%)** — 1931: (77.9 − 164.6)/164.6 ≈ −52.7%. 2008: (8776.4 − 13261.8)/13261.8 ≈ −33.8%. The 1931 change is bigger in magnitude.

Problem 43 **1800–1810 (≈ 38.5%) is bigger** — 1800–1810: 2/5.2 ≈ 38.5%. 1950–1960: 28/151.3 ≈ 18.5%. The 1800–1810 change is bigger.

Problem 44 **5 to 10 (100% vs 66.7%)** — 5 → 10: 5/5 = 100%. 30 → 50: 20/30 ≈ 66.7%. The change from 5 to 10 is bigger.

Problem 45 **$100,000 to $500,000 (400% vs 2.5%)** — $100,000 → $500,000: 400,000/100,000 = 400%. $20,000,000 → $20,500,000: 500,000/20,000,000 = 2.5%.

Problem 46 **a. 100%** — RC = 1000/1000 = 1 = 100%.

b. −50% — RC = −1000/2000 = −0.5 = −50%.

c. 0.1% — RC = 1000/1,000,000 = 0.001 = 0.1%.

Problem 47 **≈ −12.8%** — RC = (260.64 − 298.94)/298.94 = −38.3/298.94 ≈ −0.128 = −12.8%.

Problem 48 **≈ 4.76%** — RC = (44 − 42)/42 = 2/42 ≈ 0.0476 = 4.76%.

Problem 49

2005 **≈ 3.99%** — (198.3 − 190.7)/190.7 ≈ 3.99%.

2006 **≈ 2.08%** — (202.416 − 198.3)/198.3 ≈ 2.08%.

2007 **≈ 4.28%** — (211.08 − 202.416)/202.416 ≈ 4.28%.

2008 **≈ 0.03%** — (211.143 − 211.08)/211.08 ≈ 0.03%.

Problem 50

a. t = 0 **≈ 3.96% per year** — RRC = ((14.29 − 14.15)/0.25)/14.15 = 0.56/14.15 ≈ 3.96% per year.

a. t = 0.25 **≈ 3.36% per year** — RRC = ((14.41 − 14.29)/0.25)/14.29 = 0.48/14.29 ≈ 3.36% per year.

a. t = 0.5 **≈ −5.83% per year** — RRC = ((14.2 − 14.41)/0.25)/14.41 = −0.84/14.41 ≈ −5.83% per year.

a. t = 0.75 **≈ −3.10% per year** — RRC = ((14.09 − 14.2)/0.25)/14.2 = −0.44/14.2 ≈ −3.10% per year.

b. Yes, GDP decreased for two quarters in a row — GDP decreased from 14.41 to 14.2 and then to 14.09: two quarters in a row, so yes.

2024 Q1

a. −3 mm²/day: the wound area decreases by 3 mm² per day — ARC = (95 − 140)/(25 − 10) = −45/15 = −3 mm²/day.

b. ≈ −2.14% per day: the area decreases by about 2.14% per day — RRC = ARC/A(10) = −3/140 ≈ −0.0214 = −2.14% per day.

2022 Q1

a. ≈ 694.3 cases/year: cases increased by about 694 per year — ARC = (31860 − 27000)/(2004 − 1997) = 4860/7 ≈ 694.3 cases/year.

b. ≈ 2.57% per year: cases increased by about 2.57% per year — RRC = ARC/f(1997) = 694.3/27000 ≈ 0.0257 = 2.57% per year.

1.3.4 Increasing, Decreasing & Concavity

Ex 7

(a) Increasing: $x < 2$ and $x > 6$; decreasing: $2 < x < 6$ — The graph rises up to x = 2, falls from x = 2 to x = 6, then rises again after x = 6.

(b) Concave down: $x < 4$; concave up: $x > 4$ — The graph bends downward (∩) until about x = 4, then bends upward (∪) after x = 4.

Ex 8

i. D–E and H–I — D–E and H–I: going up while bending upward.

ii. A–B and E–F — A–B and E–F: going up while bending downward.

iii. C–D and G–H — C–D and G–H: going down while bending upward.

iv. B–C and F–G — B–C and F–G: going down while bending downward.

Ex 9 **(i) Increasing** — f(t) increases from 12.6 to 42.7 as t goes from 0 to 30, so f is increasing.
(ii) Concave up — The ARC between consecutive points is 0.1, 0.2, 0.42, 0.76, 1.92, 2.62: the rate of change is increasing, so the graph is concave up.

Problem 1 **Concave down** — Graph 1 bends downward: concave down.

Problem 2 **Concave up** — Graph 2 bends upward: concave up.

Problem 3 **Concave up** — Graph 3 bends upward: concave up.

Problem 4 **Neither** — Graph 4 is a straight line: neither concave up nor concave down.

Problem 9 **Decreasing, concave up** — w decreases from 100 to 17. The decreases get smaller (−42, −26, −8, −4, −2, −1), so the rate of change is increasing (less negative): concave up.

Problem 10
a. **D–E and H–I** — D–E and H–I.
b. **A–B and E–F** — A–B and E–F.
c. **C–D and G–H** — C–D and G–H.
d. **B–C and F–G** — B–C and F–G.

Problem 11 **Always rising; bends upward (∪) left of x = 0 and downward (∩) right of x = 0** — Increasing everywhere means it always rises. Concave up for x < 0 (∪) and concave down for x > 0 (∩): an S-shaped rising curve.

Problem 26
a. **Increasing, concave down** — The length increases, but more and more slowly: increasing and concave down.
b. **≈ 6 cm/year: the sturgeon grows about 6 cm per year** — L(5) ≈ 65 cm and L(15) ≈ 125 cm: ARC ≈ (125 − 65)/10 ≈ 6 cm/year.

Problem 38 **F: concave down; G: linear; H: concave up** — F increases by 7, 6, 5, 4, 3 (decreasing rate) → concave down. G increases by 3 each time → linear. H increases by 2, 3, 4, 5, 6 (increasing rate) → concave up.

Problem 39 **Decreasing, concave down** — The heart rate decreases. The rate of decrease grows: 9/21 ≈ 0.43 then 26/33 ≈ 0.79 beats/min per year, so it falls faster and faster → concave down.

Problem 40 **Increasing: concave up at first, then concave down, then flat** — Speeding up → distance rises faster (concave up ∪). Slowing down → rises more slowly (concave down ∩). Stopped → flat.

1.3.5 Average Velocity

Ex 10 **−60 ft/sec; negative because the grapefruit is falling** — Average velocity = (30 − 150)/(6 − 4) = −120/2 = −60 ft/sec. It is negative because the height is decreasing (the grapefruit is falling).

Problem 21 **1 m/sec** — v = (s(3) − s(0))/(3 − 0) = (4 − 1)/3 = 1 m/sec.

Problem 22 **2 m/sec** — $v = (s(3) - s(1))/(3 - 1) = (6 - 2)/2 = 2$ m/sec.

Problem 23 **≈ 10.29 cm/sec** — $v = (144 - 72)/(10 - 3) = 72/7 \approx 10.29$ cm/sec.

Problem 37

a. Concave up; no, the graph is nearly a straight line — Running faster and faster → distance grows faster → concave up. The graph is nearly a straight line, so it does not match.

b. ≈ 2.6 m/sec — The lizard ran about 2.1 m in 0.8 sec: $v \approx 2.1/0.8 \approx 2.6$ m/sec.

Problem 41

a. The secant line joining the points on the graph at t = 2 and t = 8 — The average velocity between t = 2 and t = 8 is the slope of the secant line joining the points at t = 2 and t = 8.

b. Between t = 3 and t = 6 — The graph rises much more between t = 3 and t = 6 than between t = 0 and t = 3, so the average velocity is greater there.

c. Negative — The position at t = 9 is lower than at t = 6, so the average velocity is negative.

1.3.6 Test Your Understanding

TYU 1 · P1 From the graph: P(0) = 1,000, P(4) = 2,000, P(10) ≈ 5,600. The graph is increasing and bending upward (concave up).

$$\text{ARC} = \frac{P(4) - P(0)}{4 - 0} = \frac{2000 - 1000}{4} = 250 \text{ people/year}$$

$$\text{Relative change} = \frac{P(4) - P(0)}{P(0)} = \frac{2000 - 1000}{1000} = 1 = 100\%$$

1. True — each t gives exactly one P.

2. True — P(4) = 2,000.

3. True — at t = 10, P = 5,600.

4. True — the graph goes up.

5. True — (2000 − 1000)/4 = 250.

6. True — 1000/1000 = 100%.

7. True — the vertical intercept is P(0).

8. True — P never reaches 0.

9. True — the graph is concave up.

10. False — it is concave up.

11. True — concave up → slope increasing.

TYU 1 · P2

1. False — y increases, but the graph bends downward (flattens) toward F → rate of change decreasing.

2. True — y goes down, but the graph bends upward (flattens) toward the minimum D → rate of change increasing (toward 0).

3. True — A → B rises very steeply; D → F rises gently.

4. True — the graph goes down from B to D.

TYU 1 · P6 The rate of change of t with respect to q is dt/dq, so its units are units of t ÷ units of q.

Answer: D — hour per mg

TYU · Tobacco 1 **(a)**

$$\text{ARC} = \frac{y_2 - y_1}{t_2 - t_1} = \frac{831 - 1517}{2003 - 1996} = \frac{-686}{7} = -98 \text{ millions of pounds/year}$$

Between 1996 and 2003, tobacco production decreased by 98 million pounds per year on average.

(b) Yes. The ARC was positive only between 1996 and 1997 (production increased):

$$\text{ARC} = \frac{1787 - 1517}{1997 - 1996} = 270 \text{ millions of pounds/year}$$

TYU · Tobacco 2 **(a)**

$$\text{ARC} = \frac{y_2 - y_1}{t_2 - t_1} = \frac{831 - 1517}{2003 - 1996} = \frac{-686}{7} = -98 \text{ millions of pounds/year}$$

$$\text{RRC} = \frac{\text{ARC}}{y_1} = \frac{-98}{1517} \approx -0.0646 \times 100\% = -6.46 \text{ \%/year}$$

Between 1996 and 2003, tobacco production decreased by about 6.46% per year (relative to the 1996 production).

(b) Yes. The relative rate of change is positive only where production increased, between 1996 and 1997:

$$\text{RRC} = \frac{(1787 - 1517)/(1997 - 1996)}{1517} = \frac{270}{1517} \approx 0.178 \times 100\% = 17.8 \text{ \%/year}$$

TYU · Quantity

(a) ARC = $\frac{\Delta y}{\Delta x}$ with $\Delta x = 5$ for every interval:

$$1979\text{–}1984: \frac{3 - 10}{5} = -1.4$$

$$1984\text{–}1989: \frac{0.5 - 3}{5} = -0.5$$

$$1989\text{–}1994: \frac{15 - 0.5}{5} = 2.9$$

$$1994\text{–}1999: \frac{12 - 15}{5} = -0.6$$

$$1999\text{–}2004: \frac{0.2 - 12}{5} = -2.36$$

Interval	**1979–84**	**1984–89**	**1989–94**	**1994–99**	**1999–2004**
ARC	−1.4	−0.5	2.9	−0.6	−2.36

(b) Increasing where ARC > 0: 1989–1994.

Decreasing where ARC < 0: 1979–1989 and 1994–2004.

(c) The ARC increases (−1.4 → −0.5 → 2.9) up to 1994, then decreases (2.9 → −0.6 → −2.36):

Concave up on (1979, 1994) and concave down on (1994, 2004)

1.5 Exponential Functions

In this section

1. Growth Factor & Exponential Growth 2 lecture · 6 textbook
2. Exponential Decay 1 lecture · 4 textbook
3. Linear vs. Exponential Functions 1 lecture · 11 textbook · 1 past exam
4. Finding Exponential Formulas from Data 2 lecture · 10 textbook
5. Test Your Understanding 5 questions

1.5.1 Growth Factor & Exponential Growth

Lecture Problems

Ex 1 Use the table to find a (the growth factor of Burkina Faso's population). Population of Burkina Faso (estimated) 2003–2009:

Year	Population (millions)
2003	$P_0 = 12.853$
2005	$P_2 = 13.747$
2007	$P_4 = 14.721$
2009	$P_6 = 15.757$

a = ?

Ex 2

Same as Textbook Problem 8

World population is approximately $P = 6.4(1.0126)^t$, with P in billions and t in years since 2004.

i. What is the yearly percent rate of growth of the world population?

ii. What was the world population in 2004?

iii. What does this model predict for the world population in 2010?

Textbook Problems

Problem 1 The following functions give the populations of four towns with time t in years: (i) $P = 600(1.12)^t$ (ii) $P = 1{,}000(1.03)^t$ (iii) $P = 200(1.08)^t$ (iv) $P = 900(0.90)^t$

a. Which town has the largest percent growth rate? What is the percent growth rate?

b. Which town has the largest initial population? What is that initial population?

c. Are any of the towns decreasing in size? If so, which one(s)?

Problem 2 Each of the following functions gives the amount of a substance present at time t. In each case, give the amount present initially (at t = 0), state whether the function represents exponential growth or decay, and give the percent growth or decay rate.

(a) $A = 100(1.07)^t$

(b) $A = 5.3(1.054)^t$

(c) $A = 3500(0.93)^t$

(d) $A = 12(0.88)^t$

Problem 8

Same as Lecture Example 2

World population is approximately $P = 6.4(1.0126)^t$, with P in billions and t in years since 2004.

a. What is the yearly percent rate of growth of the world population?

b. What was the world population in 2004? What does this model predict for the world population in 2010?

c. Use part (b) to find the average rate of change of the world population between 2004 and 2010.

Problem 11 The consumer price index (CPI) for a given year is the amount of money in that year that has the same purchasing power as $100 in 1983. At the start of 2009, the CPI was 211.

Write a formula for the CPI as a function of t, years after 2009, assuming that the CPI increases by 2.8% every year.

Problem 13

(a) Make a table of values for $y = e^x$ using x = 0, 1, 2, 3.

(b) Plot the points found in part (a). Does the graph look like an exponential growth or decay function?

(c) Make a table of values for $y = e^{-x}$ using x = 0, 1, 2, 3.

(d) Plot the points found in part (c). Does the graph look like an exponential growth or decay function?

Problem 14 Graph $y = 100e^{-0.4x}$.

Describe what you see.

1.5.2 Exponential Decay

Lecture Problems

Ex 3

Same as Textbook Problem 9

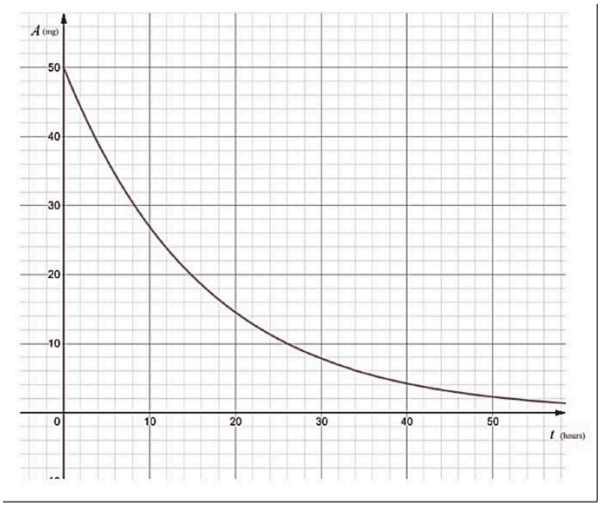
A 50 mg dose of quinine is given to a patient to prevent malaria. Quinine leaves the body at a rate of 6% per hour.

i. Find a formula for the amount, A (in mg), of quinine in the body t hours after the dose is given.

ii. How much quinine is in the body after 24 hours?

iii. Graph A as a function of t. Which description matches the graph?

iv. Use the graph to estimate when 5 mg of quinine remains.



Textbook Problems

Problem 9

Same as Lecture Example 3

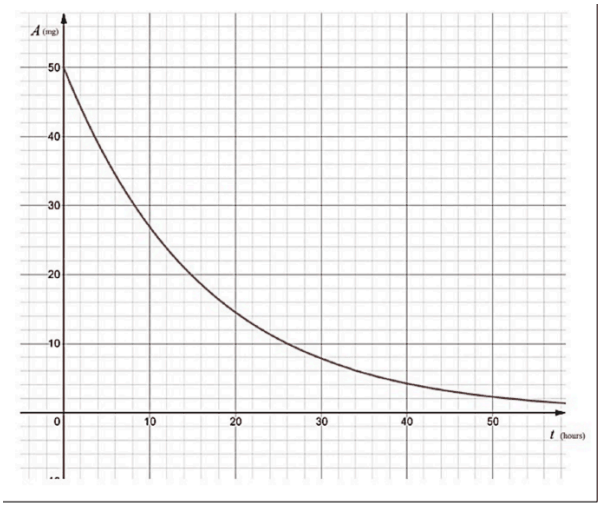
A 50 mg dose of quinine is given to a patient to prevent malaria. Quinine leaves the body at a rate of 6% per hour.

a. Find a formula for the amount, A (in mg), of quinine in the body t hours after the dose is given.

b. How much quinine is in the body after 24 hours?

c. Graph A as a function of t. Which description matches the graph?

d. Use the graph to estimate when 5 mg of quinine remains.



Problem 12 During the 1980s, Costa Rica had the highest deforestation rate in the world, at 2.9% per year. (This is the rate at which land covered by forests is shrinking.)

Assuming the rate continues, what percent of the land in Costa Rica covered by forests in 1980 will be forested in 2015?

Problem 28

(a) Niki invested $10,000 in the stock market. The investment was a loser, declining in value 10% per year each year for 10 years. How much was the investment worth after 10 years?

(b) After 10 years, the stock began to gain value at 10% per year. After how long will the investment regain its initial value ($10,000)?

Problem 29 A photocopy machine can reduce copies to 80% of their original size. By copying an already reduced copy, further reductions can be made.

a. If a page is reduced to 80%, what percent enlargement is needed to return it to its original size?

b. Estimate the number of times in succession that a page must be copied to make the final copy less than 15% of the size of the original.

1.5.3 Linear vs. Exponential Functions

Lecture Problems

Ex 4

Similar problem (different numbers): Past Exam 2022 Question 2

The amount of adrenaline in the body can change rapidly. Suppose the initial amount is 15 mg. Find a formula for A, the amount in mg, at a time t minutes later if A is:

(a) Increasing by 0.4 mg per minute.

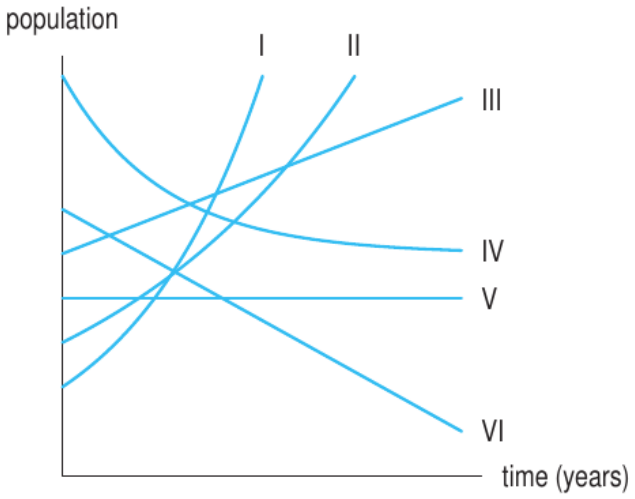
(b) Decreasing by 0.4 mg per minute.

(c) Increasing by 3% per minute.

(d) Decreasing by 3% per minute.

Textbook Problems

Problem 3 Figure 1.62 shows graphs of several cities' populations against time. Match each of the following descriptions to a graph and write a description to match each of the remaining graphs.



a. The population increased at 5% per year.

b. The population increased at 8% per year.

c. The population increased by 5000 people per year.

d. The population was stable.

IV Write a description for the remaining graph IV.

VI Write a description for the remaining graph VI.

Problem 4 A town has a population of 1000 people at time t = 0. In each of the following cases, write a formula for the population, P, of the town as a function of year t.

a. The population increases by 50 people a year.

b. The population increases by 5% a year.

Problem 5 The gross domestic product, G, of Switzerland was 310 billion dollars in 2007. Give a formula for G (in billions of dollars) t years after 2007 if G increases by

(a) 3% per year

(b) 8 billion dollars per year

Problem 6 A product costs $80 today. How much will the product cost in t days if the price is reduced by

(a) $4 a day

(b) 5% a day

Problem 7 An air-freshener starts with 30 grams and evaporates. In each of the following cases, write a formula for the quantity, Q grams, of air-freshener remaining t days after the start and sketch a graph of the function. The decrease is:

(a) 2 grams a day

(b) 12% a day

Problem 22 Find a formula for the number of zebra mussels in a bay as a function of the number of years since 2003, given that there were 2700 at the start of 2003 and 3186 at the start of 2004.

a. Assume that the number of zebra mussels is growing linearly. Give units for the slope of the line and interpret it in terms of zebra mussels.

b. Assume that the number of zebra mussels is growing exponentially. What is the percent rate of growth of the zebra mussel population?

Problem 23 Worldwide, wind energy generating capacity, W, was 40,300 megawatts in 2003 and 121,188 megawatts in 2008.

a. Use the values given to write W, in megawatts, as a linear function of t, the number of years since 2003.

b. Use the values given to write W as an exponential function of t.

c. Graph the functions found in parts (a) and (b) on the same axes. Which statement is correct?

Problem 24 Use the table.

x	f(x)	g(x)	h(x)
−2	12	16	37
−1	17	24	34
0	20	36	31
1	21	54	28
2	18	81	25

(a) Which (if any) of the functions in the table could be linear? Find formulas for those functions.

(b) Which (if any) of these functions could be exponential? Find formulas for those functions.

Problem 25 Determine whether each of the following tables of values could correspond to a linear function, an exponential function, or neither. For each table of values that could correspond to a linear or an exponential function, find a formula for the function.

(a)

x	0	1	2	3
f(x)	10.5	12.7	18.9	36.7

(b)

t	−1	0	1	2
s(t)	50.2	30.12	18.072	10.8432

(c)

u	0	2	4	6
g(u)	27	24	21	18

Problem 26 Table 1.32: Soybean production, in millions of tons.

Year	2000	2001	2002
Production	161.0	170.3	180.2

Year	2003	2004	2005
Production	190.7	201.8	213.5

(a) Could the data on annual world soybean production correspond to a linear function or an exponential function? If so, which?

(b) Find a formula for P, world soybean production in millions of tons, as a function of time, t, in years since 2000.

(c) What is the annual percent increase in soybean production?

Problem 27 The 2004 US presidential debates questioned whether the minimum wage has kept pace with inflation. In 1938, the minimum wage was 25¢; in 2004, it was $5.15. During the same period, inflation averaged 4.3%.

Decide the question using this information.

Past Exam Questions

2022 Q2

Similar problem (different numbers): Lecture Example 4

The amount of adrenaline in the body can change rapidly. Suppose that the initial amount of adrenaline is 10 mg. In each of the following cases, find a formula for the amount A (in mg) after t minutes:

(a) Adrenaline is decreasing by 0.6 mg per minute.

(b) Adrenaline is increasing by 10% per minute.

1.5.4 Finding Exponential Formulas from Data

Lecture Problems

Ex 5 Wolves were once common in the western US. By the 1990s, the wolf population in Wyoming had been wiped out by hunters and wolves put on the endangered species list. In 1995, wolves were reintroduced to Wyoming from Canada. Starting with 14 wolves, their number increased to 207 wolves in 2012. Assuming the Wyoming wolf population was growing exponentially, find a function of the form $P = P_0a^t$, where P is the population t years after 1995. What is the annual percent growth rate?

P = ? and r = ?

Ex 6 Suppose Q = f(t) is an exponential function of t. If f(20) = 88.2 and f(23) = 91.4:

a. Find the base.

---

b. Find the percent growth rate.

---

c. Evaluate f(25).

Textbook Problems

Problem 10 The Hershey Company is the largest US producer of chocolate. In 2008, annual net sales were 5.1 billion dollars and were increasing at a continuous rate of 4.3% per year.

a. Write a formula for annual net sales, S, as a function of time, t, in years since 2008.

---

b. Estimate annual net sales in 2015.

---

c. Use a graph or trial and error to estimate the year in which annual net sales are expected to pass 8 billion dollars.

Problem 15 For Problems 15–16, find a possible formula for the function represented by the data.

x	0	1	2	3
f(x)	4.30	6.02	8.43	11.80

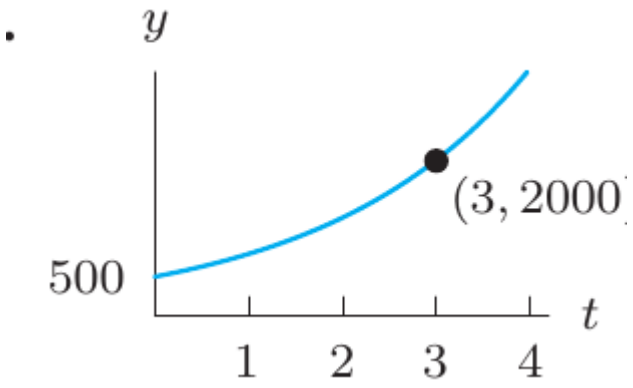
f(x) = ?

Problem 16 Find a possible formula for the function represented by the data.

t	0	1	2	3
g(t)	5.50	4.40	3.52	2.82

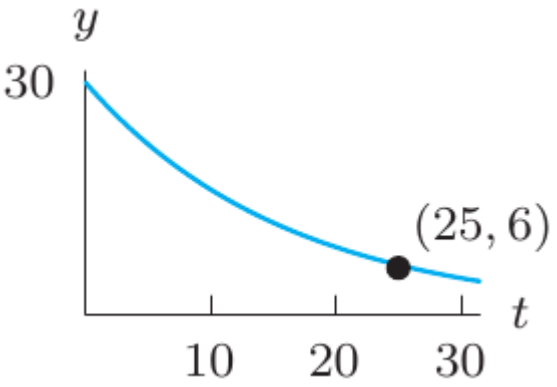
g(t) = ?

Problem 17 Give a possible formula for the function in the figure.



y = ?

Problem 18 Give a possible formula for the function in the figure.



y = ?

Problem 19 If the world's population increased exponentially from 4.453 billion in 1980 to 5.937 billion in 1998 and continued to increase at the same percentage rate between 1998 and 2008,

Calculate what the world's population would have been in 2008. How does this compare to the actual population of 6.771 billion?

Problem 20 The number of passengers using a railway fell from 190,205 to 174,989 during a 5-year period.

Find the annual percentage decrease over this period.

Problem 21 The company that produces Cliffs Notes (abridged versions of classic literature) was started in 1958 with $4000 and sold in 1998 for $14,000,000.

Find the annual percent increase in the value of this company over the 40 years.

Problem 30 Whooping cough was thought to have been almost wiped out by vaccinations. It is now known that the vaccination wears off, leading to an increase in the number of cases, w, from 1248 in 1981 to 18,957 in 2004.

a. With t in years since 1980, find an exponential function that fits this data.

b. What does your answer to part (a) give as the average annual percent growth rate of the number of cases?

c. On May 4, 2005, the Arizona Daily Star reported (correctly) that the number of cases had more than doubled between 2000 and 2004. Does your model confirm this report? Explain.

Problem 31 Aircraft require longer takeoff distances, called takeoff rolls, at high altitude airports because of diminished air density. The table shows how the takeoff roll for a certain light airplane depends on the airport elevation. (Takeoff rolls are also strongly influenced by air temperature; the data shown assume a temperature of 0° C.)

Elevation (ft)	Sea level	1000	2000	3000	4000
Takeoff roll (ft)	670	734	805	882	967

Determine a formula for this particular aircraft that gives the takeoff roll as an exponential function of airport elevation.

1.5.5 Test Your Understanding

Test Your Understanding

TYU 2 · P1 The amount of nicotine in the bloodstream is given by N, in mg, as a function of time, t, in hours. 2 mg of nicotine is absorbed after smoking a cigarette. Decide whether each statement is True or False.

1. If nicotine is decreasing by 1 mg per hour, then $N = -2t - 1$
2. If nicotine is decreasing by 0.5 mg per hour, then $N = -0.5t + 2$
3. If nicotine is decreasing by 50% per hour, then $N = 50(2)^t$
4. If nicotine is decreasing by 25% per hour, then $N = 2(0.75)^t$

TYU 2 · P2 The following table gives values of three functions f, g and h.

x	**1**	**3**	**5**	**7**	**9**
f(x)	1	3	9	27	81
g(x)	12	16	20	24	28
h(x)	14	11	8	5	2

a. Which of the following functions could be **linear**?

A. f(x)
B. g(x)
C. h(x)
D. g(x) and h(x)
E. None of the above

b. Which of the following functions could be **exponential**?

A. f(x)
B. g(x)
C. h(x)
D. f(x) and h(x)
E. None of the above

TYU 2 · P4 A population of 100 million pathogens increases by 2% every hour. Let P be the size of this population at t hours.

a. The mathematical function that best describes this behavior is

A. Hyperbolic
B. Logistic
C. Linear
D. Exponential
E. Power

b. What is the best formula to describe this model?

A. $P = 100(1.2)^t$
B. $P = 100(0.8)^t$
C. $P = 100(0.98)^t$
D. $P = 100(1.02)^t$
E. $P = 2t + 100$

TYU 2 · P5 The curve below describes the population size, y, after x years.

a. Find a formula for y

A. $y = 2t + 2$

B. $y = 4(2)^{t/2}$

C. $y = 16t / (1 + t)$

D. $y = 2(2)^{t}$

E. $y = 16 / (1 + e^{-2t})$

b. What is the population size after 5 years?

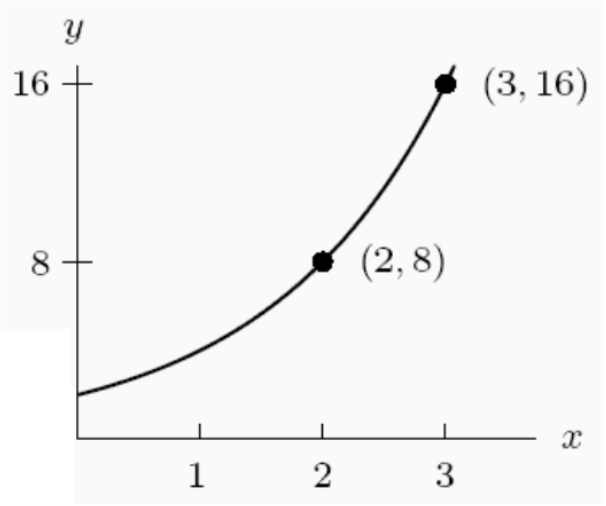
A. 2

B. 8

C. 32

D. 64

E. 128



TYU 3 · P2 The following table gives values of three functions f, g and h.

x	**1**	**3**	**5**	**7**	**9**
f(x)	1	3	9	27	81
g(x)	12	16	20	24	28
h(x)	32	16	8	4	2

a. Which of the following functions could be **exponential**?

A. f(x)
B. g(x)
C. h(x)
D. f(x) and h(x)
E. None of the above

b. What is the best formula to describe the function **h(x)**?

A. $P = 32(2)^t$
B. $P = 32(0.5)^t$
C. $P = 32(1/\sqrt{2})^t$
D. $P = 32\sqrt{2}(1/\sqrt{2})^t$
E. $P = \sqrt{2}t + 32$

Answers · Section 1.5

1.5.1 Growth Factor & Exponential Growth

Ex 1 **a = 1.034** — Solution:

$\Delta t = 2$ (2003 → 2005 → 2007 → 2009)

$P_2/P_0 = 13.747/12.853 = 1.0695$

$P_4/P_2 = 14.721/13.747 = 1.0695$

$a = (P_{n+\Delta t} / P_n)^{1/\Delta t} = (1.0695)^{1/2} = 1.034$

Another solution:

$P_t = P_0 a^t \Rightarrow P_t = 12.853a^t$

$t = 2$: $P_2 = 12.853a^2$

$13.747 = 12.853a^2$ (÷ 12.853)

$\sqrt{(a^2)} = \sqrt{1.0695}$

$a = 1.034$

Ex 2 **i. r = 1.26% per year** — $a = 1.0126$ (growth factor, $a > 1$)

$a = 1 + r$

growth rate $r = a - 1 = 1.0126 - 1 = 0.0126$ (× 100%)

$r = 1.26\%$

ii. P_0 = 6.4 billion — 2004 ⇒ $t = 0$

$P_0 = 6.4$ billion

OR: $P_0 = 6.4(1.0126)^0 = 6.4(1) = 6.4$ billion

iii. 6.9 billion — 2010 ⇒ $t = 6$

$P = 6.4(1.0126)^6 = 6.9$ billions

Problem 1 **a. (i), 12%** — Growth factors: 1.12, 1.03, 1.08, 0.90. The largest is 1.12 → town (i), r = 12%.

b. (ii), 1,000 — The initial population is the number in front (t = 0): the largest is 1,000 → town (ii).

c. Yes, (iv) — Only (iv) has a base less than 1 (0.90), so it decreases (10% per year).

Problem 2 **(a) Initial 100; growth; 7%** — $A_0 = 100$; $a = 1.07 > 1$ → growth; r = 7%.

(b) Initial 5.3; growth; 5.4% — $A_0 = 5.3$; $a = 1.054$ → growth; r = 5.4%.

(c) Initial 3500; decay; 7% — $A_0 = 3500$; $a = 0.93 < 1$ → decay; rate = 1 − 0.93 = 7%.

(d) Initial 12; decay; 12% — $A_0 = 12$; $a = 0.88$ → decay; rate = 1 − 0.88 = 12%.

Problem 8 **a. 1.26% per year** — $r = 1.0126 - 1 = 0.0126 = 1.26\%$ per year.

b. 6.4 billion; ≈ 6.9 billion — $t = 0$ → 6.4 billion; $t = 6$ → $6.4(1.0126)^6 \approx 6.9$ billion.

c. ≈ 0.083 billion people/year — ARC = (6.899 − 6.4)/(2010 − 2004) ≈ 0.499/6 ≈ 0.083 billion people/year.

Problem 11 **CPI = 211(1.028)^t** — $P_0 = 211$, $a = 1 + 0.028 = 1.028$: $CPI = 211(1.028)^t$.

Problem 13 **(a) 1, 2.718, 7.389, 20.086** — $e^0 = 1$, $e^1 \approx 2.718$, $e^2 \approx 7.389$, $e^3 \approx 20.086$.

(b) Growth — The values increase faster and faster: exponential growth.

(c) 1, 0.368, 0.135, 0.050 — $e^0 = 1$, $e^{-1} \approx 0.368$, $e^{-2} \approx 0.135$, $e^{-3} \approx 0.050$.

(d) Decay — The values decrease toward 0: exponential decay.

Problem 14 **Starts at 100 and decreases, concave up, getting closer to 0 (exponential decay)** — At x = 0, y = 100. The negative exponent makes it decrease, more and more slowly, approaching the x-axis: exponential decay, concave up.

1.5.2 Exponential Decay

Ex 3 **i. $A = 50(0.94)^t$** — $P_0 = 50$

r = 6% decay rate

$a = 1 - r = 1 - 6/100 = 1 - 0.06$

$a = 0.94$

$A_t = P_0 a^t$

$A_t = 50(0.94)^t$

ii. 11.33 mg — t = 24

$A_{24} = 50(0.94)^{24} = 11.33$ mg

iii. Starts at 50, decreasing and concave up, approaching 0 — The graph starts at A = 50 (when t = 0), and it is decreasing and concave up. It gets close to 0 but does not cross the t-axis.

iv. t ≈ 37 hours — Draw a horizontal line at A = 5 until it meets the curve, then go down to the t-axis. t ≈ 37 hours

Problem 9 **a. $A = 50(0.94)^t$** — $a = 1 - 0.06 = 0.94$: $A = 50(0.94)^t$.

b. ≈ 11.3 mg — $A = 50(0.94)^{24} \approx 11.3$ mg.

c. Starts at 50, decreasing and concave up, approaching 0 — Exponential decay from 50, concave up, through (24, 11.3).

d. ≈ 37 hours — A = 5 at about t = 37 hours.

Problem 12 **≈ 35.7%** — $a = 1 - 0.029 = 0.971$; $t = 2015 - 1980 = 35$. $0.971^{35} \approx 0.357 = 35.7\%$.

Problem 28 **(a) ≈ $3,486.78** — $10{,}000(0.9)^{10} \approx \$3{,}486.78$.

(b) ≈ 11.05 years — $3{,}486.78(1.1)^t = 10{,}000 \rightarrow 1.1^t \approx 2.868 \rightarrow t = \ln 2.868 / \ln 1.1 \approx 11.05$ years.

Problem 29 **a. 25%** — $0.8 \times (1 + r) = 1 \rightarrow 1 + r = 1.25 \rightarrow$ 25% enlargement.

b. 9 times — $0.8^8 \approx 0.168$ (still more than 15%); $0.8^9 \approx 0.134 < 0.15$. So 9 copies.

1.5.3 Linear vs. Exponential Functions

Ex 4 **(a)** **$A = 0.4t + 15$** — Constant increase ⇒ linear function

$y = mx + b$, $b = 15 = A_0$

$m = 0.4$ mg/min

$A = mt + A_0$

$A = 0.4t + 15$

(b) **$A = -0.4t + 15$** — Constant decrease ⇒ linear function

$m = -0.4$ mg/min

$A = -0.4t + 15$

(c) **$A = 15(1.03)^t$** — growth rate r = 3% = 0.03 (not constant change)

⇒ exponential function

⇒ exponential growth: $a = 1 + r = 1 + 0.03 = 1.03$

⇒ $A = A_0a^t$

⇒ $A = 15(1.03)^t$

(d) **$A = 15(0.97)^t$** — decay rate r = 3% = 0.03

exponential decay: $a = 1 - r$

⇒ $a = 1 - 0.03$

$a = 0.97$

⇒ $A = A_0a^t$

$A = 15(0.97)^t$

Problem 3

a. **II** — 5% per year → exponential growth, the slower one of the two exponential curves: II.

b. **I** — 8% → the faster-growing exponential curve: I.

c. **III** — Constant increase → increasing straight line: III.

d. **V** — Stable → horizontal line: V.

IV **Decreasing by a constant percent and leveling off** — IV is a decreasing curve that levels off: decreasing by a constant percent.

VI **Decreasing by a constant number of people per year** — VI is a decreasing straight line: decreasing by a constant number of people per year.

Problem 4

a. **$P = 1000 + 50t$** — Constant increase → linear: $P = 1000 + 50t$.

b. **$P = 1000(1.05)^t$** — Constant percent → exponential: $P = 1000(1.05)^t$.

Problem 5

(a) **$G = 310(1.03)^t$** — $G = 310(1.03)^t$.

(b) **$G = 310 + 8t$** — $G = 310 + 8t$.

Problem 6

(a) **$P = 80 - 4t$** — $P = 80 - 4t$.

(b) **$P = 80(0.95)^t$** — $a = 1 - 0.05 = 0.95$: $P = 80(0.95)^t$.

Problem 7

(a) **$Q = 30 - 2t$ (a decreasing straight line)** — Constant decrease → linear: $Q = 30 - 2t$, a straight line falling from 30 to 0 at t = 15.

(b) **$Q = 30(0.88)^t$ (a decreasing curve)** — $a = 1 - 0.12 = 0.88$: $Q = 30(0.88)^t$, an exponential decay curve, concave up, approaching 0.

Problem 22

a. N = 2700 + 486t; 486 mussels/year: increases by 486 each year — m = (3186 − 2700)/1 = 486 mussels/year: N = 2700 + 486t.

b. 18% — a = 3186/2700 = 1.18 → r = 18% per year.

Problem 23

a. W = 40,300 + 16,177.6t — m = (121,188 − 40,300)/5 = 80,888/5 = 16,177.6 MW/year: W = 40,300 + 16,177.6t.

b. $W = 40{,}300(1.246)^t$ — $121{,}188 = 40{,}300a^5 \rightarrow a^5 \approx 3.007 \rightarrow a \approx 1.246$: $W = 40{,}300(1.246)^t$.

c. Both pass through (0, 40,300) and (5, 121,188); the exponential is below the line between them — Both functions use the same two given values, so both pass through (0, 40,300) and (5, 121,188). Between them the concave-up exponential curve lies below the straight line.

Problem 24

(a) h: h(x) = 31 − 3x — h decreases by 3 each step: linear with m = −3 and h(0) = 31 → h(x) = 31 − 3x. f and g are not linear.

(b) g: $g(x) = 36(1.5)^x$ — g: ratios 24/16 = 36/24 = 54/36 = 81/54 = 1.5, and g(0) = 36 → $g(x) = 36(1.5)^x$.

Problem 25

(a) Neither — Differences 2.2, 6.2, 17.8 (not constant); ratios 1.21, 1.49, 1.94 (not constant) → neither.

(b) Exponential: $s(t) = 30.12(0.6)^t$ — Ratios 30.12/50.2 = 18.072/30.12 = 0.6 (constant); s(0) = 30.12 → $s(t) = 30.12(0.6)^t$.

(c) Linear: g(u) = 27 − 1.5u — g decreases by 3 for every 2 units of u: m = −3/2 = −1.5 → g(u) = 27 − 1.5u.

Problem 26

(a) Exponential — Ratios: 170.3/161 ≈ 180.2/170.3 ≈ … ≈ 1.058 (constant), but differences grow (9.3, 9.9, 10.5, …) → exponential.

(b) $P = 161(1.058)^t$ — $P_0 = 161$, a ≈ 1.058: $P = 161(1.058)^t$.

(c) ≈ 5.8% — a − 1 ≈ 0.058 = 5.8% per year.

Problem 27

Yes: inflation alone gives ≈ $4.02, less than the actual $5.15 — $0.25(1.043)^{66} \approx \$4.02$ (66 years from 1938 to 2004). The actual $5.15 is more, so the minimum wage kept pace with inflation.

2022 Q2

(a) A = 10 − 0.6t — Constant decrease → linear: A = 10 − 0.6t.

(b) $A = 10(1.1)^t$ — Constant percent increase → exponential: a = 1.1, $A = 10(1.1)^t$.

1.5.4 Finding Exponential Formulas from Data

Ex 5 **$P = 14(1.172)^t$; r = 17.2%** — In 1995, (t = 0): $P_0 = 14$
In 2012, (t = 17): $P_{17} = 207$
$P_t = P_0a^t \Rightarrow P_t = 14a^t$
$t = 17 \Rightarrow 207 = 14a^{17}$ (÷ 14)
$\sqrt[17]{(a^{17})} = \sqrt[17]{(207/14)} = (207/14)^{1/17}$
a = 1.172 (exponential growth)
$P = 14(1.172)^t$
$r = a - 1 \Rightarrow r = 0.172$ (× 100%)
r = 17.2%

Ex 6 **a. a = 1.012** — $Q = Q_0a^t$
$t = 20 \rightarrow Q = 88.2 \Rightarrow 88.2 = Q_0a^{20}$
$t = 23 \rightarrow Q = 91.4 \Rightarrow 91.4 = Q_0a^{23}$
$91.4/88.2 = Q_0a^{23}/Q_0a^{20}$ (Q_0 cancels)
$a^3 = 91.4/88.2$
$\sqrt[3]{(a^3)} = \sqrt[3]{(91.4/88.2)}$
a = 1.012
(or: $a = (Q_{23}/Q_{20})^{1/(23-20)}$)
b. r = 1.2% — r = a − 1
= 1.012 − 1
= 0.012 (× 100%)
r = 1.2%
c. f(25) = 93.61 — Find Q_0: $Q_t = Q_0(1.012)^t$
t = 20: $Q_{20} = 88.2$
$\Rightarrow 88.2 = Q_0(1.012)^{20}$ (÷ $(1.012)^{20}$)
$Q_0 = 69.47$

$Q_t = 69.47(1.012)^t$
t = 25: $Q_{25} = 69.47(1.012)^{25}$
= 93.61

Problem 10 **a. S = 5.1e^(0.043t)** — Continuous rate → $S = S_0e^{(kt)}$ with k = 0.043: S = 5.1e^(0.043t).
b. ≈ 6.89 billion dollars — t = 7: S = 5.1e^(0.043·7) = 5.1e^0.301 ≈ 6.89 billion dollars.
c. 2018 — 5.1e^(0.043t) = 8 → t = ln(8/5.1)/0.043 ≈ 10.5 → in 2018.

Problem 15 **$f(x) = 4.30(1.4)^x$** — Ratios: 6.02/4.30 ≈ 8.43/6.02 ≈ 11.80/8.43 ≈ 1.4; f(0) = 4.30 → $f(x) = 4.30(1.4)^x$.

Problem 16 **$g(t) = 5.50(0.8)^t$** — Ratios: 4.40/5.50 = 3.52/4.40 ≈ 0.8; g(0) = 5.50 → $g(t) = 5.50(0.8)^t$.

Problem 17 **$y = 500(1.587)^t$** — $y = 500a^t$; $2000 = 500a^3 \rightarrow a^3 = 4 \rightarrow$ a = 4^(1/3) ≈ 1.587: $y = 500(1.587)^t$.

Problem 18 **$y = 30(0.938)^t$** — $y = 30a^t$; $6 = 30a^{25} \rightarrow a^{25} = 0.2 \rightarrow$ a = 0.2^(1/25) ≈ 0.938: $y = 30(0.938)^t$.

Problem 19 **≈ 6.97 billion, higher than the actual 6.771 billion** — $a^{18} = 5.937/4.453 \rightarrow a \approx 1.0161$. 2008: $5.937(1.0161)^{10} \approx 6.97$ billion, higher than the actual 6.771 billion.

Problem 20 **≈ 1.65% per year** — $a^5 = 174{,}989/190{,}205 \approx 0.920 \rightarrow a \approx 0.9835 \rightarrow$ decrease ≈ 1.65% per year.

Problem 21 **≈ 22.6% per year** — $a^{40} = 14{,}000{,}000/4000 = 3500 \rightarrow$ a = 3500^(1/40) ≈ 1.226 → ≈ 22.6% per year.

Problem 30

a. $w = 1109(1.126)^t$ — 1981 → t = 1, 2004 → t = 24. $a^{23} = 18{,}957/1248 \approx 15.19 \rightarrow a \approx 1.126$. $w_0 = 1248/1.126 \approx 1109$: $w = 1109(1.126)^t$.

b. ≈ 12.6% — a − 1 ≈ 0.126 = 12.6% per year.

c. No: the model gives only about a 61% increase — Over 4 years the model multiplies by $1.126^4 \approx 1.61$, only about a 61% increase, not a doubling.

Problem 31 **R = 670(1.096)^(h/1000)** — Each 1000 ft multiplies the roll by about 1.096 (734/670 ≈ 805/734 ≈ … ≈ 1.096). With h in feet: R = 670(1.096)^(h/1000).

1.5.5 Test Your Understanding

TYU 2 · P1 Initial amount N(0) = 2 mg. Linear: N = 2 − (decrease per hour)·t. Exponential: $N = 2(1 - r)^t$.

1. False — decreasing by 1 mg/hour gives N = −t + 2 (slope −1, intercept 2).

2. True — slope −0.5, initial value 2.

3. False — decreasing by 50% per hour gives $N = 2(0.5)^t$.

4. True — b = 1 − 0.25 = 0.75, initial value 2.

TYU 2 · P2 **(a)** The x-values go up by 2 each time.

g: 16 − 12 = 20 − 16 = … = 4 → constant differences → linear.

h: 11 − 14 = 8 − 11 = … = −3 → constant differences → linear.

f: differences 2, 6, 18, 54 → not linear.

Answer: D — g(x) and h(x)

(b) f: 3/1 = 9/3 = 27/9 = 81/27 = 3 → constant ratios → exponential.

h: 11/14 ≈ 0.786, 8/11 ≈ 0.727 → ratios not constant → not exponential. g: 16/12 ≈ 1.33, 20/16 = 1.25 → not exponential.

Answer: A — f(x)

TYU 2 · P4 **(a)** The population increases by a fixed percentage (2%) every hour → exponential.

Answer: D — Exponential

(b) $P = P_0(1 + r)^t$ with $P_0 = 100$ and r = 0.02:

$$P = 100(1.02)^t$$

Answer: D

TYU 2 · P5 **(a)** The curve is exponential through (2, 8) and (3, 16): $y = ab^t$.

$$b = \frac{16}{8} = 2$$

$$a(2)^2 = 8 \Rightarrow a = 2$$

$$y = 2(2)^t$$

Answer: D

(b)

$$y(5) = 2(2)^5 = 2(32) = 64$$

Answer: D — 64

TYU 3 · P2 **(a)** f: ratios 3/1 = 9/3 = … = 3 (constant). h: ratios 16/32 = 8/16 = … = 1/2 (constant). g has constant differences (linear).

Answer: D — f(x) and h(x)

(b) h is halved every 2 units of x, so per 1 unit the factor is b with $b^2 = 1/2$:

$$b = \sqrt{(1/2)} = 1/\sqrt{2}$$

$$h(1) = a(1/\sqrt{2})^1 = 32 \Rightarrow a = 32\sqrt{2}$$

$$P = 32\sqrt{2}(1/\sqrt{2})^t$$

Check: t = 3 → $32\sqrt{2} / (2\sqrt{2}) = 16$ ✓

Answer: D

1.6 The Natural Logarithm

In this section

1. The Natural Logarithm & Solving Equations 3 lecture · 16 textbook
2. Exponential Functions with Base e 1 lecture · 19 textbook
3. Continuous Growth & Decay Models 2 lecture · 6 textbook
4. Test Your Understanding 1 questions

1.6.1 The Natural Logarithm & Solving Equations

Lecture Problems

Ex 1 Find t such that $3^t = 10$.

t = ?

Ex 2 Find t such that $12 = 5e^{3t}$.

t = ?

Ex 3 If t is in years since 2000, the population of Nevada (in millions) can be modeled by the function $P = f(t) = 2.020(1.036)^t$.

After how many years will the population of Nevada reach 4 million?

Textbook Problems

Problem 1 For Problems 1–16, solve for t using natural logarithms.

$5^t = 7$

Problem 2 Solve for t using natural logarithms.

$10 = 2^t$

Problem 3 Solve for t using natural logarithms.

$2 = (1.02)^t$

Problem 4 Solve for t using natural logarithms.

$130 = 10^t$

Problem 5 Solve for t using natural logarithms.

$50 = 10 \cdot 3^t$

Problem 6 Solve for t using natural logarithms.

$100 = 25(1.5)^t$

Problem 7 Solve for t using natural logarithms.

$10 = e^t$

Problem 8 Solve for t using natural logarithms.

$5 = 2e^t$

Problem 9 Solve for t using natural logarithms.

$e^{3t} = 100$

Problem 10 Solve for t using natural logarithms.

$10 = 6e^{0.5t}$

Problem 11 Solve for t using natural logarithms.

$40 = 100e^{-0.03t}$

Problem 12 Solve for t using natural logarithms.

$a = b^t$

Problem 13 Solve for t using natural logarithms.

$B = Pe^{rt}$

Problem 14 Solve for t using natural logarithms.

$2P = Pe^{0.3t}$

Problem 15 Solve for t using natural logarithms.

$5e^{3t} = 8e^{2t}$

Problem 16 Solve for t using natural logarithms.

$7 \cdot 3^t = 5 \cdot 2^t$

1.6.2 Exponential Functions with Base e

Lecture Problems

Ex 4

i. Convert the function $P = 1000e^{0.05t}$ to the form $P = P_0a^t$.

ii. Convert the function $P = 500(1.06)^t$ to the form $P = P_0e^{kt}$.

Textbook Problems

Problem 17 The functions in Problems 17–20 represent exponential growth or decay. What is the initial quantity? What is the growth rate? State if the growth rate is continuous.

$P = 5(1.07)^t$

Problem 18 What is the initial quantity? What is the growth rate? State if the growth rate is continuous.

$P = 7.7(0.92)^t$

Problem 19 What is the initial quantity? What is the growth rate? State if the growth rate is continuous.

$P = 15e^{-0.06t}$

Problem 20 What is the initial quantity? What is the growth rate? State if the growth rate is continuous.

$P = 3.2e^{0.03t}$

Problem 23 Write the functions in Problems 23–26 in the form $P = P_0a^t$. Which represent exponential growth and which represent exponential decay?

$P = 15e^{0.25t}$

Problem 24 Write the function in the form $P = P_0a^t$. Does it represent exponential growth or decay?

$P = 2e^{-0.5t}$

Problem 25 Write the function in the form $P = P_0a^t$. Does it represent exponential growth or decay?

$P = P_0e^{0.2t}$

Problem 26 Write the function in the form $P = P_0a^t$. Does it represent exponential growth or decay?

$P = 7e^{-\pi t}$

Problem 27 In Problems 27–30, put the functions in the form $P = P_0e^{kt}$.

$P = 15(1.5)^t$

Problem 28 Put the function in the form $P = P_0e^{kt}$.

$P = 10(1.7)^t$

Problem 29 Put the function in the form $P = P_0e^{kt}$.

$P = 174(0.9)^t$

Problem 30 Put the function in the form $P = P_0e^{kt}$.

$P = 4(0.55)^t$

Problem 31 The population of the world can be represented by $P = 6.4(1.0126)^t$, where P is in billions of people and t is years since 2004.

Find a formula for the population of the world using a continuous growth rate.

Problem 34

(a) What is the continuous percent growth rate for $P = 100e^{0.06t}$, with time, t, in years?

(b) Write this function in the form $P = P_0a^t$. What is the annual percent growth rate?

Problem 35

(a) What is the annual percent decay rate for $P = 25(0.88)^t$, with time, t, in years?

(b) Write this function in the form $P = P_0e^{kt}$. What is the continuous percent decay rate?

Problem 36 The gross world product is $W = 32.4(1.036)^t$, where W is in trillions of dollars and t is years since 2001.

Find a formula for gross world product using a continuous growth rate.

Problem 37 The population, P, in millions, of Nicaragua was 5.4 million in 2004 and growing at an annual rate of 3.4%. Let t be time in years since 2004.

a. Express P as a function in the form $P = P_0a^t$.

---

b. Express P as an exponential function using base e.

---

c. Compare the annual and continuous growth rates.

Problem 38

What annual percent growth rate is equivalent to a continuous percent growth rate of 8%?

Problem 39

What continuous percent growth rate is equivalent to an annual percent growth rate of 10%?

1.6.3 Continuous Growth & Decay Models

Lecture Problems

Ex 5 Sketch graphs of

i. $P = e^{0.5t}$, a continuous growth rate of 50%

---

ii. $Q = 5e^{-0.2t}$, a continuous decay rate of 20%.

Ex 6

Same as Textbook Problem 21

The following formulas give the populations of four different towns, A, B, C, and D, with t in years from now.
A: $P_A = 600e^{0.08t}$; B: $P_B = 1000e^{-0.02t}$; C: $P_C = 1200e^{0.03t}$; D: $P_D = 900e^{0.12t}$

a. Which town is growing fastest (that is, has the largest percentage growth rate)?

b. Which town is the largest now?

c. Are any of the towns decreasing in size? If so, which one(s)?

Textbook Problems

Problem 21

Same as Lecture Example 6

The following formulas give the populations of four different towns, A, B, C, and D, with t in years from now.
A: $P_A = 600e^{0.08t}$; B: $P_B = 1000e^{-0.02t}$; C: $P_C = 1200e^{0.03t}$; D: $P_D = 900e^{0.12t}$

a. Which town is growing fastest (that is, has the largest percentage growth rate)?

b. Which town is the largest now?

c. Are any of the towns decreasing in size? If so, which one(s)?

Problem 22 A city's population is 1000 and growing at 5% a year.

a. (i) Find a formula for the population at time t years from now assuming that the 5% per year is an annual rate.

a. (ii) Find a formula assuming that the 5% per year is a continuous annual rate.

b. (i) Estimate the population of the city in 10 years with the annual rate.

b. (ii) Estimate the population of the city in 10 years with the continuous rate.

Problem 32 A fishery stocks a pond with 1000 young trout. The number of trout t years later is given by $P(t) = 1000e^{-0.5t}$.

a. How many trout are left after six months? After 1 year?

b. Find P(3) and interpret it in terms of trout.

c. At what time are there 100 trout left?

d. Graph the number of trout against time, and describe how the population is changing. What might be causing this?

Problem 33 During a recession a firm's revenue declines continuously so that the revenue, R (measured in millions of dollars), in t years' time is given by $R = 5e^{-0.15t}$.

a. Calculate the current revenue and the revenue in two years' time.

b. After how many years will the revenue decline to $2.7 million?

Problem 40 The population of a city is 50,000 in 2008 and is growing at a continuous yearly rate of 4.5%.

a. Give the population of the city as a function of the number of years since 2008. Sketch a graph of the population against time.

b. What will be the city's population in the year 2018?

c. Calculate the time for the population of the city to reach 100,000 (the doubling time).

Problem 41 In 2000, there were about 213 million vehicles (cars and trucks) and about 281 million people in the US. The number of vehicles has been growing at 4% a year, while the population has been growing at 1% a year.

If the growth rates remain constant, when is there, on average, one vehicle per person?

1.6.4 Test Your Understanding

Test Your Understanding

TYU 3 · P3 The curve below describes the population size, y, after x years.

a. Find a formula for y

A. $y = 4(2)^t$

B. $y = 4(2)^{t/2}$

C. $y = 4(0.5)^t$

D. $y = 2(2)^t$

E. $y = 2(4)^t$

b. Another equivalent formula for y is ……

A. $y = 4e^{0.693t}$

B. $y = 4e^{0.347t}$

C. $y = 4e^{-0.693t}$

D. $y = 2e^{0.693t}$

E. $y = 2e^{1.386t}$

c. The doubling-time of this population is ……

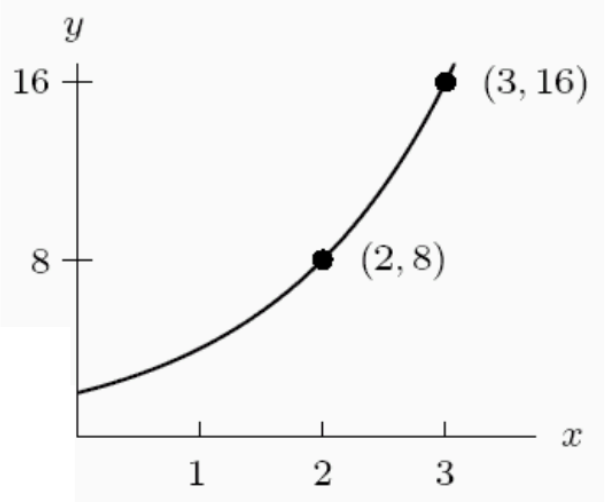
A. 1

B. ln 2 / 2

C. ln 2 / 4

D. 2

E. There is no doubling-time



Answers · Section 1.6

1.6.1 The Natural Logarithm & Solving Equations

Ex 1 **t = 2.096** — $\ln(3^t) = \ln 10$
$t \ln(3) = \ln(10)$ ($\div \ln(3)$)
$t = \ln(10) / \ln(3) = 2.096$

Ex 2 **t = 0.292** — $12 = 5e^{3t}$ ($\div 5$)
$\ln e^{3t} = \ln(12/5)$
$3t \cdot \ln e = \ln(12/5)$ ($\ln e = 1$)
$3t = \ln(12/5)$ ($\div 3$)
$t = 0.292$ ($t = 0.2918$)

Ex 3 **t = 19.32 years** — Find t, P = 4 millions
$4 = 2.020(1.036)^t$ ($\div 2.020$)
$\ln(1.036)^t = \ln(4/2.020)$
$t \ln(1.036) = \ln(4/2.020)$ ($\div \ln(1.036)$)
$t = 19.32$ years ($t = 19.317$)
It means after 2019, the population will reach 4 millions.

Problem 1 **t ≈ 1.209** — $t \ln 5 = \ln 7 \rightarrow t = \ln 7 / \ln 5 \approx 1.209$.

Problem 2 **t ≈ 3.322** — $t = \ln 10 / \ln 2 \approx 3.322$.

Problem 3 **t ≈ 35.00** — $t = \ln 2 / \ln 1.02 \approx 35.00$.

Problem 4 **t ≈ 2.114** — $t = \ln 130 / \ln 10 \approx 2.114$.

Problem 5 **t ≈ 1.465** — $5 = 3^t \rightarrow t = \ln 5 / \ln 3 \approx 1.465$.

Problem 6 **t ≈ 3.419** — $4 = 1.5^t \rightarrow t = \ln 4 / \ln 1.5 \approx 3.419$.

Problem 7 **t ≈ 2.303** — $t = \ln 10 \approx 2.303$.

Problem 8 **t ≈ 0.916** — $2.5 = e^t \rightarrow t = \ln 2.5 \approx 0.916$.

Problem 9 **t ≈ 1.535** — $3t = \ln 100 \rightarrow t = \ln 100 / 3 \approx 1.535$.

Problem 10 **t ≈ 1.022** — $10/6 = e^{0.5t} \rightarrow 0.5t = \ln(1.667) \rightarrow t \approx 1.022$.

Problem 11 **t ≈ 30.54** — $0.4 = e^{-0.03t} \rightarrow -0.03t = \ln 0.4 \rightarrow t = \ln 0.4 / (-0.03) \approx 30.54$.

Problem 12 **t = ln a / ln b** — $\ln a = t \ln b \rightarrow t = \ln a / \ln b$.

Problem 13 **t = ln(B/P) / r** — $B/P = e^{rt} \rightarrow rt = \ln(B/P) \rightarrow t = \ln(B/P) / r$.

Problem 14 **t ≈ 2.31** — $2 = e^{0.3t} \rightarrow t = \ln 2 / 0.3 \approx 2.31$.

Problem 15 **t ≈ 0.470** — $e^{3t}/e^{2t} = 8/5 \rightarrow e^t = 1.6 \rightarrow t = \ln 1.6 \approx 0.470$.

Problem 16 **t ≈ −0.830** — $3^t/2^t = 5/7 \rightarrow (1.5)^t = 5/7 \rightarrow t = \ln(5/7)/\ln 1.5 \approx -0.830$.

1.6.2 Exponential Functions with Base e

Ex 4

i. **$P = 1000(1.051)^t$** — Given $P = P_0e^{kt}$

$P_0 = 1000$, $k = 0.05$

$a = e^k = e^{0.05}$

$a = 1.051$

$P = 1000(1.051)^t$

ii. **$P = 500e^{0.058t}$** — Given $P = P_0a^t$

$P_0 = 500$, $a = 1.06$

$k = \ln(1.06) = 0.0583$

$P = 500e^{0.058t}$

Problem 17 **Initial 5; growth 7%; not continuous** — $P_0 = 5$; $a = 1.07 \rightarrow$ growth rate 7% per year (annual, not continuous).

Problem 18 **Initial 7.7; decay 8%; not continuous** — $P_0 = 7.7$; $a = 0.92 \rightarrow$ decay rate $1 - 0.92 = 8\%$ (not continuous).

Problem 19 **Initial 15; decay 6%; continuous** — $P_0 = 15$; $k = -0.06 \rightarrow$ continuous decay rate 6%.

Problem 20 **Initial 3.2; growth 3%; continuous** — $P_0 = 3.2$; $k = 0.03 \rightarrow$ continuous growth rate 3%.

Problem 23 **$P = 15(1.284)^t$; growth** — $a = e^{0.25} \approx 1.284 > 1 \rightarrow$ growth.

Problem 24 **$P = 2(0.607)^t$; decay** — $a = e^{-0.5} \approx 0.607 < 1 \rightarrow$ decay.

Problem 25 **$P = P_0(1.221)^t$; growth** — $a = e^{0.2} \approx 1.221 \rightarrow$ growth.

Problem 26 **$P = 7(0.0432)^t$; decay** — $a = e^{-\pi} \approx 0.0432 \rightarrow$ decay.

Problem 27 **$P = 15e^{0.405t}$** — $k = \ln 1.5 \approx 0.405 \rightarrow P = 15e^{0.405t}$.

Problem 28 **$P = 10e^{0.531t}$** — $k = \ln 1.7 \approx 0.531 \rightarrow P = 10e^{0.531t}$.

Problem 29 **$P = 174e^{-0.105t}$** — $k = \ln 0.9 \approx -0.105 \rightarrow P = 174e^{-0.105t}$.

Problem 30 **$P = 4e^{-0.598t}$** — $k = \ln 0.55 \approx -0.598 \rightarrow P = 4e^{-0.598t}$.

Problem 31 **$P = 6.4e^{0.0125t}$** — $k = \ln 1.0126 \approx 0.0125 \rightarrow P = 6.4e^{0.0125t}$.

Problem 34 **(a) 6%** — k = 0.06 → continuous growth rate 6% per year.

(b) $P = 100(1.0618)^t$; 6.18% — $a = e^{0.06} \approx 1.0618$ → $P = 100(1.0618)^t$; annual rate ≈ 6.18%.

Problem 35 **(a) 12%** — Decay rate = 1 − 0.88 = 0.12 = 12% per year.

(b) $P = 25e^{-0.128t}$; 12.8% — $k = \ln 0.88 \approx -0.128$ → $P = 25e^{-0.128t}$; continuous decay rate ≈ 12.8%.

Problem 36 **$W = 32.4e^{0.0354t}$** — $k = \ln 1.036 \approx 0.0354$ → $W = 32.4e^{0.0354t}$.

Problem 37 **a. $P = 5.4(1.034)^t$** — $a = 1 + 0.034 = 1.034$ → $P = 5.4(1.034)^t$.

b. $P = 5.4e^{0.0334t}$ — $k = \ln 1.034 \approx 0.0334$ → $P = 5.4e^{0.0334t}$.

c. The continuous rate is slightly less than the annual rate — $k = \ln 1.034 \approx 0.0334$, so the continuous rate (3.34%) is slightly smaller than the annual rate (3.4%).

Problem 38 **≈ 8.33%** — $a = e^{0.08} \approx 1.0833$ → annual rate ≈ 8.33%.

Problem 39 **≈ 9.53%** — $k = \ln 1.1 \approx 0.0953$ → ≈ 9.53%.

1.6.3 Continuous Growth & Decay Models

Ex 5 **i. Starts at 1, increasing and concave up** — $t = 0 \Rightarrow P_0 = 1$

$k = 0.5$ (+) ⇒ increasing

$P_0 > 0$ ⇒ concave up

Same shape as the previous exponential growth curves: increasing and concave up.

ii. Starts at 5, decreasing and concave up — $t = 0 \Rightarrow P_0 = 5e^0 = 5$

$k = -0.2$ ⇒ decreasing

$P_0 > 0$ ⇒ concave up

Same shape as other exponential decay functions.

Ex 6 **a. Town D** — A: $P_0 = 600$, $k = 0.08$ · B: $P_0 = 1000$, $k = -0.02$ · C: $P_0 = 1200$, $k = 0.03$ · D: $P_0 = 900$, $k = 0.12$

Town D, because k is largest.

b. Town C — Town C, because it has largest P_0.

c. Town B — Town B, because $k < 0$.

Problem 21 **a. D (12%)** — Town D, because its k = 0.12 is the largest.

b. C (1200) — Town C, because it has the largest P_0 (1200).

c. Yes, B — Town B, because its $k = -0.02 < 0$.

Problem 22 **a. (i) $P = 1000(1.05)^t$** — Annual rate: $a = 1.05$ → $P = 1000(1.05)^t$.

a. (ii) $P = 1000e^{0.05t}$ — Continuous rate: $k = 0.05$ → $P = 1000e^{0.05t}$.

b. (i) ≈ 1629 — $1000(1.05)^{10} \approx 1629$.

b. (ii) ≈ 1649 — $1000e^{0.5} \approx 1649$.

Problem 32

a. **≈ 779; ≈ 607** — Six months → $t = 0.5$: $1000e^{-0.25} \approx 779$. One year → $t = 1$: $1000e^{-0.5} \approx 607$.

b. **P(3) ≈ 223: about 223 trout are left after 3 years** — $P(3) = 1000e^{-1.5} \approx 223$ trout remain after 3 years.

c. **≈ 4.6 years** — $100 = 1000e^{-0.5t} \rightarrow 0.1 = e^{-0.5t} \rightarrow t = \ln 0.1 / (-0.5) \approx 4.6$ years.

d. **Decreasing and leveling off (e.g., fishing or deaths)** — The graph decreases from 1000 and levels off toward 0 (concave up). Possible causes: fishing or deaths.

Problem 33

a. **$5 million; ≈ $3.70 million** — Now ($t = 0$): $R = 5$. $t = 2$: $R = 5e^{-0.3} \approx \$3.70$ million.

b. **≈ 4.1 years** — $2.7 = 5e^{-0.15t} \rightarrow 0.54 = e^{-0.15t} \rightarrow t = \ln 0.54 / (-0.15) \approx 4.1$ years.

Problem 40

a. **$P = 50{,}000e^{0.045t}$ (increasing, concave up)** — Continuous rate $k = 0.045$: $P = 50{,}000e^{0.045t}$, an increasing, concave-up curve starting at (0, 50,000).

b. **≈ 78,416** — $t = 10$: $50{,}000e^{0.45} \approx 78{,}416$.

c. **≈ 15.4 years** — $100{,}000 = 50{,}000e^{0.045t} \rightarrow 2 = e^{0.045t} \rightarrow t = \ln 2 / 0.045 \approx 15.4$ years.

Problem 41

t ≈ 9.5 years (between 2009 and 2010) — $213(1.04)^t = 281(1.01)^t \rightarrow (1.04/1.01)^t = 281/213 \rightarrow t = \ln(1.319)/\ln(1.0297) \approx 9.5$ years.

1.6.4 Test Your Understanding

TYU 3 · P3

(a) Through (2, 8) and (3, 16):

$$b = \frac{16}{8} = 2, \quad a(2)^2 = 8 \Rightarrow a = 2$$

$$y = 2(2)^t$$

Answer: D

(b)

$$k = \ln b = \ln 2 \approx 0.693 \Rightarrow y = 2e^{0.693t}$$

Answer: D

(c) $b = 2$ means the population doubles every 1 year:

$$2(2)^t = 4 \Rightarrow t = \frac{\ln 2}{0.693} = 1 \text{ year}$$

Answer: A — 1

1.7 Exponential Growth and Decay

In this section

1. Exponential Growth & Decay with Base e 2 lecture · 9 textbook
2. Doubling Time & Half-Life 6 lecture · 15 textbook · 3 past exam
3. Compound Interest, Present & Future Value 11 textbook
4. Test Your Understanding 6 questions

1.7.1 Exponential Growth & Decay with Base e

Lecture Problems

Ex 1 The Environmental Protection Agency (EPA) recently investigated a spill of radioactive iodine. The radiation level at the site was about 2.4 millirems/hour (four times the maximum acceptable limit of 0.6 millirems/hour), so the EPA ordered an evacuation of the surrounding area. The level of radiation from an iodine source decays at a continuous hourly rate of $k = -0.004$.

(a) What was the level of radiation 24 hours later?

(b) Find the number of hours until the level of radiation reached the maximum acceptable limit, and the inhabitants could return.

Ex 2 The population of Kenya was 19.5 million in 1984 and 39.0 million in 2009.

Assuming the population increases exponentially, find a formula for the population of Kenya as a function of time.

Textbook Problems

Problem 1 World wind energy generating capacity, W, was 18,000 megawatts in 2000 and has been increasing at a continuous rate of approximately 27% per year. Assume this rate continues.

a. Give a formula for W, in megawatts, as a function of time, t, in years since 2000.

b. When is wind capacity predicted to pass 250,000 megawatts?

Problem 14 The exponential function $y(x) = Ce^{\alpha x}$ satisfies the conditions $y(0) = 2$ and $y(1) = 1$.

Find the constants C and α. What is $y(2)$?

Problem 15 Air pressure, P, decreases exponentially with the height, h, in meters above sea level: $P = P_0 e^{-0.00012h}$, where P_0 is the air pressure at sea level.

a. At the top of Mount McKinley, height 6194 meters (about 20,320 feet), what is the air pressure, as a percent of the pressure at sea level?

---

b. The maximum cruising altitude of an ordinary commercial jet is around 12,000 meters (about 39,000 feet). At that height, what is the air pressure, as a percent of the sea level value?

Problem 17 A firm decides to increase output at a constant relative rate from its current level of 20,000 to 30,000 units during the next five years.

Calculate the annual percent rate of increase required to achieve this growth.

Problem 20 The number of people living with HIV infections increased worldwide approximately exponentially from 2.5 million in 1985 to 37.8 million in 2003. (HIV is the virus that causes AIDS.)

a. Give a formula for the number of HIV infections, H, (in millions) as a function of years, t, since 1985. Use the form $H = H_0 e^{kt}$. Graph this function.

---

b. What was the yearly continuous percent change in the number of HIV infections between 1985 and 2003?

Problem 22 An exponentially growing animal population numbers 500 at time $t = 0$; two years later, it is 1500.

Find a formula for the size of the population in t years and find the size of the population at $t = 5$.

Problem 26 In 1923, koalas were introduced on Kangaroo Island off the coast of Australia. In 1996, the population was 5000. By 2005, the population had grown to 27,000, prompting a debate on how to control their growth and avoid koalas dying of starvation.

Assuming exponential growth, find the (continuous) rate of growth of the koala population between 1996 and 2005. Find a formula for the population as a function of the number of years since 1996, and estimate the population in the year 2020.

Problem 27 The total world marine catch in 1950 was 17 million tons and in 2001 was 99 million tons.

If the marine catch is increasing exponentially, find the (continuous) rate of increase. Use it to predict the total world marine catch in the year 2020.

Problem 30 In 2004, the world's population was 6.4 billion, and the population was projected to reach 8.5 billion by the year 2030.

What annual growth rate is projected?

1.7.2 Doubling Time & Half-Life

Lecture Problems

Ex 1 (Doubling Time & Half-Life) Let $Q = Q_0 a^t$, $a > 1$.

Show that the doubling time for Q is $t = \ln 2 / \ln a$. Which derivation is correct?

Ex 2 (Doubling Time & Half-Life)

Same as Textbook Problem 7

Find the doubling time of a quantity that is increasing by 7% per year.

$t = ?$

Ex 3 (Doubling Time & Half-Life) Let $Q = Q_0 a^t$, $0 < a < 1$.

Show that the half-life for Q is $t = -\ln 2 / \ln a$. Which derivation is correct?

Ex 4 (Doubling Time & Half-Life) The release of chlorofluorocarbons used in air conditioners and household sprays (hair spray, shaving cream, etc.) destroys the ozone in the upper atmosphere. The quantity of ozone, Q, is decaying exponentially at a continuous rate of 0.25% per year.

What is the half-life of ozone? In other words, at this rate, how long will it take for half the ozone to disappear?

Ex 5 (Doubling Time & Half-Life)

Same as Textbook Problem 16

The antidepressant fluoxetine (or Prozac) has a half-life of about 3 days.

What percentage of a dose remains in the body after one day (t = 1)? After one week (t = 7)?

Ex 6 (Doubling Time & Half-Life)

Same as Textbook Problem 23

If the quantity of a substance decreases by 4% in 10 hours,

find its half-life.

Textbook Problems

Problem 2 The half-life of nicotine in the blood is 2 hours. A person absorbs 0.4 mg of nicotine by smoking a cigarette.

t (hours)	0	2	4	6	8	10
Nicotine (mg)	0.4	?	?	?	?	?

Table Fill in the table with the amount of nicotine remaining in the blood after t hours.

Time Estimate the length of time until the amount of nicotine is reduced to 0.04 mg.

Problem 5 If a bank pays 6% per year interest compounded continuously,

how long does it take for the balance in an account to double?

Problem 7

Same as Lecture Example 2 (Doubling Time and Half-Life)

Find the doubling time of a quantity that is increasing by 7% per year.

t = ?

Problem 10 From October 2002 to October 2006 the number N(t) of Wikipedia articles was approximated by $N(t) = N_0 e^{t/500}$, where t is the number of days after October 1, 2002.

Find the doubling time for the number of Wikipedia articles during this period.

Problem 11 A cup of coffee contains 100 mg of caffeine, which leaves the body at a continuous rate of 17% per hour.

a. Write a formula for the amount, A mg, of caffeine in the body t hours after drinking a cup of coffee.

---

b. Graph the function from part (a). Use the graph to estimate the half-life of caffeine.

---

c. Use logarithms to find the half-life of caffeine.

Problem 12 A population, currently 200, is growing at 5% per year.

a. Write a formula for the population, P, as a function of time, t, years in the future.

---

b. Graph P against t. Which description matches the graph?

---

c. Estimate the population 10 years from now.

---

d. Use the graph to estimate the doubling time of the population.

Problem 16

Same as Lecture Example 5 (Doubling Time and Half-Life)

The antidepressant fluoxetine (or Prozac) has a half-life of about 3 days.

What percentage of a dose remains in the body after one day? After one week?

Problem 18 The half-life of a radioactive substance is 12 days. There are 10.32 grams initially.

a. Write an equation for the amount, A, of the substance as a function of time.

---

b. When is the substance reduced to 1 gram?

Problem 19 One of the main contaminants of a nuclear accident, such as that at Chernobyl, is strontium-90, which decays exponentially at a rate of approximately 2.5% per year.

a. Write the percent of strontium-90 remaining, P, as a function of years, t, since the nuclear accident. [Hint: 100% of the contaminant remains at t = 0.]

---

b. Graph P against t. Which description matches the graph?

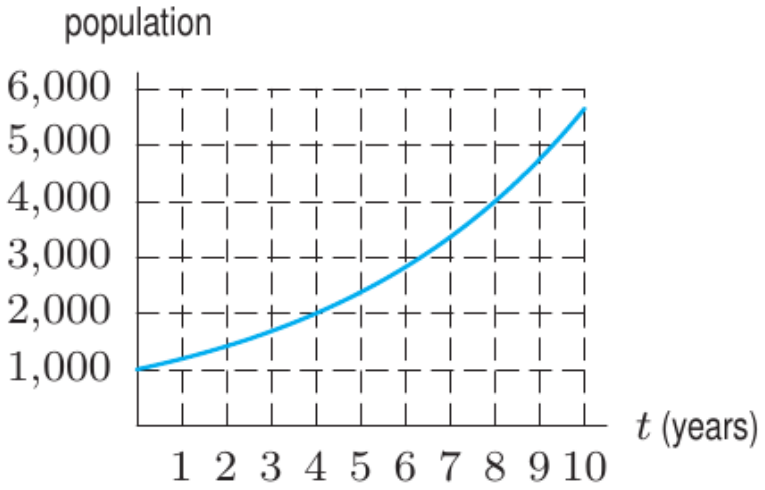
---

c. Estimate the half-life of strontium-90.

---

d. After the Chernobyl disaster, it was predicted that the region would not be safe for human habitation for 100 years. Estimate the percent of original strontium-90 remaining at this time.

Problem 21 (a) Figure 1.68 shows exponential growth.



(a) Starting at t = 0, estimate the time for the population to double.

(b) Repeat part (a), but this time start at t = 3.

(c) Pick any other value of t for the starting point. What do you notice about the doubling time?

Problem 23

Same as Lecture Example 6 (Doubling Time and Half-Life)

If the quantity of a substance decreases by 4% in 10 hours,

find its half-life.

Problem 24 Pregnant women metabolize some drugs at a slower rate than the rest of the population. The half-life of caffeine is about 4 hours for most people. In pregnant women, it is 10 hours. (This is important because caffeine, like all psychoactive drugs, crosses the placenta to the fetus.)

If a pregnant woman and her husband each have a cup of coffee containing 100 mg of caffeine at 8 am, how much caffeine does each have left in the body at 10 pm?

Problem 25 The half-life of radioactive strontium-90 is 29 years. In 1960, radioactive strontium-90 was released into the atmosphere during testing of nuclear weapons, and was absorbed into people's bones.

How many years does it take until only 10% of the original amount absorbed remains?

Problem 28 Find the doubling time of an investment which is earning 8% interest per year.

t = ?

Problem 31 A picture supposedly painted by Vermeer (1632–1675) contains 99.5% of its carbon-14 (half-life 5730 years).

From this information decide whether the picture is a fake. Explain your reasoning.

Past Exam Questions

Past Exam (Lecture) If the quantity of a substance increases by 10% in 5 days,

find its doubling-time.

2024 Q4 Ampicillin is an antibiotic drug used to treat bacterial infections. 250 mg of Ampicillin was given to a patient initially, and the amount of the drug in the bloodstream was monitored and recorded in the table below.

t (hours)	0	1	2	3	4	5
Q (mg)	250	150	90	54	32.4	19.44

a. Suppose Q = f(t), where Q is the quantity of ampicillin, in mg, in the bloodstream at time t hours since the drug was given. Find a suitable formula for Q as a function of t.

---

b. Find the half-life of ampicillin.

---

c. When will the remaining amount of ampicillin reach 6.9984 mg?

2022 Q5 Apixaban is a drug which is used to prevent and treat serious blood clots.

If 81.23% of an Apixaban dose remains in the body after 3 hours, find the half-life of this drug.

1.7.3 Compound Interest, Present & Future Value

Textbook Problems

Problem 3 If you deposit $10,000 in an account earning interest at an 8% annual rate compounded continuously,

how much money is in the account after five years?

Problem 4 If you need $20,000 in your bank account in 6 years,

how much must be deposited now? The interest rate is 10%, compounded continuously.

Problem 6 Suppose $1000 is invested in an account paying interest at a rate of 5.5% per year. How much is in the account after 8 years if the interest is compounded

(a) Annually?

(b) Continuously?

Problem 8 If $12,000 is deposited in an account paying 8% interest per year, compounded continuously,

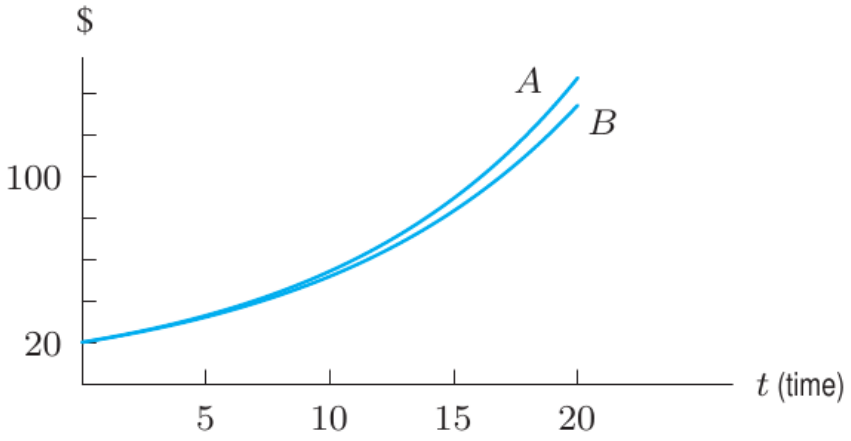
how long will it take for the balance to reach $20,000?

Problem 9 You want to invest money for your child's education in a certificate of deposit (CD). You want it to be worth $12,000 in 10 years. How much should you invest if the CD pays interest at a 9% annual rate compounded

(a) Annually?

(b) Continuously?

Problem 13 Figure 1.67 shows the balances in two bank accounts. Both accounts pay the same interest rate, but one compounds continuously and the other compounds annually.



Which curve corresponds to which compounding method? What is the initial deposit in each case?

Problem 29 The island of Manhattan was sold for $24 in 1626. Suppose the money had been invested in an account which compounded interest continuously.

a. (i) How much money would be in the account in the year 2005 if the yearly interest rate was 5%?

---

a. (ii) How much money would be in the account in the year 2005 if the yearly interest rate was 7%?

---

b. If the yearly interest rate was 6%, in what year would the account be worth one million dollars?

Problem 32 Find the future value in 8 years of a $10,000 payment today, if the interest rate is 3% per year compounded continuously.

Future value = ?

Problem 33 Find the future value in 15 years of a $20,000 payment today, if the interest rate is 3.8% per year compounded continuously.

Future value = ?

Problem 34 Find the present value of an $8000 payment to be made in 5 years. The interest rate is 4% per year compounded continuously.

Present value = ?

Problem 35 Find the present value of a $20,000 payment to be made in 10 years. Assume an interest rate of 3.2% per year compounded continuously.

Present value = ?

1.7.4 Test Your Understanding

Test Your Understanding

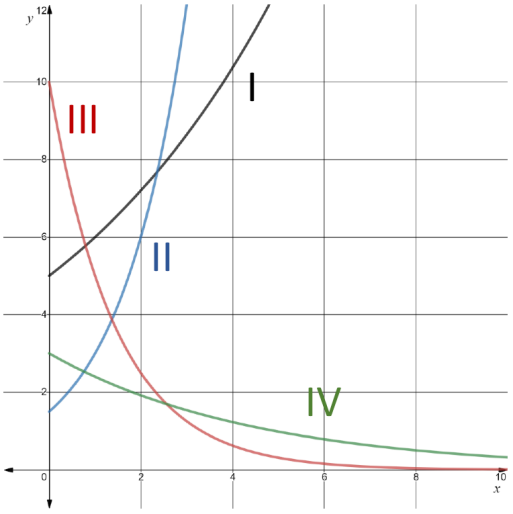
TYU 3 · P1 The amount of nicotine in mg in a person's bloodstream after t hours is given by $N = 50e^{-0.25t}$. Decide whether each statement is True or False.

1. The initial amount of nicotine is 25 mg
2. The amount of nicotine is decreasing by 0.25 mg per hour
3. The amount of nicotine is decreasing by 50 mg per hour
4. The amount of nicotine is increasing continuously by 25% per hour
5. The amount of nicotine is decreasing continuously by 25% per hour
6. The amount of nicotine cannot have a doubling time
7. The amount of nicotine has a variable half-life
8. The half-life of the amount of nicotine is ln(2) / 0.25 hours
9. 30 mg of nicotine is removed from the bloodstream after $\frac{1}{0.25}\ln(\frac{2}{5})$ hours

---

TYU 3 · P4 Consider the graph of the following functions. Decide whether each statement is True or False.

1. The curves I and II have the same percent growth rates
2. The curves III and IV have different percent decay rates
3. Curve III is decaying faster than curve IV
4. Curve II is growing faster than curve I
5. The doubling-time of curve II is smaller than that of curve I
6. The half-life of curve IV is greater than that of curve III



TYU 4 · P1 Corona virus spreads in a high school. The total number of students, N, who got infected with the virus at time t (in days) can be described by the formula $N = L / (1 + Ce^{-kt})$.

a. If the fastest increase in the number of infected students occurs when there are 10 infections, what would be the value of L?

A. 10
B. 20
C. 25
D. 50
E. 100

b. If the initial number of infected students was one, what would be the value of C?

A. 19
B. 24
C. 49
D. 99
E. 9

c. What happens to the number of un-infected students ($L - N_0$) after a very long time?

A. Stays the same
B. Grows in a linear fashion
C. Decays in a linear fashion
D. Grows in an exponential fashion
E. Decays in an exponential fashion

d. What happens to the percent growth rate during the entire growth period?

A. Remains constant
B. Grows in a linear fashion
C. Decays in a linear fashion
D. Grows in an exponential fashion
E. Decays in an exponential fashion

TYU 4 · P2 Flu virus spreads in a high school. The total number of students, N, who got infected with the virus at time t (in days) can be described by the formula $N = L / (1 + Ce^{-kt})$.

a. After a very long time, approximately 5000 people will become infected. What is L?

A. 2500
B. 5000
C. 3500
D. 1500
E. 0

b. In the early stages, the number of infected people is increasing approximately exponentially with a continuous growth rate of 1.78%. What is k?

A. 0.178
B. 0.0178
C. −0.0178
D. 0.0139
E. −0.0139

c. Initially 10 people were infected. Use your answers to the previous questions to find C.

A. 0
B. 49
C. 499
D. 4999
E. 4.9

TYU 4 · P3 The concentration of a drug in the bloodstream C (in ng/ml) is at t hours after administration is given by $C(t) = 2te^{-0.5t}$. The minimum effective concentration is 1 ng/ml.

a. The maximum absorption rate occurs at …… hour(s).

A. 0.5
B. 0
C. 1
D. 2
E. 4

b. The drug reaches a peak concentration after …… hour(s).

A. 0.5
B. 0
C. 1
D. 2
E. 4

c. The drug decays approximately exponentially after …… hour(s).

A. 0.5
B. 0
C. 1
D. 2
E. 4

d. The clearance rate of this drug is

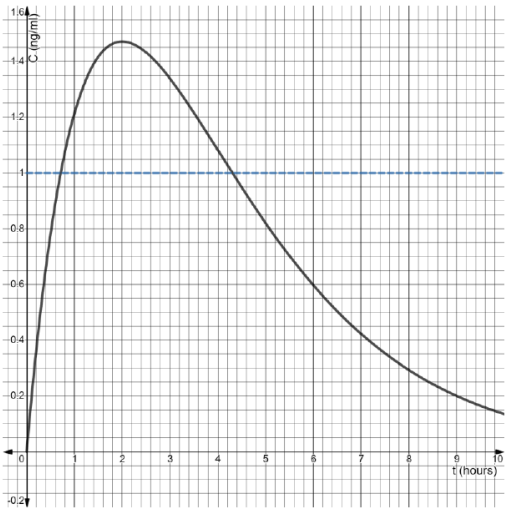
A. −50%
B. −5%
C. −200%
D. 5%
E. 50%

e. The second dose of the drug is better administered after …… hour(s).

A. 0
B. 0.7
C. 1
D. 2
E. 4.3

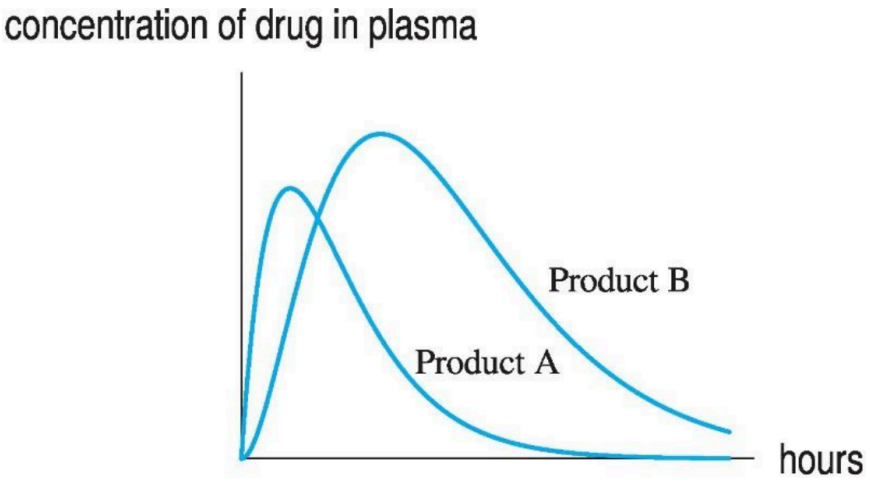
f. The length of the duration of effectiveness for this drug is ……. hour(s).

A. 0
B. 3.6
C. 1
D. 2
E. 5



TYU 4 · P4 The following graph describes the concentration of two drug products, A and B, in the bloodstream of a patient as a function of time, t, in hours after administering the drug. Decide whether each statement is True or False.

1. Product A is absorbed more quickly than Product B
2. Product A is cleared out of the bloodstream more slowly than Product B.
3. Product A has a higher peak concentration than that of Product B.



Answers · Section 1.7

1.7.1 Exponential Growth & Decay with Base e

Ex 1 **(a) P = 2.18 millirem/h** — $P_0 = 2.4$, $k = -0.004$

$P = P_0e^{kt} \Rightarrow P = 2.4e^{-0.004t}$

a) $t = 24$, find P

$P = 2.4e^{(-0.004)(24)}$

$= 2.18$ millirem/h

(b) t = 346.6 hours ≈ 15 days — b) $P = 0.6$, find t

$0.6 = 2.4e^{-0.004t}$ $(\div 2.4)$

$\ln e^{-0.004t} = \ln(0.6/2.4)$ $[(-0.004t)\ln(e), \ln e = 1]$

$-0.004t = \ln(0.6/2.4)$ $(\div -0.004)$

$t = \ln(0.6/2.4) / (-0.004)$

$t = 346.6$ hours $(\div 24) \approx 15$ days

After 15 days, the radiation level will reach the maximum accepted level.

Ex 2 **$P = 19.5e^{0.028t}$** — 1984 $\Rightarrow t = 0$, $P_0 = 19.5$

2009 $\Rightarrow t = 25$, $P = 39.0$

$P = P_0e^{kt}$, find k

$P = 19.5e^{kt}$

$t = 25 \Rightarrow 39 = 19.5e^{k(25)}$ $(\div 19.5)$

$\ln e^{25k} = \ln(39/19.5)$

$25k = \ln(39/19.5)$ $(\div 25)$

$k = \ln(39/19.5) / 25$

$k = 0.0277$ $(\times 100\%) = 2.77\%$ per year (continuous growth rate)

$P = 19.5e^{0.028t}$

To find a: $a = e^k \Rightarrow a = e^{0.0277} = 1.028$

To find r: $a > 1 \Rightarrow$ exponential growth

$a = 1 + r \Rightarrow r = a - 1$ (growth rate)

$r = 1.028 - 1 = 0.028$ $(\times 100\%) = 2.8\%$ per year

Problem 1 **a. $W = 18{,}000e^{0.27t}$** — Continuous rate → $W = 18{,}000e^{0.27t}$.

b. t ≈ 9.7 (during 2009) — $250{,}000 = 18{,}000e^{0.27t}$ → $t = \ln(250/18)/0.27 \approx 9.7$ → during 2009.

Problem 14 **$C = 2$, $\alpha \approx -0.693$, $y(2) = 0.5$** — $y(0) = C = 2$. $2e^{\alpha} = 1 \rightarrow \alpha = \ln 0.5 \approx -0.693$. $y(2) = 2e^{2\alpha} = 2(0.5)^2 = 0.5$.

Problem 15 **a. ≈ 47.6%** — $e^{-0.00012(6194)} = e^{-0.743} \approx 0.476 = 47.6\%$.

b. ≈ 23.7% — $e^{-0.00012(12{,}000)} = e^{-1.44} \approx 0.237 = 23.7\%$.

Problem 17 **≈ 8.45% per year** — $20{,}000a^5 = 30{,}000 \rightarrow a^5 = 1.5 \rightarrow a = 1.5^{\wedge}(1/5) \approx 1.0845 \rightarrow \approx 8.45\%$ per year.

Problem 20 **a. $H = 2.5e^{0.151t}$ (increasing, concave up)** — $37.8 = 2.5e^{18k} \rightarrow k = \ln(15.12)/18 \approx 0.151 \rightarrow H = 2.5e^{0.151t}$.

b. ≈ 15.1% per year — $k \approx 0.151 \rightarrow \approx 15.1\%$ per year (continuous).

Problem 22 **$P = 500(1.732)^t$; P(5) ≈ 7794** — $500a^2 = 1500 \rightarrow a^2 = 3 \rightarrow a = \sqrt{3} \approx 1.732$. $P(5) = 500(1.732)^5 \approx 7794$.

Problem 26 **$k \approx 0.187$ (18.7%); $P = 5000e^{0.187t}$; 2020: ≈ 449,000 koalas** — $27{,}000 = 5000e^{9k} \rightarrow k = \ln 5.4/9 \approx 0.187$. 2020 → $t = 24$: $5000e^{0.187(24)} \approx 449{,}000$.

Problem 27 **$k \approx 0.0345$ (3.45%); 2020: ≈ 191 million tons** — $99 = 17e^{51k} \rightarrow k = \ln(99/17)/51 \approx 0.0345$. 2020 → $t = 70$: $17e^{0.0345(70)} \approx 191$ million tons.

Problem 30 **≈ 1.1% per year** — $6.4a^{26} = 8.5 \rightarrow a = (8.5/6.4)^{\wedge}(1/26) \approx 1.011 \rightarrow \approx 1.1\%$ per year.

1.7.2 Doubling Time & Half-Life

Ex 1 (Doubling Time & Half-Life) **$t = \ln(2)/\ln(a) = \ln(2)/k$** — Doubling time ⇛ $t = ?$ for $Q = 2Q_0$

$2Q_0 = Q_0a^t$ ($\div Q_0$) — solve for t

$\ln a^t = \ln 2$

$t \ln a = \ln 2$ ($\div \ln a$)

$t = \ln(2)/\ln(a) = \ln(2)/k$

Ex 2 (Doubling Time & Half-Life) **t = 10.24 years** — $r = 7\% = 0.07$ per year (growth rate)

$a = 1 + r = 1.07$

Doubling time $t = \ln(2)/\ln(a)$

$= \ln(2)/\ln(1.07)$

$t = 10.24$ years

Ex 3 (Doubling Time & Half-Life) **$t = -\ln 2/\ln a = -\ln 2/k$** — Half-life is $t = ?$ when $Q = ½Q_0$

$½Q_0 = Q_0a^t$ ($\div Q_0$) — solve for t

$\ln a^t = \ln ½$

$t \ln(a) = \ln(½)$ ($\div \ln a$)

$t = \ln(½)/\ln(a)$

$\ln(½) = \ln(1) - \ln(2) = 0 - \ln(2) = -\ln(2)$

$t = -\ln 2/\ln a = -\ln 2/k$

Ex 4 (Doubling Time & Half-Life) **t = 277.26 years** — $k = -0.25\% = -0.0025$

$Q = Q_0e^{kt} = Q_0e^{-0.0025t}$

Half life $t = -\ln 2/k = -\ln 2/(-0.0025)$

$t = 277.26$ years

Half of the ozone will disappear after 277 years.

Ex 5 (Doubling Time & Half-Life) **79% after 1 day; about 20% after 7 days** — Half life: $t = -\ln 2/k$
$\Rightarrow 3 = -\ln 2/k \Rightarrow 3k = -\ln 2\ (\div 3) \Rightarrow k = -\ln 2/3$
$\Rightarrow k = -0.231\ (\times 100\%) = -23.1\%$ per day
$\Rightarrow Q = Q_0e^{kt} \Rightarrow Q = Q_0e^{-0.231t}$

After 1 day $\Rightarrow t = 1$
$Q = Q_0e^{(-0.231)(1)} = 0.794Q_0\ (\times 100\%) = 79.4\%\ Q_0 \approx 79\%$
It means about 79% of the dose will remain in the body after 1 day.
(The doctor's notes write 0.795; the calculator gives 0.7937.)

After 7 days $\Rightarrow t = 7$
$Q = Q_0e^{(-0.231)(7)} = 0.198Q_0\ (\times 100\%) = 19.8\%\ Q_0$
After 7 days, about 20% of the dose will remain in the body.

Ex 6 (Doubling Time & Half-Life) **$t \approx 173$ hours** — Solution 1:
$Q = Q_0e^{kt}$
Since Q_0 is decreased by 4% $\Rightarrow$ 96% remains, therefore $Q = (96/100)Q_0$
$\Rightarrow 0.96Q_0 = Q_0e^{k\cdot 10}$ (find k)
$\Rightarrow k \approx -0.004$
Now, $t = -\ln 2/k \Rightarrow t = -\ln 2/(-0.004) \Rightarrow t \approx 173$ hours

Solution 2:
$r = 4/100 = 0.04$ in 10 hours
$r = 0.04/10 = 0.004$ per hour
Since the quantity is decreased: $a = 1 - r = 1 - 0.004 \Rightarrow a = 0.996$
Now, half life $t = -\ln 2/\ln a \Rightarrow t = -\ln 2/\ln 0.996 \Rightarrow t \approx 173$ hours

The half-time of the substance is about 173 hours.

Problem 2 **Table 0.4, 0.2, 0.1, 0.05, 0.025, 0.0125** — Every 2 hours the amount halves: 0.4 → 0.2 → 0.1 → 0.05 → 0.025 → 0.0125 mg.
Time ≈ 6.6 hours — 0.04 is between 0.05 (t = 6) and 0.025 (t = 8). Exactly: 0.4(0.5)^(t/2) = 0.04 → t = 2 ln 10 / ln 2 ≈ 6.6 hours.

Problem 5 **≈ 11.55 years** — $t = \ln 2 / 0.06 \approx 11.55$ years.

Problem 7 **≈ 10.24 years** — $t = \ln 2 / \ln 1.07 \approx 10.24$ years.

Problem 10 **≈ 346.6 days** — $k = 1/500 = 0.002 \rightarrow t = \ln 2/0.002 = 500 \ln 2 \approx 346.6$ days.

Problem 11 **a. $A = 100e^{-0.17t}$** — Continuous decay at 17% → $A = 100e^{-0.17t}$.
b. Decreasing, concave-up curve from 100; A = 50 at about t ≈ 4 hours — The graph is an exponential decay from 100; it reaches 50 at about t ≈ 4 hours.
c. ≈ 4.08 hours — $t = \ln 2 / 0.17 \approx 4.08$ hours.

Problem 12

a. **$P = 200(1.05)^t$** — $P = 200(1.05)^t$.

b. **Starts at 200, increasing and concave up** — Exponential growth: starts at 200, increasing, concave up.

c. **≈ 326** — $200(1.05)^{10} \approx 326$.

d. **≈ 14 years** — P = 400 at about t ≈ 14.2 years (ln 2 / ln 1.05).

Problem 16

After 1 day ≈ 79%; after 1 week ≈ 20% — (0.5)^(1/3) ≈ 0.79 = 79%; (0.5)^(7/3) ≈ 0.20 = 20%.

Problem 18

a. **$A = 10.32e^{-0.0578t}$** — $k = -\ln 2/12 \approx -0.0578 \rightarrow A = 10.32e^{-0.0578t}$.

b. **≈ 40.4 days** — $1 = 10.32e^{-0.0578t} \rightarrow t = \ln(1/10.32)/(-0.0578) \approx 40.4$ days.

Problem 19

a. **$P = 100(0.975)^t$** — $a = 1 - 0.025 = 0.975 \rightarrow P = 100(0.975)^t$.

b. **Starts at 100, decreasing and concave up** — Exponential decay: starts at 100, decreasing, concave up, approaching 0.

c. **≈ 27.4 years** — $0.975^t = 0.5 \rightarrow t = \ln 0.5/\ln 0.975 \approx 27.4$ years.

d. **≈ 7.95%** — $100(0.975)^{100} \approx 7.95\%$.

Problem 21

(a) **≈ 4 years** — At t = 0 the population is 1000; it reaches 2000 at about t = 4.

(b) **≈ 4 years** — At t = 3 the population is about 1700; it reaches about 3400 at t ≈ 7, so again ≈ 4 years.

(c) **It is the same no matter where you start** — For exponential growth the doubling time is constant: ≈ 4 years from any starting point.

Problem 23

≈ 170 hours — $0.96 = e^{10k} \rightarrow k = \ln(0.96)/10 \approx -0.00408 \rightarrow t = -\ln 2/k \approx 170$ hours.

Problem 24

Husband ≈ 8.8 mg; pregnant woman ≈ 37.9 mg — 14 hours. Husband: 100(0.5)^(14/4) ≈ 8.8 mg. Pregnant woman: 100(0.5)^(14/10) ≈ 37.9 mg.

Problem 25

≈ 96.3 years — $k = -\ln 2/29 \approx -0.0239$. $0.1 = e^{kt} \rightarrow t = \ln 0.1/k \approx 96.3$ years.

Problem 28

≈ 9.01 years — t = ln 2 / ln 1.08 ≈ 9.01 years.

Problem 31

Fake: its age is only about 41 years — $k = -\ln 2/5730$. $0.995 = e^{kt} \rightarrow t = \ln 0.995/k \approx 41$ years. A real Vermeer would be over 300 years old, so the picture is a fake.

Past Exam (Lecture)

t = 35 days — r = 10% in 5 days (÷ 5)
⇒ r = 2% per day, r = 0.02
Increasing ⇒ exponential growth
⇒ a = 1 + r ⇒ a = 1.02
Doubling time t = ln 2/ln a
= ln 2/ln(1.02)
= 35 days

2024 Q4

a. **$Q = 250(0.6)^t$** — Ratios $150/250 = 90/150 = 0.6 \rightarrow Q = 250(0.6)^t$.

b. **≈ 1.36 hours** — $t = -\ln 2 / \ln 0.6 \approx 1.36$ hours.

c. **7 hours** — $250(0.6)^t = 6.9984 \rightarrow 0.6^t = 0.027994 \rightarrow t = 7$ hours.

2022 Q5

≈ 10 hours — $0.8123 = e^{3k} \rightarrow k = \ln(0.8123)/3 \approx -0.0693 \rightarrow t = -\ln 2/k \approx 10$ hours.

1.7.3 Compound Interest, Present & Future Value

Problem 3

≈ $14,918.25 — $B = 10{,}000e^{0.08(5)} = 10{,}000e^{0.4} \approx \$14{,}918.25$.

Problem 4

≈ $10,976.23 — $20{,}000 = Pe^{0.1(6)} \rightarrow P = 20{,}000e^{-0.6} \approx \$10{,}976.23$.

Problem 6

(a) **≈ $1,534.69** — $1000(1.055)^8 \approx \$1{,}534.69$.

(b) **≈ $1,552.71** — $1000e^{0.055(8)} = 1000e^{0.44} \approx \$1{,}552.71$.

Problem 8

≈ 6.39 years — $t = \ln(20{,}000/12{,}000)/0.08 \approx 6.39$ years.

Problem 9

(a) **≈ $5,068.93** — $12{,}000/(1.09)^{10} \approx \$5{,}068.93$.

(b) **≈ $4,878.84** — $12{,}000e^{-0.9} \approx \$4{,}878.84$.

Problem 13

A: continuous; B: annual. Initial deposit: $20 each — Continuous compounding grows a little faster, so the higher curve A is continuous and B is annual. Both start at $20.

Problem 29

a. (i) **≈ $4.07 × 10⁹** — $t = 379$: $24e^{0.05(379)} \approx \$4.07 \times 10^9$.

a. (ii) **≈ $7.98 × 10¹²** — $24e^{0.07(379)} \approx \$7.98 \times 10^{12}$.

b. **1803** — $10^6 = 24e^{0.06t} \rightarrow t = \ln(10^6/24)/0.06 \approx 177.3 \rightarrow 1626 + 177 \approx 1803$.

Problem 32

≈ $12,712.49 — $10{,}000e^{0.03(8)} = 10{,}000e^{0.24} \approx \$12{,}712.49$.

Problem 33

≈ $35,365.34 — $20{,}000e^{0.038(15)} = 20{,}000e^{0.57} \approx \$35{,}365.34$. (Note: the booklet key writes 20,000e^0.057 ≈ $21,173.12, which uses 0.038 × 1.5 by mistake; the correct exponent is 0.038 × 15 = 0.57.)

Problem 34

≈ $6,549.85 — $8000e^{-0.04(5)} = 8000e^{-0.2} \approx \$6{,}549.85$.

Problem 35

≈ $14,522.98 — $20{,}000e^{-0.32} \approx \$14{,}522.98$.

1.7.4 Test Your Understanding

TYU 3 · P1 $N = N_0 e^{kt}$ with $N_0 = 50$ mg and $k = -0.25$ (continuous decay of 25% per hour).

$$\text{Half-life: } 50e^{-0.25t} = 25 \Rightarrow t = \frac{\ln(0.5)}{-0.25} = \frac{\ln 2}{0.25} \approx 2.77 \text{ hours}$$

$$30 \text{ mg removed} \Rightarrow 20 \text{ mg remain: } 50e^{-0.25t} = 20 \Rightarrow t = \frac{\ln(2/5)}{-0.25} = \frac{1}{0.25}\ln\left(\frac{5}{2}\right) \approx 3.67 \text{ hours}$$

1. False — the initial amount is $N(0) = 50$ mg.

2. False — the decrease is 25% per hour (continuous), not a constant 0.25 mg.

3. False — the decrease is not a constant amount; 50 is the initial amount.

4. False — $k = -0.25 < 0$, so it is decreasing.

5. True — $k = -0.25 \rightarrow$ continuous decay rate of 25% per hour.

6. True — it is decaying, so it has a half-life, not a doubling time.

7. False — exponential decay has a constant half-life.

8. True — $t = \ln 2 / 0.25 \approx 2.77$ hours.

9. False — $t = (1/0.25)\ln(5/2) \approx 3.67$ hours; $(1/0.25)\ln(2/5)$ is negative.

TYU 3 · P4 From the graph: I goes from 5 to 10 in about 3.8 units (slow growth); II goes from 1.5 to 6 in 2 units (doubles about every 1 unit, fast growth). III drops from 10 to about 2.5 by $x = 2$ (fast decay); IV drops slowly from 3.

1. False — II grows much faster than I, so their percent growth rates are different.

2. True — III decays much faster than IV.

3. True — III falls steeply toward 0 while IV falls slowly.

4. True — II is steeper and overtakes I.

5. True — II grows faster, so it doubles in less time.

6. True — IV decays more slowly, so it takes longer to halve.

TYU 4 · P1 **(a)** In a logistic model the fastest increase occurs at N = L/2:

$$\frac{L}{2} = 10 \Rightarrow L = 20$$

Answer: B — 20

(b) At t = 0, N = 1:

$$1 = \frac{20}{1 + C} \Rightarrow 1 + C = 20 \Rightarrow C = 19$$

Answer: A — 19

(c) For large t, N approaches L and the gap $L - N \approx LCe^{-kt}$ shrinks toward 0 exponentially.

Answer: E — Decays in an exponential fashion

(d) The percent (relative) growth rate of a logistic model is $k(1 - N/L)$: it starts near k and decreases linearly (in N) down to 0 as $N \to L$.

Answer: C — Decays in a linear fashion (the doctor's answer)

TYU 4 · P2 **(a)** L is the limiting value (carrying capacity) reached after a very long time.

Answer: B — 5000

(b) Early on, $N \approx (L/C)e^{kt}$, so k is the continuous growth rate: k = 1.78% = 0.0178.

Answer: B — 0.0178

(c) At t = 0, N = 10:

$$10 = \frac{5000}{1 + C} \Rightarrow 1 + C = 500 \Rightarrow C = 499$$

Answer: C — 499

TYU 4 · P3 $C(t) = 2te^{-0.5t}$ is a surge function $y = ate^{-bt}$ with $a = 2$, $b = 0.5$.

(a) The graph is steepest right at the start ($t = 0$), so the absorption rate is maximum at $t = 0$.

Answer: B — 0

(b) Peak of a surge function at $t = 1/b$:

$$t = \frac{1}{0.5} = 2 \text{ hours}$$

Answer: D — 2

(c) After the inflection point $t = 2/b$ the curve behaves like exponential decay:

$$t = \frac{2}{0.5} = 4 \text{ hours}$$

Answer: E — 4

(d) The decay factor is $e^{-0.5t}$ → continuous clearance rate $k = -0.5 = -50\%$ per hour.

Answer: A — −50%

(e) Solve $C(t) = 1$ (from the graph or numerically): $t \approx 0.7$ and $t \approx 4.3$ hours. The concentration drops below the minimum effective level at $t \approx 4.3$ hours, so the second dose is given then.

Answer: E — 4.3

(f) The drug is effective while $C \geq 1$:

$$4.3 - 0.7 = 3.6 \text{ hours}$$

Answer: B — 3.6

TYU 4 · P4 **1. True** — A rises more steeply and reaches its peak earlier.

2. False — A drops to near zero sooner, so it is cleared more quickly.

3. False — B's peak is higher than A's.

1.8 New Functions From Old

In this section

1. Composite Functions 6 lecture · 27 textbook
2. Stretches, Shifts & Reflections 4 lecture · 16 textbook · 1 past exam
3. Test Your Understanding 1 questions

1.8.1 Composite Functions

Lecture Problems

Ex 1 If $f(x) = 2x / (3x - 7)$ and $g(x) = x + 1$,

find $f(g(x))$.

Ex 2 If $f(t) = t^2$ and $g(t) = t + 2$, find

(a) $f(t + 1)$

(b) $f(t) + 3$

(c) $f(t + h)$

(d) $f(g(t))$

(e) $g(f(t))$

Ex 3 If $f(x) = e^x$ and $g(x) = 5x + 1$, find

(a) $f(g(x))$

(b) $g(f(x))$

Ex 4 Using the given table, find

x	0	1	2	3
f(x)	3	1	−1	−3
g(x)	0	2	4	6

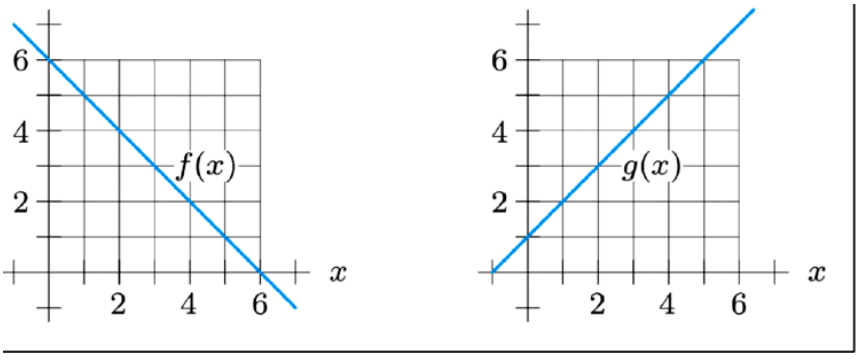
(a) $g(f(0))$

(b) $f(g(0))$

(c) $g(f(1))$

(d) $f(g(1))$

Ex 5 Use the given graphs to estimate the value of $f(g(1))$.



$f(g(1)) = ?$

Ex 6 Use a new variable u for the inside function to express each of the following as a composite function:

(a) $y = \ln(3t)$

(b) $w = 5(2r + 3)^2$

(c) $P = e^{-0.03t}$

Textbook Problems

Problem 1 For $g(x) = x^2 + 2x + 3$, find and simplify:

(a) $g(2 + h)$

(b) $g(2)$

(c) $g(2 + h) - g(2)$

Problem 2 If $f(x) = x^2 + 1$, find and simplify:

(a) $f(t + 1)$

(b) $f(t^2 + 1)$

(c) $f(2)$

(d) $2f(t)$

(e) $[f(t)]^2 + 1$

Problem 3 For the functions f and g in Problems 3–6, find: (a) f(g(1)) (b) g(f(1)) (c) f(g(x)) (d) g(f(x)) (e) f(t)g(t). $f(x) = x^2$, $g(x) = x + 1$

(a) $f(g(1))$

(b) $g(f(1))$

(c) $f(g(x))$

(d) $g(f(x))$

(e) $f(t)g(t)$

Problem 4 Find: (a) f(g(1)) (b) g(f(1)) (c) f(g(x)) (d) g(f(x)) (e) f(t)g(t). $f(x) = \sqrt{x + 4}$, $g(x) = x^2$

(a) f(g(1))

(b) g(f(1))

(c) f(g(x))

(d) g(f(x))

(e) f(t)g(t)

Problem 5 Find: (a) f(g(1)) (b) g(f(1)) (c) f(g(x)) (d) g(f(x)) (e) f(t)g(t). $f(x) = e^x$, $g(x) = x^2$

(a) f(g(1))

(b) g(f(1))

(c) f(g(x))

(d) g(f(x))

(e) f(t)g(t)

Problem 6 Find: (a) f(g(1)) (b) g(f(1)) (c) f(g(x)) (d) g(f(x)) (e) f(t)g(t). f(x) = 1/x, g(x) = 3x + 4

(a) f(g(1))

(b) g(f(1))

(c) f(g(x))

(d) g(f(x))

(e) f(t)g(t)

Problem 7 Let $f(x) = x^2$ and $g(x) = 3x - 1$. Find the following:

(a) f(2) + g(2)

(b) f(2) · g(2)

(c) f(g(2))

(d) g(f(2))

Problem 8 In Problems 8–10, find the following: (a) f(g(x)) (b) g(f(x)) (c) f(f(x)). $f(x) = 2x^2$ and $g(x) = x + 3$

(a) f(g(x))

(b) g(f(x))

(c) f(f(x))

Problem 9 Find: (a) f(g(x)) (b) g(f(x)) (c) f(f(x)). $f(x) = 2x + 3$ and $g(x) = 5x^2$

(a) f(g(x))

(b) g(f(x))

(c) f(f(x))

Problem 10 Find: (a) f(g(x)) (b) g(f(x)) (c) f(f(x)). $f(x) = x^2 + 1$ and $g(x) = \ln x$

(a) f(g(x))

(b) g(f(x))

(c) f(f(x))

Problem 11 Use Table 1.35 to find:

x	0	1	2	3	4	5
f(x)	10	6	3	4	7	11
g(x)	2	3	5	8	12	15

(a) f(g(0))

(b) f(g(1))

(c) f(g(2))

(d) g(f(2))

(e) g(f(3))

Problem 12 Make a table of values for each of the following functions using Table 1.35 ("—" = not defined from the table):

x	0	1	2	3	4	5
f(x)	10	6	3	4	7	11
g(x)	2	3	5	8	12	15

(a) $f(x) + 3$ for $x = 0, 1, 2, 3, 4, 5$

(b) $f(x - 2)$ for $x = 0, 1, 2, 3, 4, 5$

(c) $5g(x)$ for $x = 0, 1, 2, 3, 4, 5$

(d) $-f(x) + 2$ for $x = 0, 1, 2, 3, 4, 5$

(e) $g(x - 3)$ for $x = 0, 1, 2, 3, 4, 5$

(f) $f(x) + g(x)$ for $x = 0, 1, 2, 3, 4, 5$

Problem 13 Use the variable u for the inside function to express each of the following as a composite function:

(a) $y = 2^{3x-1}$

(b) $P = \sqrt{(5t^2 + 10)}$

(c) $w = 2 \ln(3r + 4)$

Problem 14 Use the variable u for the inside function to express each of the following as a composite function:

(a) $y = (5t^2 - 2)^6$

(b) $P = 12e^{-0.6t}$

(c) $C = 12 \ln(q^3 + 1)$

Problem 15 Simplify the quantities in Problems 15–18 using $m(z) = z^2$.

$m(z + 1) - m(z)$

Problem 16 Simplify using $m(z) = z^2$.

$m(z + h) - m(z)$

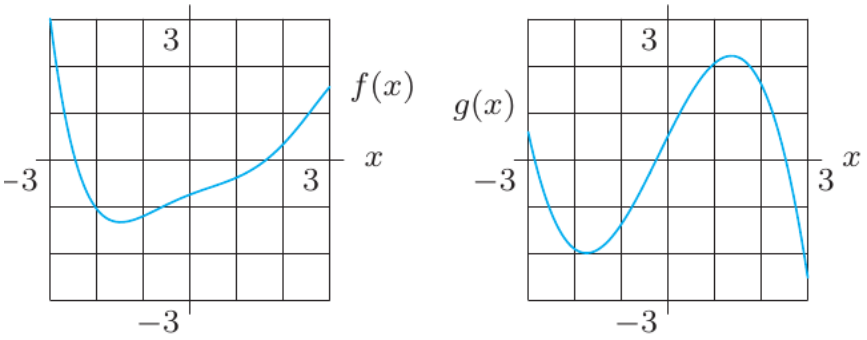
Problem 17 Simplify using $m(z) = z^2$.

$m(z) - m(z - h)$

Problem 18 Simplify using $m(z) = z^2$.

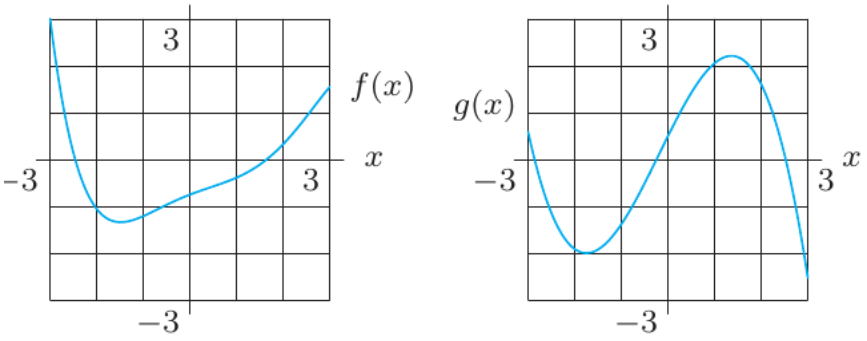
$m(z + h) - m(z - h)$

Problem 19 For Problems 19–21, use the graphs in Figure 1.77.



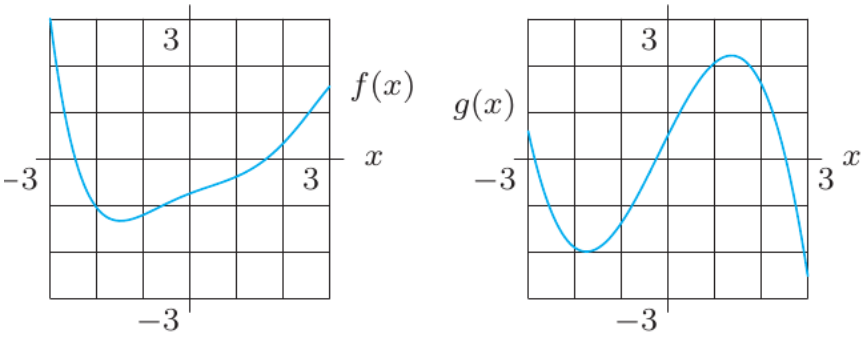
Estimate f(g(1)).

Problem 20 Use the graphs in Figure 1.77.



Estimate g(f(2)).

Problem 21 Use the graphs in Figure 1.77.



Estimate f(f(1)).

Problem 22 Using Table 1.36, create a table of values for f(g(x)) and for g(f(x)).

x	−3	−2	−1	0	1	2	3
f(x)	0	1	2	3	2	1	0
g(x)	3	2	2	0	−2	−2	−3

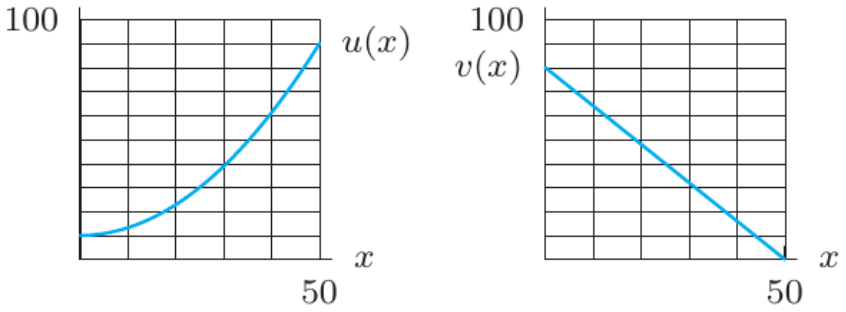
f(g(x)) Values of f(g(x)) for x = −3, −2, −1, 0, 1, 2, 3

g(f(x)) Values of g(f(x)) for x = −3, −2, −1, 0, 1, 2, 3

Problem 23 A tree of height y meters has, on average, B branches, where B = y − 1. Each branch has, on average, n leaves where $n = 2B^2 - B$.

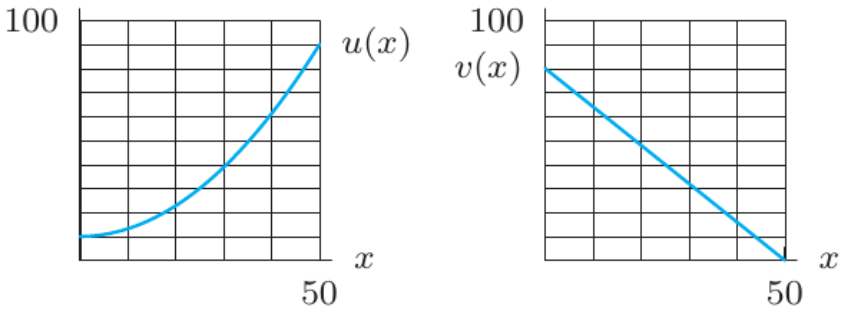
Find the average number of leaves of a tree as a function of height.

Problem 24 In Problems 24–27, use Figure 1.78 to estimate the function value or explain why it cannot be done.



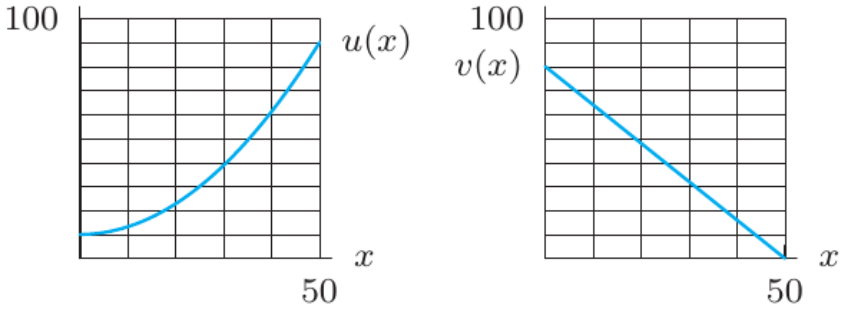
u(v(10))

Problem 25 Use Figure 1.78 to estimate the function value or explain why it cannot be done.



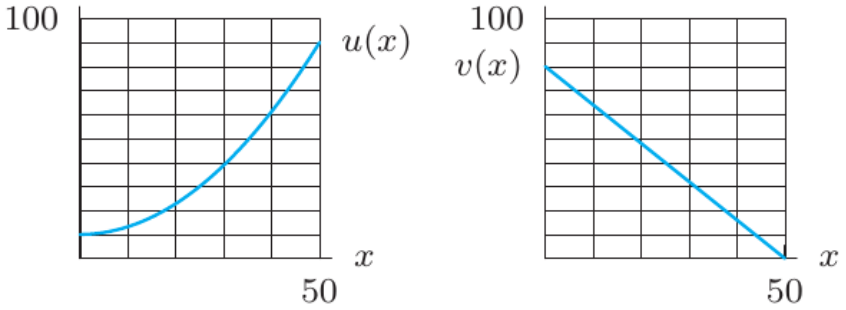
u(v(40))

Problem 26 Use Figure 1.78 to estimate the function value or explain why it cannot be done.



v(u(10))

Problem 27 Use Figure 1.78 to estimate the function value or explain why it cannot be done.



v(u(40))

1.8.2 Stretches, Shifts & Reflections

Lecture Problems

Ex 7 Sketch the graph of the functions:

(a) $y = x^2$

(b) $y = -x^2$

Ex 8

a) Using Desmos, sketch the graph of $y = \sqrt{(9 - x^2)}$.

b) Sketch the graph of $y = -\sqrt{(9 - x^2)}$.

c) Sketch the graph of $y = 2\sqrt{(9 - x^2)}$.

d) Sketch the graph of $y = (1/3)\sqrt{(9 - x^2)}$.

Ex 9 Sketch the graph of each of the following functions:

a) $y = e^x + 2$

b) $y = \ln(x - 2)$

Ex 10 Sketch the graph of each of the following functions:

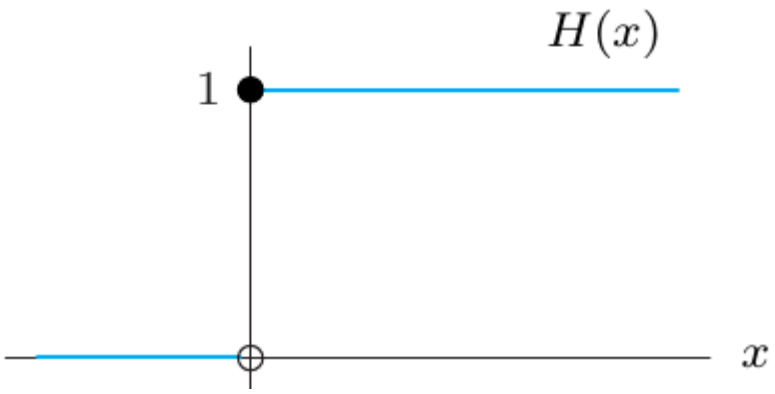
a) $y = x^2 + 3$

b) $y = (x + 3)^2$

c) $y = (x - 2)^2$

Textbook Problems

Problem 28 The Heaviside step function, H, is graphed in Figure 1.79. Graph the following functions.



(a) Graph $y = 2H(x)$. Which description is correct?

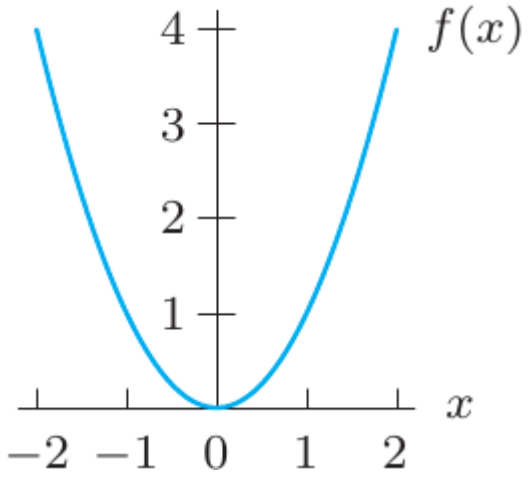
(b) Graph $y = H(x) + 1$. Which description is correct?

(c) Graph $y = H(x + 1)$. Which description is correct?

(d) Graph $y = -H(x)$. Which description is correct?

Problem 29 For the functions f in Exercises 29–31, graph: (a) f(x + 2) (b) f(x − 1) (c) f(x) − 4 (d) f(x + 1) + 3 (e) 3f(x) (f) −f(x) + 1

29.



(a) Graph y = f(x + 2). Which description is correct?

(b) Graph y = f(x − 1). Which description is correct?

(c) Graph y = f(x) − 4. Which description is correct?

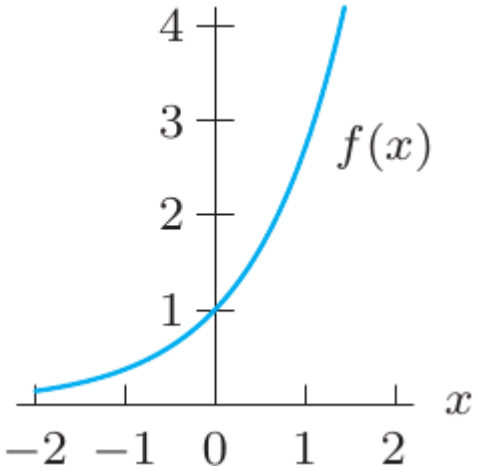
(d) Graph y = f(x + 1) + 3. Which description is correct?

(e) Graph y = 3f(x). Which description is correct?

(f) Graph y = −f(x) + 1. Which description is correct?

Problem 30 Graph: (a) f(x + 2) (b) f(x − 1) (c) f(x) − 4 (d) f(x + 1) + 3 (e) 3f(x) (f) −f(x) + 1

30.



(a) Graph y = f(x + 2). Which description is correct?

(b) Graph y = f(x − 1). Which description is correct?

(c) Graph y = f(x) − 4. Which description is correct?

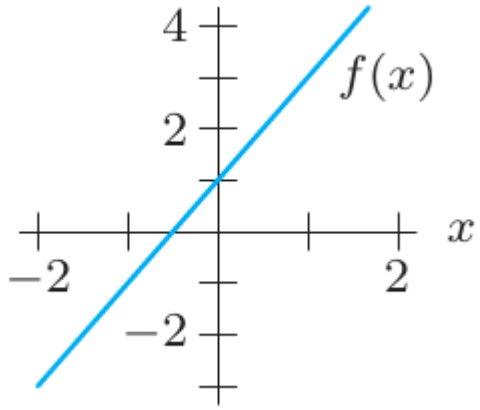
(d) Graph y = f(x + 1) + 3. Which description is correct?

(e) Graph y = 3f(x). Which description is correct?

(f) Graph y = −f(x) + 1. Which description is correct?

Problem 31 Graph: (a) f(x + 2) (b) f(x − 1) (c) f(x) − 4 (d) f(x + 1) + 3 (e) 3f(x) (f) −f(x) + 1

31.



(a) Graph y = f(x + 2). Which description is correct?

(b) Graph y = f(x − 1). Which description is correct?

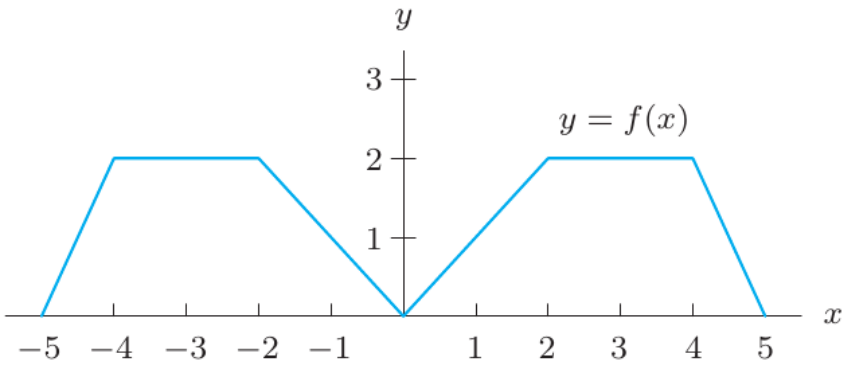
(c) Graph y = f(x) − 4. Which description is correct?

(d) Graph y = f(x + 1) + 3. Which description is correct?

(e) Graph y = 3f(x). Which description is correct?

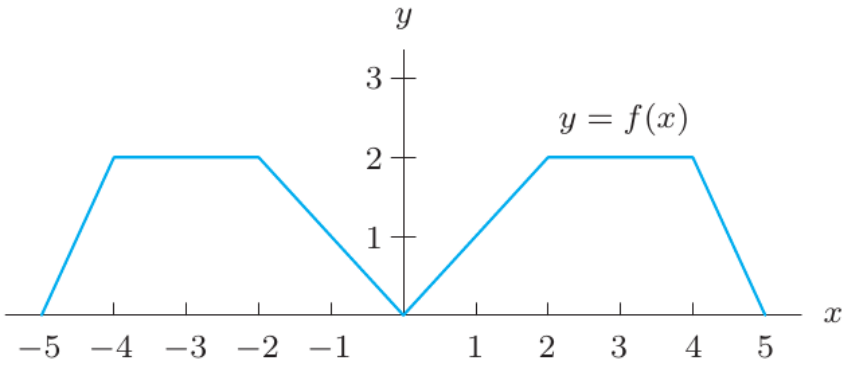
(f) Graph y = −f(x) + 1. Which description is correct?

Problem 32 Graph the functions in Problems 32–37 using Figure 1.80.



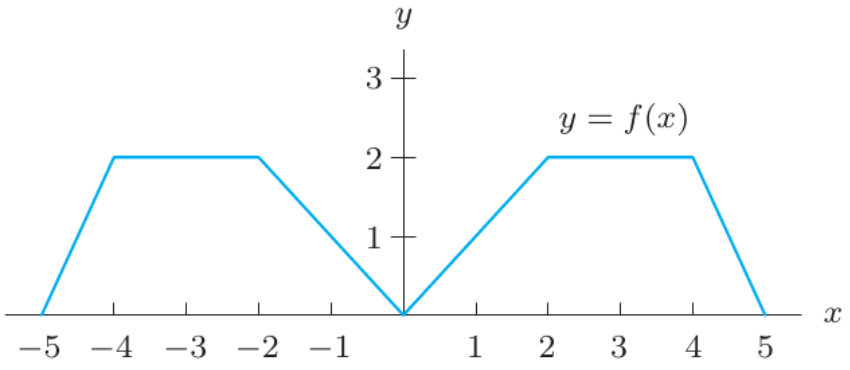
Graph y = f(x) + 2. Which description is correct?

Problem 33 Graph the function using Figure 1.80.



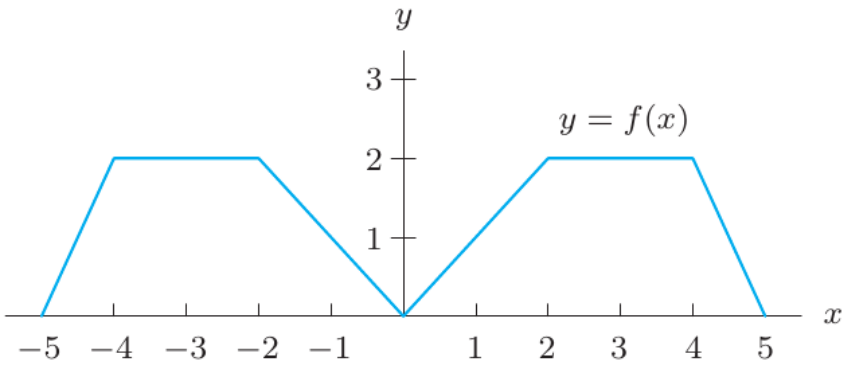
Graph y = 2f(x). Which description is correct?

Problem 34 Graph the function using Figure 1.80.



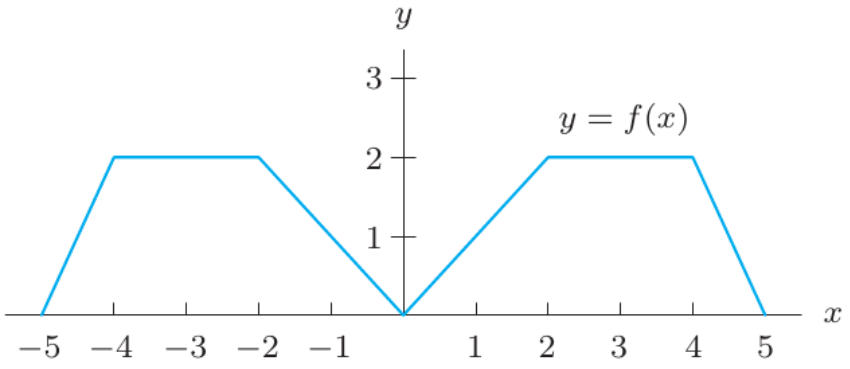
Graph y = f(x − 1). Which description is correct?

Problem 35 Graph the function using Figure 1.80.



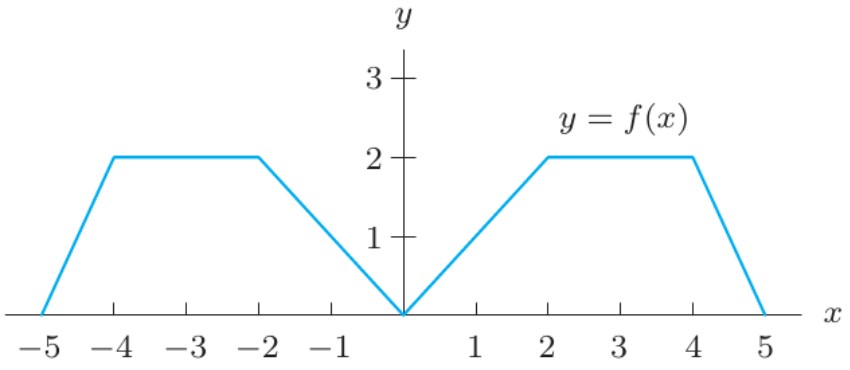
Graph y = −3f(x). Which description is correct?

Problem 36 Graph the function using Figure 1.80.



Graph y = 2f(x) − 1. Which description is correct?

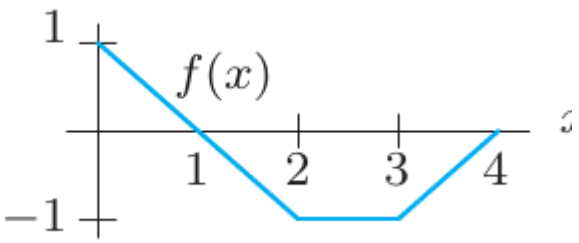
Problem 37 Graph the function using Figure 1.80.



Graph y = 2 − f(x). Which description is correct?

Problem 38 For the functions f(x) in Problems 38–41, graph: (a) y = f(x) + 2 (b) y = f(x − 1) (c) y = 3f(x) (d) y = −f(x)

38.



(a) Graph y = f(x) + 2. Which description is correct?

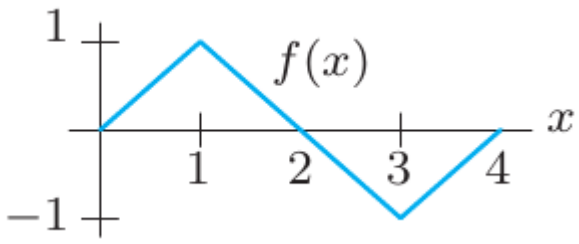
(b) Graph y = f(x − 1). Which description is correct?

(c) Graph y = 3f(x). Which description is correct?

(d) Graph y = −f(x). Which description is correct?

Problem 39 Graph: (a) y = f(x) + 2 (b) y = f(x − 1) (c) y = 3f(x) (d) y = −f(x)

39.



(a) Graph y = f(x) + 2. Which description is correct?

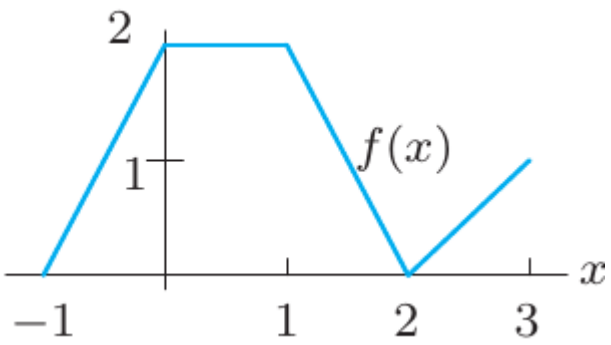
(b) Graph y = f(x − 1). Which description is correct?

(c) Graph y = 3f(x). Which description is correct?

(d) Graph y = −f(x). Which description is correct?

Problem 40 Graph: (a) y = f(x) + 2 (b) y = f(x − 1) (c) y = 3f(x) (d) y = −f(x)

40.



(a) Graph y = f(x) + 2. Which description is correct?

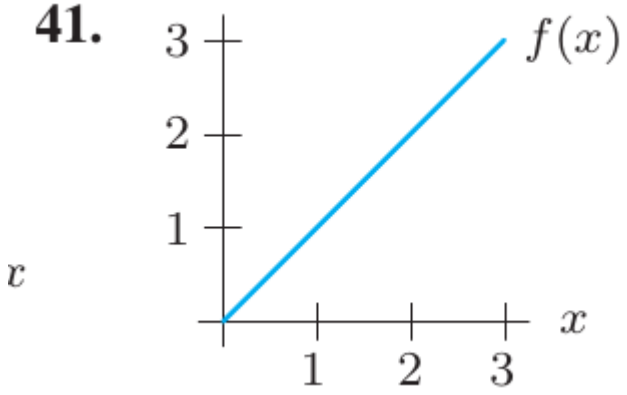
(b) Graph y = f(x − 1). Which description is correct?

(c) Graph y = 3f(x). Which description is correct?

(d) Graph y = −f(x). Which description is correct?

Problem 41 Graph: (a) y = f(x) + 2 (b) y = f(x − 1) (c) y = 3f(x) (d) y = −f(x)

41.



(a) Graph y = f(x) + 2. Which description is correct?

(b) Graph y = f(x − 1). Which description is correct?

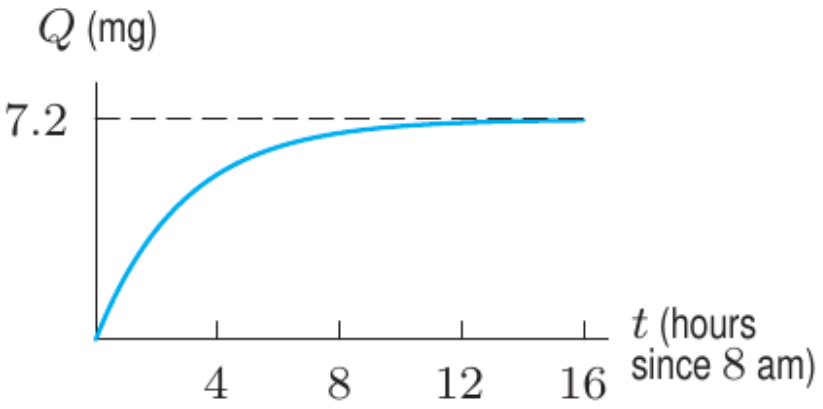
(c) Graph y = 3f(x). Which description is correct?

(d) Graph y = −f(x). Which description is correct?

Problem 42

Same as Past Exam 2024 Question 5

Morphine, a pain-relieving drug, is administered to a patient intravenously starting at 8 am. The drug saturation curve Q = f(t) in Figure 1.81 gives the quantity, Q, of morphine in the blood t hours after 8 am.



a. Draw the drug saturation curve Q = g(t) if the IV line is started at noon instead of 8 am. Which description is correct?

b. Is g one of the following transformations of f: vertical shift, vertical stretch, horizontal shift, horizontal stretch? If so, which?

c. Write g(t) in terms of the function f.

Problem 43

(a) Write an equation for a graph obtained by vertically stretching the graph of $y = x^2$ by a factor of 2, followed by a vertical upward shift of 1 unit.

(b) What is the equation if the order of the transformations (stretching and shifting) in part (a) is interchanged?

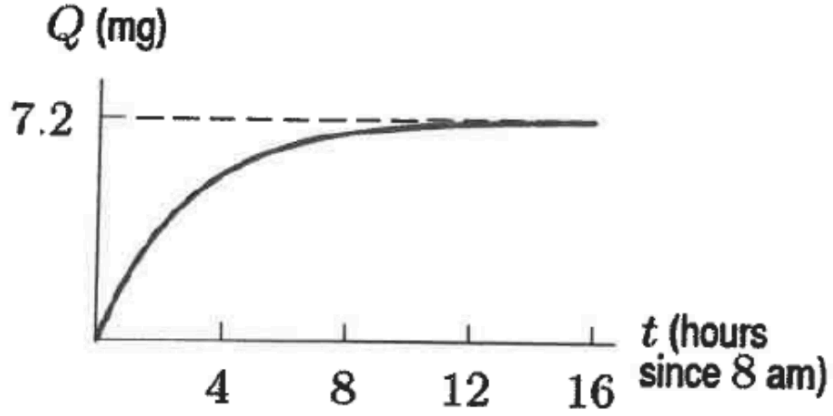
(c) Are the two graphs the same? Explain the effect of reversing the order of transformations.

Past Exam Questions

2024 Q5

Same as Textbook Problem 42

A pain-relieving drug, called Morphine, is administered to a patient intravenously starting at 8 am. The drug saturation curve Q = f(t) in the figure below gives the quantity, Q, of morphine in the blood t hours after 8 am.



a. Sketch a new drug saturation curve Q = g(t) if the drug is administered at noon (12 pm) instead of 8 am. Which description is correct?

---

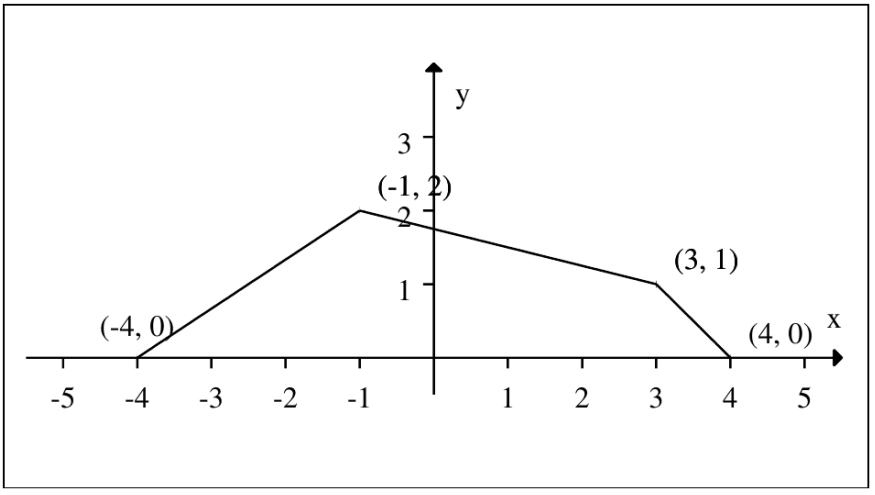
b. Write g(t) in terms of the function f(t).

1.8.3 Test Your Understanding

Test Your Understanding

TYU 5 · P1 Use the function y = f(x) in the graph to complete the table of values.

x	−4	−1	3	4
f(x)				
f(x) + 4				
− f(x)				
− f(x) + 8				
f(x + 1)				



Answers · Section 1.8

1.8.1 Composite Functions

Ex 1 **$(2x + 2)/(3x - 4)$** — $f(g(x)) = f(x + 1) = 2(x + 1)/(3(x + 1) - 7) = (2x + 2)/(3x - 4)$.

Ex 2
(a) **$(t + 1)^2$** — $f(t + 1) = (t + 1)^2$.
(b) **$t^2 + 3$** — $f(t) + 3 = t^2 + 3$.
(c) **$(t + h)^2$** — $f(t + h) = (t + h)^2$.
(d) **$(t + 2)^2$** — $f(g(t)) = f(t + 2) = (t + 2)^2$.
(e) **$t^2 + 2$** — $g(f(t)) = g(t^2) = t^2 + 2$.

Ex 3
(a) **e^{5x+1}** — $f(g(x)) = f(5x + 1) = e^{5x+1}$.
(b) **$5e^x + 1$** — $g(f(x)) = g(e^x) = 5e^x + 1$.

Ex 4
(a) **6** — $f(0) = 3 \rightarrow g(3) = 6$.
(b) **3** — $g(0) = 0 \rightarrow f(0) = 3$.
(c) **2** — $f(1) = 1 \rightarrow g(1) = 2$.
(d) **-1** — $g(1) = 2 \rightarrow f(2) = -1$.

Ex 5 **4** — From the graph $g(1) = 2$; then $f(2) = 4$. So $f(g(1)) = 4$.

Ex 6
(a) **$u = 3t, y = \ln u$** — Inside function $u = 3t$; outside $y = \ln u$.
(b) **$u = 2r + 3, w = 5u^2$** — $u = 2r + 3$; $w = 5u^2$.
(c) **$u = -0.03t, P = e^u$** — $u = -0.03t$; $P = e^u$.

Problem 1
(a) **$h^2 + 6h + 11$** — $(2 + h)^2 + 2(2 + h) + 3 = 4 + 4h + h^2 + 4 + 2h + 3 = h^2 + 6h + 11$.
(b) **11** — $4 + 4 + 3 = 11$.
(c) **$h^2 + 6h$** — $h^2 + 6h + 11 - 11 = h^2 + 6h$.

Problem 2
(a) **$t^2 + 2t + 2$** — $(t + 1)^2 + 1 = t^2 + 2t + 2$.
(b) **$t^4 + 2t^2 + 2$** — $(t^2 + 1)^2 + 1 = t^4 + 2t^2 + 2$.
(c) **5** — $4 + 1 = 5$.
(d) **$2t^2 + 2$** — $2(t^2 + 1) = 2t^2 + 2$.
(e) **$t^4 + 2t^2 + 2$** — $(t^2 + 1)^2 + 1 = t^4 + 2t^2 + 2$.

Problem 3
(a) **4** — $g(1) = 2 \rightarrow f(2) = 4$.
(b) **2** — $f(1) = 1 \rightarrow g(1) = 2$.
(c) **$(x + 1)^2$** — $f(x + 1) = (x + 1)^2$.
(d) **$x^2 + 1$** — $g(x^2) = x^2 + 1$.
(e) **$t^3 + t^2$** — $t^2(t + 1) = t^3 + t^2$.

Problem 4

(a) $\sqrt{5}$ — $g(1) = 1 \to f(1) = \sqrt{5}$.

(b) 5 — $f(1) = \sqrt{5} \to g(\sqrt{5}) = 5$.

(c) $\sqrt{x^2 + 4}$ — $f(x^2) = \sqrt{x^2 + 4}$.

(d) $x + 4$ — $g(\sqrt{x + 4}) = (\sqrt{x + 4})^2 = x + 4$.

(e) $t^2\sqrt{t + 4}$ — $t^2 \cdot \sqrt{t + 4}$.

Problem 5

(a) e — $g(1) = 1 \to f(1) = e$.

(b) e^2 — $f(1) = e \to g(e) = e^2$.

(c) e^{x^2} — $f(x^2) = e^{x^2}$.

(d) e^{2x} — $g(e^x) = (e^x)^2 = e^{2x}$.

(e) t^2e^t — $t^2 \cdot e^t$.

Problem 6

(a) $1/7$ — $g(1) = 7 \to f(7) = 1/7$.

(b) 7 — $f(1) = 1 \to g(1) = 7$.

(c) $1/(3x + 4)$ — $f(3x + 4) = 1/(3x + 4)$.

(d) $3/x + 4$ — $g(1/x) = 3/x + 4$.

(e) $(3t + 4)/t$ — $(1/t)(3t + 4) = (3t + 4)/t$.

Problem 7

(a) 9 — $4 + 5 = 9$.

(b) 20 — $4 \times 5 = 20$.

(c) 25 — $g(2) = 5 \to f(5) = 25$.

(d) 11 — $f(2) = 4 \to g(4) = 11$.

Problem 8

(a) $2x^2 + 12x + 18$ — $2(x + 3)^2 = 2x^2 + 12x + 18$.

(b) $2x^2 + 3$ — $2x^2 + 3$.

(c) $8x^4$ — $2(2x^2)^2 = 2(4x^4) = 8x^4$.

Problem 9

(a) $10x^2 + 3$ — $2(5x^2) + 3 = 10x^2 + 3$.

(b) $20x^2 + 60x + 45$ — $5(2x + 3)^2 = 20x^2 + 60x + 45$.

(c) $4x + 9$ — $2(2x + 3) + 3 = 4x + 9$.

Problem 10

(a) $(\ln x)^2 + 1$ — $f(\ln x) = (\ln x)^2 + 1$.

(b) $\ln(x^2 + 1)$ — $g(x^2 + 1) = \ln(x^2 + 1)$.

(c) $x^4 + 2x^2 + 2$ — $(x^2 + 1)^2 + 1 = x^4 + 2x^2 + 2$.

Problem 11

(a) 3 — $g(0) = 2 \to f(2) = 3$.

(b) 4 — $g(1) = 3 \to f(3) = 4$.

(c) 11 — $g(2) = 5 \to f(5) = 11$.

(d) 8 — $f(2) = 3 \to g(3) = 8$.

(e) 12 — $f(3) = 4 \to g(4) = 12$.

Problem 12 **(a) 13, 9, 6, 7, 10, 14** — Add 3 to each f value.

(b) —, —, 10, 6, 3, 4 — f(x − 2): x = 2 → f(0) = 10, x = 3 → f(1) = 6, x = 4 → f(2) = 3, x = 5 → f(3) = 4; x = 0, 1 are not defined.

(c) 10, 15, 25, 40, 60, 75 — Multiply each g value by 5.

(d) −8, −4, −1, −2, −5, −9 — −10 + 2 = −8, −6 + 2 = −4, −3 + 2 = −1, −4 + 2 = −2, −7 + 2 = −5, −11 + 2 = −9.

(e) —, —, —, 2, 3, 5 — g(x − 3): x = 3 → g(0) = 2, x = 4 → g(1) = 3, x = 5 → g(2) = 5.

(f) 12, 9, 8, 12, 19, 26 — 10 + 2, 6 + 3, 3 + 5, 4 + 8, 7 + 12, 11 + 15.

Problem 13 **(a) $u = 3x - 1$, $y = 2^u$** — $u = 3x - 1$; $y = 2^u$.

(b) $u = 5t^2 + 10$, $P = \sqrt{u}$ — $u = 5t^2 + 10$; $P = \sqrt{u}$.

(c) $u = 3r + 4$, $w = 2 \ln u$ — $u = 3r + 4$; $w = 2 \ln u$.

Problem 14 **(a) $u = 5t^2 - 2$, $y = u^6$** — $u = 5t^2 - 2$; $y = u^6$.

(b) $u = -0.6t$, $P = 12e^u$ — $u = -0.6t$; $P = 12e^u$.

(c) $u = q^3 + 1$, $C = 12 \ln u$ — $u = q^3 + 1$; $C = 12 \ln u$.

Problem 15 **$2z + 1$** — $(z + 1)^2 - z^2 = z^2 + 2z + 1 - z^2 = 2z + 1$.

Problem 16 **$2zh + h^2$** — $(z + h)^2 - z^2 = 2zh + h^2$.

Problem 17 **$2zh - h^2$** — $z^2 - (z^2 - 2zh + h^2) = 2zh - h^2$.

Problem 18 **$4zh$** — $(z^2 + 2zh + h^2) - (z^2 - 2zh + h^2) = 4zh$.

Problem 19 **≈ 0.3** — $g(1) \approx 2 \rightarrow f(2) \approx 0.3$. So $f(g(1)) \approx 0.3$.

Problem 20 **≈ 1.1** — $f(2) \approx 0.3 \rightarrow g(0.3) \approx 1.1$. So $g(f(2)) \approx 1.1$.

Problem 21 **≈ −0.9** — $f(1) \approx -0.4 \rightarrow f(-0.4) \approx -0.9$. So $f(f(1)) \approx -0.9$.

Problem 22 **f(g(x)) 0, 1, 1, 3, 1, 1, 0** — $g(-3) = 3 \rightarrow f(3) = 0$; $g(-2) = 2 \rightarrow f(2) = 1$; $g(-1) = 2 \rightarrow 1$; $g(0) = 0 \rightarrow 3$; $g(1) = -2 \rightarrow 1$; $g(2) = -2 \rightarrow 1$; $g(3) = -3 \rightarrow 0$.

g(f(x)) 0, −2, −2, −3, −2, −2, 0 — $f(-3) = 0 \rightarrow g(0) = 0$; $f(-2) = 1 \rightarrow g(1) = -2$; $f(-1) = 2 \rightarrow g(2) = -2$; $f(0) = 3 \rightarrow g(3) = -3$; then symmetric: −2, −2, 0.

Problem 23 **Leaves $= 2(y - 1)^3 - (y - 1)^2$** — Leaves = n · B = $(2B^2 - B)B = 2B^3 - B^2$, with $B = y - 1$: Leaves = $2(y - 1)^3 - (y - 1)^2$.

Problem 24 **Cannot be done** — $v(10) \approx 60$, but u is only graphed for $0 \le x \le 50$, so u(60) cannot be found from the graph.

Problem 25 **≈ 18** — $v(40) \approx 16 \rightarrow u(16) \approx 18$.

Problem 26 **≈ 59** — $u(10) \approx 13 \rightarrow v(13) \approx 59$.

Problem 27 **Cannot be done** — u(40) ≈ 62, but v is only graphed for $0 \le x \le 50$, so v(62) cannot be found from the graph.

1.8.2 Stretches, Shifts & Reflections

Ex 7 **(a) A parabola opening upward with vertex (0, 0)** — $y = x^2$ is a U-shaped parabola opening upward, vertex at the origin.

(b) A parabola opening downward with vertex (0, 0) — $y = -x^2$ is the reflection of $y = x^2$ across the x-axis: opening downward, vertex (0, 0).

Ex 8 **a) The upper half of a circle of radius 3, from (−3, 0) to (3, 0) with top (0, 3)** — $y = \sqrt{(9 - x^2)} \ge 0$ is the upper semicircle of radius 3 centered at the origin.

b) The lower half of the circle, with bottom (0, −3) — The minus sign reflects the upper semicircle across the x-axis: the lower semicircle, lowest point (0, −3).

c) The upper semicircle stretched vertically: from (−3, 0) to (3, 0) with top (0, 6) — Multiplying by 2 stretches vertically by 2: same ends (±3, 0), top at (0, 6).

d) The upper semicircle compressed vertically: from (−3, 0) to (3, 0) with top (0, 1) — Multiplying by 1/3 compresses vertically: ends (±3, 0), top at (0, 1).

Ex 9 **a) The graph of e^x shifted 2 up: passes (0, 3), horizontal asymptote y = 2** — Adding 2 outside shifts e^x up 2: (0, 1) → (0, 3) and the asymptote $y = 0 \to y = 2$.

b) The graph of ln x shifted 2 right: passes (3, 0), vertical asymptote x = 2 — $x - 2$ inside shifts ln x right 2: (1, 0) → (3, 0) and the asymptote $x = 0 \to x = 2$.

Ex 10 **a) Parabola shifted 3 up: vertex (0, 3)** — Outside + 3: up 3, vertex (0, 3).

b) Parabola shifted 3 left: vertex (−3, 0) — Inside + 3: left 3, vertex (−3, 0).

c) Parabola shifted 2 right: vertex (2, 0) — Inside − 2: right 2, vertex (2, 0).

Problem 28 **(a) Stretch vertically by 2; the point (0, 1) moves to (0, 2)** — $y = 2H(x)$ has the form $y = A \cdot H(x - h) + k$ with A = 2, h = 0, k = 0. Stretch vertically by 2. A point (x, y) on the graph of H moves to $(x + h, A \cdot y + k)$: (0, 1) → (0, 2).

(b) Shift 1 unit up; the point (0, 1) moves to (0, 2) — $y = H(x) + 1$ has the form $y = A \cdot H(x - h) + k$ with A = 1, h = 0, k = 1. Shift 1 unit up. A point (x, y) on the graph of H moves to $(x + h, A \cdot y + k)$: (0, 1) → (0, 2).

(c) Shift 1 unit left; the point (0, 1) moves to (−1, 1) — $y = H(x + 1)$ has the form $y = A \cdot H(x - h) + k$ with A = 1, h = −1, k = 0. Shift 1 unit left. A point (x, y) on the graph of H moves to $(x + h, A \cdot y + k)$: (0, 1) → (−1, 1).

(d) Reflect across the x-axis; the point (0, 1) moves to (0, −1) — $y = -H(x)$ has the form $y = A \cdot H(x - h) + k$ with A = −1, h = 0, k = 0. Reflect across the x-axis. A point (x, y) on the graph of H moves to $(x + h, A \cdot y + k)$: (0, 1) → (0, −1).

Problem 29

(a) Shift 2 units left; the point (2, 4) moves to (0, 4) — $y = f(x + 2)$ has the form $y = A\cdot f(x - h) + k$ with $A = 1$, $h = -2$, $k = 0$. Shift 2 units left. A point (x, y) on the graph of f moves to $(x + h, A\cdot y + k)$: $(2, 4) \to (0, 4)$.

(b) Shift 1 unit right; the point (2, 4) moves to (3, 4) — $y = f(x - 1)$ has the form $y = A\cdot f(x - h) + k$ with $A = 1$, $h = 1$, $k = 0$. Shift 1 unit right. A point (x, y) on the graph of f moves to $(x + h, A\cdot y + k)$: $(2, 4) \to (3, 4)$.

(c) Shift 4 units down; the point (2, 4) moves to (2, 0) — $y = f(x) - 4$ has the form $y = A\cdot f(x - h) + k$ with $A = 1$, $h = 0$, $k = -4$. Shift 4 units down. A point (x, y) on the graph of f moves to $(x + h, A\cdot y + k)$: $(2, 4) \to (2, 0)$.

(d) Shift 1 unit left, then shift 3 units up; the point (2, 4) moves to (1, 7) — $y = f(x + 1) + 3$ has the form $y = A\cdot f(x - h) + k$ with $A = 1$, $h = -1$, $k = 3$. Shift 1 unit left, then shift 3 units up. A point (x, y) on the graph of f moves to $(x + h, A\cdot y + k)$: $(2, 4) \to (1, 7)$.

(e) Stretch vertically by 3; the point (2, 4) moves to (2, 12) — $y = 3f(x)$ has the form $y = A\cdot f(x - h) + k$ with $A = 3$, $h = 0$, $k = 0$. Stretch vertically by 3. A point (x, y) on the graph of f moves to $(x + h, A\cdot y + k)$: $(2, 4) \to (2, 12)$.

(f) Reflect across the x-axis, then shift 1 unit up; the point (2, 4) moves to (2, −3) — $y = -f(x) + 1$ has the form $y = A\cdot f(x - h) + k$ with $A = -1$, $h = 0$, $k = 1$. Reflect across the x-axis, then shift 1 unit up. A point (x, y) on the graph of f moves to $(x + h, A\cdot y + k)$: $(2, 4) \to (2, -3)$.

Problem 30

(a) Shift 2 units left; the point (0, 1) moves to (−2, 1) — $y = f(x + 2)$ has the form $y = A\cdot f(x - h) + k$ with $A = 1$, $h = -2$, $k = 0$. Shift 2 units left. A point (x, y) on the graph of f moves to $(x + h, A\cdot y + k)$: $(0, 1) \to (-2, 1)$.

(b) Shift 1 unit right; the point (0, 1) moves to (1, 1) — $y = f(x - 1)$ has the form $y = A\cdot f(x - h) + k$ with $A = 1$, $h = 1$, $k = 0$. Shift 1 unit right. A point (x, y) on the graph of f moves to $(x + h, A\cdot y + k)$: $(0, 1) \to (1, 1)$.

(c) Shift 4 units down; the point (0, 1) moves to (0, −3) — $y = f(x) - 4$ has the form $y = A\cdot f(x - h) + k$ with $A = 1$, $h = 0$, $k = -4$. Shift 4 units down. A point (x, y) on the graph of f moves to $(x + h, A\cdot y + k)$: $(0, 1) \to (0, -3)$.

(d) Shift 1 unit left, then shift 3 units up; the point (0, 1) moves to (−1, 4) — $y = f(x + 1) + 3$ has the form $y = A\cdot f(x - h) + k$ with $A = 1$, $h = -1$, $k = 3$. Shift 1 unit left, then shift 3 units up. A point (x, y) on the graph of f moves to $(x + h, A\cdot y + k)$: $(0, 1) \to (-1, 4)$.

(e) Stretch vertically by 3; the point (0, 1) moves to (0, 3) — $y = 3f(x)$ has the form $y = A\cdot f(x - h) + k$ with $A = 3$, $h = 0$, $k = 0$. Stretch vertically by 3. A point (x, y) on the graph of f moves to $(x + h, A\cdot y + k)$: $(0, 1) \to (0, 3)$.

(f) Reflect across the x-axis, then shift 1 unit up; the point (0, 1) moves to (0, 0) — $y = -f(x) + 1$ has the form $y = A\cdot f(x - h) + k$ with $A = -1$, $h = 0$, $k = 1$. Reflect across the x-axis, then shift 1 unit up. A point (x, y) on the graph of f moves to $(x + h, A\cdot y + k)$: $(0, 1) \to (0, 0)$.

Problem 31

(a) Shift 2 units left; the point (0, 1) moves to (−2, 1) — $y = f(x + 2)$ has the form $y = A \cdot f(x - h) + k$ with $A = 1$, $h = -2$, $k = 0$. Shift 2 units left. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(0, 1) \to (-2, 1)$.

(b) Shift 1 unit right; the point (0, 1) moves to (1, 1) — $y = f(x - 1)$ has the form $y = A \cdot f(x - h) + k$ with $A = 1$, $h = 1$, $k = 0$. Shift 1 unit right. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(0, 1) \to (1, 1)$.

(c) Shift 4 units down; the point (0, 1) moves to (0, −3) — $y = f(x) - 4$ has the form $y = A \cdot f(x - h) + k$ with $A = 1$, $h = 0$, $k = -4$. Shift 4 units down. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(0, 1) \to (0, -3)$.

(d) Shift 1 unit left, then shift 3 units up; the point (0, 1) moves to (−1, 4) — $y = f(x + 1) + 3$ has the form $y = A \cdot f(x - h) + k$ with $A = 1$, $h = -1$, $k = 3$. Shift 1 unit left, then shift 3 units up. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(0, 1) \to (-1, 4)$.

(e) Stretch vertically by 3; the point (0, 1) moves to (0, 3) — $y = 3f(x)$ has the form $y = A \cdot f(x - h) + k$ with $A = 3$, $h = 0$, $k = 0$. Stretch vertically by 3. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(0, 1) \to (0, 3)$.

(f) Reflect across the x-axis, then shift 1 unit up; the point (0, 1) moves to (0, 0) — $y = -f(x) + 1$ has the form $y = A \cdot f(x - h) + k$ with $A = -1$, $h = 0$, $k = 1$. Reflect across the x-axis, then shift 1 unit up. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(0, 1) \to (0, 0)$.

Problem 32

Shift 2 units up; the point (2, 2) moves to (2, 4) — $y = f(x) + 2$ has the form $y = A \cdot f(x - h) + k$ with $A = 1$, $h = 0$, $k = 2$. Shift 2 units up. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(2, 2) \to (2, 4)$.

Problem 33

Stretch vertically by 2; the point (2, 2) moves to (2, 4) — $y = 2f(x)$ has the form $y = A \cdot f(x - h) + k$ with $A = 2$, $h = 0$, $k = 0$. Stretch vertically by 2. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(2, 2) \to (2, 4)$.

Problem 34

Shift 1 unit right; the point (2, 2) moves to (3, 2) — $y = f(x - 1)$ has the form $y = A \cdot f(x - h) + k$ with $A = 1$, $h = 1$, $k = 0$. Shift 1 unit right. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(2, 2) \to (3, 2)$.

Problem 35

Stretch vertically by 3, then reflect across the x-axis; the point (2, 2) moves to (2, −6) — $y = -3f(x)$ has the form $y = A \cdot f(x - h) + k$ with $A = -3$, $h = 0$, $k = 0$. Stretch vertically by 3, then reflect across the x-axis. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(2, 2) \to (2, -6)$.

Problem 36

Stretch vertically by 2, then shift 1 unit down; the point (2, 2) moves to (2, 3) — $y = 2f(x) - 1$ has the form $y = A \cdot f(x - h) + k$ with $A = 2$, $h = 0$, $k = -1$. Stretch vertically by 2, then shift 1 unit down. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(2, 2) \to (2, 3)$.

Problem 37

Reflect across the x-axis, then shift 2 units up; the point (2, 2) moves to (2, 0) — $y = 2 - f(x)$ has the form $y = A \cdot f(x - h) + k$ with $A = -1$, $h = 0$, $k = 2$. Reflect across the x-axis, then shift 2 units up. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(2, 2) \to (2, 0)$.

Problem 38

(a) Shift 2 units up; the point (0, 1) moves to (0, 3) — $y = f(x) + 2$ has the form $y = A \cdot f(x - h) + k$ with $A = 1$, $h = 0$, $k = 2$. Shift 2 units up. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(0, 1) \to (0, 3)$.

(b) Shift 1 unit right; the point (0, 1) moves to (1, 1) — $y = f(x - 1)$ has the form $y = A \cdot f(x - h) + k$ with $A = 1$, $h = 1$, $k = 0$. Shift 1 unit right. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(0, 1) \to (1, 1)$.

(c) Stretch vertically by 3; the point (0, 1) moves to (0, 3) — $y = 3f(x)$ has the form $y = A \cdot f(x - h) + k$ with $A = 3$, $h = 0$, $k = 0$. Stretch vertically by 3. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(0, 1) \to (0, 3)$.

(d) Reflect across the x-axis; the point (0, 1) moves to (0, −1) — $y = -f(x)$ has the form $y = A \cdot f(x - h) + k$ with $A = -1$, $h = 0$, $k = 0$. Reflect across the x-axis. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(0, 1) \to (0, -1)$.

Problem 39

(a) Shift 2 units up; the point (1, 1) moves to (1, 3) — $y = f(x) + 2$ has the form $y = A \cdot f(x - h) + k$ with $A = 1$, $h = 0$, $k = 2$. Shift 2 units up. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(1, 1) \to (1, 3)$.

(b) Shift 1 unit right; the point (1, 1) moves to (2, 1) — $y = f(x - 1)$ has the form $y = A \cdot f(x - h) + k$ with $A = 1$, $h = 1$, $k = 0$. Shift 1 unit right. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(1, 1) \to (2, 1)$.

(c) Stretch vertically by 3; the point (1, 1) moves to (1, 3) — $y = 3f(x)$ has the form $y = A \cdot f(x - h) + k$ with $A = 3$, $h = 0$, $k = 0$. Stretch vertically by 3. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(1, 1) \to (1, 3)$.

(d) Reflect across the x-axis; the point (1, 1) moves to (1, −1) — $y = -f(x)$ has the form $y = A \cdot f(x - h) + k$ with $A = -1$, $h = 0$, $k = 0$. Reflect across the x-axis. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(1, 1) \to (1, -1)$.

Problem 40

(a) Shift 2 units up; the point (0, 2) moves to (0, 4) — $y = f(x) + 2$ has the form $y = A \cdot f(x - h) + k$ with $A = 1$, $h = 0$, $k = 2$. Shift 2 units up. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(0, 2) \to (0, 4)$.

(b) Shift 1 unit right; the point (0, 2) moves to (1, 2) — $y = f(x - 1)$ has the form $y = A \cdot f(x - h) + k$ with $A = 1$, $h = 1$, $k = 0$. Shift 1 unit right. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(0, 2) \to (1, 2)$.

(c) Stretch vertically by 3; the point (0, 2) moves to (0, 6) — $y = 3f(x)$ has the form $y = A \cdot f(x - h) + k$ with $A = 3$, $h = 0$, $k = 0$. Stretch vertically by 3. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(0, 2) \to (0, 6)$.

(d) Reflect across the x-axis; the point (0, 2) moves to (0, −2) — $y = -f(x)$ has the form $y = A \cdot f(x - h) + k$ with $A = -1$, $h = 0$, $k = 0$. Reflect across the x-axis. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(0, 2) \to (0, -2)$.

Problem 41

(a) Shift 2 units up; the point (3, 3) moves to (3, 5) — $y = f(x) + 2$ has the form $y = A \cdot f(x - h) + k$ with $A = 1$, $h = 0$, $k = 2$. Shift 2 units up. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(3, 3) \to (3, 5)$.

(b) Shift 1 unit right; the point (3, 3) moves to (4, 3) — $y = f(x - 1)$ has the form $y = A \cdot f(x - h) + k$ with $A = 1$, $h = 1$, $k = 0$. Shift 1 unit right. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(3, 3) \to (4, 3)$.

(c) Stretch vertically by 3; the point (3, 3) moves to (3, 9) — $y = 3f(x)$ has the form $y = A \cdot f(x - h) + k$ with $A = 3$, $h = 0$, $k = 0$. Stretch vertically by 3. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(3, 3) \to (3, 9)$.

(d) Reflect across the x-axis; the point (3, 3) moves to (3, −3) — $y = -f(x)$ has the form $y = A \cdot f(x - h) + k$ with $A = -1$, $h = 0$, $k = 0$. Reflect across the x-axis. A point (x, y) on the graph of f moves to $(x + h, A \cdot y + k)$: $(3, 3) \to (3, -3)$.

Problem 42

a. The same curve shifted 4 hours to the right: Q = 0 until t = 4, then rising toward 7.2 — Starting at noon means everything happens 4 hours later: the curve is shifted 4 units to the right (it starts rising at $t = 4$).

b. Horizontal shift — It is a horizontal shift (4 hours to the right).

c. g(t) = f(t − 4) — Shift right 4: $g(t) = f(t - 4)$.

Problem 43

(a) $y = 2x^2 + 1$ — Stretch: $2x^2$. Then up 1: $y = 2x^2 + 1$.

(b) $y = 2(x^2 + 1) = 2x^2 + 2$ — Shift first: $x^2 + 1$. Then stretch by 2: $y = 2(x^2 + 1) = 2x^2 + 2$.

(c) No: $2x^2 + 1$ has vertex (0, 1) and $2x^2 + 2$ has vertex (0, 2); the order matters — $2x^2 + 1 \neq 2x^2 + 2$. When the shift is done first, the stretch also doubles the shift, so the order matters.

2024 Q5

a. The same curve shifted 4 hours to the right: Q = 0 until t = 4, then rising toward 7.2 — Noon is 4 hours after 8 am: the curve shifts 4 units right.

b. g(t) = f(t − 4) — $g(t) = f(t - 4)$.

1.8.3 Test Your Understanding

TYU 5 · P1 From the graph: f(−4) = 0, f(−1) = 2, f(3) = 1, f(4) = 0.

f(x + 1) needs f(−3), f(0) and f(4). On the segment from (−4, 0) to (−1, 2) the slope is 2/3; on the segment from (−1, 2) to (3, 1) the slope is −1/4:

$$f(-3) = 0 + \frac{2}{3}(1) = \frac{2}{3} \approx 0.67$$

$$f(0) = 2 - \frac{1}{4}(1) = 1.75$$

$$f(4) = 0$$

x	**−4**	**−1**	**3**	**4**
f(x)	0	2	1	0
f(x) + 4	4	6	5	4
− f(x)	0	−2	−1	0
− f(x) + 8	8	6	7	8
f(x + 1)	2/3 ≈ 0.67	1.75	0	—

(The handwritten key reads f(−3) from the graph as ≈ 0.5; the exact value from the line is 2/3.)

1.9 Proportionality and Power Functions

In this section

1. Proportionality 4 lecture · 14 textbook · 1 past exam
2. Power Functions 1 lecture · 16 textbook
3. Test Your Understanding 2 questions

1.9.1 Proportionality

Lecture Problems

Ex 1 The heart mass of a mammal is proportional to its body mass.

i. Write a formula for heart mass, H, as a function of body mass, B.

ii. A human with a body mass of 70 kilograms has a heart mass of 0.42 kilograms. Use this information to find the constant of proportionality.

iii. Estimate the heart mass of a horse with a body mass of 650 kg.

Ex 2

Same as Textbook Problem 22 (similar problem with different numbers: Past Exam 2022 Question 4)

The blood mass of a mammal is proportional to its body mass. A rhinoceros with body mass 3000 kilograms has blood mass of 150 kilograms.

i. Find a formula for the blood mass of a mammal as a function of the body mass.

ii. Estimate the blood mass of a human with body mass 70 kilograms.

Ex 3 The period of a pendulum, T, is the amount of time required for the pendulum to make one complete swing. For small swings, the period, T, is approximately proportional to the square root of l, the pendulum's length.

i. Write a formula for period, T, as a function of pendulum's length, l.

ii. Calculate the proportionality constant k for a 4 meter pendulum with a period of 10 seconds.

iii. Estimate the period T for a 0.25 meter pendulum.

Ex 5 If N is the average number of species found on an island and A is the area of the island, observations have shown that N is approximately proportional to the cube root of A.

Write a formula for N as a function of A and describe the shape of the graph of this function.

Textbook Problems

Problem 13 In Problems 13–16, write a formula representing the function.

The strength, S, of a beam is proportional to the square of its thickness, h.

Problem 14 Write a formula representing the function.

The energy, E, expended by a swimming dolphin is proportional to the cube of the speed, v, of the dolphin.

Problem 15 Write a formula representing the function.

The average velocity, v, for a trip over a fixed distance, d, is inversely proportional to the time of travel, t.

Problem 16 Write a formula representing the function.

The gravitational force, F, between two bodies is inversely proportional to the square of the distance d between them.

Problem 17 The number of species of lizards, N, found on an island off Baja California is proportional to the fourth root of the area, A, of the island.

Write a formula for N as a function of A. Graph this function. Is it increasing or decreasing? Is the graph concave up or concave down? What does this tell you about lizards and island area?

Problem 18 The surface area of a mammal, S, satisfies the equation $S = kM^{2/3}$, where M is the body mass, and the constant of proportionality k depends on the body shape of the mammal. A human of body mass 70 kilograms has surface area 18,600 cm^2.

Find the constant of proportionality for humans. Find the surface area of a human with body mass 60 kilograms.

Problem 20 Kleiber's Law states that the metabolic needs (such as calorie requirements) of a mammal are proportional to its body weight raised to the 0.75 power. Surprisingly, the daily diets of mammals conform to this relation well. Assuming Kleiber's Law holds:

a. Write a formula for C, daily calorie consumption, as a function of body weight, W.

---

b. Sketch a graph of this function. Which description is correct?

---

c. If a human weighing 150 pounds needs to consume 1800 calories a day, estimate the daily calorie requirement of a horse weighing 700 lbs and of a rabbit weighing 9 lbs.

---

d. On a per-pound basis, which animal requires more calories: a mouse or an elephant?

Problem 21 Allometry is the study of the relative size of different parts of a body as a consequence of growth. In this problem, you will check the accuracy of an allometric equation: the weight of a fish is proportional to the cube of its length. Table 1.37 relates the weight, y, in gm, of plaice (a type of fish) to its length, x, in cm.

x	33.5	34.5	35.5	36.5	37.5	38.5
y	332	363	391	419	455	500

x	39.5	40.5	41.5	42.5	43.5
y	538	574	623	674	724

Does this data support the hypothesis that (approximately) $y = kx^3$? If so, estimate the constant of proportionality, k.

Problem 22

Same as Lecture Example 2 (similar problem with different numbers: Past Exam 2022 Question 4)

The blood mass of a mammal is proportional to its body mass. A rhinoceros with body mass 3000 kilograms has blood mass of 150 kilograms.

Find a formula for the blood mass of a mammal as a function of the body mass and estimate the blood mass of a human with body mass 70 kilograms.

Problem 23 Biologists estimate that the number of animal species of a certain body length is inversely proportional to the square of the body length.

Write a formula for the number of animal species, N, of a certain body length as a function of the length, L. Are there more species at large lengths or at small lengths? Explain.

Problem 24 The specific heat, s, of an element is the number of calories of heat required to raise the temperature of one gram of the element by one degree Celsius.

Element	Li	Mg	Al	Fe	Ag	Pb	Hg
w	6.9	24.3	27.0	55.8	107.9	207.2	200.6
s	0.92	0.25	0.21	0.11	0.056	0.031	0.033

Use the table to decide if s is proportional or inversely proportional to the atomic weight, w, of the element. If so, find the constant of proportionality.

Problem 25 The circulation time of a mammal (that is, the average time it takes for all the blood in the body to circulate once and return to the heart) is proportional to the fourth root of the body mass of the mammal.

a. Write a formula for the circulation time, T, in terms of the body mass, B.

b. If an elephant of body mass 5230 kilograms has a circulation time of 148 seconds, find the constant of proportionality.

c. What is the circulation time of a human with body mass 70 kilograms?

Problem 27 The infrastructure needs of a region (for example, the number of miles of electrical cable, the number of miles of roads, the number of gas stations) depend on its population. Cities enjoy economies of scale. For example, the number of gas stations is proportional to the population raised to the power of 0.77.

a. Write a formula for the number, N, of gas stations in a city as a function of the population, P, of the city.

b. If city A is 10 times bigger than city B, how do their number of gas stations compare?

c. Which is expected to have more gas stations per person, a town of 10,000 people or a city of 500,000 people?

Problem 28 According to the National Association of Realtors, the minimum annual gross income, m, in thousands of dollars, needed to obtain a 30-year home loan of A thousand dollars at 9% is given in Table 1.38. Note that the larger the loan, the greater the income needed. Of course, not every mortgage is financed at 9%. In fact, excepting for slight variations, mortgage interest rates are generally determined not by individual banks but by the economy as a whole. The minimum annual gross income, m, in thousands of dollars, needed for a home loan of $100,000 at various interest rates, r, is given in Table 1.39. Note that obtaining a loan at a time when interest rates are high requires a larger income.

Table 1.38

A	50	75	100	150	200
m	17.242	25.863	34.484	51.726	68.968

Table 1.39

r	8	9	10	11	12
m	31.447	34.484	37.611	40.814	44.084

a. Is the size of the loan, A, proportional to the minimum annual gross income, m?

b. Is the percentage rate, r, proportional to the minimum annual gross income, m?

Past Exam Questions

2022 Q4

Similar problem (different numbers): Lecture Example 2 and Textbook Problem 22

The blood mass of a mammal is proportional to its body mass. A horse with body mass 600 kilograms has a blood mass of 30 kilograms.

a. Find a formula for the blood mass of a mammal, B, as a function of the body mass, M.

b. Estimate the blood mass of a human with body mass 80 kilograms.

1.9.2 Power Functions

Lecture Problems

Ex 4 Which of the following are power functions? For those which are, write the function in the form $y = kx^p$, and give the coefficient k and the exponent p.

a) $y = 5 / x^3$

b) $y = 2 / (3x)$

c) $y = 5x^2 / 2$

d) $y = 5.2^x$

e) $y = 3\sqrt{x}$

f) $y = (3x^2)^3$

Textbook Problems

Problem 1 In Problems 1–12, determine whether or not the function is a power function. If it is a power function, write it in the form $y = kx^p$ and give the values of k and p.

$y = 5\sqrt{x}$

Problem 2 Is it a power function? If so, write it as $y = kx^p$ and give k and p.

$y = 3 / x^2$

Problem 3 Is it a power function? If so, write it as $y = kx^p$ and give k and p.

$y = 2^x$

Problem 4 Is it a power function? If so, write it as $y = kx^p$ and give k and p.

$y = 3 / (8x)$

Problem 5 Is it a power function? If so, write it as $y = kx^p$ and give k and p.

$y = (3x^5)^2$

Problem 6 Is it a power function? If so, write it as $y = kx^p$ and give k and p.

$y = 5 / (2\sqrt{x})$

Problem 7 Is it a power function? If so, write it as $y = kx^p$ and give k and p.

$y = 3 \cdot 5^x$

Problem 8 Is it a power function? If so, write it as $y = kx^p$ and give k and p.

$y = 2x^2 / 10$

Problem 9 Is it a power function? If so, write it as $y = kx^p$ and give k and p.

$y = 8 / x$

Problem 10 Is it a power function? If so, write it as $y = kx^p$ and give k and p.

$y = (5x)^3$

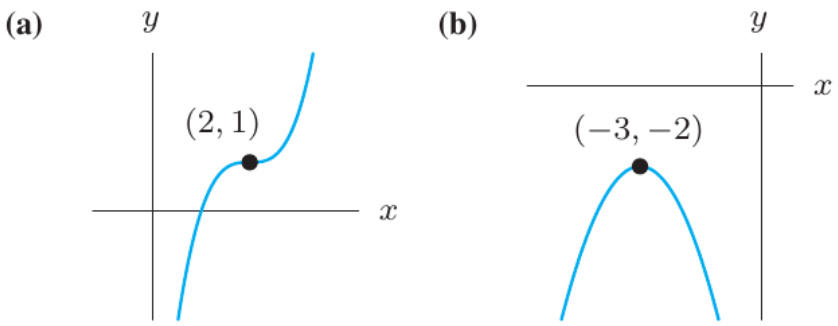
Problem 11 Is it a power function? If so, write it as $y = kx^p$ and give k and p.

$y = 3x^2 + 4$

Problem 12 Is it a power function? If so, write it as $y = kx^p$ and give k and p.

$y = x / 5$

Problem 19 Use shifts of power functions to find a possible formula for each of the graphs:



(a) Graph (a)

(b) Graph (b)

Problem 26 The DuBois formula relates a person's surface area s, in m^2, to weight w, in kg, and height h, in cm, by $s = 0.01w^{0.25}h^{0.75}$.

a. What is the surface area of a person who weighs 65 kg and is 160 cm tall?

---

b. What is the weight of a person whose height is 180 cm and who has a surface area of 1.5 m^2?

---

c. For people of fixed weight 70 kg, solve for h as a function of s. Simplify your answer.

Problem 29 A standard tone of 20,000 dynes/cm² (about the loudness of a rock band) is assigned a value of 10. A subject listened to other sounds, such as a light whisper, normal conversation, thunder, a jet plane at takeoff, and so on. In each case, the subject was asked to judge the loudness and assign it a number relative to 10, the value of the standard tone. This is a "judgment of magnitude" experiment. The power law $J = al^{0.3}$ was found to model the situation well, where l is the actual loudness (measured in dynes/cm²) and J is the judged loudness.

a. What is the value of a?

b. What is the judged loudness if the actual loudness is 0.2 dynes/cm² (normal conversation)?

c. What is the actual loudness if judged loudness is 20?

Problem 30 A sporting goods wholesaler finds that when the price of a product is $25, the company sells 500 units per week. When the price is $30, the number sold per week decreases to 460 units.

Find the demand, q, as a function of price, p, assuming that the demand curve is linear.

1.9.3 Test Your Understanding

Test Your Understanding

TYU 5 · P2 The following table gives the values for y = f(x).

x	1	2	3	4
y	3	3/2	1	3/4

Which of the following statements best describes the relationship between y and x?

A. y is inversely proportional to x
B. y is proportional to x^2
C. y is proportional to x
D. y is inversely proportional to x^2
E. None of the above

TYU 5 · P3 The number of species, S, on an island is proportional to the square root of the area, A, of the island. An island which has an area of 25 square miles contains 30 species.

a. Which of the following statements best describes the relationship between S and A?

A. S is inversely proportional to A
B. S is proportional to A^2
C. S is proportional to A
D. S is inversely proportional to $A^{1/2}$
E. S is proportional to $A^{1/2}$

b. The best function describing the number of species as a function of the area of an island is

A. Surge
B. Power
C. Exponential
D. Logistic
E. Periodic

c. How many species would be expected on an island of 9 square miles?

A. 6
B. 36
C. 18
D. 0
E. 9

Answers · Section 1.9

1.9.1 Proportionality

Ex 1 **i. H = kB** — Proportional → H = kB.
ii. k = 0.006 — 0.42 = 70k → k = 0.42/70 = 0.006.
iii. 3.9 kg — H = 0.006(650) = 3.9 kg.

Ex 2 **i. B = 0.05M** — 150 = 3000k → k = 0.05 → B = 0.05M.
ii. 3.5 kg — B = 0.05(70) = 3.5 kg.

Ex 3 **i. T = k√l** — Proportional to the square root → T = k√l.
ii. k = 5 — 10 = k√4 = 2k → k = 5.
iii. T = 2.5 seconds — T = 5√0.25 = 5(0.5) = 2.5 seconds.

Ex 5 **$N = kA^{1/3}$; increasing and concave down** — Cube root → $N = kA^{1/3}$. Since $0 < 1/3 < 1$, the graph is increasing and concave down.

Problem 13 **$S = kh^2$** — $S = kh^2$.

Problem 14 **$E = kv^3$** — $E = kv^3$.

Problem 15 **v = k/t (k = d)** — Inversely proportional → v = k/t, where k = d.

Problem 16 **$F = k/d^2$** — $F = k/d^2$.

Problem 17 **$N = kA^{1/4}$; increasing, concave down: more area gives more species, but gains slow down** — $N = kA^{1/4}$. Since $0 < 1/4 < 1$, it is increasing and concave down: bigger islands have more lizard species, but the increase slows down as the area grows.

Problem 18 **k ≈ 1095; S(60) ≈ 16,783 cm²** — $k = 18{,}600/70^{2/3} \approx 1095$. $S(60) = 1095 \cdot 60^{2/3} \approx 16{,}783$ cm².

Problem 20
a. $C = kW^{0.75}$ — $C = kW^{0.75}$.
b. Increasing and concave down, starting at the origin — $0 < 0.75 < 1$, so the graph rises from the origin and is concave down.
c. Horse ≈ 5715 calories; rabbit ≈ 218 calories — $k = 1800/150^{0.75} \approx 42.0$. Horse: $42.0(700)^{0.75} \approx 5715$. Rabbit: $42.0(9)^{0.75} \approx 218$.
d. The mouse — Calories per pound = $kW^{0.75}/W = kW^{-0.25}$, which is larger for small W: the mouse.

Problem 21 **Yes, k ≈ 0.0087** — y/x^3 for each column is about 0.0086–0.0088 (nearly constant), so yes, with k ≈ 0.0087.

Problem 22 **B = 0.05M; B(70) = 3.5 kg** — k = 150/3000 = 0.05 → B = 0.05M; B(70) = 3.5 kg.

Problem 23 **$N = k/L^2$; more species at small lengths** — $N = k/L^2$. As L increases, N decreases, so there are more species at small lengths.

Problem 24 **Inversely proportional, $k \approx 6$** — $s \cdot w \approx 6.35, 6.08, 5.67, 6.14, 6.04, 6.42, 6.62$ — roughly constant ≈ 6. So $s \approx 6/w$: inversely proportional with $k \approx 6$.

Problem 25

a. **$T = kB^{1/4}$** — $T = kB^{1/4}$.

b. **$k \approx 17.4$** — $k = 148/5230^{1/4} \approx 148/8.504 \approx 17.4$.

c. **≈ 50.3 seconds** — $T = 17.4 \cdot 70^{1/4} \approx 17.4 \cdot 2.893 \approx 50.3$ seconds.

Problem 27

a. **$N = kP^{0.77}$** — $N = kP^{0.77}$.

b. **City A has ≈ 5.9 times as many** — $k(10P)^{0.77} = 10^{0.77} \cdot kP^{0.77} \approx 5.9 \cdot kP^{0.77}$.

c. **The town of 10,000** — Per person: $N/P = kP^{-0.23}$, which is larger for smaller P: the town.

Problem 28

a. **Yes ($A = 2.9$m)** — $A/m = 50/17.242 = 75/25.863 = \ldots \approx 2.9$ (constant), so yes: $A \approx 2.9$m.

b. **No** — r/m = 0.254, 0.261, 0.266, 0.270, 0.272 — not constant, so no.

2022 Q4

a. **$B = 0.05M$** — $30 = 600k \rightarrow k = 0.05 \rightarrow B = 0.05M$.

b. **4 kg** — $B = 0.05(80) = 4$ kg.

1.9.2 Power Functions

Ex 4

a) **Yes: $y = 5x^{-3}$; $k = 5$, $p = -3$** — $5/x^3 = 5x^{-3}$: $k = 5$, $p = -3$.

b) **Yes: $y = (2/3)x^{-1}$; $k = 2/3$, $p = -1$** — $2/(3x) = (2/3)x^{-1}$.

c) **Yes: $y = (5/2)x^2$; $k = 5/2$, $p = 2$** — $5x^2/2 = (5/2)x^2$.

d) **Not a power function** — The variable is in the exponent: it is an exponential function, not a power function.

e) **Yes: $y = 3x^{1/2}$; $k = 3$, $p = 1/2$** — $3\sqrt{x} = 3x^{1/2}$.

f) **Yes: $y = 27x^6$; $k = 27$, $p = 6$** — $(3x^2)^3 = 27x^6$.

Problem 1 **Yes: $y = 5x^{1/2}$; $k = 5$, $p = 1/2$** — $5\sqrt{x} = 5x^{1/2}$.

Problem 2 **Yes: $y = 3x^{-2}$; $k = 3$, $p = -2$** — $3/x^2 = 3x^{-2}$.

Problem 3 **Not a power function** — The variable is in the exponent: exponential, not a power function.

Problem 4 **Yes: $y = (3/8)x^{-1}$; $k = 3/8$, $p = -1$** — $3/(8x) = (3/8)x^{-1}$.

Problem 5 **Yes: $y = 9x^{1^0}$; $k = 9$, $p = 10$** — $(3x^5)^2 = 9x^{1^0}$.

Problem 6 **Yes: $y = 2.5x^{-1/2}$; $k = 5/2$, $p = -1/2$** — $5/(2\sqrt{x}) = (5/2)x^{-1/2}$.

Problem 7 **Not a power function** — The variable is in the exponent: exponential, not a power function.

Problem 8 **Yes: $y = 0.2x^2$; $k = 1/5$, $p = 2$** — $2x^2/10 = (1/5)x^2$.

Problem 9 **Yes: $y = 8x^{-1}$; $k = 8$, $p = -1$** — $8/x = 8x^{-1}$.

Problem 10 **Yes: $y = 125x^3$; $k = 125$, $p = 3$** — $(5x)^3 = 125x^3$.

Problem 11 **Not a power function** — The "+ 4" means it is not of the form kx^p.

Problem 12 **Yes: $y = 0.2x^1$; $k = 1/5$, $p = 1$** — $x/5 = (1/5)x^1$.

Problem 19 **(a)** **$y = (x - 2)^3 + 1$** — A cubic shape moved so its center point is (2, 1): $y = (x - 2)^3 + 1$.
(b) **$y = -(x + 3)^2 - 2$** — A downward parabola with vertex $(-3, -2)$: $y = -(x + 3)^2 - 2$.

Problem 26 **a.** **$\approx 1.28\ m^2$** — $0.01(65)^{0.25}(160)^{0.75} \approx 0.01(2.839)(45.0) \approx 1.28\ m^2$.
b. **≈ 86.8 kg** — $w^{0.25} = 1.5/(0.01 \cdot 180^{0.75}) \rightarrow w = (1.5/(0.01 \cdot 180^{0.75}))^4 \approx 86.8$ kg.
c. **$h \approx 112.6\ s^{4/3}$** — $h^{0.75} = s/(0.01 \cdot 70^{0.25}) \rightarrow h = (s/(0.01 \cdot 70^{0.25}))^{4/3} \approx 112.6\ s^{4/3}$.

Problem 29 **a.** **$a \approx 0.512$** — $10 = a(20{,}000)^{0.3} \rightarrow a = 10/20{,}000^{0.3} \approx 0.512$.
b. **$J \approx 0.32$** — $J = 0.512(0.2)^{0.3} \approx 0.32$.
c. **$l \approx 201{,}600$ dynes/cm²** — $l = (20/0.512)^{1/0.3} \approx 201{,}600$ dynes/cm².

Problem 30 **$q = 700 - 8p$** — $m = (460 - 500)/(30 - 25) = -8$. $q - 500 = -8(p - 25) \rightarrow q = 700 - 8p$.

1.9.3 Test Your Understanding

TYU 5 · P2 Check the product xy:

$$1(3) = 2(3/2) = 3(1) = 4(3/4) = 3$$

xy is constant, so $y = 3/x$.
Answer: A — y is inversely proportional to x

TYU 5 · P3 **(a)** $S = k\sqrt{A} = kA^{1/2}$.
Answer: E — S is proportional to $A^{1/2}$
(b) $S = kA^{1/2}$ has the form $y = kx^p$.
Answer: B — Power
(c)

$$30 = k\sqrt{25} = 5k \Rightarrow k = 6$$

$$S = 6\sqrt{9} = 6(3) = 18 \text{ species}$$

Answer: C — 18

1.10 Periodic Functions

In this section

1. Periodic Data & Graphs 1 lecture · 6 textbook
2. Sine & Cosine: Amplitude, Period & Shifts 4 lecture · 8 textbook
3. Finding Formulas for Periodic Functions 2 lecture · 18 textbook · 3 past exam
4. Test Your Understanding 1 questions

1.10.1 Periodic Data & Graphs

Lecture Problems

Ex 1 Figure 1.89 shows the temperature in an unopened "closed" freezer.

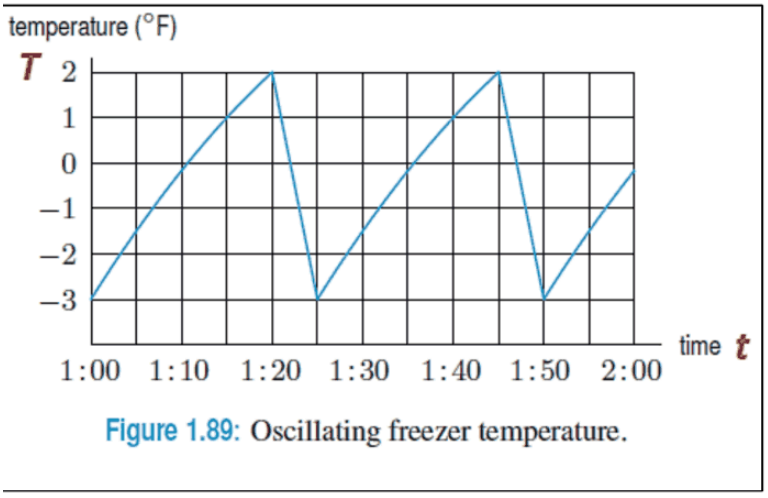


Figure 1.89: Oscillating freezer temperature.

i. Estimate the amplitude and the period.

ii. Estimate the temperature in the freezer at 12:30.

iii. Estimate the temperature in the freezer at 2:45.

Textbook Problems

Problem 1 A graduate student in environmental science studied the temperature fluctuations of a river. Figure 1.99 shows the temperature of the river (in °C) every hour, with hour 0 being midnight of the first day.

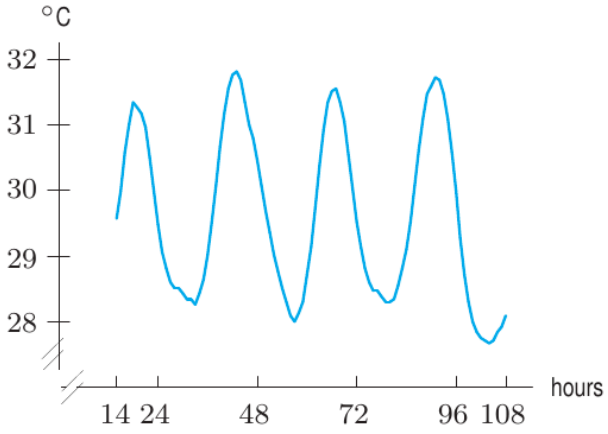


Figure 1.99

a. Explain why a periodic function could be used to model these data.

b. Approximately when does the maximum occur? The minimum? Why does this make sense?

c. What is the period for these data? What is the amplitude?

Problem 2 Figure 1.100 shows quarterly beer production during the period 1997 to 1999. Quarter 1 reflects production during the first three months of the year, etc.

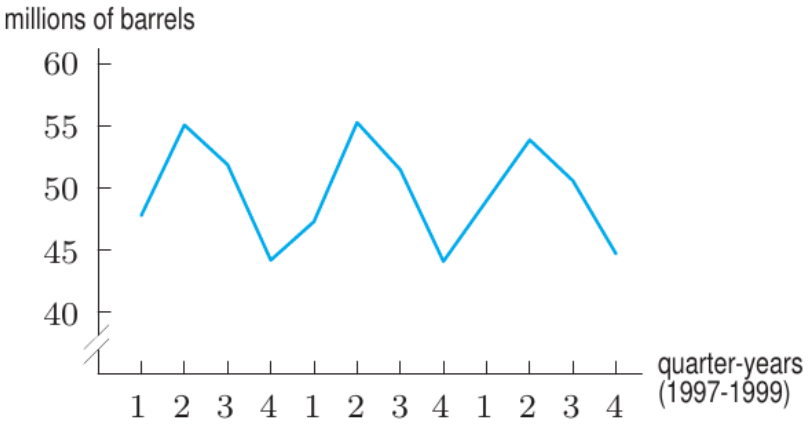


Figure 1.100

a. Explain why a periodic function should be used to model these data.

b. Approximately when does the maximum occur? The minimum? Why does this make sense?

c. What are the period and amplitude for these data?

Problem 3 Sketch a possible graph of sales of sunscreen in the northeastern US over a 3-year period, as a function of months since January 1 of the first year.

Explain why your graph should be periodic. What is the period?

Problem 10 Values of a function are given in the following table.

t	20	25	30	35	40
f(t)	1.8	1.4	1.7	2.3	2.0

t	45	50	55	60
f(t)	1.8	1.4	1.7	2.3

Period/amplitude Explain why this function appears to be periodic. Approximately what are the period and amplitude of the function?

Values Assuming that the function is periodic, estimate its value at t = 15, at t = 75, and at t = 135.

Problem 12 Figure 1.101 shows the levels of the hormones estrogen and progesterone during the monthly ovarian cycles in females.

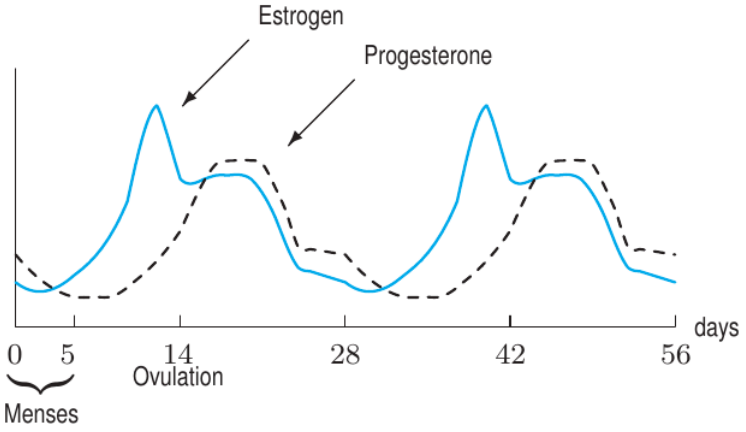


Figure 1.101

Is the level of both hormones periodic? What is the period in each case? Approximately when in the monthly cycle is estrogen at a peak? Approximately when is progesterone at a peak?

Problem 31 Table 1.40 gives values for g(t), a periodic function.

t	0	2	4	6	8	10	12	14
g(t)	14	19	17	15	13	11	14	19

t	16	18	20	22	24	26	28
g(t)	17	15	13	11	14	19	17

a. Estimate the period and amplitude for this function.

b. Estimate g(34) and g(60).

1.10.2 Sine & Cosine: Amplitude, Period & Shifts

Lecture Problems

Ex 2 Use a graph of y = 3 sin 2t to estimate the amplitude and period of this function.

Amplitude and period?

Ex 3 Use the graph of y = sin t to sketch the graphs of

i. y = 2 sin t

ii. y = sin t + 3

Ex 4 Explain how the graphs of each of the following functions differ from the graph of $y = \sin t$.

(a) $y = 6 \sin t$

(b) $y = 5 + \sin t$

(c) $y = \sin(t + \pi/2)$

Ex 5

(a) Graph $y = A \sin t$ for several positive values of A. Describe the effect of A on the graph.

(b) Graph $y = \sin(Bt)$ for several positive values of B. Describe the effect of B on the graph.

Textbook Problems

Problem 4 For Problems 4–9, sketch graphs of the functions. What are their amplitudes and periods?

$y = 3 \sin x$

Problem 5 Sketch the graph. What are its amplitude and period?

$y = 3 \sin 2x$

Problem 6 Sketch the graph. What are its amplitude and period?

$y = -3 \sin 2\theta$

Problem 7 Sketch the graph. What are its amplitude and period?

$y = 4 \cos 2x$

Problem 8 Sketch the graph. What are its amplitude and period?

$y = 4\cos(\frac{1}{2}t)$

Problem 9 Sketch the graph. What are its amplitude and period?

$y = 5 - \sin 2t$

Problem 14 A person breathes in and out every three seconds. The volume of air in the person's lungs varies between a minimum of 2 liters and a maximum of 4 liters.

Which of the following is the best formula for the volume of air in the person's lungs as a function of time?

Problem 32 The Bay of Fundy in Canada has the largest tides in the world. The difference between low and high water levels is 15 meters (nearly 50 feet). At a particular point the depth of the water, y meters, is given as a function of time, t, in hours since midnight by $y = D + A\cos(B(t - C))$.

a. What is the physical meaning of D?

b. What is the value of A?

c. What is the value of B? Assume the time between successive high tides is 12.4 hours.

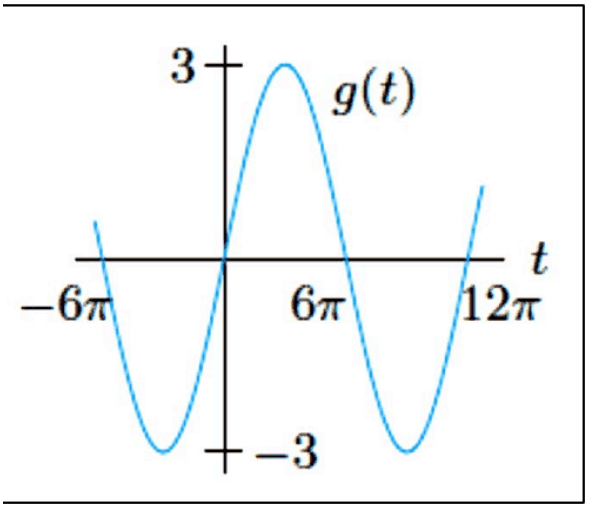
d. What is the physical meaning of C?

1.10.3 Finding Formulas for Periodic Functions

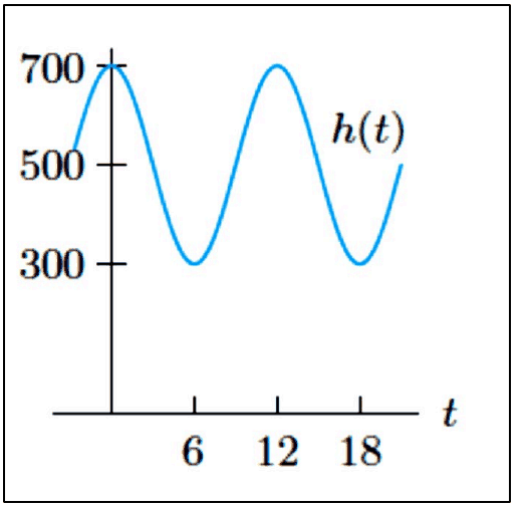
Lecture Problems

Ex 6 Find possible formulas for the following periodic functions:

i. g(t)



ii. h(t)



Ex 7

Same as Textbook Problem 11

The following table shows values of a periodic function f(x). The maximum value attained by the function is 5.

x	0	2	4	6	8	10	12
f(x)	5	0	−5	0	5	0	−5

i. What is the amplitude of the function?

---

ii. What is the period of the function?

---

iii. Find a formula for this periodic function.

Textbook Problems

Problem 11

Same as Lecture Example 7

The following table shows values of a periodic function f(x). The maximum value attained by the function is 5.

x	0	2	4	6	8	10	12
f(x)	5	0	−5	0	5	0	−5

a. What is the amplitude of the function?

---

b. What is the period of the function?

---

c. Find a formula for this periodic function.

Problem 13 Average daily high temperatures in Ottawa, the capital of Canada, range from a low of −6° Celsius on January 1 to a high of 26° Celsius on July 1 six months later. See Figure 1.102.

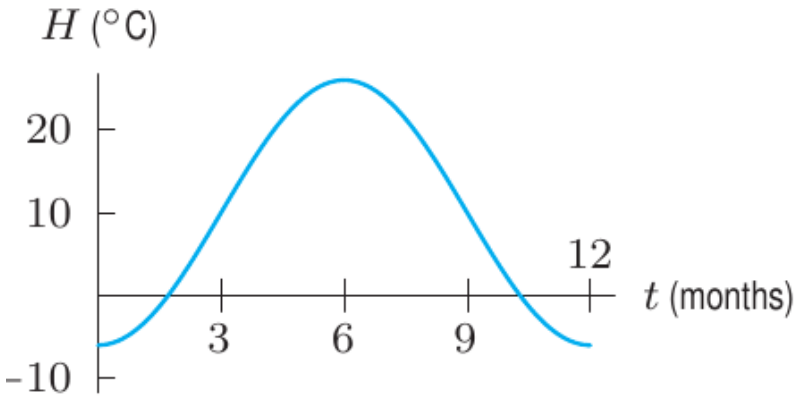


Figure 1.102

Find a formula for H, the average daily high temperature in Ottawa, in °C, as a function of t, the number of months since January 1.

Problem 15 Delta Cephei is one of the most visible stars in the night sky. Its brightness has periods of 5.4 days, the average brightness is 4.0 and its brightness varies by ±0.35.

Find a formula that models the brightness of Delta Cephei as a function of time, t, with t = 0 at peak brightness.

Problem 16 Most breeding birds in the northeast US migrate elsewhere during the winter. The number of bird species in an Ohio forest preserve oscillates between a high of 28 in June and a low of 10 in December.

a. Graph the number of bird species as a function of t, the number of months since June (at least three years). Which description is correct?

---

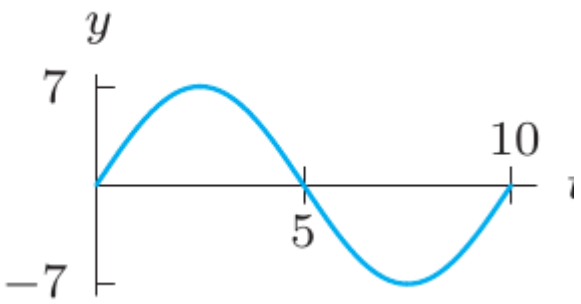
b. What are the amplitude and period of this function?

---

c. Find a formula for the number of bird species, B, as a function of the number of months, t, since June.

Problem 17 For Problems 17–28, find a possible formula for each graph.

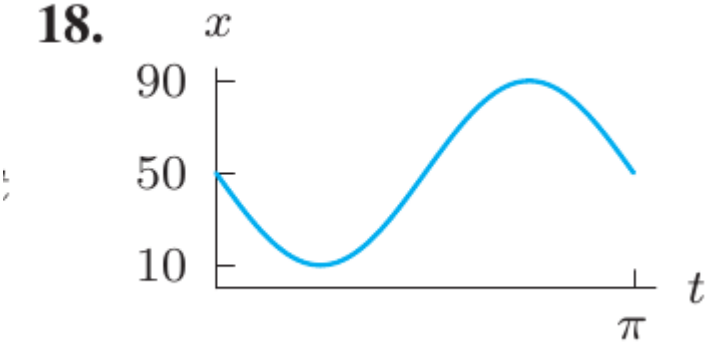
17.



Formula = ?

Problem 18 Find a possible formula for the graph.

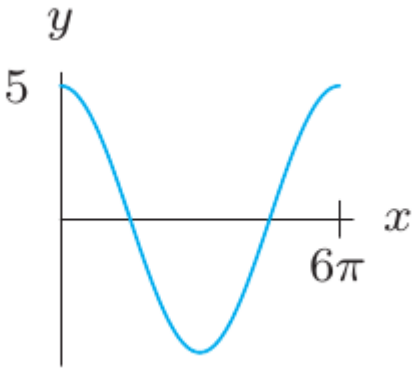
18.



Formula = ?

Problem 19 Find a possible formula for the graph.

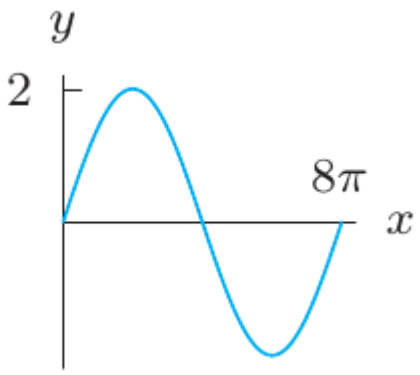
19.



Formula = ?

Problem 20 Find a possible formula for the graph.

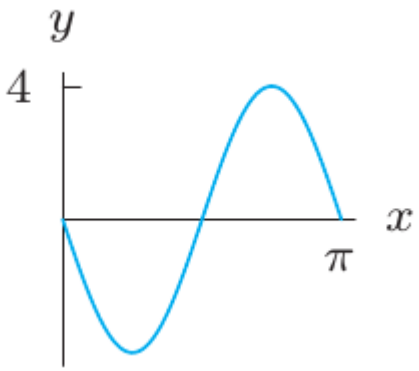
20.



Formula = ?

Problem 21 Find a possible formula for the graph.

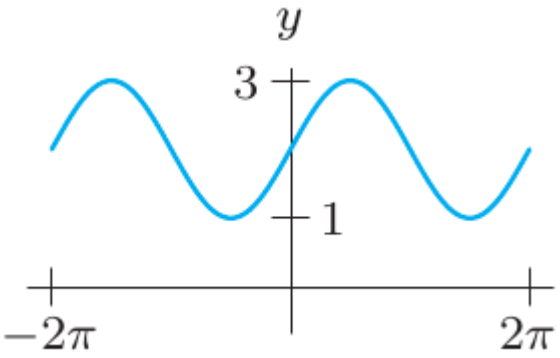
21.



Formula = ?

Problem 22 Find a possible formula for the graph.

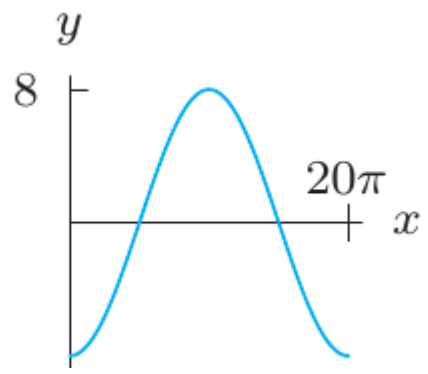
22.



Formula = ?

Problem 23 Find a possible formula for the graph.

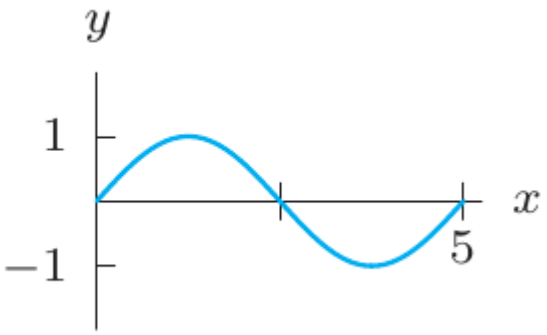
23.



Formula = ?

Problem 24 Find a possible formula for the graph.

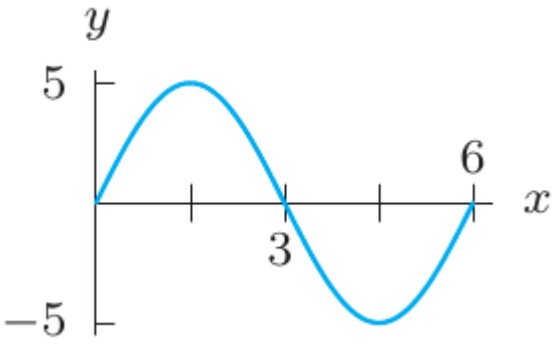
24.



Formula = ?

Problem 25 Find a possible formula for the graph.

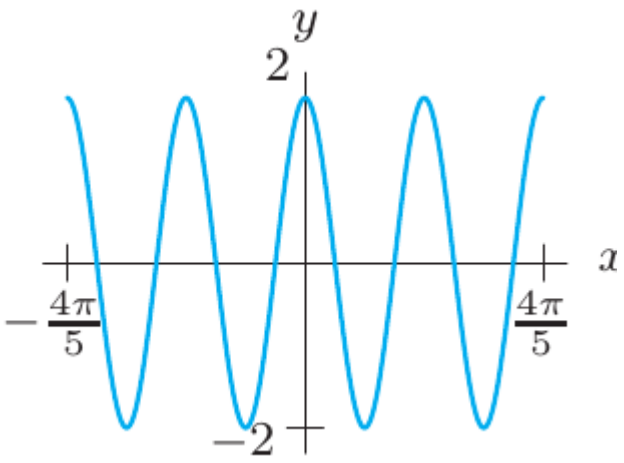
25.



Formula = ?

Problem 26 Find a possible formula for the graph.

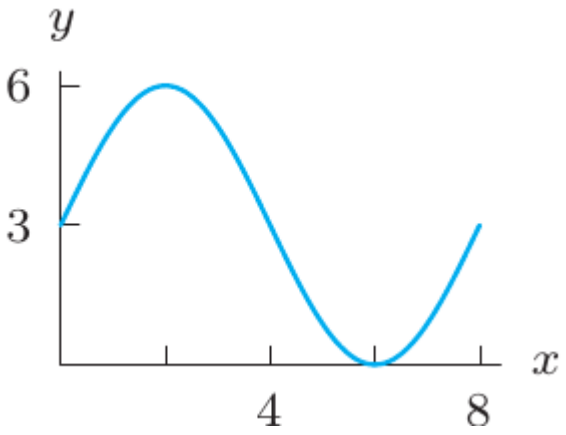
26.



Formula = ?

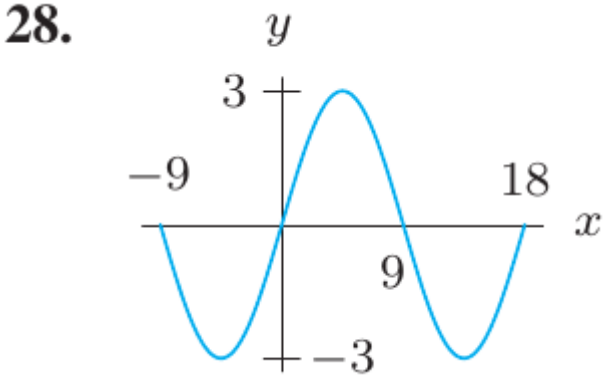
Problem 27 Find a possible formula for the graph.

27.



Formula = ?

Problem 28 Find a possible formula for the graph.



Formula = ?

Problem 29 The depth of water in a tank oscillates once every 6 hours. If the smallest depth is 5.5 feet and the largest depth is 8.5 feet,

find a possible formula for the depth in terms of time in hours.

Problem 30 The desert temperature, H, oscillates daily between 40°F at 5 am and 80°F at 5 pm.

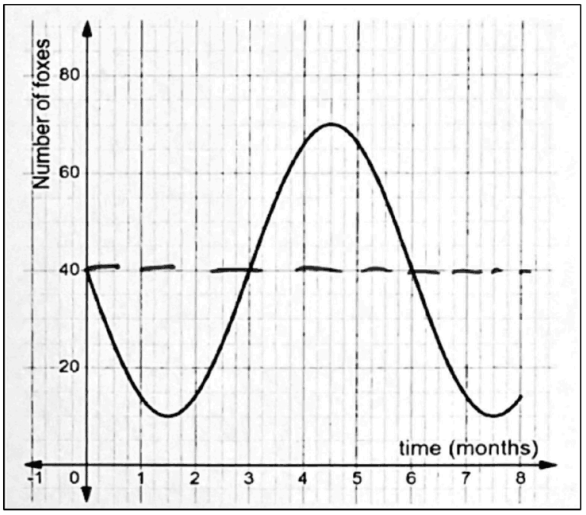
Write a possible formula for H in terms of t, measured in hours from 5 am.

Past Exam Questions

Past Exam (Lecture)

Same as Past Exam 2022 Question 6

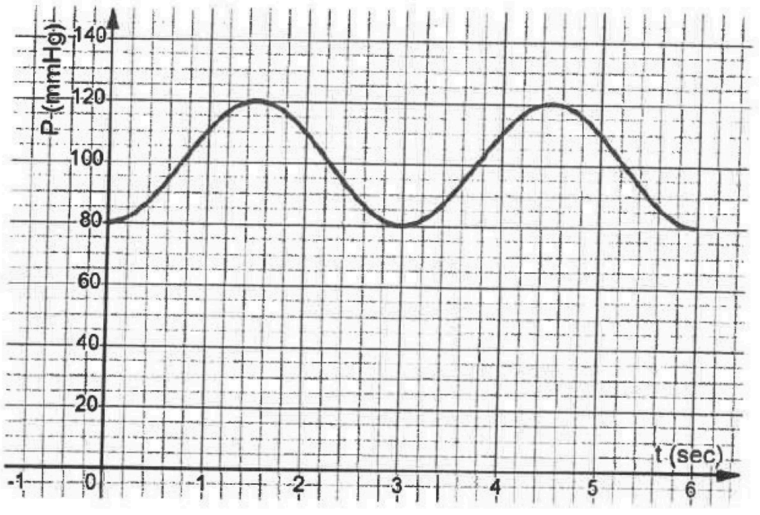
The number of foxes is a periodic function of time as shown in the figure below.



i. Find a possible formula for the number of foxes, N, in terms of time t (in months).

ii. Predict the number of foxes after 24 months.

2024 Q6 The figure below shows the blood pressure, P (in mmHg), as a function of time, t (in seconds).

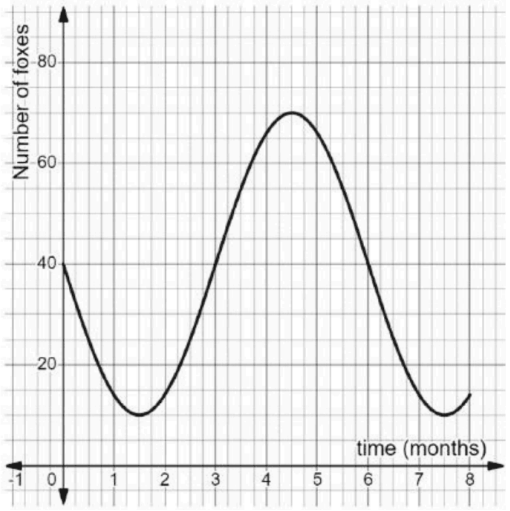


Find a possible formula for the blood pressure in terms of time.

2022 Q6

Same as Lecture Exam Question

The number of foxes is a periodic function of time as shown in the figure below.



a. Find a possible formula for the number of foxes, N, in terms of time t (in months).

b. Predict the number of foxes after 24 months.

1.10.4 Test Your Understanding

Test Your Understanding

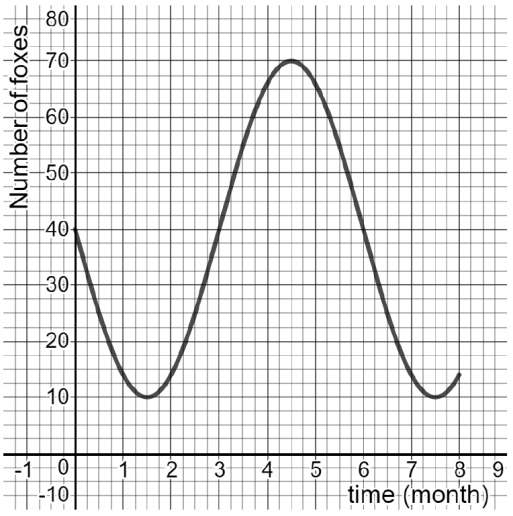
TYU 5 · P4 The population of foxes is observed to oscillate with time as shown in the graph below.

a. The best function modeling the population of foxes is

A. Surge
B. Power
C. Exponential
D. Logistic
E. Periodic

b. Which of the following formulas best describes this function?

A. $y = -30\cos(\pi x/6) + 40$
B. $y = -70\cos(\pi x/6) + 30$
C. $y = -30\cos(\pi x/3) + 40$
D. $y = -30\sin(\pi x/3) + 40$
E. $y = -30xe^{-6x} + 30$



Answers · Section 1.10

1.10.1 Periodic Data & Graphs

Ex 1 **i. Amplitude 2.5°F, period 25 minutes** — Temperature goes between −3°F and 2°F: amplitude = (2 − (−3))/2 = 2.5°F. One cycle (e.g. 1:20 to 1:45) takes 25 minutes.

ii. ≈ 2°F — Go back from 1:20 (a peak, 2°F) by whole periods: 1:20 − 2(25 min) = 12:30 is also a peak: ≈ 2°F.

iii. ≈ −1.5°F — 2:45 − 2(25 min) = 1:55, where the graph is at about −1.5°F.

Problem 1 **a. The pattern repeats every day** — The temperature rises and falls in the same pattern every day.

b. Maximum ≈ 8 pm; minimum ≈ 9 am (water warms during the day and cools at night) — Peaks are around hours 20, 44, 68, 92 (≈ 8 pm); lows around 9 am. The water warms during the day and cools at night.

c. Period ≈ 24 hours; amplitude ≈ 1.75°C — The pattern repeats every ≈ 24 hours; max ≈ 31.75, min ≈ 28.25 → amplitude ≈ 1.75°C.

Problem 2 **a. The pattern repeats every year** — The same pattern repeats every year.

b. Maximum in quarter 2; minimum in quarter 4 — Each year production peaks in quarter 2 (spring/early summer, to supply summer demand) and is lowest in quarter 4.

c. Period 4 quarters (1 year); amplitude ≈ 5.5 million barrels — Repeats every 4 quarters; max ≈ 55, min ≈ 44 → amplitude ≈ 5.5 million barrels.

Problem 3 **Sales repeat each year (high in summer, low in winter); period 12 months** — Sunscreen sales rise every summer and fall every winter, repeating each year: period 12 months.

Problem 10 **Period/amplitude** **The values repeat every 25 units: period ≈ 25; amplitude ≈ 0.45** — f(20) = f(45) = 1.8, f(25) = f(50) = 1.4, … so period ≈ 25. Max 2.3, min 1.4 → amplitude ≈ 0.45.

Values **f(15) ≈ 2.0, f(75) ≈ 1.4, f(135) ≈ 2.3** — 15 + 25 = 40 → 2.0; 75 − 25 = 50 → 1.4; 135 − 100 = 35 → 2.3.

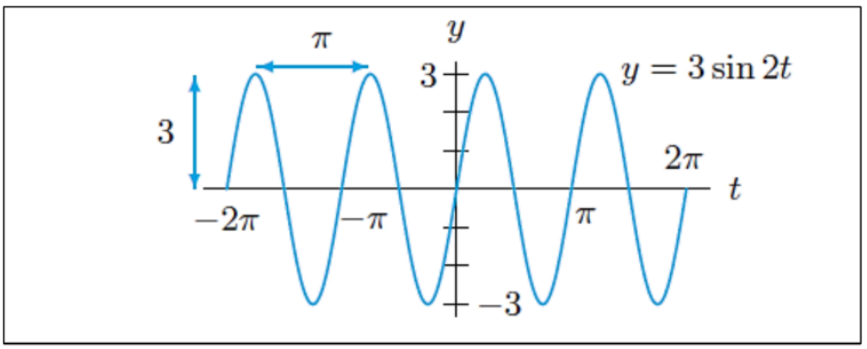
Problem 12 **Yes; period 28 days. Estrogen peaks around day 12; progesterone around day 20** — Both repeat every 28 days. Estrogen peaks around day 12 (before ovulation) and progesterone around day 20.

Problem 31 **a. Period 12; amplitude 4** — g(0) = g(12) = g(24) = 14 and the pattern repeats: period 12. Max 19, min 11 → amplitude 4.

b. g(34) = 11, g(60) = 14 — g(34) = g(34 − 24) = g(10) = 11; g(60) = g(0) = 14.

1.10.2 Sine & Cosine: Amplitude, Period & Shifts

Ex 2 **Amplitude 3, period π** — The graph goes from −3 to 3 (amplitude 3) and repeats every π (period 2π/2 = π).



Ex 3 **i. The sine wave stretched vertically: between −2 and 2, period 2π** — Multiplying by 2 doubles the amplitude: it oscillates between −2 and 2 with the same period 2π.

ii. The sine wave shifted up 3: between 2 and 4, period 2π — Adding 3 shifts the graph up 3: it oscillates between 2 and 4, period 2π.

Ex 4 **(a) Amplitude is 6 instead of 1 (vertical stretch); same period** — y = 6 sin t has amplitude 6 (it oscillates between −6 and 6); the period is still 2π.

(b) Shifted up 5 (oscillates between 4 and 6); same amplitude and period — Adding 5 moves the midline from 0 to 5: the graph oscillates between 4 and 6.

(c) Shifted π/2 to the left (it is the graph of cos t) — t + π/2 inside shifts the graph π/2 units left; sin(t + π/2) = cos t.

Ex 5 **(a) A is the amplitude: larger A gives a taller wave; the period stays 2π** — For y = A sin t the wave goes between −A and A: A is the amplitude (larger A → taller wave). The period does not change.

(b) Period = 2π/B: larger B gives a shorter period (more waves) — For y = sin(Bt) the period is 2π/B: larger B squeezes the wave (shorter period); the amplitude stays 1.

Problem 4 **Amplitude 3, period 2π** — A = 3; period 2π/1 = 2π.

Problem 5 **Amplitude 3, period π** — A = 3; period 2π/2 = π.

Problem 6 **Amplitude 3, period π** — Amplitude |−3| = 3 (the minus sign reflects the graph); period 2π/2 = π.

Problem 7 **Amplitude 4, period π** — A = 4; period 2π/2 = π.

Problem 8 **Amplitude 4, period 4π** — A = 4; period 2π/(½) = 4π.

Problem 9 **Amplitude 1, period π** — Amplitude |−1| = 1 (midline y = 5); period 2π/2 = π.

Problem 14 **y = 3 + sin(2πt/3)** — Midline (2 + 4)/2 = 3, amplitude 1, period 3 → B = 2π/3: y = 3 + sin(2πt/3). (Option (b) in the booklet.)

Problem 32

a. **The average water depth** — D is the midline: the average water depth.

b. **A = 7.5 m** — The difference between high and low is 15 m, so A = 15/2 = 7.5 m.

c. **B ≈ 0.507** — Period 12.4 = 2π/B → B = 2π/12.4 ≈ 0.507.

d. **The time of a high tide** — cos is maximum when t − C = 0, so C is the time of a high tide.

1.10.3 Finding Formulas for Periodic Functions

Ex 6

i. **g(t) = 3 sin(t/6)** — Starts at 0 going up → sine. Amplitude 3. Period 12π → B = 2π/12π = 1/6. g(t) = 3 sin(t/6).

ii. **h(t) = 500 + 200 cos(πt/6)** — Starts at the maximum → cosine. Midline 500, amplitude 200, period 12 → B = π/6. h(t) = 500 + 200 cos(πt/6).

Ex 7

i. **5** — The function goes between −5 and 5: amplitude 5.

ii. **8** — f(0) = 5 and f(8) = 5 again (one full cycle): period 8.

iii. **f(x) = 5 cos(πx/4)** — Starts at the maximum → cosine. A = 5, period 8 → B = 2π/8 = π/4. f(x) = 5 cos(πx/4).

Problem 11

a. **5** — The function goes between −5 and 5: amplitude 5.

b. **8** — f(0) = 5 and f(8) = 5 again (one full cycle): period 8.

c. **f(x) = 5 cos(πx/4)** — Starts at the maximum → cosine. A = 5, period 8 → B = 2π/8 = π/4. f(x) = 5 cos(πx/4).

Problem 13

H = 10 − 16 cos(πt/6) — Midline (26 + (−6))/2 = 10, amplitude 16, period 12 → B = π/6. Minimum at t = 0 → −cos. H = 10 − 16 cos(πt/6).

Problem 15

B = 4.0 + 0.35 cos(2πt/5.4) — Peak at t = 0 → cosine. Midline 4.0, amplitude 0.35, period 5.4 → B = 2π/5.4.

Problem 16

a. **Starts at 28 (t = 0), falls to 10 at t = 6, back to 28 at t = 12, repeating** — June is t = 0 (high 28), December t = 6 (low 10), June again t = 12 (28).

b. **Amplitude 9; period 12 months** — Amplitude (28 − 10)/2 = 9; period 12 months.

c. **B = 19 + 9 cos(πt/6)** — Midline 19, amplitude 9, period 12 → B = π/6; maximum at t = 0 → cosine.

Problem 17

y = 7 sin(πt/5) — Starts at 0 going up → sine; amplitude 7; period 10 → B = π/5.

Problem 18

x = 50 − 40 sin(2t) — Midline 50, amplitude 40; starts at the midline going down → −sin; period π → B = 2.

Problem 19

y = 5 cos(x/3) — Starts at the maximum → cosine; amplitude 5; period 6π → B = 1/3.

Problem 20

y = 2 sin(x/4) — Starts at 0 going up → sine; amplitude 2; period 8π → B = 1/4.

Problem 21

y = −4 sin(2x) — Starts at 0 going down → −sin; amplitude 4; period π → B = 2.

Problem 22 **$y = 2 + \sin x$** — Midline 2, amplitude 1 (between 1 and 3); period 2π; at 0 on the midline going up → sine.

Problem 23 **$y = -8\cos(x/10)$** — Starts at the minimum -8 → $-\cos$; amplitude 8; period 20π → $B = 1/10$.

Problem 24 **$y = \sin(2\pi x/5)$** — Starts at 0 going up → sine; amplitude 1; period 5 → $B = 2\pi/5$.

Problem 25 **$y = 5\sin(\pi x/3)$** — Starts at 0 going up → sine; amplitude 5; period 6 → $B = \pi/3$.

Problem 26 **$y = 2\cos(5x)$** — Maximum at 0 → cosine; amplitude 2; two periods in $4\pi/5$ → period $2\pi/5$ → $B = 5$.

Problem 27 **$y = 3 + 3\sin(\pi x/4)$** — Midline 3, amplitude 3 (between 0 and 6); starts at the midline going up → sine; period 8 → $B = \pi/4$.

Problem 28 **$y = 3\sin(\pi x/9)$** — At 0 on the midline going up → sine; amplitude 3; period 18 → $B = \pi/9$.

Problem 29 **$D = 7 + 1.5\sin(\pi t/3)$** — Midline $(5.5 + 8.5)/2 = 7$, amplitude 1.5, period 6 → $B = 2\pi/6 = \pi/3$.

Problem 30 **$H = 60 - 20\cos(\pi t/12)$** — Midline 60, amplitude 20, period 24 → $B = \pi/12$. Minimum at $t = 0$ (5 am) → $-\cos$.

Past Exam (Lecture)

i. **$N = 40 - 30\sin(\pi t/3)$** — Midline 40 (between 10 and 70), amplitude 30. It starts at 40 and goes down first → $-\sin$. Period 6 months → $B = 2\pi/6 = \pi/3$. $N = 40 - 30\sin(\pi t/3)$.

ii. **40 foxes** — $N(24) = 40 - 30\sin(8\pi) = 40 - 0 = 40$ foxes (24 months is exactly 4 periods).

2024 Q6 **$P = 100 - 20\cos(2\pi t/3)$** — Between 80 and 120: midline 100, amplitude 20. Minimum at $t = 0$ → $-\cos$. Period 3 seconds → $B = 2\pi/3$.

2022 Q6

a. **$N = 40 - 30\sin(\pi t/3)$** — Midline 40 (between 10 and 70), amplitude 30. It starts at 40 and goes down first → $-\sin$. Period 6 months → $B = 2\pi/6 = \pi/3$. $N = 40 - 30\sin(\pi t/3)$.

b. **40 foxes** — $N(24) = 40 - 30\sin(8\pi) = 40 - 0 = 40$ foxes (24 months is exactly 4 periods).

1.10.4 Test Your Understanding

TYU 5 · P4

(a) The graph repeats (oscillates).

Answer: E — Periodic

(b) From the graph: maximum 70, minimum 10.

$$\text{Midline} = \frac{70 + 10}{2} = 40, \quad \text{Amplitude} = \frac{70 - 10}{2} = 30$$

Minimum at x = 1.5 and next minimum at x = 7.5, so the period is 6:

$$B = \frac{2\pi}{6} = \frac{\pi}{3}$$

At x = 0, y = 40 (the midline) and the graph goes down first → −sin.

$$y = -30\sin\left(\frac{\pi}{3}x\right) + 40$$

Answer: D

2.1 Instantaneous Rate of Change

In this section

1. Instantaneous Velocity & Rate of Change 4 lecture · 14 textbook · 2 past exam
2. The Derivative from Graphs 3 lecture · 10 textbook
3. Test Your Understanding 1 questions

2.1.1 Instantaneous Velocity & Rate of Change

Lecture Problems

Intro We throw a grapefruit straight upward into the air. Table 2.1 gives its height, y, at time t.

t (sec)	0	1	2	3	4	5	6
y = s(t) (feet)	6	90	142	162	150	106	30

What is the velocity of the grapefruit at exactly t = 1?

Ex 1 The quantity (mg) of a drug in the blood at time t (minutes) is given by $Q = 25(0.8)^t$.

Estimate the rate of change of the quantity at t = 3 and interpret your answer.

Ex 2 Estimate $f'(2)$ if $f(x) = x^3$.

$f'(2) \approx$?

Ex 6 The total acreage of farms in the US has decreased since 1980. See Table 2.2.

Year	1980	1985	1990	1995	2000
Farm land (million acres)	1039	1012	987	963	945

a. What was the average rate of change in farm land between 1980 and 2000?

---

b. Estimate $f'(1995)$ (average of the left and right rates) and interpret your answer in terms of farm land.

Textbook Problems

Problem 3 In a time of t seconds, a particle moves a distance of s meters from its starting point, where $s = 3t^2$.

a. Find the average velocity between $t = 1$ and $t = 1 + h$ if: (i) $h = 0.1$, (ii) $h = 0.01$, (iii) $h = 0.001$.

b. Use your answers to part (a) to estimate the instantaneous velocity of the particle at time $t = 1$.

Problem 4 Find the average velocity over the interval $0 \le t \le 0.8$, and estimate the velocity at $t = 0.2$ of a car whose position, s, is given by the following table.

t (sec)	0	0.2	0.4	0.6	0.8	1.0
s (ft)	0	0.5	1.8	3.8	6.5	9.6

Average Average velocity over $0 \le t \le 0.8$?

At t = 0.2 Estimate the velocity at $t = 0.2$ (average of the left and right rates).

Problem 5 The distance (in feet) of an object from a point is given by $s(t) = t^2$, where time t is in seconds.

a. What is the average velocity of the object between $t = 3$ and $t = 5$?

b. By using smaller and smaller intervals around 3, estimate the instantaneous velocity at time $t = 3$.

Problem 7 The size, S, of a tumor (in cubic millimeters) is given by $S = 2^t$, where t is the number of months since the tumor was discovered. Give units with your answers.

a. What is the total change in the size of the tumor during the first six months?

b. What is the average rate of change in the size of the tumor during the first six months?

c. Estimate the rate at which the tumor is growing at t = 6. (Use smaller and smaller intervals.)

Problem 8 Use Table 2.3.

Year	1975	1980	1985	1990
Population (billions)	4.08	4.45	4.84	5.27

Year	1995	2000	2005
Population (billions)	5.68	6.07	6.45

(a) Find the average rate of change in the world's population between 1975 and 2005. Give units.

(b) If P = f(t) with t in years, estimate $f'(2005)$ and give units.

Problem 9 Let $f(x) = 5^x$.

Use a small interval to estimate $f'(2)$. Now improve your accuracy by estimating $f'(2)$ again, using an even smaller interval.

Problem 10

(a) Let $g(t) = (0.8)^t$. Use a graph to determine whether $g'(2)$ is positive, negative, or zero.

(b) Use a small interval to estimate $g'(2)$.

Problem 12 Estimate $f'(2)$ for $f(x) = 3^x$. Explain your reasoning.

$f'(2) \approx$?

Problem 14 The following table gives the percent of the US population living in urban areas as a function of year.

Year	1800	1830	1860	1890	1920
Percent	6.0	9.0	19.8	35.1	51.2

Year	1950	1980	1990	2000
Percent	64.0	73.7	75.2	79.0

a. Find the average rate of change of the percent of the population living in urban areas between 1890 and 1990.

---

b. Estimate the rate at which this percent is increasing at the year 1990.

---

c. Estimate the rate of change of this function for the year 1830 and explain what it is telling you.

---

d. Is this function increasing or decreasing?

Problem 16 Table 2.4 gives $P = f(t)$, the percent of households in the US with cable television t years since 1990.

t (years since 1990)	0	2	4	6	8	10	12
P (% with cable)	59.0	61.5	63.4	66.7	67.4	67.8	68.9

a. Does $f'(6)$ appear to be positive or negative? What does this tell you about the percent of households with cable television?

---

b. Estimate $f'(2)$. Estimate $f'(10)$. Explain what each is telling you, in terms of cable television.

Problem 17 Estimate $P'(0)$ if $P(t) = 200(1.05)^t$. Explain how you obtained your answer.

$P'(0) \approx$?

Problem 22 Estimate the instantaneous rate of change of the function $f(x) = x \ln x$ at $x = 1$ and at $x = 2$.

What do these values suggest about the concavity of the graph between 1 and 2?

Problem 23 On October 17, 2006, in an article called "US Population Reaches 300 Million," the BBC reported that the US gains 1 person every 11 seconds. If $f(t)$ is the US population in millions t years after October 17, 2006,

find $f(0)$ and $f'(0)$.

Problem 24 The following table shows the number of hours worked in a week, $f(t)$, hourly earnings, $g(t)$, in dollars, and weekly earnings, $h(t)$, in dollars, of production workers as functions of t, the year.

t	1970	1975	1980	1985	1990	1995	2000
f(t)	37.0	36.0	35.2	34.9	34.3	34.3	34.3
g(t)	3.40	4.73	6.84	8.73	10.09	11.64	14.00
h(t)	125.80	170.28	240.77	304.68	349.29	399.53	480.41

a. Indicate whether each of the following derivatives is positive, negative, or zero: $f'(t)$, $g'(t)$, $h'(t)$. Interpret each answer.

---

b. (i) Estimate $f'(1970)$ and $f'(1995)$.

---

b. (ii) Estimate $g'(1970)$ and $g'(1995)$.

---

b. (iii) Estimate $h'(1970)$ and $h'(1995)$.

Past Exam Questions

Second Spring 2021 Q3 The table below shows experimental data involving malarial parasites. The number of parasites per microliter of blood, N, is a function of time t, measured in days.

t	1	2	3	4	5	6
N	228	2,357	12,750	26,661	372,331	2,217,441

a. Estimate the rate of change of N at t = 3 (average of the left and right rates).

b. Interpret the value obtained in Part (a) using the appropriate units.

Midterm-B Q3 A post-meal blood-glucose monitoring session at a Kuwaiti outpatient clinic generated the dataset in Table 1 below. This information is further used to generate the average rate of change (ARC) over each of the time intervals in Table 2.

Table 1

Time, t (min)	0	30	60	90	120	150	180	210
Glucose, G (mg/dL)	92	135	J	155	120	95	88	96

Table 2

Time interval	0–30	30–60	60–90	90–120
ARC	1.433	1.100	−0.433	−1.167

Time interval	120–150	150–180	180–210
ARC	−0.833	−0.233	0.267

a. Find the time interval(s) where the graph of G is concave up and concave down.

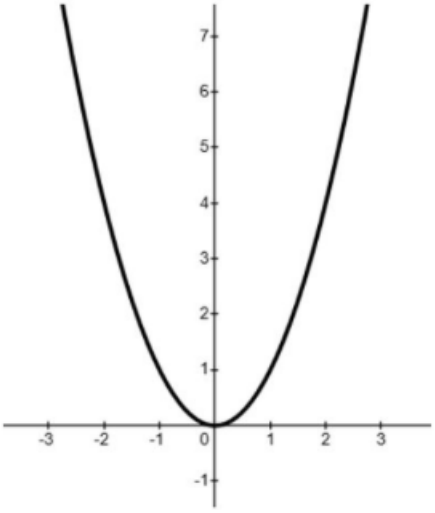
b. Estimate the instantaneous rate of change IRC at t = 60 (average of the left and right rates). Give its units and interpret it.

c. Compute the missing quantity J in Table 1.

2.1.2 The Derivative from Graphs

Lecture Problems

Ex 3 Use the graph of $f(x) = x^2$ to determine whether each of the following quantities is positive, negative or zero.



a. $f'(1)$

b. $f'(-1)$

c. $f'(2)$

d. $f'(0)$

Ex 4 Estimate the derivative of $y = 2^x$ at $x = 0$ graphically and numerically.

$f'(0) \approx ?$

Ex 5 The graph of a function y = f(x) is shown in Figure 2.7. Indicate whether each of the following quantities is positive or negative, and illustrate your answers graphically.

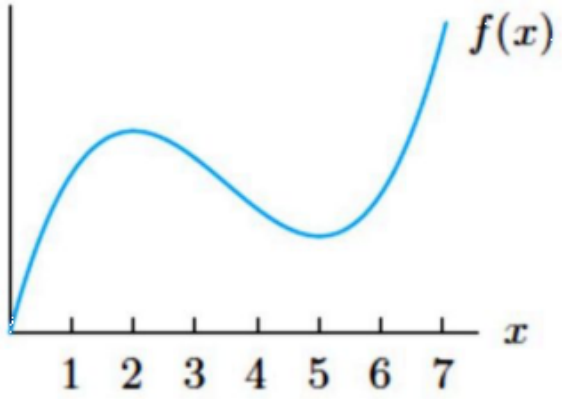


Figure 2.7

i. f′(1)

ii. (f(3) − f(1))/(3 − 1)

iii. f(4) − f(2)

Textbook Problems

Problem 1 Use the graph in Figure 2.7 to decide if each of the following quantities is positive, negative or approximately zero. Illustrate your answers graphically.

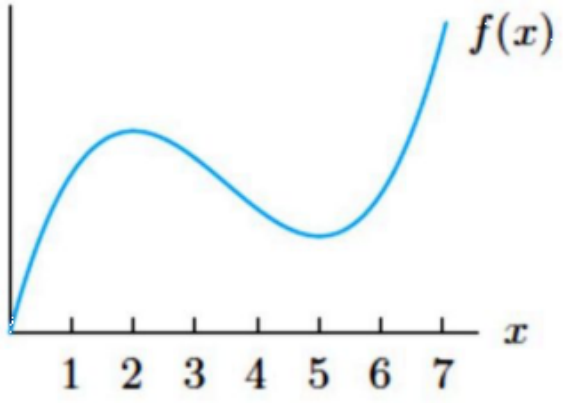


Figure 2.7

a. The average rate of change of f(x) between x = 3 and x = 7.

b. The instantaneous rate of change of f(x) at x = 3.

Problem 2 Figure 2.11 shows N = f(t), the number of farms in the US between 1930 and 2000 as a function of year, t.

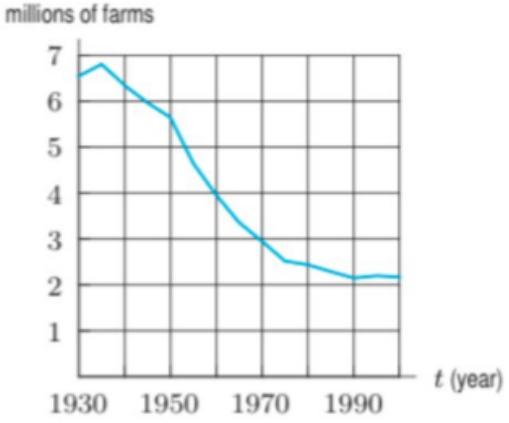


Figure 2.11

a. Is f′(1950) positive or negative? What does this tell you about the number of farms?

b. Which is more negative: f′(1960) or f′(1980)? Explain.

Problem 6 Figure 2.12 shows the cost, y = f(x), of manufacturing x kilograms of a chemical.

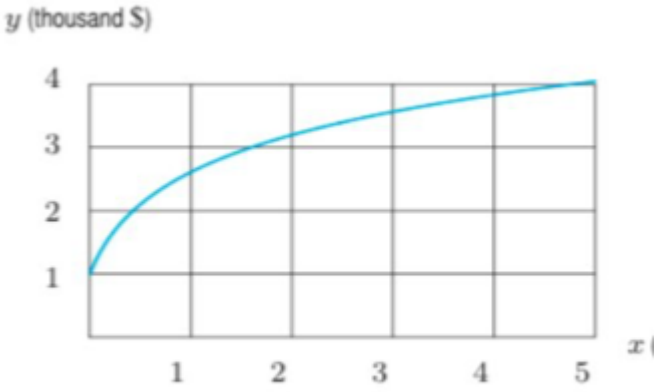


Figure 2.12

a. Is the average rate of change of the cost greater between x = 0 and x = 3, or between x = 3 and x = 5? Explain your answer graphically.

b. Is the instantaneous rate of change of the cost of producing x kilograms greater at x = 1 or at x = 4? Explain your answer graphically.

c. What are the units of these rates of change?

Problem 11 The function f is given in Figure 2.13.

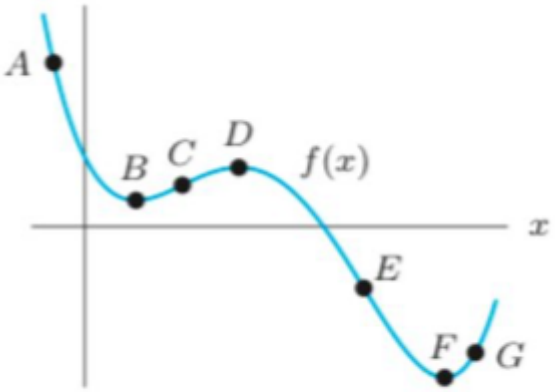
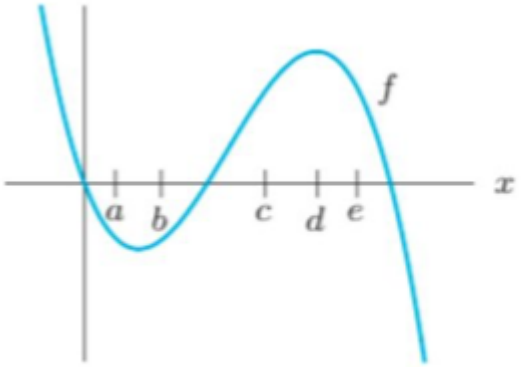


Figure 2.13

(a) At which of the labeled points is f′(x) positive? Negative? Zero?

(b) At which labeled point is f′ largest? At which labeled point is f′ most negative?

Problem 13 Figure 2.14 shows the graph of f. Match the derivatives in the table with the points a, b, c, d, e.



x	$f'(x)$
	0
	0.5
	2
	−0.5
	−2

Figure 2.14

Point a f′(a) = ?

Point b f′(b) = ?

Point c f′(c) = ?

Point d f′(d) = ?

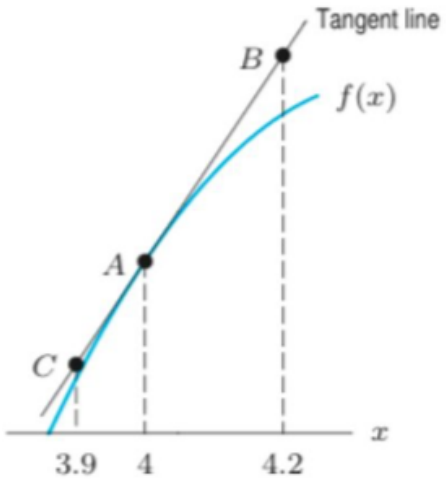
Point e f′(e) = ?

Problem 15 Graph $f(x) = x^2$ and $g(x) = x^2 + 3$ on the same axes.

(a) What can you say about the slopes of the tangent lines to the two graphs at the point x = 0? x = 1? x = 2? x = a, where a is any value?

(b) Explain why adding a constant to any function will not change the value of the derivative at any point.

Problem 18 The function in Figure 2.15 has f(4) = 25 and f′(4) = 1.5.



Find the coordinates of the points A, B, C.

Problem 19 Use Figure 2.16 to fill in the blanks in the following statements about the function g at point B.

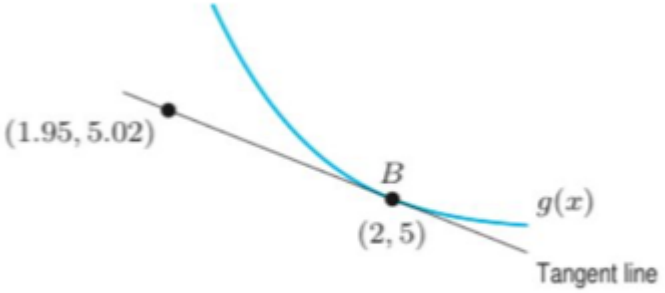


Figure 2.16

(a) g(__) = __

(b) g′(__) = __

Problem 20 Show how to represent the following on Figure 2.17.

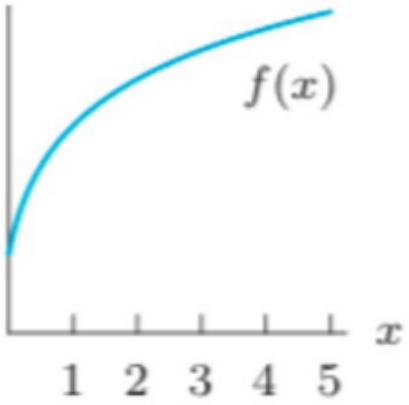


Figure 2.17

(a) f(4)

(b) f(4) − f(2)

(c) (f(5) − f(2))/(5 − 2)

(d) f′(3)

Problem 21 For each of the following pairs of numbers, use Figure 2.17 to decide which is larger. Explain your answer.

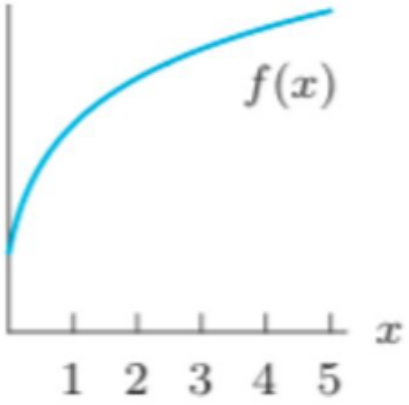


Figure 2.17

a. f(3) or f(4)?

b. f(3) − f(2) or f(2) − f(1)?

c. (f(2) − f(1))/(2 − 1) or (f(3) − f(1))/(3 − 1)?

d. f′(1) or f′(4)?

2.1.3 Test Your Understanding

Test Your Understanding

TYU 6 · P4 The following table describes the size of a population.

t	**0**	**2**	**4**	**6**	**8**	**10**
P	150	200	300	600	700	900

a. What is the best estimate for the instantaneous rate of change at t = 4?

A. 50
B. 75
C. 100
D. 150
E. 200

b. What is the relative rate of change at t = 4?

A. 17 %
B. 25 %
C. 33 %
D. 50 %
E. 67 %

c. What is the best estimate for the instantaneous rate of change at t = 10?

A. 0
B. 50
C. 100
D. 150
E. 200

Answers · Section 2.1

2.1.1 Instantaneous Velocity & Rate of Change

Intro **About 68 ft/sec** — Average velocity on $0 \le t \le 1$: $(90 - 6)/1 = 84$ ft/sec; on $1 \le t \le 2$: $(142 - 90)/1 = 52$ ft/sec, so $52 \le v \le 84$. Taking smaller and smaller intervals around $t = 1$, the average velocities on both sides get closer to 68 ft/sec. The instantaneous velocity at $t = 1$ is about 68 ft/sec.

Ex 1 **≈ −2.85 mg/min: after 3 minutes, the drug is decreasing by about 2.85 mg per minute** — ARC on $3 \le t \le 3.01$: $(25(0.8)^{3.01} - 25(0.8)^3)/0.01 \approx -2.85$ mg/min. After 3 minutes, the quantity of the drug is decreasing by about 2.85 mg per minute.

Ex 2 **≈ 12** — Using $2 \le x \le 2.001$: $((2.001)^3 - 2^3)/0.001 = (8.012006 - 8)/0.001 \approx 12.006 \approx 12$. So $f'(2) = 12$.

Ex 6

a. **−4.7 million acres/year** — ARC $= (945 - 1039)/(2000 - 1980) = -94/20 = -4.7$ million acres/year.

b. **≈ −4.2 million acres/year: farm land was decreasing by about 4.2 million acres per year** — Right slope: $(945 - 963)/5 = -3.6$. Left slope: $(963 - 987)/5 = -4.8$. Average: $(-3.6 + (-4.8))/2 = -4.2$ million acres/year.

Problem 3

a. **6.3, 6.03, 6.003 m/sec** — $(3(1.1)^2 - 3)/0.1 = 6.3$; $(3(1.01)^2 - 3)/0.01 = 6.03$; $(3(1.001)^2 - 3)/0.001 = 6.003$ m/sec.

b. **≈ 6 m/sec** — As h gets smaller, the average velocity gets closer to 6: instantaneous velocity ≈ 6 m/sec.

Problem 4

Average **8.125 ft/sec** — $(6.5 - 0)/(0.8 - 0) = 8.125$ ft/sec.

At t = 0.2 **≈ 4.5 ft/sec** — Right: $(1.8 - 0.5)/0.2 = 6.5$; left: $(0.5 - 0)/0.2 = 2.5$. Average ≈ $(6.5 + 2.5)/2 = 4.5$ ft/sec.

Problem 5

a. **8 ft/sec** — $(25 - 9)/(5 - 3) = 16/2 = 8$ ft/sec.

b. **≈ 6 ft/sec** — $3 \le t \le 3.1$: 6.1; $3 \le t \le 3.01$: 6.01; $3 \le t \le 3.001$: 6.001 → ≈ 6 ft/sec.

Problem 7

a. **63 mm³** — $S(6) - S(0) = 2^6 - 2^0 = 64 - 1 = 63$ mm³.

b. **10.5 mm³/month** — $63/6 = 10.5$ mm³/month.

c. **≈ 44.4 mm³/month** — $6 \le t \le 6.01$: ≈ 44.52; $6 \le t \le 6.001$: ≈ 44.38 → ≈ 44.4 mm³/month.

Problem 8

(a) **≈ 0.079 billion people/year** — $(6.45 - 4.08)/(2005 - 1975) = 2.37/30 \approx 0.079$ billion people/year.

(b) **≈ 0.076 billion people/year** — No data after 2005, so use the left interval: $(6.45 - 6.07)/5 = 0.076$ billion people/year.

Problem 9 **≈ 40.56 (h = 0.01), then ≈ 40.27 (h = 0.001): f′(2) ≈ 40.3** — $(5^{2.01} - 25)/0.01 \approx 40.56$; $(5^{2.001} - 25)/0.001 \approx 40.27$. So $f'(2) \approx 40.3$.

Problem 10

(a) Negative — Since 0.8 < 1, g is decreasing (exponential decay), so g′(2) is negative.

(b) ≈ −0.143 — ((0.8)^2.01 − (0.8)²)/0.01 = (0.638573 − 0.64)/0.01 ≈ −0.143.

Problem 12

≈ 9.89 — f′(2) is the instantaneous rate of change at x = 2. Using 2 ≤ x ≤ 2.001: (3^2.001 − 9)/0.001 ≈ 9.89.

Problem 14

a. ≈ 0.401 percentage points/year — (75.2 − 35.1)/(1990 − 1890) = 40.1/100 = 0.401 percentage points/year.

b. ≈ 0.38 percentage points/year — Using the interval to the right: (79.0 − 75.2)/10 = 0.38 percentage points/year.

c. ≈ 0.36 percentage points/year: in 1830 the urban percent was increasing by about 0.36 points per year — Using the interval to the right: (19.8 − 9.0)/30 = 10.8/30 = 0.36 percentage points/year.

d. Increasing — The percent keeps getting larger: increasing.

Problem 16

a. Positive: in 1996 the percent with cable was increasing — Around t = 6 the values increase (63.4 → 66.7 → 67.4), so f′(6) is positive: the percent with cable was increasing in 1996.

b. f′(2) ≈ 0.95 and f′(10) ≈ 0.55 percentage points/year: the increase was slowing down — f′(2) ≈ (63.4 − 61.5)/2 = 0.95; f′(10) ≈ (68.9 − 67.8)/2 = 0.55 percentage points per year. In 1992 the percent was increasing by about 0.95 points per year; in 2000 by about 0.55.

Problem 17

≈ 9.76 — Average rate over 0 ≤ t ≤ 0.001: (200(1.05)^0.001 − 200)/0.001 ≈ 9.76.

Problem 22

f′(1) ≈ 1.0, f′(2) ≈ 1.69: the derivative is increasing, so the graph is concave up — f′(1) ≈ (1.001 ln 1.001 − 0)/0.001 ≈ 1.0; f′(2) ≈ (2.001 ln 2.001 − 2 ln 2)/0.001 ≈ 1.69. The rate is larger at x = 2, so the graph is concave up between 1 and 2.

Problem 23

f(0) = 300 million; f′(0) ≈ 2.87 million people per year — f(0) = 300 (million). One year has 365 × 24 × 60 × 60 = 31,536,000 seconds, so people per year = 31,536,000/11 ≈ 2,866,909 → f′(0) ≈ 2.87 million people per year.

Problem 24

a. f′ negative (then zero from 1990); g′ positive; h′ positive — Hours worked decreased, then stayed constant from 1990 (f′ negative, then zero). Hourly and weekly earnings increased (g′, h′ positive).

b. (i) f′(1970) ≈ −0.2 and f′(1995) ≈ 0 hours per week per year — f′(1970) ≈ (36.0 − 37.0)/5 = −0.2; f′(1995) ≈ (34.3 − 34.3)/5 = 0 hours per week per year.

b. (ii) g′(1970) ≈ 0.266 and g′(1995) ≈ 0.472 dollars per hour per year — g′(1970) ≈ (4.73 − 3.40)/5 = 0.266; g′(1995) ≈ (14.00 − 11.64)/5 = 0.472 dollars per hour per year.

b. (iii) h′(1970) ≈ 8.90 and h′(1995) ≈ 16.18 dollars per week per year — h′(1970) ≈ (170.28 − 125.80)/5 ≈ 8.90; h′(1995) ≈ (480.41 − 399.53)/5 ≈ 16.18 dollars per week per year.

Second Spring 2021 Q3

a. ≈ 12,152 — $ARC^{+} = (26{,}661 - 12{,}750)/1 = 13{,}911$; $ARC^{-} = (12{,}750 - 2{,}357)/1 = 10{,}393$. $N'(3) \approx (13{,}911 + 10{,}393)/2 = 12{,}152$.

b. On day 3, the number of parasites is increasing by about 12,152 parasites per microliter of blood per day — Units: parasites per microliter of blood per day. On day 3, the number of parasites is increasing by about 12,152 parasites per microliter per day.

Midterm-B Q3

a. Concave down on (0, 120); concave up on (120, 210) — The ARC values decrease from 1.433 down to −1.167 (up to t = 120) → concave down on (0, 120). Then they increase from −1.167 up to 0.267 → concave up on (120, 210).

b. ≈ 0.33 mg/dL per minute: 60 minutes after the meal, glucose is increasing by about 0.33 mg/dL per minute — $IRC(60) \approx \frac{1}{2}(ARC^{-} + ARC^{+}) = \frac{1}{2}(1.100 + (-0.433)) = 0.667/2 \approx 0.33$ mg/dL per minute.

c. J = 168 mg/dL — Using the ARC on 30–60: $1.100 = (J - 135)/30 \rightarrow J = 30(1.1) + 135 = 168$ mg/dL. (Check with 60–90: $-0.433 = (155 - J)/30 \rightarrow J \approx 168$ ✓.)

2.1.2 The Derivative from Graphs

Ex 3

a. Positive — At x = 1 the graph is increasing → f′(1) is positive. (Numerically, $((1.001)^2 - 1^2)/0.001 = 2.001$, so $f'(1) \approx 2$.)

b. Negative — At x = −1 the graph is decreasing → f′(−1) is negative.

c. Positive — At x = 2 the graph is increasing (steeper than at x = 1) → f′(2) is positive.

d. Zero — At x = 0 the tangent line is horizontal → f′(0) = 0.

Ex 4

≈ 0.693 (graphically about 0.67) — Graphically: the tangent line passes through (0, 1) and about (0.3, 1.2): slope ≈ 0.2/0.3 ≈ 0.67. Numerically: (2^0.001 − 2°)/0.001 ≈ 0.693. So f′(0) ≈ 0.693.

Ex 5

i. Positive — f′(1) is the slope of the tangent line at x = 1. The graph is increasing at x = 1 → positive.

ii. Positive — This is the slope of the secant line joining x = 1 and x = 3. Since f(3) > f(1), the secant line goes up → positive.

iii. Negative — f(4) − f(2) is the change in height from x = 2 to x = 4. Since f(4) < f(2) (the graph goes down from 2 to 4) → negative.

Problem 1

a. Positive — The ARC is the slope of the secant line from x = 3 to x = 7. Since f(7) > f(3), the secant line goes up → positive.

b. Negative — The slope of the tangent line at x = 3: the graph is decreasing at x = 3 → f′(3) is negative.

Problem 2

a. Negative: in 1950 the number of farms was decreasing — The graph is decreasing at t = 1950 → f′(1950) is negative: the number of farms was decreasing.

b. f′(1960), because the graph is steeper at 1960 — The graph goes down faster (is steeper) at 1960 than at 1980, so f′(1960) is more negative.

Problem 6

a. Between x = 0 and x = 3 (≈ 0.83 vs ≈ 0.25): its secant line is steeper — f(0) ≈ 1, f(3) ≈ 3.5, f(5) ≈ 4. 0–3: (3.5 − 1)/3 ≈ 0.83; 3–5: (4 − 3.5)/2 = 0.25. The secant line on 0–3 is steeper (concave down graph).

b. At x = 1: the graph is steeper there — The tangent line is steeper at x = 1 than at x = 4, so the instantaneous rate of change is greater at x = 1.

c. Thousand dollars per kilogram — Units of y over units of x: thousand dollars per kilogram.

Problem 11

(a) Positive at C and G; negative at A and E; zero at B, D and F — $f' > 0$ where the graph is increasing (C, G); $f' < 0$ where it is decreasing (A, E); $f' = 0$ at the horizontal tangents B (minimum), D (maximum), F (minimum).

(b) Largest at G; most negative at A — f' is largest at G (going up most steeply) and most negative at A (going down most steeply).

Problem 13

Point a **−0.5** — a is just before the minimum (going down slowly) → −0.5.

Point b **0.5** — b is just after the minimum (going up slowly) → 0.5.

Point c **2** — c is where the graph is going up steeply → 2.

Point d **0** — d is at the maximum → $f' = 0$.

Point e **−2** — e is where the graph is going down steeply → −2.

Problem 15

(a) They are equal at every x: the tangent lines are parallel — g is f shifted up by 3, so at each x the tangent lines are parallel: at x = 0 both slopes are 0; at x = 1 both are 2; at x = 2 both are 4; at x = a both are 2a.

(b) The graph only shifts up or down, so its slopes do not change — Adding a constant shifts the graph vertically without changing its shape, so the slope (the derivative) at any point stays the same.

Problem 18

A = (4, 25), B = (4.2, 25.3), C = (3.9, 24.85) — A is on the graph at x = 4: (4, 25). B and C are on the tangent line (slope 1.5): B: y = 25 + 1.5(0.2) = 25.3 → (4.2, 25.3); C: y = 25 + 1.5(−0.1) = 24.85 → (3.9, 24.85).

Problem 19

(a) g(2) = 5 — B = (2, 5) is on the graph of g: g(2) = 5.

(b) g′(2) = −0.4 — g′(2) is the slope of the tangent line through (1.95, 5.02) and (2, 5): (5 − 5.02)/(2 − 1.95) = −0.02/0.05 = −0.4.

Problem 20

(a) The height of the graph at x = 4 — f(4) is the vertical distance from the x-axis to the graph at x = 4.

(b) The vertical distance between the heights of the graph at x = 4 and x = 2 — f(4) − f(2) is the difference between the heights at x = 4 and x = 2.

(c) The slope of the secant line joining the points at x = 2 and x = 5 — It is the slope of the secant line joining the points on the graph at x = 2 and x = 5.

(d) The slope of the tangent line to the graph at x = 3 — f′(3) is the slope of the tangent line at x = 3.

Problem 21

a. f(4), because f is increasing — The graph is increasing, so f(4) > f(3).

b. f(2) − f(1), because the graph rises more between x = 1 and x = 2 (concave down) — The graph is concave down (rises more and more slowly), so it rises more between 1 and 2 than between 2 and 3.

c. (f(2) − f(1))/(2 − 1): its secant line is steeper because the graph is concave down — The secant line from x = 1 to x = 2 is steeper than the one from x = 1 to x = 3, because the graph is concave down.

d. f′(1), because the graph is steeper at x = 1 — The graph is steeper at x = 1 than at x = 4, so f′(1) is larger.

2.1.3 Test Your Understanding

TYU 6 · P4

(a) Average of the slopes to the left and to the right of t = 4:

$$\text{ARC}^- = \frac{300 - 200}{4 - 2} = 50, \quad \text{ARC}^+ = \frac{600 - 300}{6 - 4} = 150$$

$$P'(4) \approx \frac{50 + 150}{2} = 100$$

Answer: C (100)

(b)

$$\text{Relative rate} = \frac{P'(4)}{P(4)} = \frac{100}{300} \approx 0.33 = 33\ \%$$

Answer: C (33 %)

(c) t = 10 is the last value in the table, so we can only use the slope to the left:

$$P'(10) \approx \frac{900 - 700}{10 - 8} = \frac{200}{2} = 100$$

Answer: C (100)

2.2 The Derivative Function

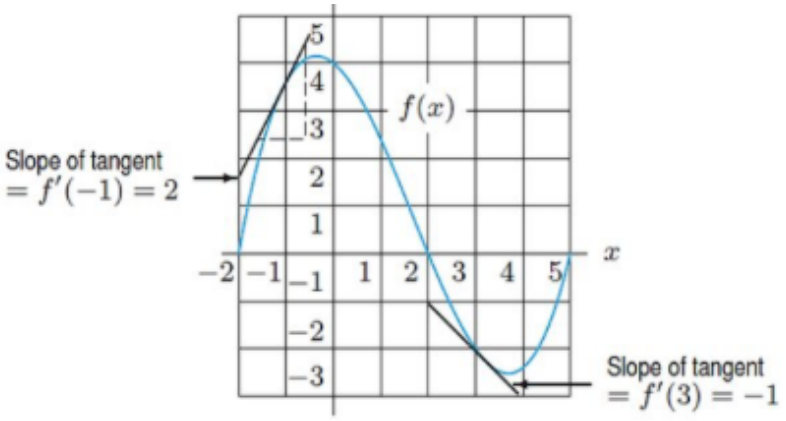
In this section

1. Graphing the Derivative from the Graph of f 4 lecture · 21 textbook · 1 past exam
2. The Derivative from Tables and Formulas 3 lecture · 5 textbook · 1 past exam
3. Interpreting f′ and Sketching f from f′ 5 textbook · 1 past exam

2.2.1 Graphing the Derivative from the Graph of f

Lecture Problems

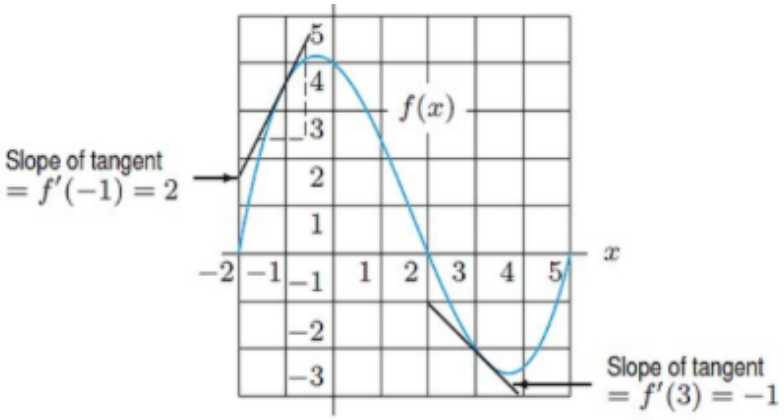
Ex 1 Estimate the derivative of the function f(x) graphed in Figure 2.18 at x = −2, −1, 0, 1, 2, 3, 4, 5. Then graph f′(x).



i. f′(−2), f′(−1), f′(0), f′(1), f′(2), f′(3), f′(4), f′(5) ≈ ?

ii. What does the graph of f′(x) look like?

Ex 2 Plot the values of the derivative function calculated in Example 1. Compare the graphs of f′ and f.

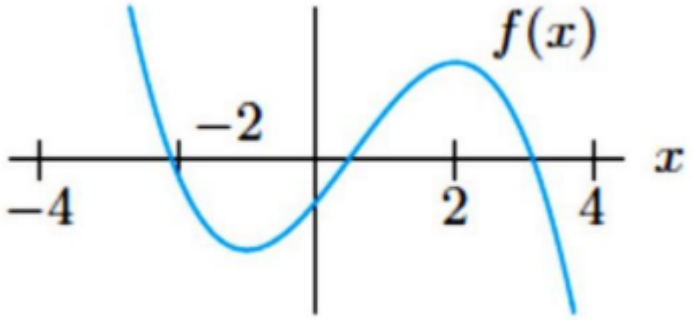


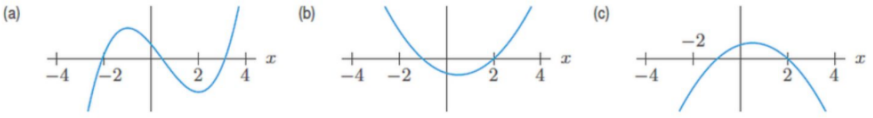
a. Where f is increasing, the graph of f′ is …

b. Where f is decreasing, the graph of f′ is …

c. At the points where f has a maximum or a minimum, …

Ex 3 The graph of f is in Figure 2.21. Which of the graphs (a)–(c) is a graph of the derivative, f′?





Which graph is f′?

Ex 7 In the graph of f given below (Figure 2.23), at which of the labeled x-values is:

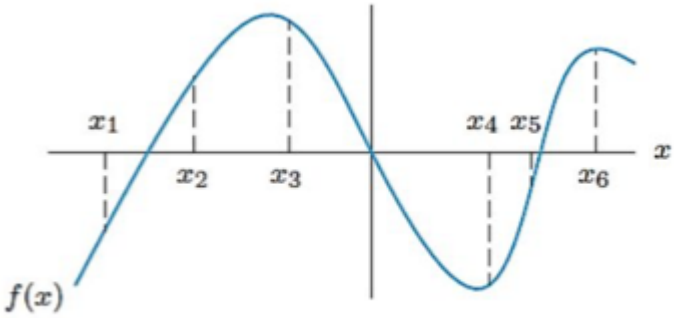


Figure 2.23

a. f(x) greatest?

b. f′(x) greatest?

c. f(x) least?

d. f′(x) least?

Textbook Problems

Problem 1 The graph of f(x) is given in Figure 2.22. Draw tangent lines to the graph at x = −2, x = −1, x = 0, and x = 2. Estimate f′(−2), f′(−1), f′(0), and f′(2).

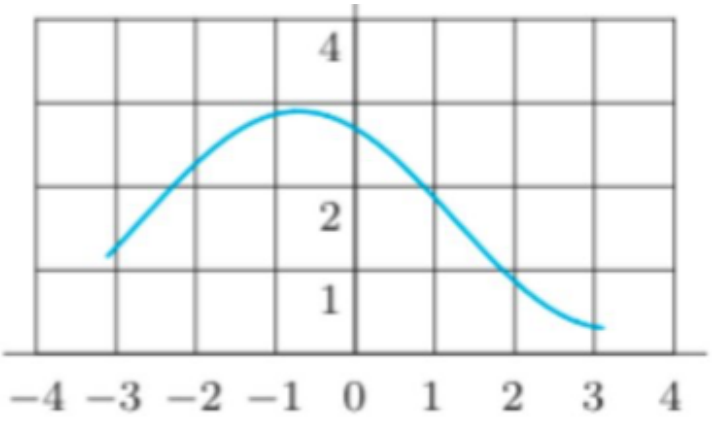
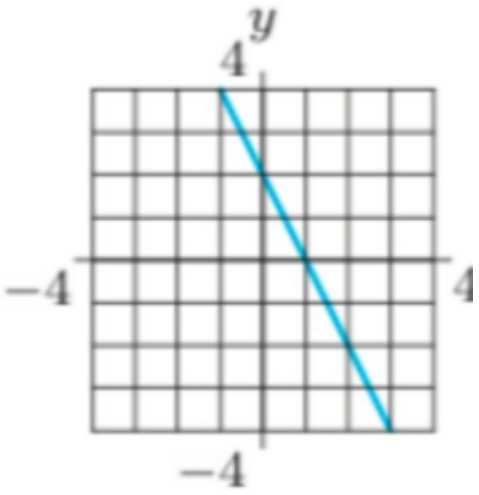


Figure 2.22

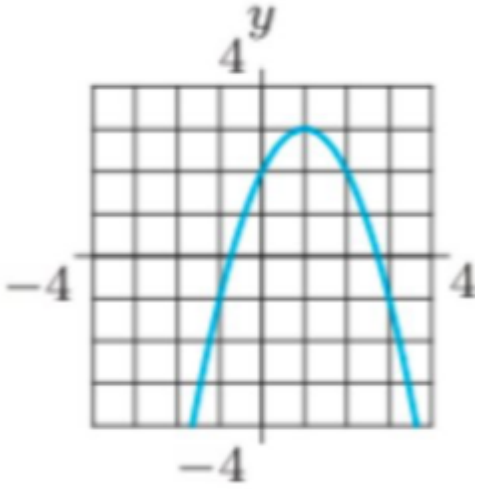
f′(−2), f′(−1), f′(0), f′(2) ≈ ?

Problem 2 Graph the derivative of the given function.



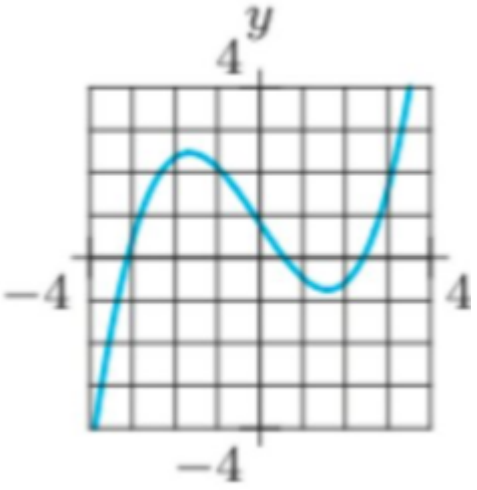
What is the graph of f′?

Problem 3 Graph the derivative of the given function.



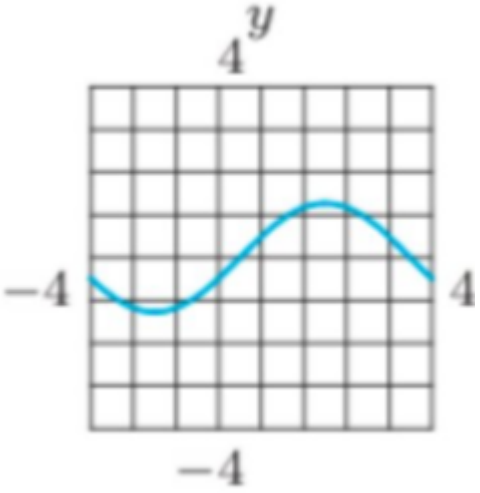
What is the graph of f′?

Problem 4 Graph the derivative of the given function.



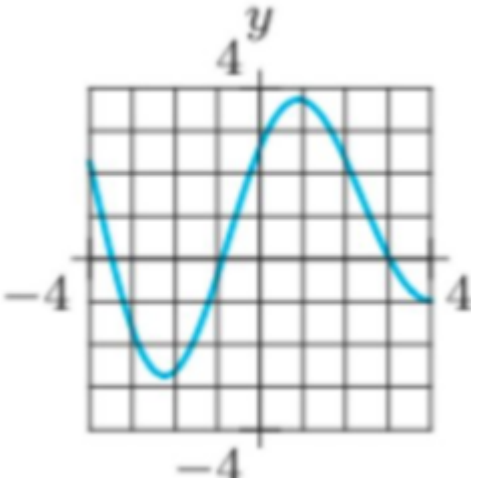
What is the graph of f′?

Problem 5 Graph the derivative of the given function.



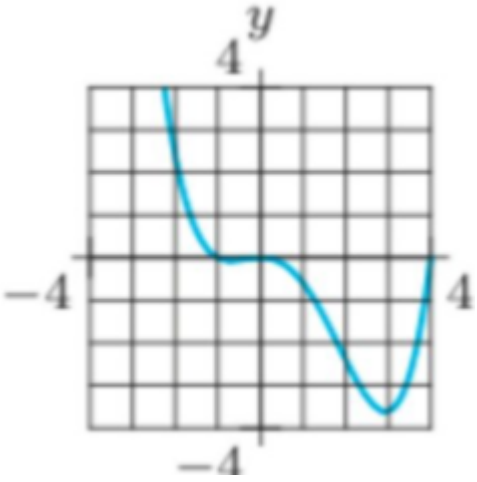
What is the graph of f′?

Problem 6 Graph the derivative of the given function.



What is the graph of f′?

Problem 7 Graph the derivative of the given function.



What is the graph of f′?

Problem 10 In the graph of f in Figure 2.23, at which of the labeled x-values is

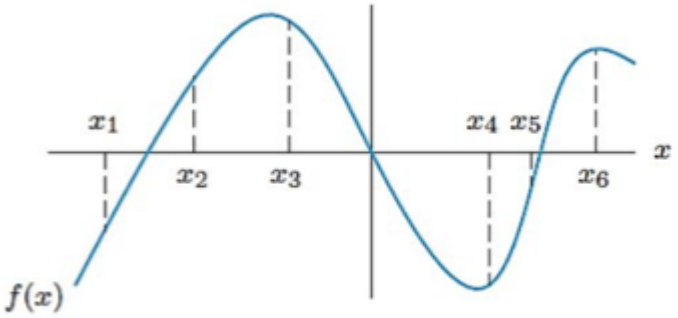


Figure 2.23

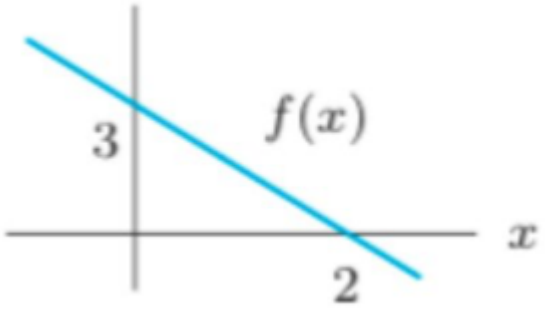
a. f(x) greatest?

b. f(x) least?

c. f′(x) greatest?

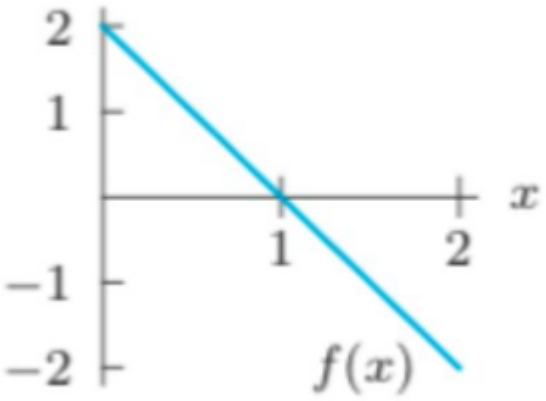
d. f′(x) least?

Problem 11 Sketch the graph of the derivative of the function shown. Be sure your sketch is consistent with the important features of the graph of the original function.



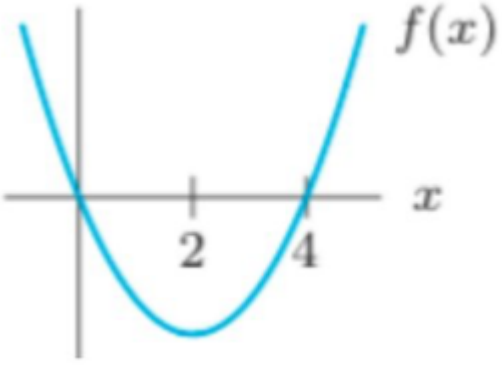
What is the graph of f′?

Problem 12 Sketch the graph of the derivative of the function shown. Be sure your sketch is consistent with the important features of the graph of the original function.



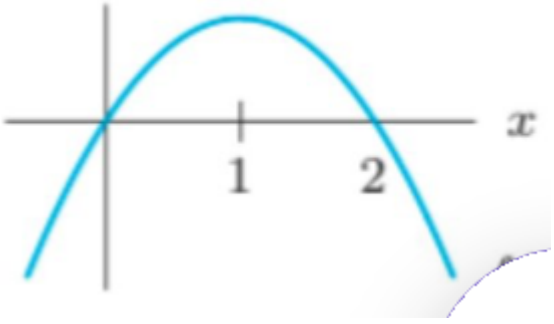
What is the graph of f′?

Problem 13 Sketch the graph of the derivative of the function shown. Be sure your sketch is consistent with the important features of the graph of the original function.



What is the graph of f′?

Problem 14 Sketch the graph of the derivative of the function shown. Be sure your sketch is consistent with the important features of the graph of the original function.



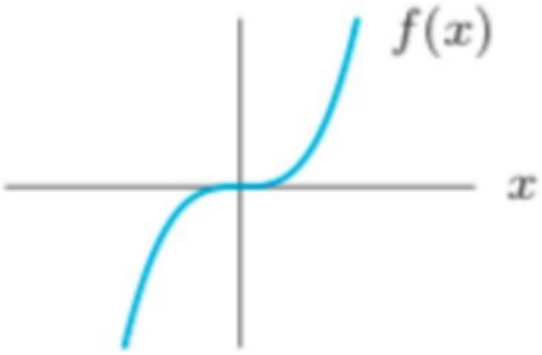
What is the graph of f′?

Problem 15 Sketch the graph of the derivative of the function shown. Be sure your sketch is consistent with the important features of the graph of the original function.



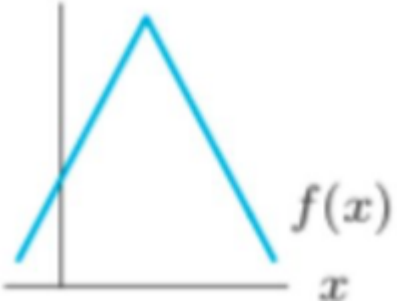
What is the graph of f′?

Problem 16 Sketch the graph of the derivative of the function shown. Be sure your sketch is consistent with the important features of the graph of the original function.



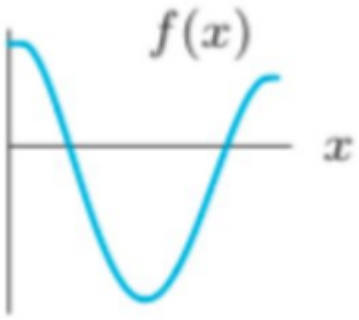
What is the graph of f′?

Problem 17 Sketch the graph of the derivative of the function shown. Be sure your sketch is consistent with the important features of the graph of the original function.



What is the graph of f′?

Problem 18 Sketch the graph of the derivative of the function shown. Be sure your sketch is consistent with the important features of the graph of the original function.



What is the graph of f′?

Problem 21 Match the function with one of the derivatives in Figure 2.24.

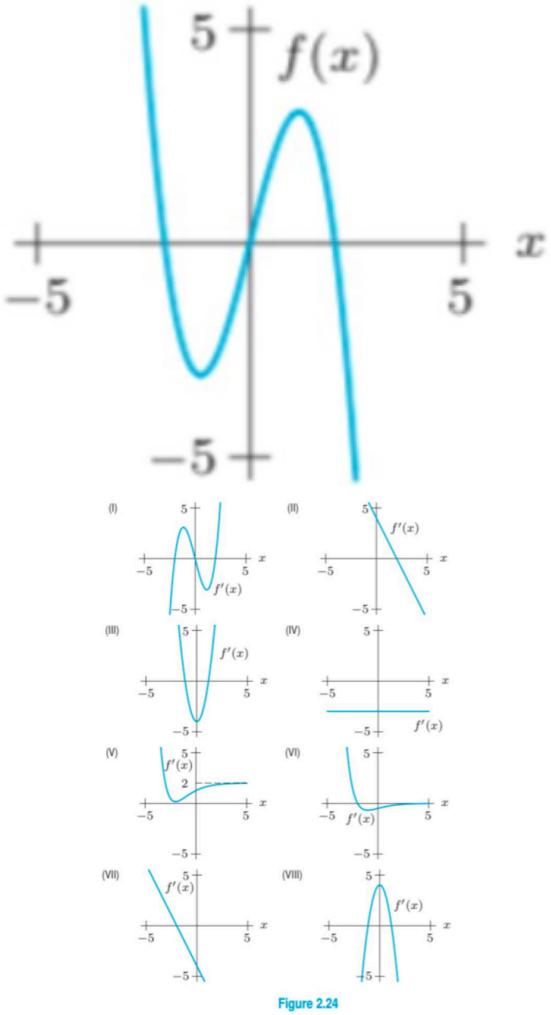


Figure 2.24

Which graph is f′?

Problem 22 Match the function with one of the derivatives in Figure 2.24. (Figure 2.24 is given in Problem 21.)

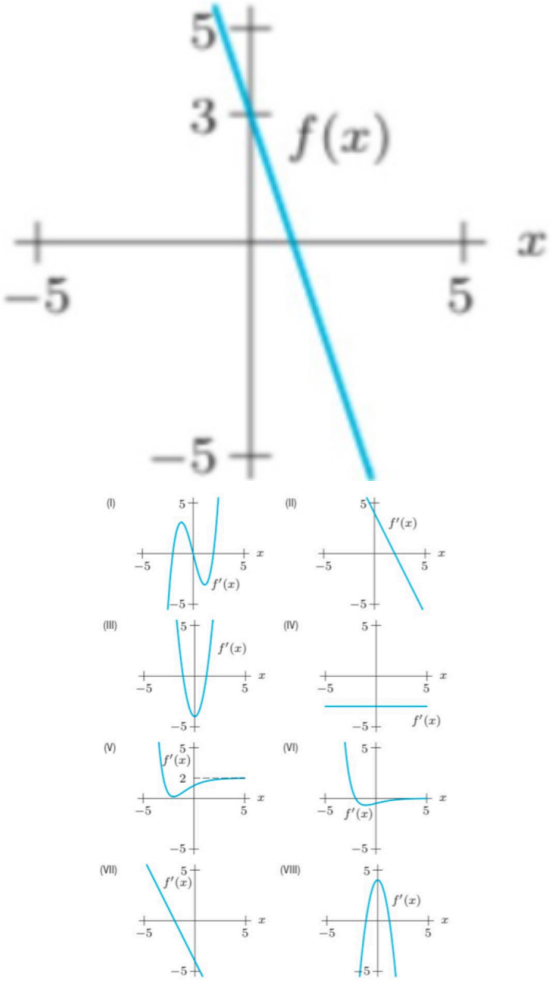
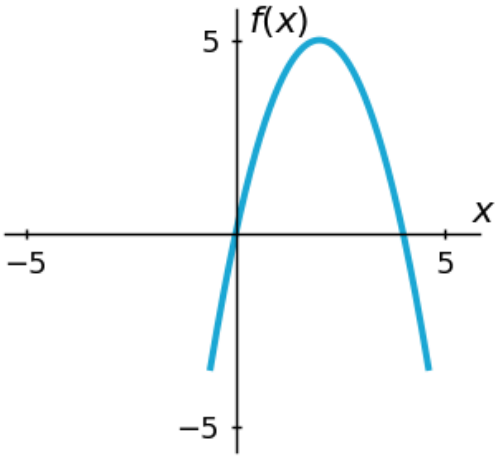


Figure 2.24

Which graph is f'?

Problem 23 Match the function with one of the derivatives in Figure 2.24. (Figure 2.24 is given in Problem 21.)



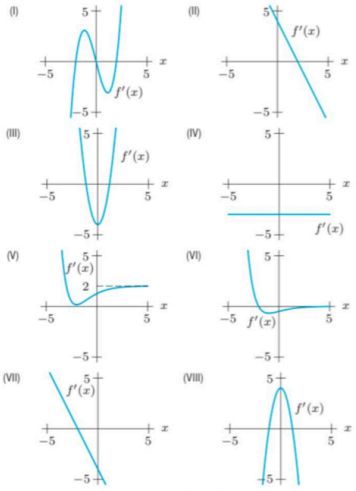
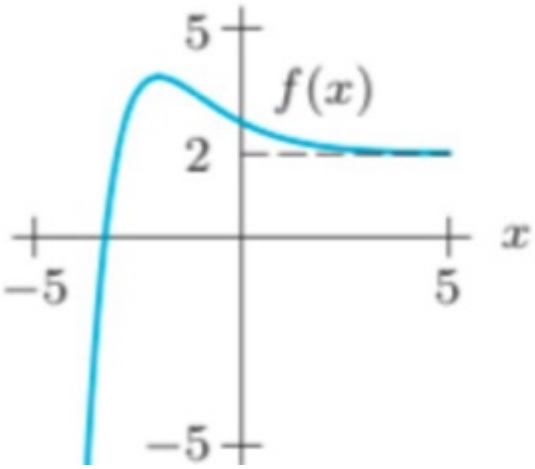


Figure 2.24

Which graph is f′?

Problem 24 Match the function with one of the derivatives in Figure 2.24. (Figure 2.24 is given in Problem 21.)



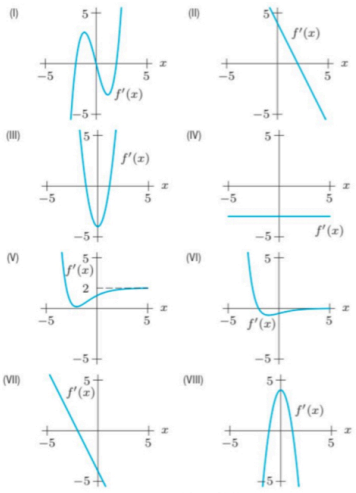
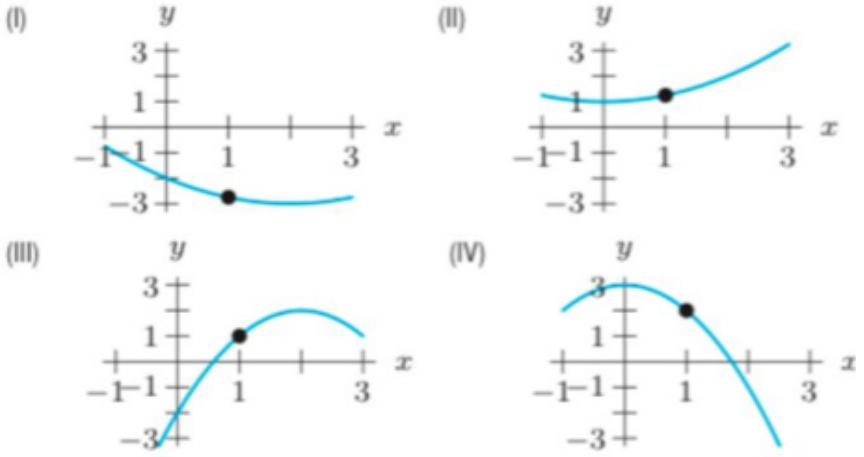


Figure 2.24

Which graph is f′?

Problem 30 Match each of the following descriptions of a function with one of the following graphs.



a. $f'(1) > 0$ and f' is always decreasing

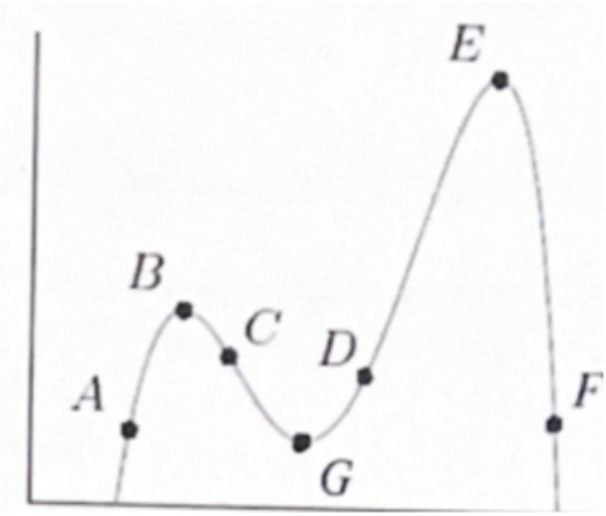
b. $f'(1) > 0$ and f' is always increasing

c. $f'(1) < 0$ and f' is always decreasing

d. $f'(1) < 0$ and f' is always increasing

Past Exam Questions

Midterm-B Q7 Consider the function f(x) shown in the figure below.



a. (i) At which of the labeled point(s) is the derivative f′(x) positive?

---

a. (ii) At which labeled point(s) is f′(x) negative?

---

a. (iii) At which labeled point(s) is f′(x) zero?

---

a. (iv) At which labeled point is f′(x) largest?

---

a. (v) At which labeled point is f′(x) most negative?

---

b. Which describes the graph of the derivative f′(x)?

2.2.2 The Derivative from Tables and Formulas

Lecture Problems

Ex 4 Table 2.6 gives values of c(t), the concentration (mg/cc) of a drug in the bloodstream at time t (min). Construct a table of estimated values for c′(t), the rate of change of c(t) with respect to t.

t (min)	0	0.1	0.2	0.3	0.4	0.5
c(t) (mg/cc)	0.84	0.89	0.94	0.98	1.00	1.00

t (min)	0.6	0.7	0.8	0.9	1.0
c(t) (mg/cc)	0.97	0.90	0.79	0.63	0.41

a. Using the average rate of change to the right ($\Delta t = 0.1$), estimate $c'(0)$, $c'(0.1)$, $c'(0.2)$.

b. Using the same method, estimate $c'(0.6)$ and $c'(0.9)$.

c. What does the table of c' tell you about the concentration?

d. Improve the estimate of $c'(0.2)$ by averaging the slopes to the left and right.

Ex 5 Guess a formula for the derivative of $f(x) = x^2$.

$f'(x) = ?$

Ex 6 Let $R(x) = 100(1.1)^x$. Using slopes to the left and right of 0, estimate $R'(0)$.

$R'(0) \approx ?$

Textbook Problems

Problem 8 Given the numerical values shown, find approximate values for the derivative of f(x) at each of the x-values given. Where is the rate of change of f(x) positive? Where is it negative? Where does the rate of change of f(x) seem to be greatest?

x	0	1	2	3	4	5	6	7	8
f(x)	18	13	10	9	9	11	15	21	30

a. Estimate $f'(0)$, $f'(1)$ and $f'(8)$.

b. Where is the rate of change of f(x) positive? Where is it negative?

c. Where does the rate of change of f(x) seem to be greatest?

Problem 9 Find approximate values for $f'(x)$ at each of the x-values given in the following table.

x	0	5	10	15	20
f(x)	100	70	55	46	40

$f'(0)$, $f'(5)$, $f'(10)$, $f'(15)$, $f'(20) \approx$?

Problem 20 Values of x and g(x) are given in the table. For what value of x does $g'(x)$ appear to be closest to 3?

x	2.7	3.2	3.7	4.2	4.7	5.2	5.7	6.2
g(x)	3.4	4.4	5.0	5.4	6.0	7.4	9.0	11.0

$g'(x)$ is closest to 3 at x = ?

Problem 28

a. Let $f(x) = \ln x$. Use small intervals to estimate $f'(1)$, $f'(2)$, $f'(3)$, $f'(4)$, and $f'(5)$.

b. Use your answers to part (a) to guess a formula for the derivative of $f(x) = \ln x$.

Problem 29 Suppose $f(x) = \frac{1}{3}x^3$. Estimate $f'(2)$, $f'(3)$, and $f'(4)$. What do you notice? Can you guess a formula for $f'(x)$?

$f'(2), f'(3), f'(4) \approx$? and $f'(x) =$?

Past Exam Questions

Final 2023 Q4 Coal gas is produced at a gasworks. Pollutants in the gas are removed by scrubbers, which become less and less efficient as time goes on. At the start of each month the rate, R(t), at which pollutants are escaping (in tons/month) into the gas is measured in the following table.

t (month)	0	1	2	3	4	5	6
R(t)	5	7	8	10	13	16	20

(a) Estimate $R'(3)$ and provide units with your answer.

(a) meaning Interpret your answer.

2.2.3 Interpreting f′ and Sketching f from f′

Textbook Problems

Problem 19 A city grew in population throughout the 1980s. The population was at its largest in 1990, and then shrank throughout the 1990s. Let P = f(t) represent the population of the city t years since 1980. Sketch graphs of f(t) and f′(t), labeling the units on the axes.

a. Which describes f′(t)?

b. What are the units on the axes?

Problem 25 Draw a possible graph of y = f(x) given the following information about its derivative.

- f′(x) > 0 for x < −1
- f′(x) < 0 for x > −1
- f′(x) = 0 at x = −1

Which describes the graph of f?

Problem 26 Draw the graph of a continuous function y = f(x) that satisfies the following three conditions.

- f′(x) > 0 for 1 < x < 3
- f′(x) < 0 for x < 1 and x > 3
- f′(x) = 0 at x = 1 and x = 3

Which describes the graph of f?

Problem 27 A vehicle moving along a straight road has distance f(t) from its starting point at time t. Which of the graphs in Figure 2.25 could be f′(t) for the following scenarios? (Assume the scales on the vertical axes are all the same.)

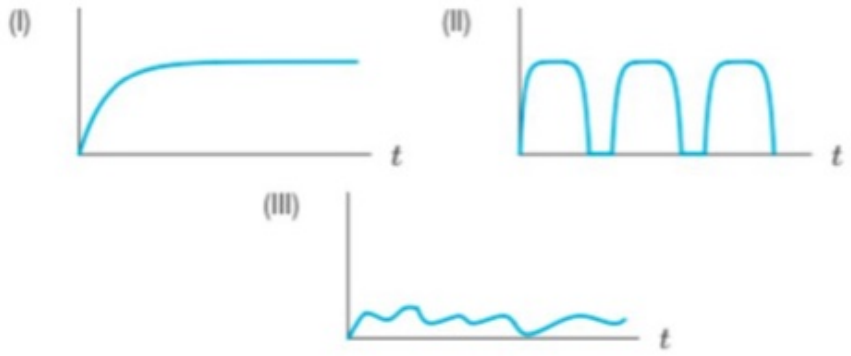


Figure 2.25

a. A bus on a popular route, with no traffic

b. A car with no traffic and all green lights

c. A car in heavy traffic conditions

Problem 31 A child inflates a balloon, admires it for a while and then lets the air out at a constant rate. If V(t) gives the volume of the balloon at time t, then Figure 2.26 shows V′(t) as a function of t. At what time does the child:

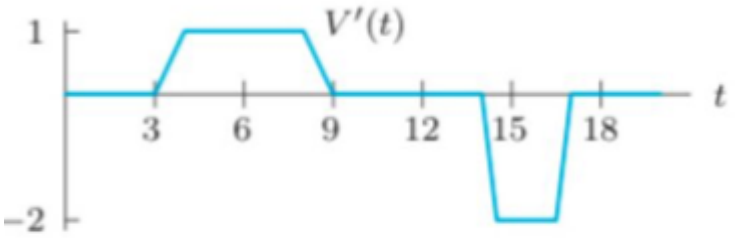


Figure 2.26

a. Begin to inflate the balloon?

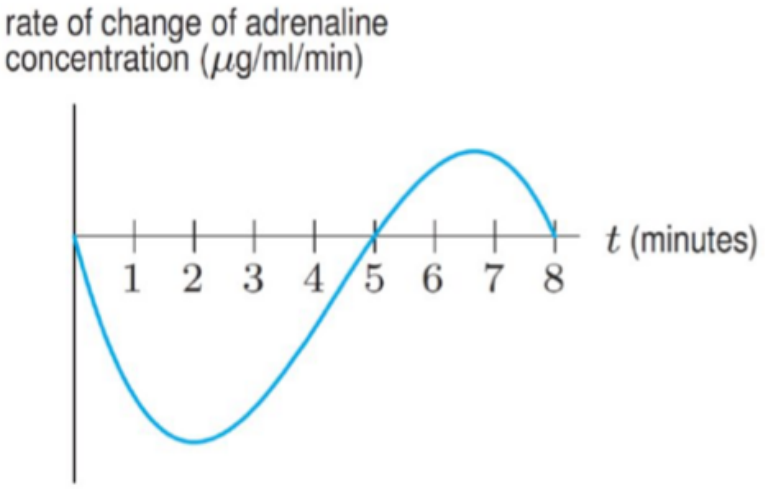
b. Finish inflating the balloon?

c. Begin to let the air out?

d. What would the graph of V′(t) look like if the child had alternated between pinching and releasing the open end of the balloon, instead of letting the air out at a constant rate?

Past Exam Questions

Final 2023 Q3 The figure below shows the rate of change of the concentration of adrenaline, in µg/ml per minute, in a person's body. Sketch a graph of the concentration of adrenaline, in µg/ml, in the body as a function of time, in minutes.



a. Where is the concentration increasing and where is it decreasing?

b. What happens to the concentration at t = 5?

c. Where is the graph of the concentration concave up and concave down?

d. Where are the largest values of the concentration?

Answers · Section 2.2

2.2.1 Graphing the Derivative from the Graph of f

Ex 1 **i. 6, 2, −1, −2, −2, −1, 1, 4** — $f'(x)$ = the slope of the tangent line at each point. From the tangent lines drawn on the graph: $f'(-2) \approx 6$, $f'(-1) \approx 2$, $f'(0) \approx -1$, $f'(1) \approx -2$, $f'(2) \approx -2$, $f'(3) \approx -1$, $f'(4) \approx 1$, $f'(5) \approx 4$. (For example, the tangent line at $x = -1$ has slope 2 and at $x = 3$ has slope -1.)

ii. A U-shaped curve: positive, then negative (between about $x = -0.5$ and $x = 3.5$), then positive again — Plot the points $(x, f'(x))$ from part (i) and join them with a smooth curve: it starts high (6), goes below the x-axis (down to −2), then rises again (to 4).

Ex 2 **a. Above the x-axis (f' positive)** — Where f is increasing, f' is positive (the graph of f' is above the x-axis).

b. Below the x-axis (f' negative) — Where f is decreasing, f' is negative (the graph of f' is below the x-axis).

c. $f' = 0$ (the graph of f' crosses the x-axis) — At a maximum or minimum the tangent is horizontal, so $f' = 0$: the graph of f' crosses the x-axis.

Ex 3 **(c)** — For $x < -2$, f is decreasing → $f' < 0$. For $-2 < x < 2$, f is increasing → $f' > 0$. For $x > 2$, f is decreasing → $f' < 0$. At $x = -2$ and $x = 2$ (minimum and maximum), $f' = 0$. The only graph that is negative, then positive between −2 and 2, then negative is graph (c).

Ex 7 **a. x_3** — $f(x)$ is greatest at x_3 (the highest point of the graph).

b. x_5 — $f'(x)$ is greatest at x_5 (the graph is steepest going up there).

c. x_4 — $f(x)$ is least at x_4 (the lowest point of the graph).

d. x_3 — x_3 is just past the top, so the graph is already going down there and $f'(x_3) < 0$. At every other labeled point $f' \geq 0$ (at x_6, the top of the right peak, $f' = 0$). So f' is least at x_3.

Problem 1 **1, 0, −0.5, −1** — $x = -2$: tangent through $(-2, 2.3)$ and $(-1, 3.3)$: slope ≈ 1. $x = -1$: top of the graph, horizontal tangent: $f'(-1) \approx 0$. $x = 0$: tangent through $(0, 2.7)$ and $(1, 2.2)$: slope ≈ −0.5. $x = 2$: tangent through $(2, 0.9)$ and $(3, -0.1)$: slope ≈ −1. (Answers close to these are acceptable.)

Problem 2 **A horizontal line at $y = -2$** — f is a straight line through $(0, 2)$ and $(1, 0)$, so its slope is the same everywhere: $m = (0 - 2)/(1 - 0) = -2$. So $f'(x) = -2$ for all x: a horizontal line at $y = -2$.

Problem 3 **A decreasing straight line through (1, 0): $f'(x) \approx -2x + 2$** — f is increasing for $x < 1$ → $f' > 0$; maximum at $x = 1$ → $f'(1) = 0$; decreasing for $x > 1$ → $f' < 0$. From the tangent lines, $f'(0) \approx 2$ and $f'(2) \approx -2$. f is a parabola, so f' is a decreasing straight line through (1, 0): $f'(x) \approx -2x + 2$.

Problem 4 **An upward parabola crossing the x-axis at $x \approx -1.6$ and $x \approx 1.6$** — f is increasing for $x < -1.6$ → $f' > 0$; maximum at $x \approx -1.6$ → $f' = 0$; decreasing for $-1.6 < x < 1.6$ → $f' < 0$; minimum at $x \approx 1.6$ → $f' = 0$; increasing for $x > 1.6$ → $f' > 0$. f is steepest going down at about $x = 0$ ($f'(0) \approx -1.5$) and very steep going up at both ends. So f' looks like an upward parabola crossing the x-axis at $x \approx -1.6$ and 1.6.

Problem 5 **Negative, 0 at $x \approx -2.5$, positive (largest ≈ 1 near $x = -0.5$), 0 at $x \approx 1.5$, then negative** — f is decreasing for $x < -2.5 \rightarrow f' < 0$; minimum at $x \approx -2.5 \rightarrow f' = 0$; increasing for $-2.5 < x < 1.5 \rightarrow f' > 0$; maximum at $x \approx 1.5 \rightarrow f' = 0$; decreasing for $x > 1.5 \rightarrow f' < 0$. f rises most steeply at about $x = -0.5$, so f' is largest there ($f' \approx 1$).

Problem 6 **Negative, 0 at $x \approx -2.2$, positive (peak ≈ 3 near $x = -0.7$), 0 at $x \approx 1$, negative (≈ -3 near $x = 2.5$), then close to 0** — f is decreasing for $x < -2.2 \rightarrow f' < 0$; minimum at $x \approx -2.2 \rightarrow f' = 0$; increasing for $-2.2 < x < 1 \rightarrow f' > 0$; maximum at $x \approx 1 \rightarrow f' = 0$; decreasing for $x > 1 \rightarrow f' < 0$, and near $x = 4$ the graph levels off, so f' gets close to 0. f rises most steeply at about $x = -0.7$ ($f' \approx 3$) and falls most steeply at about $x = 2.5$ ($f' \approx -3$).

Problem 7 **Negative for $x < 3$ (but touching 0 at $x \approx 0$), 0 at $x = 3$, then positive** — f is decreasing for $x < 3$, except that it levels off at $x \approx 0$ (horizontal tangent) $\rightarrow f'(x) < 0$, and f' touches 0 at $x \approx 0$ without changing sign. f has a minimum at $x = 3 \rightarrow f'(3) = 0$. f is increasing for $x > 3 \rightarrow f' > 0$. f is very steep going down on the left (f' very negative) and very steep going up on the right (f' large).

Problem 10

a. $\mathbf{x_3}$ — f(x) is greatest at x_3 (the highest point of the graph).

b. $\mathbf{x_4}$ — f(x) is least at x_4 (the lowest point of the graph).

c. $\mathbf{x_5}$ — $f'(x)$ is greatest at x_5 (the graph is steepest going up).

d. $\mathbf{x_3}$ — x_3 is just past the top, so the graph is already going down and $f'(x_3) < 0$. At every other labeled point $f' \geq 0$.

Problem 11 **A horizontal line at $y = -1.5$** — f is a straight line through (0, 3) and (2, 0): $m = (0 - 3)/(2 - 0) = -1.5$. So $f'(x) = -1.5$ for all x: a horizontal line at $y = -1.5$.

Problem 12 **A horizontal line at $y = -2$** — f is a straight line through (0, 2) and (1, 0): $m = (0 - 2)/(1 - 0) = -2$. So $f'(x) = -2$ for all x: a horizontal line at $y = -2$.

Problem 13 **An increasing straight line crossing the x-axis at $x = 2$** — f is decreasing for $x < 2 \rightarrow f' < 0$; minimum at $x = 2 \rightarrow f'(2) = 0$; increasing for $x > 2 \rightarrow f' > 0$. f is a parabola, so f' is an increasing straight line crossing the x-axis at $x = 2$.

Problem 14 **A decreasing straight line crossing the x-axis at $x = 1$** — f is increasing for $x < 1 \rightarrow f' > 0$; maximum at $x = 1 \rightarrow f'(1) = 0$; decreasing for $x > 1 \rightarrow f' < 0$. f is a parabola, so f' is a decreasing straight line crossing the x-axis at $x = 1$.

Problem 15 **It crosses the x-axis three times ($x \approx -1.5$, 0, 1.5) with signs −, +, −, +** — f is decreasing, then has a minimum at $x \approx -1.5 \rightarrow f' < 0$, then $f' = 0$. f is increasing up to a maximum at $x = 0 \rightarrow f' > 0$, then $f'(0) = 0$. f is decreasing to a minimum at $x \approx 1.5 \rightarrow f' < 0$, then $f' = 0$. f is increasing after $x \approx 1.5 \rightarrow f' > 0$. So f' crosses the x-axis three times with signs −, +, −, +.

Problem 16 **U-shaped, always ≥ 0, touching the x-axis at $x = 0$** — f is increasing everywhere $\rightarrow f'(x) \geq 0$. At $x = 0$ the graph is flat (horizontal tangent) $\rightarrow f'(0) = 0$. The graph gets steeper as we move away from 0 in both directions, so f' gets larger: f' touches the x-axis at 0 and is U-shaped.

Problem 17 **A positive constant (horizontal line), then a negative constant; not defined at the peak** — f is made of two straight lines. On the left piece, f is increasing with a constant slope → f′ is a positive constant. On the right piece, f is decreasing with a constant slope → f′ is a negative constant. At the peak (the corner) the slope changes suddenly, so f′ is not defined there (open circles).

Problem 18 **Negative, 0 at the minimum, then positive but getting smaller near the right end** — f is decreasing at first → $f' < 0$. f has a minimum → $f' = 0$ there. f is increasing after the minimum → $f' > 0$. Near the right end, the graph of f levels off (it rises more slowly), so f′ is positive but getting smaller.

Problem 21 **(VIII)** — f is decreasing, has a minimum at about $x = -1.3$, is increasing, has a maximum at about $x = 1$, then is decreasing. So f′ is negative, then 0, positive, 0, then negative again: f′ is positive only on a short interval in the middle. This matches graph (VIII), a narrow downward parabola that is positive only near $x = 0$.

Problem 22 **(IV)** — f is a decreasing straight line, so its slope is a negative constant (about −3). f′ is a horizontal line below the x-axis → graph (IV).

Problem 23 **(II)** — f is a downward parabola crossing the x-axis at about $x = 0$ and $x = 4$: increasing for $x < 2$, maximum at $x \approx 2$, decreasing for $x > 2$. So f′ is a decreasing straight line crossing the x-axis at $x \approx 2$ → graph (II). (Graph (VII) crosses at a negative x, so it does not match.)

Problem 24 **(VI)** — f increases very steeply, has a maximum at about $x = -2$, then decreases and levels off toward $y = 2$. So f′ is large and positive, crosses 0 at about $x = -2$, is slightly negative, and then approaches 0 (not 2!) → graph (VI).

Problem 30

a. **(III)** — $f'(1) > 0$ means the graph is going up at $x = 1$. f′ always decreasing means the graph bends downward (concave down) → graph (III).

b. **(II)** — Going up at $x = 1$ and bending upward (concave up) → graph (II).

c. **(IV)** — Going down at $x = 1$ and bending downward (concave down) → graph (IV).

d. **(I)** — Going down at $x = 1$ and bending upward (concave up) → graph (I).

Midterm-B Q7

a. (i) **A, D** — f′ is the slope at each point: positive at A and D (the graph is going up).

a. (ii) **C, F** — Negative at C and F (the graph is going down).

a. (iii) **B, E, G** — B and E are maximums and G is a minimum (horizontal tangent) → $f' = 0$.

a. (iv) **A** — A is on the steepest rising part → f′ is largest at A.

a. (v) **F** — F is on the steepest falling part → f′ is most negative at F.

b. **Positive before B, 0 at B, negative between B and G, 0 at G, positive between G and E, 0 at E, then very negative at F** — f′ is positive before B, crosses 0 at B, is negative between B and G, crosses 0 at G, is positive between G and E, crosses 0 at E, then is very negative at F.

2.2.2 The Derivative from Tables and Formulas

Ex 4 **a. 0.5, 0.5, 0.4** — $c'(0) \approx (0.89 - 0.84)/0.1 = 0.5$; $c'(0.1) \approx (0.94 - 0.89)/0.1 = 0.5$; $c'(0.2) \approx (0.98 - 0.94)/0.1 = 0.4$.

b. −0.7 and −2.2 — $c'(0.6) \approx (0.90 - 0.97)/0.1 = -0.7$; $c'(0.9) \approx (0.41 - 0.63)/0.1 = -2.2$.

c. $c' > 0$ for t = 0 to 0.3 (c increasing), $c'(0.4) \approx 0$; $c' < 0$ for t = 0.5 to 0.9 (c decreasing) — c': 0.5, 0.5, 0.4, 0.2, 0.0, −0.3, −0.7, −1.1, −1.6, −2.2. The derivative has small positive values up until t = 0.3, about 0 at t = 0.4, then it gets more and more negative: c is increasing from t = 0 to 0.4 and decreasing from t = 0.5 to 0.9.

d. ≈ 0.45 — $c'(0.2) \approx \frac{1}{2}((0.94 - 0.89)/0.1 + (0.98 - 0.94)/0.1) = (0.5 + 0.4)/2 = 0.45$.

Ex 5 **2x** — Using difference quotients: $f'(1) \approx ((1.001)^2 - 1^2)/0.001 = 2$; $f'(2) \approx ((2.001)^2 - 2^2)/0.001 = 4$; $f'(3) \approx ((3.001)^2 - 3^2)/0.001 = 6$. $f'(1) = 2$, $f'(2) = 4$, $f'(3) = 6$ suggest that $f'(x) = 2x$.

Ex 6 **≈ 9.53** — Left ($-0.001 \le x \le 0$): (100(1.1)^o − 100(1.1)^−0.001)/0.001 ≈ 9.53. Right ($0 \le x \le 0.001$): (100(1.1)^0.001 − 100(1.1)^o)/0.001 ≈ 9.53. Average of the two slopes: $R'(0) \approx 9.53$.

Problem 8 **a. −5, −3, 9** — Using the average rate of change to the right ($\Delta x = 1$): $f'(0) \approx (13 - 18)/1 = -5$; $f'(1) \approx (10 - 13)/1 = -3$. At x = 8 there is no value to the right, so we use the interval to the left: $f'(8) \approx (30 - 21)/1 = 9$.

b. Negative for $0 \le x < 3$ and positive for $x > 3$ ($f'(3) \approx 0$) — f': −5, −3, −1, 0, 2, 4, 6, 9, 9. The rate of change is negative for $0 \le x < 3$ and positive for $x > 3$ ($f'(3) \approx 0$).

c. At the right end, near x = 7 to 8 ($f' \approx 9$) — The rate of change seems to be greatest at the right end, near x = 7 to 8 ($f' \approx 9$).

Problem 9 **−6, −3, −1.8, −1.2, −1.2** — Right interval ($\Delta x = 5$): $f'(0) \approx (70 - 100)/5 = -6$; $f'(5) \approx (55 - 70)/5 = -3$; $f'(10) \approx (46 - 55)/5 = -1.8$; $f'(15) \approx (40 - 46)/5 = -1.2$. At x = 20 we use the interval to the left: $f'(20) \approx (40 - 46)/5 = -1.2$.

Problem 20 **5.2** — Slopes on each interval ($\Delta x = 0.5$): 2.0, 1.2, 0.8, 1.2, 2.8, 3.2, 4.0. At x = 5.2, the slope to the left is (7.4 − 6.0)/0.5 = 2.8 and the slope to the right is (9.0 − 7.4)/0.5 = 3.2, so $g'(5.2) \approx (2.8 + 3.2)/2 = 3$.

Problem 28 **a. ≈ 1, 0.5, 0.333, 0.25, 0.2** — Using intervals of length 0.001 to the right: $f'(1) \approx (\ln 1.001 - \ln 1)/0.001 \approx 1$; $f'(2) \approx 0.5$; $f'(3) \approx 0.333$; $f'(4) \approx 0.25$; $f'(5) \approx 0.2$.

b. $f'(x) = 1/x$ — The values are 1, 1/2, 1/3, 1/4, 1/5, which suggests $f'(x) = 1/x$.

Problem 29 **≈ 4, 9, 16; $f'(x) = x^2$** — Using intervals of length 0.001 to the right: $f'(2) \approx (\frac{1}{3}(2.001)^3 - \frac{1}{3}(2)^3)/0.001 \approx 4.002 \approx 4$; $f'(3) \approx 9.003 \approx 9$; $f'(4) \approx 16.004 \approx 16$. We notice $f'(2) \approx 2^2$, $f'(3) \approx 3^2$, $f'(4) \approx 4^2$. This suggests $f'(x) = x^2$.

Final 2023 Q4 **(a) ≈ 2.5 (tons/month) per month** — Slope to the right: (R(4) − R(3))/1 = (13 − 10)/1 = 3. Slope to the left: (R(3) − R(2))/1 = (10 − 8)/1 = 2. Average: R′(3) ≈ (3 + 2)/2 = 2.5 (tons/month) per month.

(a) meaning **At the start of the 3rd month, the rate at which pollutants escape is increasing by about 2.5 tons/month every month** — R′(3) ≈ 2.5 means: at the start of the 3rd month, the rate at which pollutants escape is increasing by about 2.5 tons/month every month.

2.2.3 Interpreting f′ and Sketching f from f′

Problem 19 **a. f′(t) > 0 for 0 < t < 10, f′(10) = 0, and f′(t) < 0 for 10 < t < 20** — f(t) is increasing for 0 < t < 10 (the 1980s), has a maximum at t = 10 (1990), and is decreasing for 10 < t < 20 (the 1990s). So f′(t) > 0 for 0 < t < 10, f′(10) = 0, and f′(t) < 0 for 10 < t < 20.

b. t in years; f(t) in people; f′(t) in people per year — Units: t in years (since 1980); f(t) in people; f′(t) in people per year.

Problem 25 **Increasing for x < −1, a maximum at x = −1, decreasing for x > −1** — f′(x) > 0 for x < −1 → f is increasing for x < −1. f′(x) < 0 for x > −1 → f is decreasing for x > −1. f′(−1) = 0 → horizontal tangent at x = −1. So f has a maximum at x = −1 (like a downward parabola).

Problem 26 **Decreasing for x < 1, a minimum at x = 1, increasing for 1 < x < 3, a maximum at x = 3, decreasing for x > 3** — f′ < 0 for x < 1 → f is decreasing for x < 1. f′ > 0 for 1 < x < 3 → f is increasing between 1 and 3. f′ < 0 for x > 3 → f is decreasing for x > 3. So f has a minimum at x = 1 and a maximum at x = 3 (horizontal tangents at both).

Problem 27 **a. (II)** — f′(t) is the velocity of the vehicle. A bus stops at many bus stops: it speeds up, drives, slows down to 0, stops, and repeats → graph (II).

b. (I) — A car with no traffic and all green lights speeds up and then drives at a constant velocity → graph (I).

c. (III) — A car in heavy traffic moves slowly, speeding up and slowing down all the time → graph (III) (small, wavy velocity).

Problem 31 **a. t = 3** — V′(t) is the rate of change of the volume: V′ > 0 when air goes in. The child begins to inflate the balloon when V′ becomes positive: t = 3.

b. t = 9 — The child finishes inflating when V′ comes back down to 0: t = 9.

c. t ≈ 14 — The child begins to let the air out when V′ becomes negative: t ≈ 14.

d. After t ≈ 14 it would jump back and forth between 0 and a negative value — While pinching, no air goes out → V′ = 0. While releasing, air goes out → V′ is negative. So after t ≈ 14 the graph of V′ would jump back and forth between 0 and a negative value.

Final 2023 Q3

a. Decreasing on (0, 5) and increasing on (5, 8) — For $0 < t < 5$: $C'(t) < 0$ → the concentration is decreasing. For $5 < t < 8$: $C'(t) > 0$ → the concentration is increasing.

b. It has a local minimum — At $t = 5$: $C'(5) = 0$ and C' changes from negative to positive → the concentration has a minimum at $t = 5$.

c. Concave down on (0, 2), concave up on (2, 6.7), concave down on (6.7, 8) — C' is decreasing for $0 < t < 2$ → concave down; C' is increasing for $2 < t < 6.7$ → concave up; C' is decreasing for $6.7 < t < 8$ → concave down. Inflection points of C at $t \approx 2$ and $t \approx 6.7$ (where C' has its lowest and highest points).

d. At the endpoints t = 0 and t = 8 — C is decreasing on (0, 5) and increasing on (5, 8) with a minimum at $t = 5$, so the largest values are at the endpoints $t = 0$ and $t = 8$. (At $t = 0$ and $t = 8$, $C' = 0$ → horizontal tangent.)

2.3 Interpretations of the Derivative

In this section

1. Units and Meaning of the Derivative 4 lecture · 34 textbook · 4 past exam
2. Using the Derivative to Estimate Values 2 lecture · 7 textbook · 2 past exam
3. Relative Rate of Change 2 lecture · 5 textbook · 2 past exam
4. Test Your Understanding 1 questions

2.3.1 Units and Meaning of the Derivative

Lecture Problems

Ex 1 The cost of extracting T tons of ore from a copper mine is C = f(T) dollars. What does it mean to say that f′(2000) = 100?

Meaning of f′(2000) = 100?

Ex 2 If q = f(p) gives the number of thousands of tons of zinc produced when the price is p dollars per ton, then

i. What are the units of dq/dp?

ii. What is the meaning of dq/dp = 0.2 at p = 900?

Ex 3 The time, L (in hours), that a drug stays in a person's system is a function of the quantity administered, q, in mg, so L = f(q).

a. Interpret the statement f(10) = 6. Give units for the numbers 10 and 6.

b. Write the derivative of L = f(q) in Leibniz notation. If f′(10) = 0.5, what are the units of the 0.5?

c. Interpret the statement f′(10) = 0.5 in terms of dose and duration.

Ex 4 If the velocity of a body at time t seconds is measured in meters/sec, what are the units of the acceleration?

Units of acceleration?

Textbook Problems

Problem 1 Write the Leibniz notation for the derivative of the given function and include units.
The cost, C, of a steak, in dollars, is a function of the weight, W, of the steak, in pounds.

Leibniz notation and units?

Problem 2 Write the Leibniz notation for the derivative of the given function and include units.
The distance to the ground, D, in feet, of a skydiver is a function of the time t in minutes since the skydiver jumped out of the airplane.

Leibniz notation and units?

Problem 3 Write the Leibniz notation for the derivative of the given function and include units.
An employee's pay, P, in dollars, for a week is a function of the number of hours worked, H.

Leibniz notation and units?

Problem 4 Write the Leibniz notation for the derivative of the given function and include units.
The number, N, of gallons of gas left in a gas tank is a function of the distance, D, in miles, the car has been driven.

Leibniz notation and units?

Problem 5 The cost, C = f(w), in dollars of buying a chemical is a function of the weight bought, w, in pounds.

a. In the statement f(12) = 5, what are the units of the 12? Of the 5? What is this saying about the cost?

b. Do you expect the derivative f′ to be positive or negative? Why?

c. In the statement f′(12) = 0.4, what are the units of the 12? Of the 0.4? What is this saying about the cost?

Problem 6 The time for a chemical reaction, T (in minutes), is a function of the amount of catalyst present, a (in milliliters), so T = f(a).

a. If f(5) = 18, what are the units of 5? Of 18? What does this tell us about the reaction?

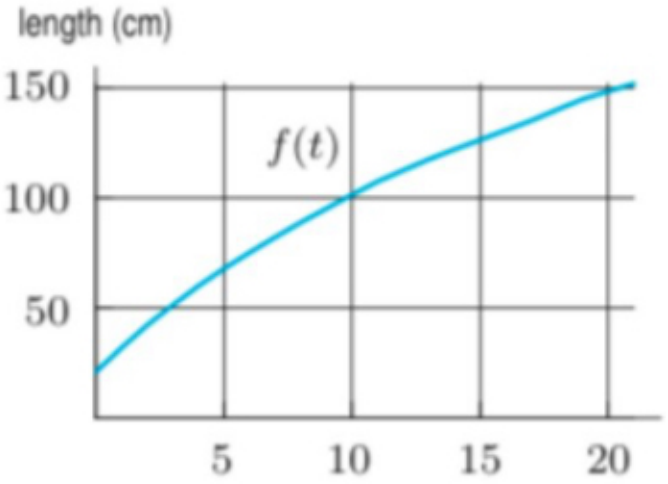
b. If f′(5) = −3, what are the units of 5? Of −3? What does this statement tell us?

Problem 7 An economist is interested in how the price of a certain item affects its sales. At a price of $p, a quantity, q, of the item is sold. If q = f(p), explain the meaning of each of the following statements:

a. f(150) = 2000

b. f′(150) = −25

Problem 8 Figure 2.27 shows the length, L, in cm, of a sturgeon (a type of fish) as a function of the time, t, in years. Estimate f′(10). Give units and interpret your answer.



f′(10) ≈ ?

Problem 9 The temperature, T, in degrees Fahrenheit, of a cold yam placed in a hot oven is given by T = f(t), where t is the time in minutes since the yam was put in the oven.

a. What is the sign of f′(t)? Why?

---

b. What are the units of f′(20)? What is the practical meaning of f′(20) = 2?

Problem 10 On May 9, 2007, CBS Evening News had a 4.3 point rating. (Ratings measure the number of viewers.) News executives estimated that a 0.1 drop in the ratings for the CBS Evening News corresponds to a $5.5 million drop in revenue. Express this information as a derivative. Specify the function, the variables, the units, and the point at which the derivative is evaluated.

Which statement is correct?

Problem 11 When you breathe, a muscle (called the diaphragm) reduces the pressure around your lungs and they expand to fill with air. The table shows the volume of a lung as a function of the reduction in pressure from the diaphragm. Pulmonologists (lung doctors) define the *compliance* of the lung as the derivative of this function.

Pressure reduction (cm of water)	0	5	10	15	20	25	30
Volume (liters)	0.20	0.29	0.49	0.70	0.86	0.95	1.00

a. What are the units of compliance?

b. Estimate the maximum compliance of the lung.

c. Explain why the compliance gets small when the lung is nearly full (around 1 liter).

Problem 12 Meteorologists define the temperature lapse rate to be $-dT/dz$ where T is the air temperature in Celsius at altitude z kilometers above the ground.

a. What are the units of the lapse rate?

b. What is the practical meaning of a lapse rate of 6.5?

Problem 13 Investing \$1000 at an annual interest rate of r%, compounded continuously, for 10 years gives you a balance of \$B, where $B = g(r)$. Give a financial interpretation of the statements:

a. $g(5) \approx 1649$

b. $g'(5) \approx 165$. What are the units of $g'(5)$?

Problem 14 Let $f(x)$ be the elevation in feet of the Mississippi River x miles from its source. What are the units of $f'(x)$? What can you say about the sign of $f'(x)$?

Units and sign of $f'(x)$?

Problem 15 The average weight, W, in pounds, of an adult is a function, W = f(c), of the average number of Calories per day, c, consumed.

a. Interpret the statements f(1800) = 155 and f′(2000) = 0 in terms of diet and weight.

b. What are the units of f′(c) = dW/dc?

Problem 16 The cost, C (in dollars), to produce g gallons of a chemical can be expressed as C = f(g). Using units, explain the meaning of the following statements in terms of the chemical:

a. f(200) = 1300

b. f′(200) = 6

Problem 17 The weight, W, in lbs, of a child is a function of its age, a, in years, so W = f(a).

a. Do you expect f′(a) to be positive or negative? Why?

b. What does f(8) = 45 tell you? Give units for the numbers 8 and 45.

c. What are the units of f′(a)? Explain what f′(a) tells you in terms of age and weight.

d. What does f′(8) = 4 tell you about age and weight?

e. As a increases, do you expect f′(a) to increase or decrease? Explain.

Problem 18 Let G be annual US government purchases, T be annual US tax revenues, and Y be annual US output of all goods and services. All three quantities are given in dollars. Interpret the statements about the two derivatives, called fiscal policy multipliers.

a. $dY/dG = 0.60$

b. $dY/dT = -0.26$

Problem 19 A recent study reports that men who retired late developed Alzheimer's at a later stage than those who stopped work earlier. Each additional year of employment was associated with about a six-week later age of onset. Express these results as a statement about the derivative of a function. State clearly what function you use, including the units of the dependent and independent variables.

Which statement is correct?

Problem 20 The thickness, P, in mm, of pelican eggshells depends on the concentration, c, of PCBs in the eggshell, measured in ppm (parts per million); that is, $P = f(c)$.

a. The derivative $f'(c)$ is negative. What does this tell you?

b. Give units and interpret $f(200) = 0.28$ and $f'(200) = -0.0005$ in terms of PCBs and eggs.

Problem 21 Problems 21–24 concern g(t) in Figure 2.28, which gives the weight of a human fetus as a function of its age.

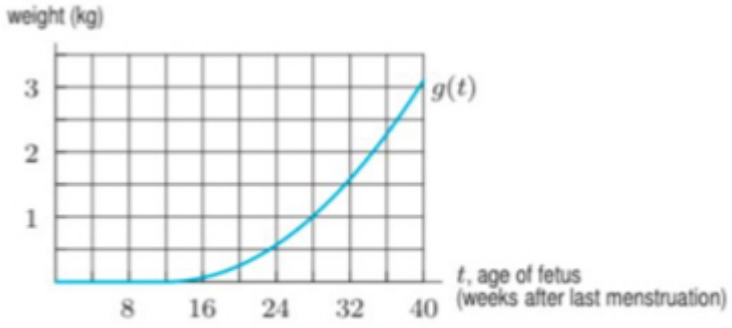


Figure 2.28

a. What are the units of g′(24)?

b. What is the biological meaning of g′(24) = 0.096?

Problem 22 Problems 21–24 concern g(t) in Figure 2.28, which gives the weight of a human fetus as a function of its age. (Figure 2.28 is given in Problem 21.)

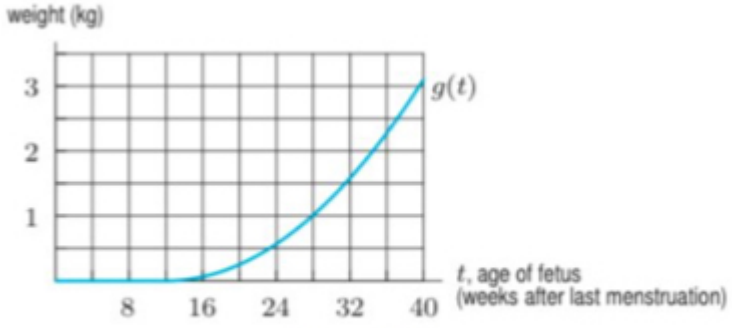


Figure 2.28

a. Which is greater, g′(20) or g′(36)?

b. What does your answer say about fetal growth?

Problem 23 Problems 21–24 concern g(t) in Figure 2.28, which gives the weight of a human fetus as a function of its age. Is the instantaneous weight growth rate greater or less than the average rate of change of weight over the 40-week period (Figure 2.28 is given in Problem 21.)

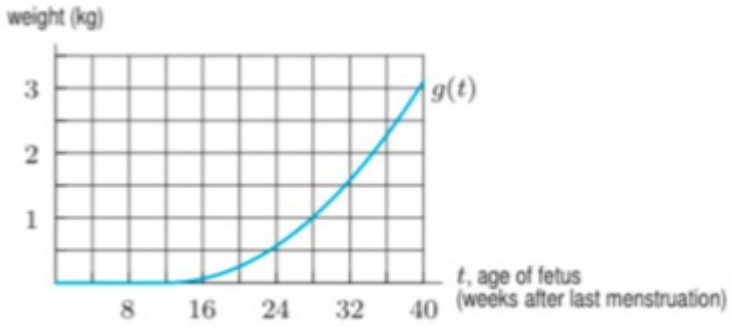


Figure 2.28

a. At week 20?

b. At week 36?

Problem 24 Problems 21–24 concern g(t) in Figure 2.28, which gives the weight of a human fetus as a function of its age. Estimate (Figure 2.28 is given in Problem 21.)

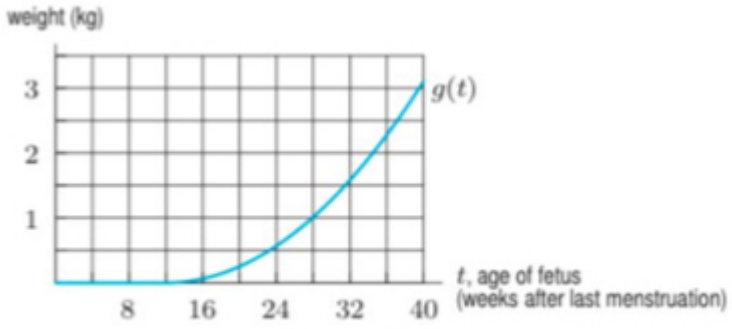


Figure 2.28

a. g′(20)

b. g′(36)

c. The average rate of change of weight for the entire 40-week gestation.

Problem 32 Figure 2.29 shows how the contraction velocity, v(x), of a muscle changes as the load on it changes.

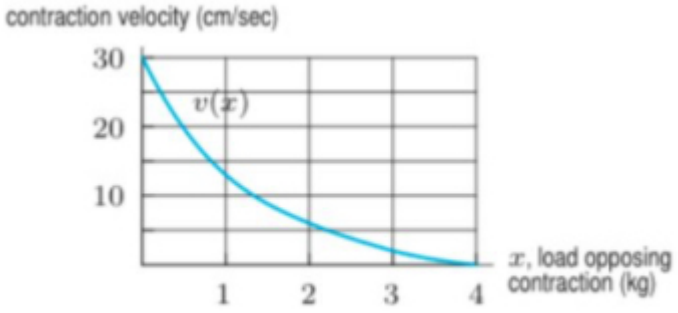


Figure 2.29

a. Find the slope of the line tangent to the graph of contraction velocity at a load of 2 kg. Give units.

b. Using your answer to part (a), estimate the change in the contraction velocity if the load is increased from 2 kg by adding 50 grams.

c. Express your answer to part (a) as a derivative of v(x).

Problem 33 Figure 2.30 shows how the pumping rate of a person's heart changes after bleeding.

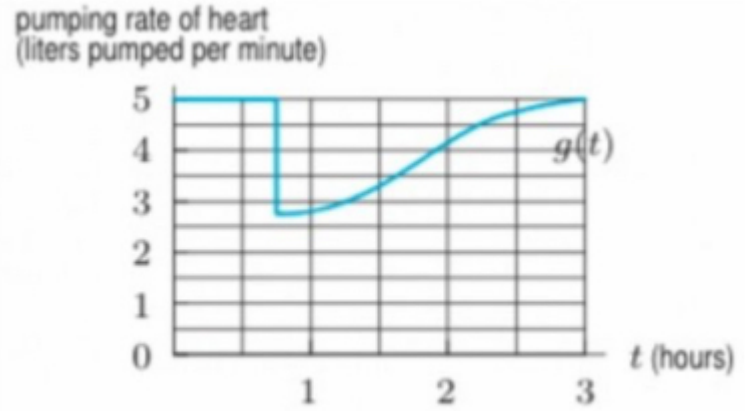


Figure 2.30

a. Find the slope of the line tangent to the graph at time 2 hours. Give units.

b. Using your answer to part (a), estimate how much the pumping rate increases during the minute beginning at time 2 hours.

c. Express your answer to part (a) as a derivative of g(t).

Problem 34 Suppose C(r) is the total cost of paying off a car loan borrowed at an annual interest rate of r%. What are the units of C′(r)? What is the practical meaning of C′(r)? What is its sign?

Which is correct?

Problem 35 A company's revenue from car sales, C (in thousands of dollars), is a function of advertising expenditure, a, in thousands of dollars, so C = f(a).

a. What does the company hope is true about the sign of f′?

b. What does f′(100) = 2 mean in practical terms? How about f′(100) = 0.5?

c. The company plans to spend about $100,000 on advertising. If f′(100) = 2, should it spend more or less? What if f′(100) = 0.5?

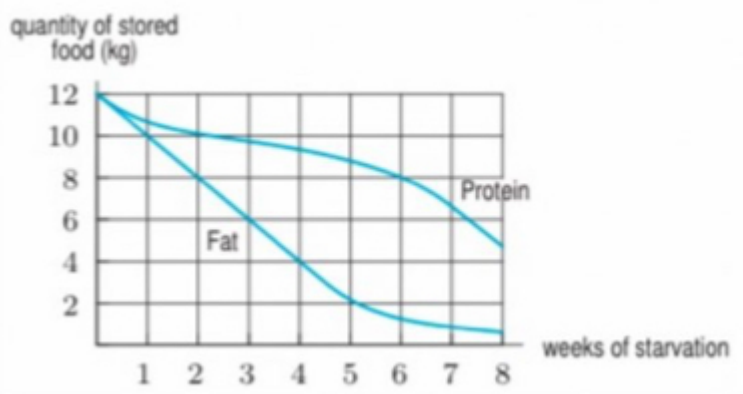
Problem 36 A person with a certain liver disease first exhibits larger and larger concentrations of certain enzymes (called SGOT and SGPT) in the blood. As the disease progresses, the concentration of these enzymes drops, first to the predisease level and eventually to zero (when almost all of the liver cells have died). Monitoring the levels of these enzymes allows doctors to track the progress of a patient with this disease. If C = f(t) is the concentration of the enzymes in the blood as a function of time,

a. Which is a possible graph of C = f(t)?

b. Where is $f' > 0$ and where is $f' < 0$?

c. What does f′(t) represent, in practical terms?

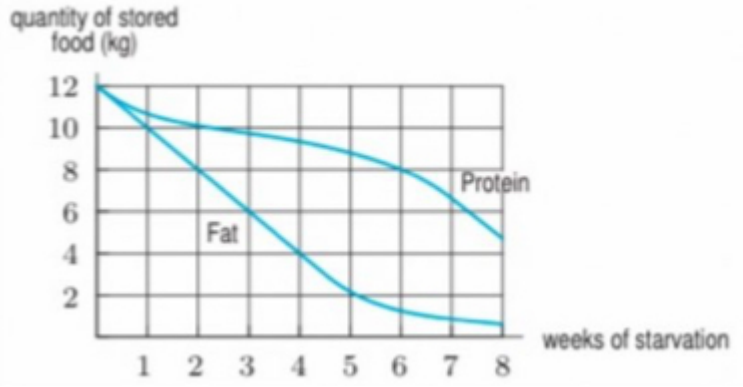
Problem 37 Problems 37–41 refer to Figure 2.31, which shows the depletion of food stores in the human body during starvation. Which is being consumed at a greater rate, fat or protein, during the



a. Third week?

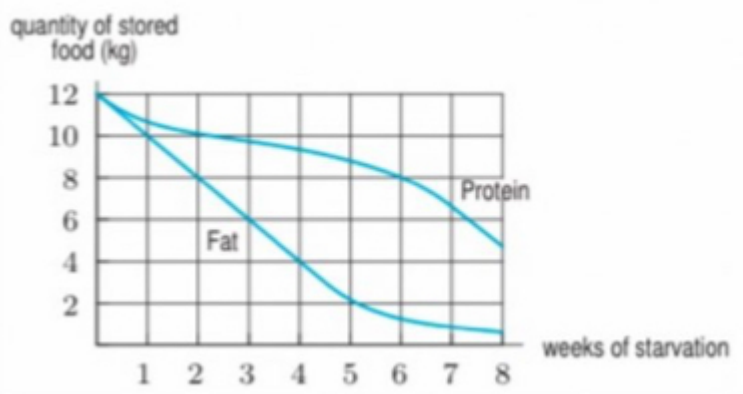
b. Seventh week?

Problem 38 Problems 37–41 refer to Figure 2.31, which shows the depletion of food stores in the human body during starvation. The fat storage graph is linear for the first four weeks. What does this tell you about the use of stored fat? (Figure 2.31 is given in Problem 37.)



What does it tell you?

Problem 39 Problems 37–41 refer to Figure 2.31, which shows the depletion of food stores in the human body during starvation. Estimate the rate of fat consumption after (Figure 2.31 is given in Problem 37.)

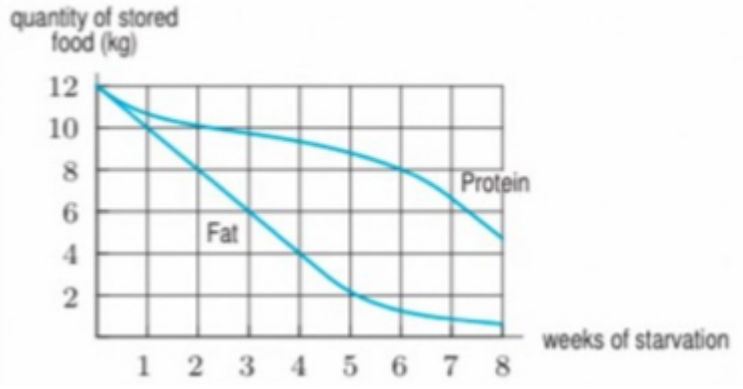


a. 3 weeks

b. 6 weeks

c. 8 weeks

Problem 40 Problems 37–41 refer to Figure 2.31, which shows the depletion of food stores in the human body during starvation. What seems to happen during the sixth week? Why do you think this happens? (Figure 2.31 is given in Problem 37.)



What happens?

Problem 41 Problems 37–41 refer to Figure 2.31, which shows the depletion of food stores in the human body during starvation. Figure 2.32 shows the derivatives of the protein and fat storage functions. Which graph is which? (Figure 2.31 is given in Problem 37.)

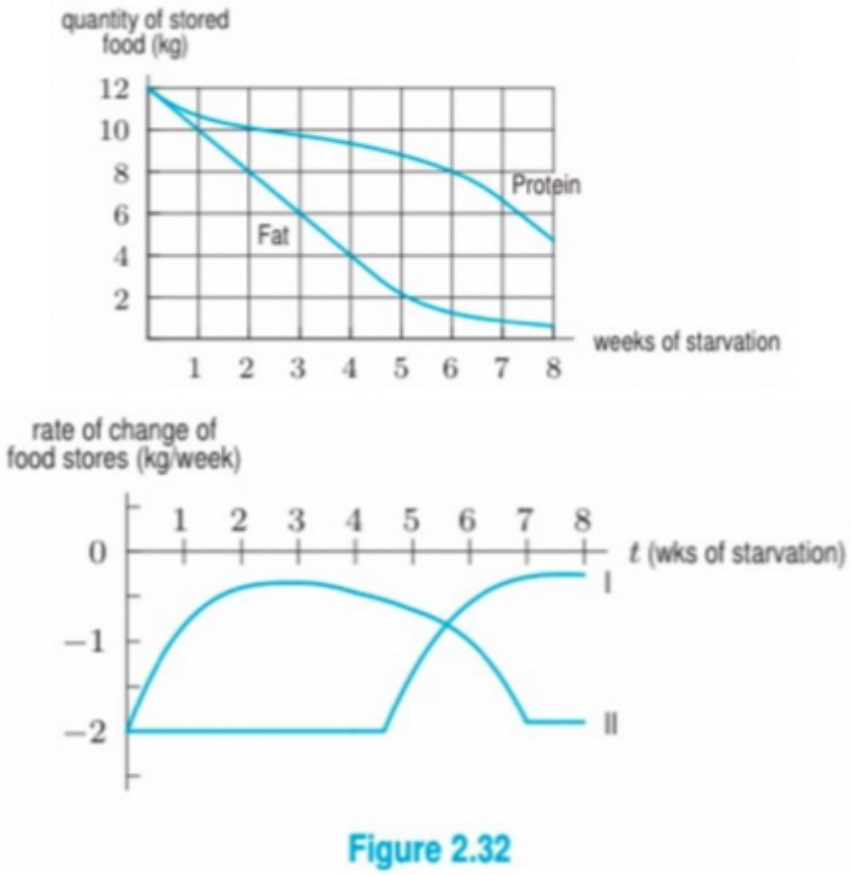


Figure 2.32

Which graph is which?

Past Exam Questions

Second 2022 Q1 When you breathe, the diaphragm muscle reduces the pressure around your lungs, and they expand to fill with air. The following table shows the volume of a lung, V, as a function of the reduction in pressure, s, from the diaphragm.

s (cm of water)	0	5	10	15	20	25	30
V (liters)	0.2	0.29	0.49	0.7	0.86	0.95	1

a. Estimate the value of dV/ds at s = 10 cm, give units and interpret it.

b. Estimate the value of dV/ds at s = 30 cm, give units and interpret it.

c. Based on your answers to Parts (a) & (b), is the rate increasing, decreasing or constant?

Final 2024 Q5 The table below shows the volume, V, of a lung as a function of the reduction in pressure, p, from the diaphragm. Lung doctors define the compliance of the lung as the derivative of this function, dV/dp.

Pressure reduction (cm of water)	0	5	10	15	20	25	30
Volume (liters)	0.20	0.29	0.49	0.70	0.86	0.95	1.00

a. What are the units of compliance?

b. Estimate the compliance at p = 15 cm of water.

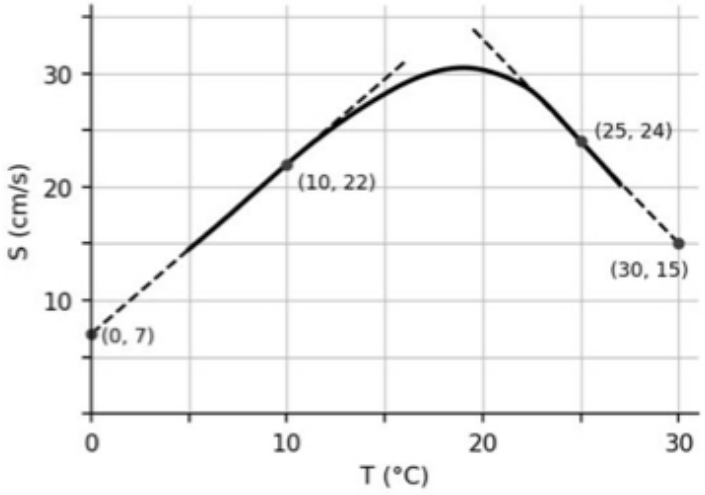
Second 2021 Q1 The table below shows values of the viral load in a patient V(t), measured in RNA copies/mL, t days after treatment was begun.

t	4	8	11	15	22
V(t)	53	18	9.4	5.2	3.6

a. Estimate and interpret the value of $V'(4)$. Provide units for your answer.

b. Estimate and interpret the value of $V'(11)$. Provide units for your answer.

Second 2021 Q2 The graph below shows the influence of the temperature, T, on the maximum sustainable swimming speed, S, of salmon fish. (The dashed lines are the tangent lines at T = 10 and T = 25.)



a. Estimate and interpret the value of $S'(10)$. Provide units with your answer.

b. Estimate and interpret the value of $S'(25)$. Provide units with your answer.

2.3.2 Using the Derivative to Estimate Values

Lecture Problems

Ex 5 If $f(2) = 4$ and $f'(2) = 1.5$, estimate $f(2.1)$. (Or equivalently: use linear approximation to estimate $f(2.1)$.)

$f(2.1) \approx$?

Ex 6 Fertilizers can improve agricultural production. A Cornell University study on maize (corn) production in Kenya found that the average value, y = f(x), in Kenyan shillings of the yearly maize production from an average plot of land is a function of the quantity, x, of fertilizer used in kilograms. (The shilling is the Kenyan unit of currency.)

i. Interpret the statements f(5) = 11,500 and f′(5) = 350.

ii. Use the statements in part (i) to estimate f(6) and f(10).

iii. The value of the derivative, f′, is increasing for x near 5. Which estimate in part (ii) is more reliable?

Textbook Problems

Problem 25 Suppose that f(t) is a function with f(25) = 3.6 and f′(25) = −0.2. Estimate f(26) and f(30).

f(26) and f(30) ≈ ?

Problem 26 Suppose that f(x) is a function with f(20) = 345 and f′(20) = 6. Estimate f(22).

f(22) ≈ ?

Problem 27 Annual net sales, in billion of dollars, for the Hershey Company, the largest US producer of chocolate, is a function S = f(t) of time, t, in years since 2000.

a. Interpret the statements f(8) = 5.1 and f′(8) = 0.22 in terms of Hershey sales.

b. Estimate f(12) and interpret it in terms of Hershey sales.

Problem 28 World meat production, M = f(t), in millions of metric tons, is a function of t, years since 2000.

a. Interpret f(5) = 249 and f′(5) = 6.5 in terms of meat production.

b. Estimate f(10) and interpret it in terms of meat production.

Problem 29 For some painkillers, the size of the dose, D, given depends on the weight of the patient, W. Thus, D = f(W), where D is in milligrams and W is in pounds.

a. Interpret the statements f(140) = 120 and f′(140) = 3 in terms of this painkiller.

b. Use the information in part (a) to estimate f(145).

Problem 30 The quantity, Q mg, of nicotine in the body t minutes after a cigarette is smoked is given by Q = f(t).

a. Interpret f(20) = 0.36 and f′(20) = −0.002 in terms of nicotine. What are the units of 20, 0.36, and −0.002?

b. Use the information in part (a) to estimate f(21) and f(30). Justify your answers.

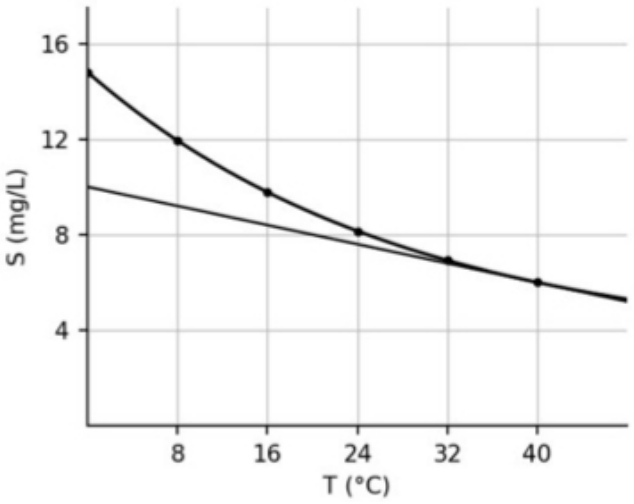
Problem 31 A mutual fund is currently valued at $80 per share and its value per share is increasing at a rate of $0.50 a day. Let V = f(t) be the value of the share t days from now.

a. Express the information given about the mutual fund in terms of f and f′.

b. Assuming that the rate of growth stays constant, estimate and interpret f(10).

Past Exam Questions

Second 2022 Q2 The quantity of oxygen that can dissolve in water depends on the temperature of the water. The following graph shows how oxygen solubility, S, varies as a function of the water temperature, T.

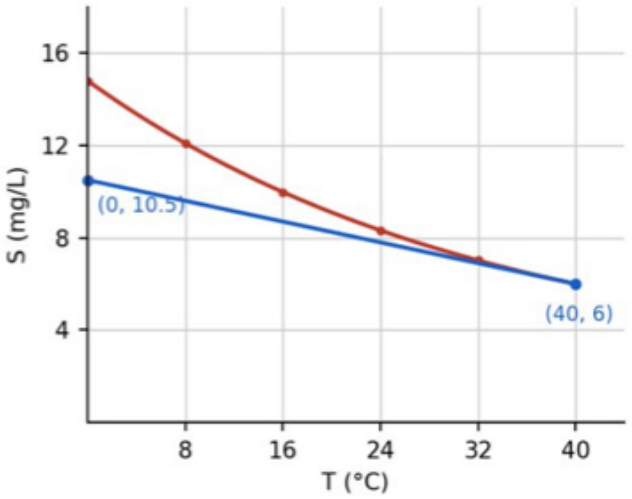


a. Write the Leibniz notation for the derivative $S'(T)$.

b. Estimate the value of $S'(40)$, give units and interpret it.

c. Use linear approximation to estimate $S(42)$.

Final Summer Q2 The quantity of oxygen that can dissolve in water depends on the temperature of the water. The following graph shows how oxygen solubility, S, varies as a function of the water temperature, T. (The blue line is the tangent line at T = 40.)



a. Write the Leibniz notation for the derivative S′(T).

b. Estimate the value of S′(40), give units and interpret it.

c. Use linear approximation to estimate S(42).

2.3.3 Relative Rate of Change

Lecture Problems

Ex 7 Annual world soybean production, W = f(t), in million tons, is a function of t years since the start of 2000.

i. Interpret the statements f(8) = 253 and f′(8) = 17 in terms of soybean production.

ii. Calculate the relative rate of change of W at t = 8; interpret it in terms of soybean production.

iii. Estimate and interpret f(9).

Ex 8 Solar photovoltaic (PV) cells are the world's fastest growing energy source. Annual production of PV cells, S, in megawatts, is approximated by $S = 277e^{0.368t}$, where t is in years since 2000. Estimate the relative rate of change of PV cell production in 2010 using

i. $\Delta t = 1$

ii. $\Delta t = 0.1$

iii. $\Delta t = 0.01$

Textbook Problems

Problem 42 The area of Brazil's rain forest, R = f(t), in million acres, is a function of the number of years, t, since 2000.

a. Interpret $f(9) = 740$ and $f'(9) = -2.7$ in terms of Brazil's rain forests.

b. Find and interpret the relative rate of change of f(t) when t = 9.

Problem 43 The number of active Facebook users hit 175 million at the end of February 2009 and 200 million at the end of April 2009. With t in months since the start of 2009, let f(t) be the number of active users in millions. Estimate f(4) and f′(4) and the relative rate of change of f at t = 4. Interpret your answers in terms of Facebook users.

f(4), f′(4) and the relative rate at t = 4?

Problem 44 The weight, w, in kilograms, of a baby is a function f(t) of her age, t, in months.

a. What does $f(2.5) = 5.67$ tell you?

b. What does $f'(2.5)/f(2.5) = 0.13$ tell you?

Problem 45 Estimate the relative rate of change of $f(t) = t^2$ at $t = 4$. Use $\Delta t = 0.01$.

Relative rate $\approx$?

Problem 46 The population, P, of a city (in thousands) at time t (in years) is $P = 700e^{0.035t}$. Estimate the relative rate of change of the population at $t = 3$ using

a. $\Delta t = 1$

---

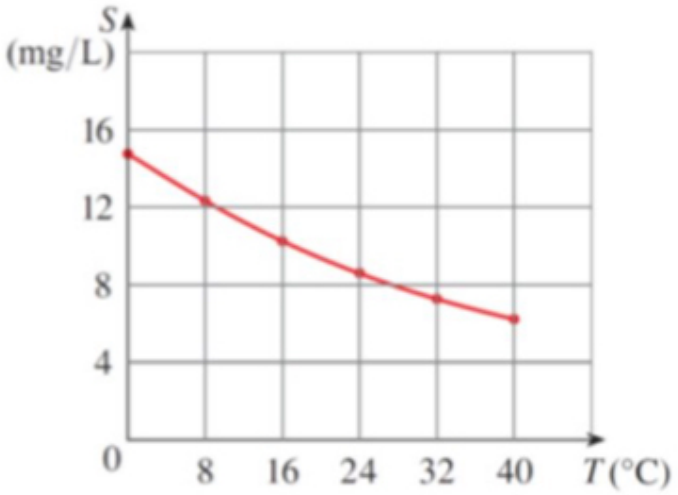
b. $\Delta t = 0.1$

---

c. $\Delta t = 0.01$

Past Exam Questions

Practice Q2 The quantity of oxygen that can dissolve in water depends on the temperature of the water. The graph below shows how oxygen solubility, S, varies as a function of the water temperature, T.



a. Estimate the value of $S'(16)$.

b. Give units to your answer in Part (a) and interpret it.

c. Estimate the relative rate of change of oxygen solubility at T = 16 °C.

Midterm-B Q8 A pharmaceutical cooling process follows a temperature curve, H (in °C), as a function of time, t (in seconds). The graph of H at t = 2 has a tangent line following the equation $y - 30 = -4(t - 2)$.

a. Compute the derivative $H'(2)$. Give its units and interpret it.

b. Compute the relative rate of change of H(t) at t = 2. Give its units and interpret it.

2.3.4 Test Your Understanding

Test Your Understanding

TYU 6 · P2 On a mountain, the number of bat species, N, is a function of the elevation, h, in feet above sea level.

a. What is the rate of change of this function?

A. $\frac{dh}{dN}$

B. $\frac{dN}{dh}$

C. dN

D. dh

E. Nh

b. What are the units of the rate of change of this function?

A. Feet per bat species.

B. Feet.

C. Bat species.

D. Bat species per feet.

E. Feet · bat species.

c. Interpret the statement $f'(600) = -2$ in terms of bat species and elevation.

A. There are 2 bat species less at an elevation of 600 feet above sea level.

B. There are 600 species of bats when the elevation above sea level is lowered by 2 feet.

C. At 600 feet above sea level, the number of bat species decreases by approximately 2 species when the elevation increases by 1 feet.

D. At 600 feet above sea level, the number of bat species increases by approximately 2 species when the elevation increases by 1 feet.

E. At 2 feet below sea level, the number of bat species increases by approximately 600 species when the elevation increases by 1 feet.

d. At every elevation level h, the instantaneous rate of change

A. is zero.

B. is negative.

C. is positive.

D. can be positive, negative or zero depending on the value of h.

E. None of the above.

e. The rate of change of N

A. is always increasing.

B. is always decreasing.

C. is always constant.

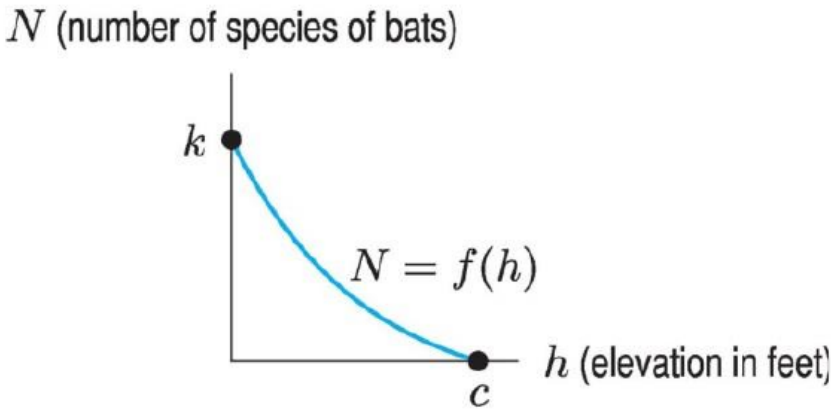
D. can be increasing, decreasing or constant depending on the value of h.

E. None of the above.

f. The instantaneous rate of change of N at h = 0 is

A. smaller than the rate at h = c.

B. greater than the rate at h = c.
C. equal to the rate at h = c.
D. both choices A and C are correct.
E. None of the above.



Answers · Section 2.3

2.3.1 Units and Meaning of the Derivative

Ex 1 **After 2000 tons have been extracted, one more ton costs about $100** — $f'(2000) = dC/dT = 100$ dollars/ton. When 2000 tons of ore have already been extracted, the cost of extracting one more ton is about $100. (You can think of dC as the extra cost of extracting an extra dT tons of ore.)

Ex 2 **i. Thousands of tons per dollar** — The units of dq/dp are the units of q over the units of p: thousands of tons per dollar.

ii. When the price is $900, the quantity produced increases by about 0.2 thousand tons for a one-dollar increase in price — dq/dp at $p = 900 = f'(900) = 0.2$ thousand tons per dollar: when the price is $900, the quantity produced increases by about 0.2 thousand tons, or 200 tons, for a one-dollar increase in price.

Ex 3 **a. 10 is in mg and 6 is in hours: a dose of 10 mg stays in the person's system for 6 hours** — $f(10) = 6$: 10 is in mg and 6 is in hours. A dose of 10 mg stays in the person's system for 6 hours.

b. $f'(q) = dL/dq$; the units of 0.5 are hours per mg — Leibniz notation: $f'(q) = dL/dq$. The units of 0.5 are hours/mg.

c. At a 10 mg dose, 1 extra mg keeps the drug in the system about 0.5 hour longer — $f'(10) = 0.5$ hours/mg means that when the dose is 10 mg, increasing the dose by 1 mg increases the time the drug stays in the system by about 0.5 hour (30 minutes).

Ex 4 **m/sec²** — Acceleration is the derivative of velocity: $a = dv/dt$. Units = (units of v) ÷ (units of t) = (m/sec)/sec = m/sec².

Problem 1 **dC/dW, in dollars per pound** — Leibniz notation: dC/dW. Units = units of C ÷ units of W = dollars per pound.

Problem 2 **dD/dt, in feet per minute** — Leibniz notation: dD/dt. Units = units of D ÷ units of t = feet per minute.

Problem 3 **dP/dH, in dollars per hour** — Leibniz notation: dP/dH. Units = units of P ÷ units of H = dollars per hour.

Problem 4 **dN/dD, in gallons per mile** — Leibniz notation: dN/dD. Units = units of N ÷ units of D = gallons per mile.

Problem 5

a. 12 is in pounds and 5 is in dollars: it costs $5 to buy 12 pounds of the chemical — 12 is in pounds and 5 is in dollars. It costs $5 to buy 12 pounds of the chemical.

b. Positive: the more chemical you buy, the more it costs — f' is positive: the more chemical you buy, the more it costs, so C increases as w increases.

c. 12 is in pounds and 0.4 is in dollars per pound: after 12 pounds, buying one more pound costs about $0.40 extra — 12 is in pounds and 0.4 is in dollars per pound. When you have already bought 12 pounds, buying one more pound costs about $0.40 extra (the 13th pound costs about 40 cents).

Problem 6

a. 5 is in milliliters and 18 is in minutes: with 5 ml of catalyst, the reaction takes 18 minutes — 5 is in milliliters and 18 is in minutes. With 5 ml of catalyst, the reaction takes 18 minutes.

b. 5 is in milliliters and −3 is in minutes per milliliter: with 5 ml, adding 1 more ml decreases the reaction time by about 3 minutes — 5 is in milliliters and −3 is in minutes per milliliter. When 5 ml of catalyst is present, adding 1 more ml decreases the reaction time by about 3 minutes.

Problem 7

a. When the price is $150, 2000 items are sold — When the price is $150, 2000 items are sold.

b. When the price is $150, increasing the price by $1 decreases the quantity sold by about 25 items — $f'(150) = -25$ items per dollar: when the price is $150, increasing the price by $1 decreases the quantity sold by about 25 items.

Problem 8

≈ 6 cm/year: when the sturgeon is 10 years old, its length is increasing at about 6 cm per year — $f'(10)$ is the slope of the tangent line at $t = 10$. Using points on either side: $f(8) \approx 89$ and $f(12) \approx 113$, so $f'(10) \approx (113 - 89)/(12 - 8) = 24/4 = 6$ cm/year. When the sturgeon is 10 years old, its length is increasing at a rate of about 6 cm per year.

Problem 9

a. Positive: the yam is getting hotter, so its temperature is increasing — $f'(t)$ is positive, because the yam is getting hotter, so its temperature is increasing.

b. °F per minute: at 20 minutes, the temperature is rising about 2°F per minute — The units of $f'(20)$ are °F per minute. $f'(20) = 2$ means that 20 minutes after the yam is put in the oven, its temperature is increasing at a rate of about 2°F per minute.

Problem 10

R = f(r), revenue (millions of dollars) as a function of rating r (points): $f'(4.3) \approx 55$ million dollars per rating point — Let $R = f(r)$ be the revenue, in millions of dollars, as a function of the rating, r (in rating points). A change of $\Delta r = -0.1$ gives $\Delta R = -5.5$, so $f'(4.3) \approx \Delta R/\Delta r = -5.5/-0.1 = 55$ million dollars per rating point. The derivative is evaluated at $r = 4.3$: when the rating is 4.3, a change of 1 rating point changes the revenue by about $55 million (in the same direction).

Problem 11

a. Liters per cm of water — Compliance = dV/dP, so its units are liters per cm of water.

b. ≈ 0.042 liters per cm of water, when the pressure reduction is between 10 and 15 cm of water — Compliance on each interval (ΔP = 5): 0.018, 0.040, 0.042, 0.032, 0.018, 0.010. For example, (0.70 − 0.49)/(15 − 10) = 0.21/5 = 0.042. The maximum compliance is about 0.042 liters per cm of water, when the pressure reduction is between 10 and 15 cm of water.

c. The lung is nearly full, so it can hardly expand any more — When the lung is nearly full, it cannot expand much more, so a further reduction in pressure adds very little volume. So dV/dP (the compliance) is small.

Problem 12

a. °C per kilometer — The units are °C per kilometer.

b. The temperature decreases by about 6.5°C for each 1 kilometer increase in altitude — −dT/dz = 6.5 means dT/dz = −6.5: the temperature decreases by about 6.5°C for each 1 kilometer increase in altitude.

Problem 13

a. $1000 at 5% (continuous) grows to about $1649 after 10 years — If $1000 is invested at 5% annual interest, compounded continuously, the balance after 10 years is about $1649.

b. Dollars per percent: at 5%, a 1% higher rate adds about $165 to the 10-year balance — The units of g′(5) are dollars per percent. At an interest rate of 5%, increasing the rate by 1% (to 6%) increases the balance after 10 years by about $165.

Problem 14

Feet per mile; negative, because a river flows downhill — The units of f′(x) are feet per mile. f′(x) is negative: a river flows downhill, so its elevation decreases as we move away from the source.

Problem 15

a. At 1800 Calories/day the average weight is 155 lb; near 2000 Calories/day, small changes in Calories barely change weight — f(1800) = 155: an adult who consumes 1800 Calories per day weighs 155 pounds on average. f′(2000) = 0: for an adult who consumes 2000 Calories per day, a small change in the number of Calories consumed does not (approximately) change the average weight.

b. Pounds per (Calorie per day) — The units of dW/dc are pounds per (Calorie per day).

Problem 16

a. It costs $1300 to produce 200 gallons of the chemical — It costs $1300 to produce 200 gallons of the chemical.

b. 6 dollars per gallon: after 200 gallons, one more gallon costs about $6 — f′(200) = 6 dollars per gallon: when 200 gallons have been produced, producing one more gallon costs about $6 (the 201st gallon costs about $6).

Problem 17

a. Positive, because children gain weight as they get older — f′(a) is positive, because children gain weight as they get older.

b. 8 is in years and 45 is in pounds: an 8-year-old child weighs 45 pounds — 8 is in years and 45 is in pounds: an 8-year-old child weighs 45 pounds.

c. Pounds per year: how fast the child is gaining weight at age a — The units of f′(a) are pounds per year. f′(a) tells us how fast the child's weight is changing (how many pounds per year it gains) at age a.

d. At age 8, the child gains about 4 pounds per year — At age 8, the child is gaining weight at a rate of about 4 pounds per year.

e. Decrease: babies gain weight fast, and the rate slows as the child ages — We expect f′(a) to decrease overall: a baby gains weight very quickly, and the rate of weight gain slows down as the child gets older (it may speed up again for a while at puberty).

Problem 18

a. Each extra $1 of government purchases raises output by about $0.60 — Increasing government purchases by $1 increases the output of goods and services by about $0.60.

b. Each extra $1 of tax revenue lowers output by about $0.26 — Increasing tax revenues by $1 decreases the output of goods and services by about $0.26.

Problem 19

A = f(y), age of onset (weeks) as a function of years of employment y (years): f′(y) = dA/dy ≈ 6 weeks per year — Let A = f(y) be the age of onset of Alzheimer's (in weeks) as a function of the number of years of employment, y (in years). Each extra year of work delays the onset by about 6 weeks, so f′(y) = dA/dy ≈ 6 weeks per year. (Equivalently, measuring A in years: f′(y) ≈ 6/52 ≈ 0.115 years per year.)

Problem 20

a. As the concentration of PCBs increases, the eggshells get thinner — As the concentration of PCBs increases, the eggshells get thinner.

b. 200 is in ppm and 0.28 in mm: at 200 ppm the eggshell is 0.28 mm thick; f′(200) = −0.0005 mm per ppm: increasing the PCBs by 1 ppm decreases the thickness by about 0.0005 mm — f(200) = 0.28: 200 is in ppm and 0.28 is in mm. When the PCB concentration is 200 ppm, the eggshell is 0.28 mm thick. f′(200) = −0.0005 mm per ppm: when the concentration is 200 ppm, increasing it by 1 ppm decreases the thickness of the eggshell by about 0.0005 mm.

Problem 21

a. kg per week — The units of g′(24) are kg per week.

b. At 24 weeks, the fetus gains about 0.096 kg per week — At 24 weeks, the fetus is gaining weight at a rate of about 0.096 kg per week (about 96 grams per week).

Problem 22

a. g′(36), because the graph is steeper at t = 36 than at t = 20 — g′(36) is greater, because the graph is steeper at t = 36 than at t = 20.

b. The fetus grows faster at 36 weeks than at 20 weeks — The fetus grows (gains weight) faster at 36 weeks than at 20 weeks.

Problem 23

a. Less: g′(20) ≈ 0.06 kg/week is less than the average 0.075 kg/week — Average rate over 40 weeks: ARC = (g(40) − g(0))/(40 − 0) ≈ (3 − 0)/40 = 0.075 kg/week. At week 20 the graph is not steep (g′(20) ≈ 0.06 kg/week), so the instantaneous rate is less than the average rate.

b. Greater: g′(36) ≈ 0.19 kg/week is greater than the average 0.075 kg/week — At week 36 the graph is steep (g′(36) ≈ 0.19 kg/week), so the instantaneous rate is greater than the average rate of 0.075 kg/week.

Problem 24

a. ≈ 0.06 kg/week — From the graph, g(16) ≈ 0.07 and g(24) ≈ 0.56: g′(20) ≈ (0.56 − 0.07)/(24 − 16) ≈ 0.49/8 ≈ 0.06 kg/week.

b. ≈ 0.19 kg/week — From the graph, g(32) ≈ 1.55 and g(40) ≈ 3.05: g′(36) ≈ (3.05 − 1.55)/(40 − 32) ≈ 1.5/8 ≈ 0.19 kg/week.

c. ≈ 0.075 kg/week — ARC = (g(40) − g(0))/(40 − 0) ≈ (3 − 0)/40 = 0.075 kg/week.

Problem 32

a. ≈ −5 cm/sec per kg — From the graph, v(1.5) ≈ 9 and v(2.5) ≈ 4: slope ≈ (4 − 9)/(2.5 − 1.5) = −5 cm/sec per kg.

b. Decreases by about 0.25 cm/sec — 50 grams = 0.05 kg: Δv ≈ (−5)(0.05) = −0.25 cm/sec. The contraction velocity decreases by about 0.25 cm/sec.

c. v′(2) ≈ −5 cm/sec per kg — v′(2) ≈ −5 cm/sec per kg.

Problem 33

a. ≈ 1.6 liters per minute per hour — From the graph, g(1.75) ≈ 3.7 and g(2.25) ≈ 4.5: slope ≈ (4.5 − 3.7)/(2.25 − 1.75) = 0.8/0.5 = 1.6 liters per minute per hour.

b. ≈ 0.027 liters per minute — One minute = 1/60 hour: Δg ≈ 1.6 × (1/60) ≈ 0.027 liters per minute. During that minute the pumping rate increases by about 0.027 liters per minute.

c. g′(2) ≈ 1.6 liters per minute per hour — g′(2) ≈ 1.6 liters per minute per hour.

Problem 34

Dollars per percent; the extra total amount you pay if the rate increases by 1%; positive — The units of C′(r) are dollars per percent. C′(r) is approximately the extra amount of money you pay (in total) if the interest rate increases by 1%, from r% to (r + 1)%. C′(r) is positive: a higher interest rate means a higher total cost.

Problem 35

a. Positive: spending more on advertising should increase revenue — The company hopes f′ is positive: spending more on advertising should increase revenue.

b. When $100,000 is spent, an extra $1000 increases revenue by about $2000 (if f′ = 2) or by only about $500 (if f′ = 0.5) — f′(100) = 2: when $100,000 is spent on advertising, spending an extra $1000 increases revenue by about $2000. f′(100) = 0.5: spending an extra $1000 increases revenue by only about $500.

c. If f′(100) = 2, spend more; if f′(100) = 0.5, spend less — If f′(100) = 2, the company should spend more than $100,000, because each extra $1000 brings in about $2000. If f′(100) = 0.5, the company should spend less, because each extra $1000 brings in only about $500.

Problem 36

a. Rises from the predisease level to a peak, then falls back and eventually to zero — The concentration starts at the normal (predisease) level, rises to a maximum, then falls back to the predisease level and eventually to zero.

b. $f' > 0$ before the maximum; $f' < 0$ after the maximum — $f' > 0$ while the concentration is rising (before the maximum); $f' < 0$ after the maximum, while the concentration drops to zero.

c. The rate of change of the enzyme concentration in the blood — $f'(t)$ is the rate at which the concentration of the enzymes in the blood is changing (in units of concentration per unit time).

Problem 37

a. Fat — Third week ($t = 2$ to 3): fat 8 → 6, a decrease of 2 kg; protein about 10.1 → 9.7, a decrease of about 0.4 kg. Fat is being consumed at a greater rate.

b. Protein — Seventh week ($t = 6$ to 7): fat about 1.2 → 0.8, a decrease of 0.4 kg; protein about 7.9 → 6.6, a decrease of 1.3 kg. Protein is being consumed at a greater rate.

Problem 38

Stored fat is used at a constant rate of about 2 kg per week — A linear graph has a constant slope: slope $= (4 - 12)/(4 - 0) = -2$ kg/week. So during the first four weeks, stored fat is used at a constant rate of about 2 kg per week.

Problem 39

a. ≈ 2 kg/week — The fat graph is a line with slope −2 for the first 4 weeks, so fat is consumed at about 2 kg/week.

b. ≈ 0.7 kg/week — From the graph, fat(5) ≈ 2.2 and fat(7) ≈ 0.8: slope ≈ $(0.8 - 2.2)/(7 - 5) = -0.7$ kg/week. Fat is consumed at about 0.7 kg/week.

c. ≈ 0.2 kg/week — From the graph, fat(7) ≈ 0.8 and fat(8) ≈ 0.6: slope ≈ $(0.6 - 0.8)/(8 - 7) = -0.2$ kg/week. Fat is consumed at about 0.2 kg/week.

Problem 40

Fat use slows and protein use speeds up: the fat is almost used up, so the body switches to protein — During the sixth week, the fat graph levels off (fat is used more slowly), while the protein graph starts to drop more steeply (protein is used faster). This happens because the stored fat is almost used up, so the body switches to using protein (muscle) as its main source of energy.

Problem 41

I is fat, II is protein — The fat graph is linear with slope −2 for the first 4 weeks, then levels off (slope close to 0). So the derivative of fat is −2 for about the first 4.5 weeks and then rises toward 0 → graph I is fat. The protein graph drops quickly at first, then slowly, and then very quickly after week 6. So its derivative starts negative, rises close to 0, then falls to −2 → graph II is protein.

Second 2022 Q1

a. ≈ 0.041 liters per cm of water: at 10 cm, the lung volume increases by about 0.041 liters for each additional 1 cm of water — Slope to the right: (0.7 − 0.49)/(15 − 10) = 0.21/5 = 0.042. Slope to the left: (0.49 − 0.29)/(10 − 5) = 0.20/5 = 0.040. dV/ds at s = 10 ≈ (0.042 + 0.040)/2 = 0.041 liters per cm of water. When the pressure reduction is 10 cm of water, the volume of the lung increases by about 0.041 liters for each additional 1 cm of water.

b. ≈ 0.01 liters per cm of water: at 30 cm, the lung volume increases by only about 0.01 liters for each additional 1 cm of water — There is no value after s = 30, so we use the interval to the left: dV/ds ≈ (1 − 0.95)/(30 − 25) = 0.05/5 = 0.01 liters per cm of water. When the pressure reduction is 30 cm of water, the volume of the lung increases by only about 0.01 liters for each additional 1 cm of water.

c. Decreasing — The rate went down from 0.041 to 0.01, so the rate is decreasing.

Final 2024 Q5

a. Liters per cm of water — Compliance = dV/dp, so its units are liters per cm of water.

b. ≈ 0.037 liters per cm of water — Slope to the right: (0.86 − 0.70)/(20 − 15) = 0.16/5 = 0.032. Slope to the left: (0.70 − 0.49)/(15 − 10) = 0.21/5 = 0.042. dV/dp at p = 15 ≈ (0.032 + 0.042)/2 = 0.037 liters per cm of water. When the pressure reduction is 15 cm of water, the lung volume increases by about 0.037 liters for each additional 1 cm of water.

Second 2021 Q1

a. ≈ −8.75 RNA copies/mL per day: after 4 days, the viral load is decreasing by about 8.75 RNA copies/mL every day — t = 4 is the first value in the table, so we use the slope to the right: V′(4) ≈ (18 − 53)/(8 − 4) = −35/4 = −8.75 RNA copies/mL per day. After 4 days of treatment, the viral load is decreasing by approximately 8.75 RNA copies/mL every day.

b. ≈ −1.96 RNA copies/mL per day: after 11 days, the viral load is decreasing by about 1.96 RNA copies/mL every day — ARC^{-} = (9.4 − 18)/(11 − 8) ≈ −2.867; ARC^{+} = (5.2 − 9.4)/(15 − 11) = −1.05. V′(11) ≈ (−2.867 + (−1.05))/2 ≈ −1.96 RNA copies/mL per day. After 11 days of treatment, the viral load is decreasing by approximately 1.96 RNA copies/mL every day.

Second 2021 Q2

a. ≈ 1.5 cm/s per °C: at 10 °C, the speed increases by about 1.5 cm/s for every 1 °C increase in temperature — S′(10) = slope of the tangent line at T = 10, which passes through (0, 7) and (10, 22): S′(10) ≈ (22 − 7)/(10 − 0) = 1.5 cm/s per °C. At 10 °C, the maximum sustainable swimming speed increases by approximately 1.5 cm/s for every 1 °C increase in temperature.

b. ≈ −1.8 cm/s per °C: at 25 °C, the speed decreases by about 1.8 cm/s for every 1 °C increase in temperature — S′(25) = slope of the tangent line at T = 25, which passes through (25, 24) and (30, 15): S′(25) ≈ (15 − 24)/(30 − 25) = −9/5 = −1.8 cm/s per °C. At 25 °C, the maximum sustainable swimming speed decreases by approximately 1.8 cm/s for every 1 °C increase in temperature.

2.3.2 Using the Derivative to Estimate Values

Ex 5 **≈ 4.15** — Linear approximation: $f(x) \approx f(a) + f'(a)(x - a)$ with $a = 2$, $x = 2.1$, $\Delta x = 0.1$: $f(2.1) \approx f(2) + f'(2)(0.1) = 4 + 1.5(0.1) = 4 + 0.15 = 4.15$.

Ex 6 **i. Using 5 kg of fertilizer, the average value is 11,500 shillings; when 5 kg is used, each extra kg increases the value by about 350 shillings** — $f(5) = 11{,}500$: using 5 kg of fertilizer, the average value of the yearly maize production is 11,500 Kenyan shillings. $f'(5) = 350$ shillings/kg: when 5 kg of fertilizer is used, each extra kilogram of fertilizer increases the value of the maize production by about 350 Kenyan shillings.

ii. $f(6) \approx 11{,}850$ and $f(10) \approx 13{,}250$ shillings — $f(x) \approx f(a) + f'(a)(x - a)$ with $a = 5$: $f(6) \approx 11{,}500 + 350(1) = 11{,}850$; $f(10) \approx 11{,}500 + 350(10 - 5) = 11{,}500 + 1750 = 13{,}250$.

iii. The estimate of $f(6)$, because 6 is close to 5 — To estimate $f(6)$ we assume that production increases at 350 shillings per kg between $x = 5$ and $x = 6$ kg. To estimate $f(10)$, we assume it continues at the same rate all the way from $x = 5$ to $x = 10$ kg. Since the derivative is increasing for x near 5, the estimate of $f(6)$ is more reliable.

Problem 25 **$f(26) \approx 3.4$ and $f(30) \approx 2.6$** — Using $f(a + \Delta t) \approx f(a) + f'(a)\Delta t$: $f(26) \approx 3.6 + (-0.2)(1) = 3.4$; $f(30) \approx 3.6 + (-0.2)(5) = 2.6$.

Problem 26 **≈ 357** — $f(22) \approx f(20) + f'(20)(22 - 20) = 345 + 6(2) = 357$.

Problem 27 **a. In 2008, sales were \$5.1 billion and were increasing at a rate of \$0.22 billion per year** — $f(8) = 5.1$: in 2008, Hershey sales were \$5.1 billion. $f'(8) = 0.22$: in 2008, sales were increasing at a rate of \$0.22 billion per year.

b. $f(12) \approx 5.98$: in 2012, sales are expected to be about \$5.98 billion — $f(12) \approx f(8) + f'(8)(12 - 8) = 5.1 + 0.22(4) = 5.98$. In 2012, Hershey sales are expected to be about \$5.98 billion.

Problem 28 **a. In 2005, world meat production was 249 million metric tons and was increasing at 6.5 million metric tons per year** — $f(5) = 249$: in 2005, world meat production was 249 million metric tons. $f'(5) = 6.5$: in 2005, meat production was increasing at a rate of 6.5 million metric tons per year.

b. $f(10) \approx 281.5$: in 2010, world meat production is expected to be about 281.5 million metric tons — $f(10) \approx f(5) + f'(5)(10 - 5) = 249 + 6.5(5) = 281.5$. In 2010, world meat production is expected to be about 281.5 million metric tons.

Problem 29 **a. A 140-pound patient should get a dose of 120 mg; for a 140-pound patient, the dose increases by about 3 mg for each additional pound** — $f(140) = 120$: a patient who weighs 140 pounds should get a dose of 120 mg. $f'(140) = 3$: for a 140-pound patient, the dose increases by about 3 mg for each additional pound of weight.

b. ≈ 135 mg — $f(145) \approx f(140) + f'(140)(145 - 140) = 120 + 3(5) = 135$ mg.

Problem 30

a. 20 min, 0.36 mg, −0.002 mg per minute: 20 minutes after the cigarette, there is 0.36 mg of nicotine, decreasing at 0.002 mg per minute — 20 is in minutes, 0.36 is in mg, and −0.002 is in mg per minute. 20 minutes after the cigarette is smoked, there is 0.36 mg of nicotine in the body, and the amount is decreasing at a rate of 0.002 mg per minute.

b. f(21) ≈ 0.358 mg and f(30) ≈ 0.34 mg; the estimate of f(21) is more reliable — Assuming the rate stays about the same: f(21) ≈ 0.36 + (−0.002)(1) = 0.358 mg; f(30) ≈ 0.36 + (−0.002)(10) = 0.34 mg. The estimate of f(21) is more reliable, because 21 is close to 20; over 10 minutes the rate of decrease may change.

Problem 31

a. f(0) = 80 and f′(0) = 0.50 — "Currently" means t = 0: f(0) = 80 and f′(0) = 0.50.

b. f(10) ≈ 85: in 10 days, the value of a share will be about $85 — f(10) ≈ f(0) + f′(0)(10) = 80 + 0.50(10) = 85. In 10 days, the value of a share will be about $85.

Second 2022 Q2

a. S′(T) = dS/dT — S′(T) = dS/dT.

b. ≈ −0.1 (mg/L) per °C: at 40 °C, the solubility decreases by about 0.1 mg/L for each 1 °C increase in temperature — S′(40) is the slope of the tangent line drawn at T = 40. The tangent line passes through (0, 10) and (40, 6): S′(40) ≈ (6 − 10)/(40 − 0) = −4/40 = −0.1 (mg/L) per °C. When the water temperature is 40 °C, the oxygen solubility decreases by about 0.1 mg/L for each 1 °C increase in temperature.

c. ≈ 5.8 mg/L — From the graph, S(40) ≈ 6: S(42) ≈ S(40) + S′(40)(42 − 40) = 6 + (−0.1)(2) = 5.8 mg/L.

Final Summer Q2

a. S′(T) = dS/dT — S′(T) = dS/dT.

b. ≈ −0.1125 (mg/L) per °C: oxygen solubility decreases by about 0.1125 mg/L when the temperature increases from 40 °C to 41 °C — S′(40) = slope of the tangent line at T = 40, which passes through (0, 10.5) and (40, 6): S′(40) ≈ (6 − 10.5)/(40 − 0) = −4.5/40 = −0.1125 (mg/L) per °C. Oxygen solubility decreases by approximately 0.1125 mg/L when the temperature increases from 40 °C to 41 °C.

c. ≈ 5.775 mg/L — S(40) = 6: S(42) ≈ S(40) + S′(40)(42 − 40) = 6 + (−0.1125)(2) = 5.775 mg/L.

2.3.3 Relative Rate of Change

Ex 7

i. At the start of 2008, production was 253 million tons and was increasing at 17 million tons per year — f(8) = 253: at the start of 2008, the annual world soybean production was 253 million tons. f′(8) = 17 million tons/year: at the start of 2008, the annual soybean production was increasing at a rate of 17 million tons per year.

ii. ≈ 0.067 = 6.7% per year: at the start of 2008, production was increasing at about 6.7% per year — RRC = f′(8)/f(8) = 17/253 ≈ 0.067 = 6.7% per year. At the start of 2008, the annual soybean production was increasing at a rate of about 6.7% per year.

iii. f(9) ≈ 270: at the start of 2009, the annual soybean production is about 270 million tons — f(9) ≈ f(8) + f′(8)(9 − 8) = 253 + 17(1) = 270 million tons. At the start of 2009, the annual world soybean production is about 270 million tons.

Ex 8 **i. ≈ 0.445 = 44.5% per year** — 2010 → t = 10: S(10) = 277e^(0.368·10) ≈ 10,982.05. RRC ≈ (S(10 + Δt) − S(10))/(Δt · S(10)). Δt = 1: S(11) ≈ 15,867.33, RRC ≈ (15,867.33 − 10,982.05)/(1 × 10,982.05) ≈ 0.445 = 44.5% per year.

ii. ≈ 0.375 = 37.5% per year — Δt = 0.1: S(10.1) ≈ 11,393.72, RRC ≈ (11,393.72 − 10,982.05)/(0.1 × 10,982.05) ≈ 0.375 = 37.5% per year.

iii. ≈ 0.369 = 36.9% per year — Δt = 0.01: S(10.01) ≈ 11,022.54, RRC ≈ (11,022.54 − 10,982.05)/(0.01 × 10,982.05) ≈ 0.369 = 36.9% per year. The relative rate of change is approximately 36.9% per year. The exponential function S = 277e^(0.368t) has a continuous rate of change of 36.8% per year for all t, which is the exact relative rate of change.

Problem 42 **a. In 2009, the area was 740 million acres and was decreasing at 2.7 million acres per year** — f(9) = 740: in 2009, the area of Brazil's rain forest was 740 million acres. f′(9) = −2.7: in 2009, the area was decreasing at a rate of 2.7 million acres per year.

b. ≈ −0.0036 = −0.36% per year: in 2009, the area was decreasing at about 0.36% per year — Relative rate of change = f′(9)/f(9) = −2.7/740 ≈ −0.0036 = −0.36% per year. In 2009, the area of Brazil's rain forest was decreasing at a rate of about 0.36% per year.

Problem 43 **f(4) = 200 million; f′(4) ≈ 12.5 million users per month; relative rate ≈ 6.25% per month** — End of February: t = 2, f(2) = 175. End of April: t = 4, f(4) = 200 million users. f′(4) ≈ (200 − 175)/(4 − 2) = 12.5 million users per month. Relative rate = f′(4)/f(4) ≈ 12.5/200 = 0.0625 = 6.25% per month. At the end of April 2009, Facebook had 200 million active users, increasing by about 12.5 million users per month, which is about 6.25% per month.

Problem 44 **a. At 2.5 months old, the baby weighs 5.67 kg** — At 2.5 months old, the baby weighs 5.67 kg.

b. At 2.5 months old, the baby's weight is increasing at a rate of about 13% per month — This is the relative rate of change: at 2.5 months old, the baby's weight is increasing at a rate of about 13% per month.

Problem 45 **≈ 0.50 = 50% per unit time** — f′(4) ≈ (f(4.01) − f(4))/0.01 = (16.0801 − 16)/0.01 = 8.01. Relative rate of change = f′(4)/f(4) ≈ 8.01/16 ≈ 0.50 = 50% per unit time.

Problem 46 **a. ≈ 0.0356 = 3.56% per year** — P(3) = 700e^0.105 ≈ 777.4974. (1/P(3)) · (P(4) − P(3))/1 = (805.1917 − 777.4974)/777.4974 ≈ 0.0356 = 3.56% per year.

b. ≈ 0.0351 = 3.51% per year — (1/777.4974) · (780.2234 − 777.4974)/0.1 ≈ 0.0351 = 3.51% per year.

c. ≈ 0.0350 = 3.50% per year — (1/777.4974) · (777.7696 − 777.4974)/0.01 ≈ 0.0350 = 3.50% per year. The relative rate of change is about 3.5% per year (the continuous growth rate of the function).

Practice Q2

a. **≈ −0.24** — From the graph: S(8) ≈ 12.3, S(16) ≈ 10.2, S(24) ≈ 8.5. ARC^{-} = (10.2 − 12.3)/(16 − 8) ≈ −0.26; ARC^{+} = (8.5 − 10.2)/(24 − 16) ≈ −0.21. S′(16) ≈ ((−0.26) + (−0.21))/2 ≈ −0.24 (about −0.25).

b. **mg/L per °C: at 16 °C, the oxygen solubility decreases by about 0.24 mg/L for each 1 °C increase in temperature** — Units: mg/L per °C. At 16 °C, the oxygen solubility decreases by about 0.24 mg/L for each 1 °C increase in the water temperature.

c. **≈ −0.023 = −2.3% per °C** — Relative rate = S′(16)/S(16) ≈ −0.24/10.2 ≈ −0.023 = −2.3% per °C. At 16 °C, the oxygen solubility decreases by about 2.3% per °C. (Reading the graph slightly differently, e.g. S′(16) ≈ −0.25 and S(16) ≈ 10, gives −2.5% per °C — both are acceptable estimates.)

Midterm-B Q8

a. **H′(2) = −4 °C per second: after 2 seconds, the temperature drops by about 4 °C for each additional second** — The tangent line at t = 2 is y − H(2) = H′(2)(t − 2). Comparing with y − 30 = −4(t − 2): H(2) = 30 °C and H′(2) = −4 °C per second. After 2 seconds of the cooling process, the temperature drops by approximately 4 °C for each additional second.

b. **≈ −0.1333 = −13.33% per second: at t = 2, the temperature is decreasing by about 13.33% of its current value per second** — Relative rate = H′(2)/H(2) = −4/30 ≈ −0.1333 = −13.33% per second. At t = 2, the temperature is decreasing by about 13.33% (of its current value) per second.

2.3.4 Test Your Understanding

TYU 6 · P2

(a) The rate of change of N with respect to h is dN/dh. **Answer: B**

(b) Units = units of N ÷ units of h = bat species per foot. **Answer: D**

(c) f′(600) = −2: at 600 feet, one more foot of elevation gives about 2 fewer bat species. **Answer: C**

(d) The graph of N is decreasing everywhere → the instantaneous rate of change is negative (the slope only approaches 0 near h = c). **Answer: B**

(e) The graph is concave up (it decreases more and more slowly), so the rate of change of N is always increasing (from very negative toward 0). **Answer: A**

(f) At h = 0 the graph is very steep going down (large negative slope); at h = c it is almost flat (slope close to 0). So the rate at h = 0 is smaller than the rate at h = c. **Answer: A**

2.4 The Second Derivative

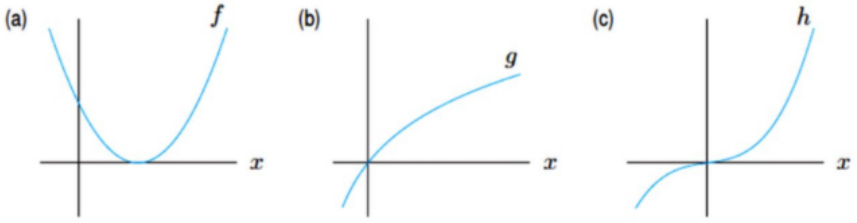
In this section

1. Concavity and the Sign of f″ from a Graph 2 lecture · 12 textbook · 4 past exam
2. The Second Derivative from Tables and Tangent Lines 6 textbook · 1 past exam
3. Interpreting the Second Derivative 1 lecture · 10 textbook
4. Test Your Understanding 1 questions

2.4.1 Concavity and the Sign of f″ from a Graph

Lecture Problems

Ex 1 For the functions whose graphs are given in the figure below, decide where their second derivatives are positive and where they are negative.



(a) f

(b) g

(c) h

Ex 2 At exactly two of the labeled points in Figure 2.40, the derivative f′ is 0; the second derivative f″ is not zero at any of the labeled points. On a copy of the table, give the signs of f, f′ and f″ at each marked point.

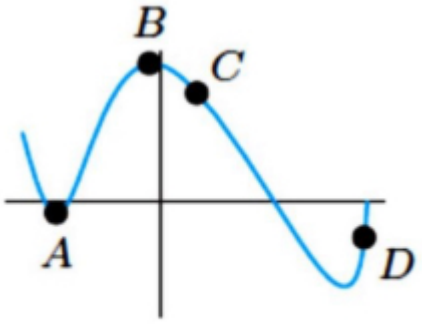


Figure 2.40

Point A Signs of f, f′, f″ at A?

Point B Signs of f, f′, f″ at B?

Point C Signs of f, f′, f″ at C?

Point D Signs of f, f′, f″ at D?

Textbook Problems

Problem 1 For the function graphed in Figure 2.38, are the following nonzero quantities positive or negative?

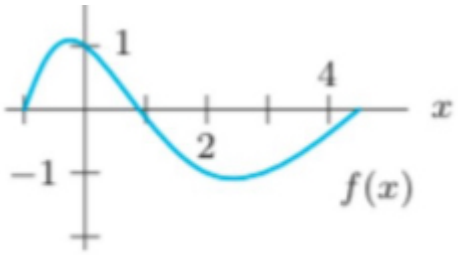


Figure 2.38

a. f(2)

b. f′(2)

c. f″(2)

Problem 2 At one of the labeled points on the graph in Figure 2.39 both dy/dx and d^2y/dx^2 are positive. Which is it?

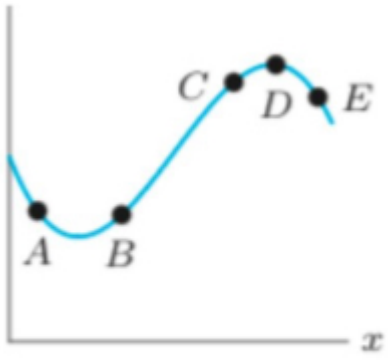
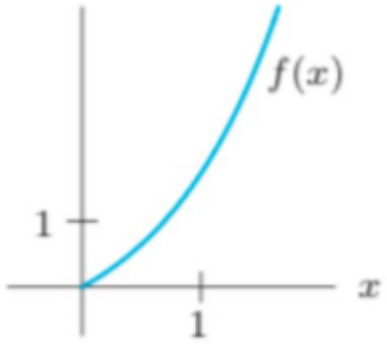


Figure 2.39

Which point?

Problem 3 Give the signs of the first and second derivatives for the following function. Each derivative is either positive everywhere, zero everywhere, or negative everywhere.



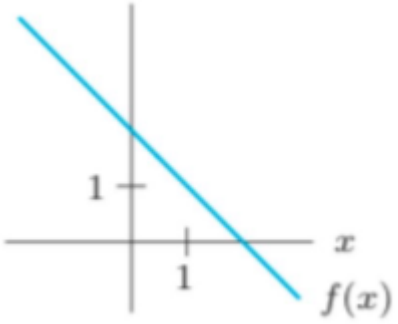
Signs of f′ and f″?

Problem 4 Give the signs of the first and second derivatives for the following function. Each derivative is either positive everywhere, zero everywhere, or negative everywhere.



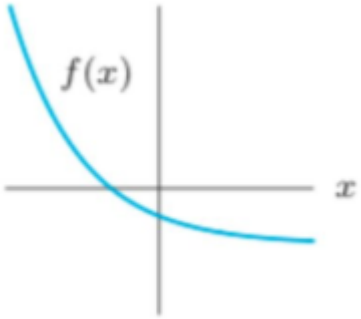
Signs of f′ and f″?

Problem 5 Give the signs of the first and second derivatives for the following function. Each derivative is either positive everywhere, zero everywhere, or negative everywhere.



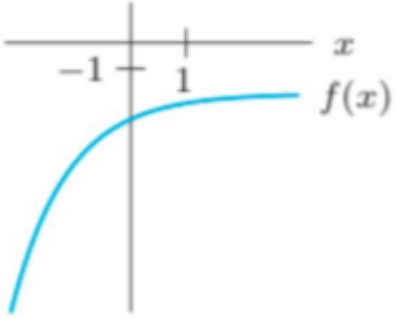
Signs of f′ and f″?

Problem 6 Give the signs of the first and second derivatives for the following function. Each derivative is either positive everywhere, zero everywhere, or negative everywhere.



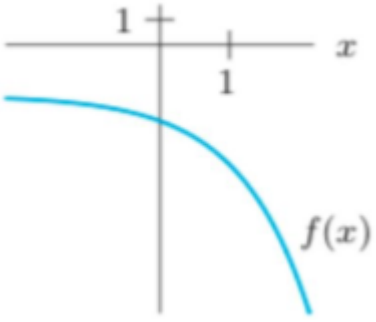
Signs of f′ and f″?

Problem 7 Give the signs of the first and second derivatives for the following function. Each derivative is either positive everywhere, zero everywhere, or negative everywhere.



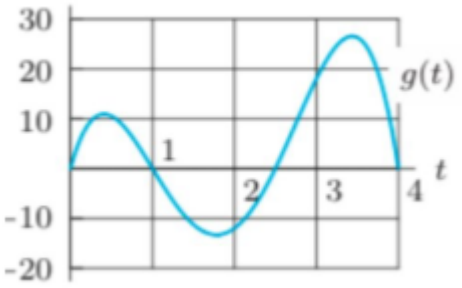
Signs of f′ and f″?

Problem 8 Give the signs of the first and second derivatives for the following function. Each derivative is either positive everywhere, zero everywhere, or negative everywhere.



Signs of f′ and f″?

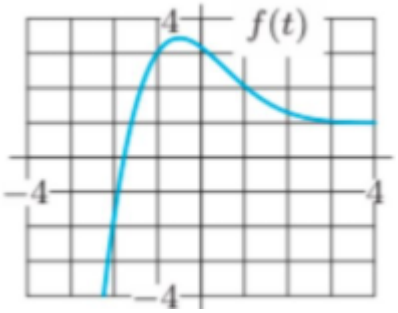
Problem 16 In Problems 16–17, use the graph given for each function.



a. Estimate the intervals on which the derivative is positive and the intervals on which the derivative is negative.

b. Estimate the intervals on which the second derivative is positive and the intervals on which it is negative.

Problem 17 In Problems 16–17, use the graph given for each function.



a. Estimate the intervals on which the derivative is positive and the intervals on which the derivative is negative.

b. Estimate the intervals on which the second derivative is positive and the intervals on which it is negative.

Problem 19 At exactly two of the labeled points in Figure 2.40, the derivative f' is 0; the second derivative f'' is not zero at any of the labeled points. On a copy of the table, give the signs of f, f', f'' at each marked point.

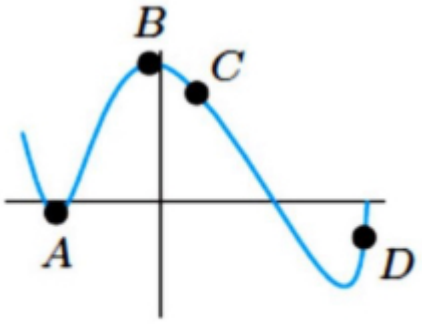


Figure 2.40

Point A Signs of f, f', f'' at A?

Point B Signs of f, f', f'' at B?

Point C Signs of f, f', f'' at C?

Point D Signs of f, f', f'' at D?

Problem 28 Each of the graphs in Figure 2.42 shows the position of a particle moving along the x-axis as a function of time, $0 \le t \le 5$. The vertical scales of the graphs are the same. During this time interval, which particle has

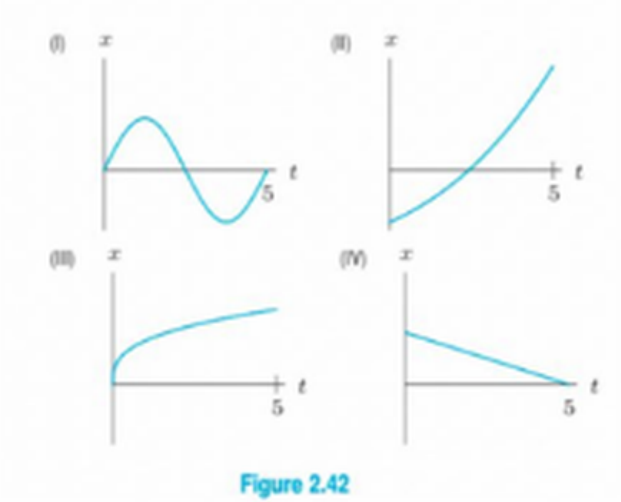


Figure 2.42

a. Constant velocity?

b. The greatest initial velocity?

c. The greatest average velocity?

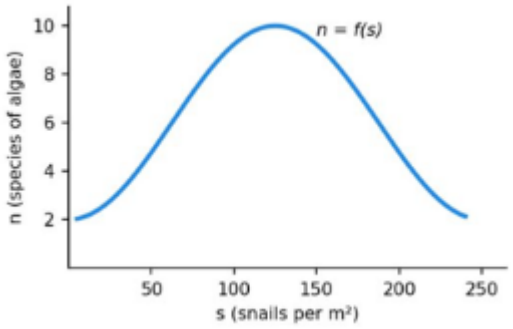
d. Zero average velocity?

e. Zero acceleration?

f. Positive acceleration throughout?

Past Exam Questions

Second 2024 Q1 In tide pools at the coast, snails eat algae. Hence, the number of species of algae, n, is a function of the density of snails, s (in snails per m^2), as described in the figure below.



a. Write the derivative of this function using Leibniz notation.

b. What are the units of the first derivative?

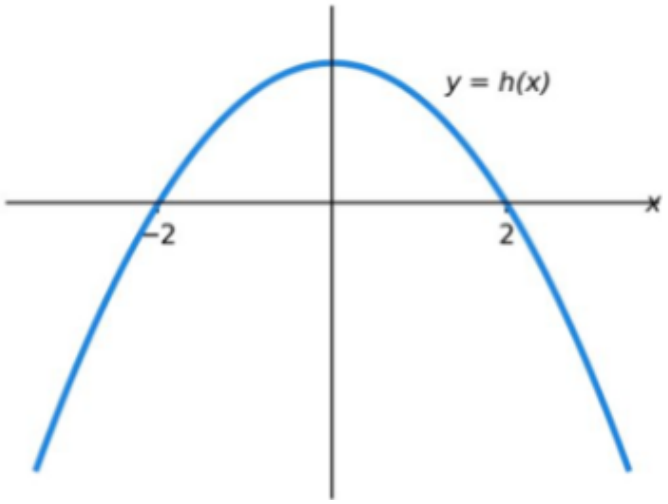
c. Interpret the statement $f'(150) = -5/100$ in terms of the number of species of algae.

d. Determine the intervals where the derivative is positive. What does that tell you about the effect of snails on algae?

e. Find all inflection points.

f. Study the concavity of this function.

First 2024 Q7 The figure below shows a graph of a function h(x).



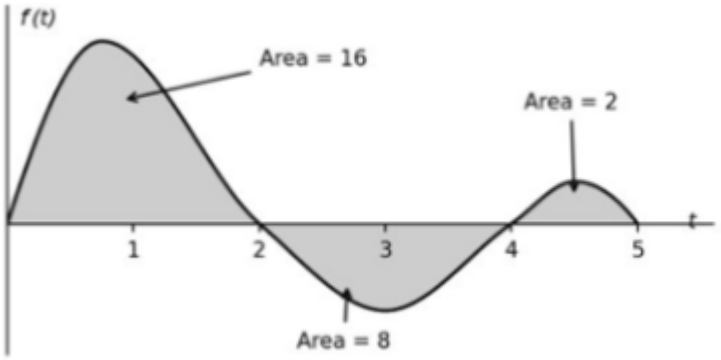
a. In what interval(s) is $h'(x) > 0$?

b. In what interval(s) is $h'(x) < 0$?

c. In what interval(s) is $h'(x)$ increasing?

d. In what interval(s) is $h''(x) < 0$?

Final 2024 Q1 Consider C = f(t), where C is the concentration of adrenaline in the body (in µg/ml) and t is the time (in minutes). The figure below shows the rate of change of the concentration of adrenaline, f′(t). (Parts (d) and (e) of this question use areas (integrals) — Chapter 5.)

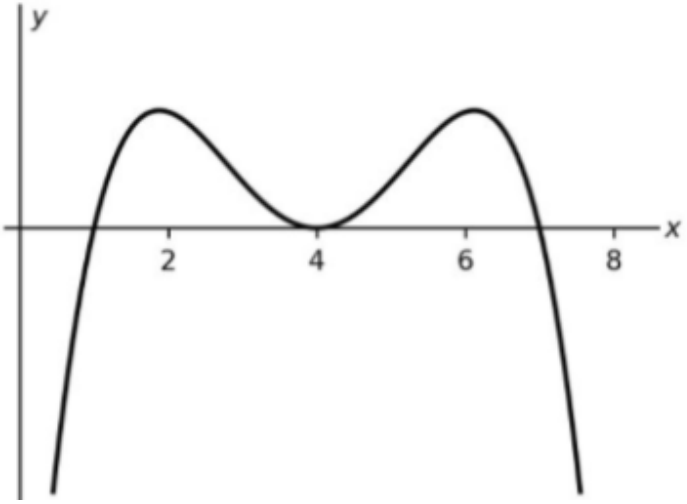


a. Find the intervals in which C is increasing and decreasing.

b. At what times does C have local maximum and minimum?

c. Find the intervals in which C is concave up and concave down.

Midterm Q4 Consider the graph of the **derivative function** $f'(x)$ defined on the interval $(0, 8)$ below.



a. Find the intervals where the function $f(x)$ is increasing and decreasing.

b. Find the x-coordinates of the local maximum and minimum points of $f(x)$.

c. Find the x-coordinates of the inflection point(s) of $f(x)$.

2.4.2 The Second Derivative from Tables and Tangent Lines

Textbook Problems

Problem 10 In Problems 10–11, use the values given for each function.

t	100	110	120	130	140
w(t)	10.7	6.3	4.2	3.5	3.3

a. Does the derivative of the function appear to be positive or negative over the given interval? Explain.

b. Does the second derivative appear to be positive or negative over the given interval? Explain.

Problem 11 In Problems 10–11, use the values given for each function.

t	0	1	2	3	4	5
s(t)	12	14	17	20	31	55

a. Does the derivative of the function appear to be positive or negative over the given interval? Explain.

b. Does the second derivative appear to be positive or negative over the given interval? Explain.

Problem 14 Values of f(t) are given in the following table.

t	0	2	4	6	8	10
f(t)	150	145	137	122	98	56

a. Does this function appear to have a positive or negative first derivative? Second derivative? Explain.

b. Estimate $f'(2)$ and $f'(8)$.

Problem 15 The table gives the number of passenger cars, C = f(t), in millions, in the US in the year t.

t	1940	1950	1960	1970	1980	1990	2000
C	27.5	40.3	61.7	89.2	121.6	133.7	133.6

a. Do $f'(t)$ and $f''(t)$ appear to be positive or negative during the period 1940–1980?

b. Estimate $f'(1975)$. Using units, interpret your answer in terms of passenger cars.

Problem 22 Sketch the graph of a function f such that $f(2) = 5$, $f'(2) = 1/2$, and $f''(2) > 0$.

Which describes the graph near $x = 2$?

Problem 23 A function f has $f(5) = 20$, $f'(5) = 2$, and $f''(x) < 0$, for $x \geq 5$. Which of the following are possible values for $f(7)$ and which are impossible?

a. 26

b. 24

c. 22

Past Exam Questions

First 2024 Q8 A quantity, $q = f(x)$, has the following features:

- $f(10) = 100$
- $f'(10) = -20$
- $f''(10) = -5$

a. Use linear approximation to estimate $f(12)$.

b. Is your answer in Part (a) an overestimate or an underestimate of the true value of $f(12)$? Justify your answer.

2.4.3 Interpreting the Second Derivative

Lecture Problems

Ex 3 A population, P, growing in a confined environment often follows a logistic growth curve, like the graph shown in Figure 2.36. Describe how the rate at which the population is increasing changes over time.

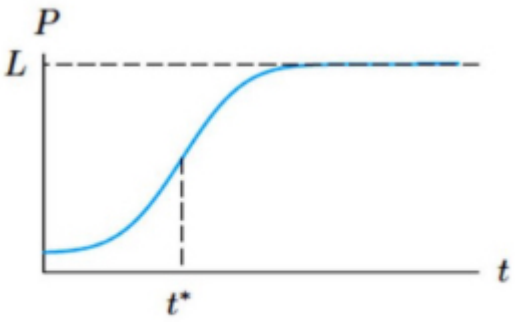


Figure 2.36: Logistic growth curve

i. What is the sign of the second derivative d^2P/dt^2?

ii. What is the practical interpretation of t* and L?

Textbook Problems

Problem 9 Graph the functions described in parts (a)–(d).

a. First and second derivatives everywhere positive.

b. Second derivative everywhere negative; first derivative everywhere positive.

c. Second derivative everywhere positive; first derivative everywhere negative.

d. First and second derivatives everywhere negative.

Problem 12 Sketch the graph of a function whose first derivative is everywhere negative and whose second derivative is positive for some x-values and negative for other x-values.

Which describes such a graph?

Problem 13 IBM-Peru uses second derivatives to assess the relative success of various advertising campaigns. They assume that all campaigns produce some increase in sales. If a graph of sales against time shows a positive second derivative during a new advertising campaign, what does this suggest to IBM management? Why? What does a negative second derivative suggest?

Which is correct?

Problem 18 Sketch a graph of a continuous function f with the following properties:
• $f'(x) > 0$ for all x
• $f''(x) < 0$ for $x < 2$ and $f''(x) > 0$ for $x > 2$.

Which describes the graph?

Problem 20 For three minutes the temperature of a feverish person has had positive first derivative and negative second derivative. Which of the following is correct?

Which is correct?

Problem 21 Yesterday's temperature at t hours past midnight was f(t) °C. At noon the temperature was 20°C. The first derivative, $f'(t)$, decreased all morning, reaching a low of 2°C/hour at noon, then increased for the rest of the day. Which one of the following must be correct?

Which must be correct?

Problem 24 An industry is being charged by the Environmental Protection Agency (EPA) with dumping unacceptable levels of toxic pollutants in a lake. Over a period of several months, an engineering firm makes daily measurements of the rate at which pollutants are being discharged into the lake. The engineers produce a graph similar to either Figure 2.41(a) or Figure 2.41(b). For each case, give an idea of what argument the EPA might make in court against the industry and of the industry's defense.

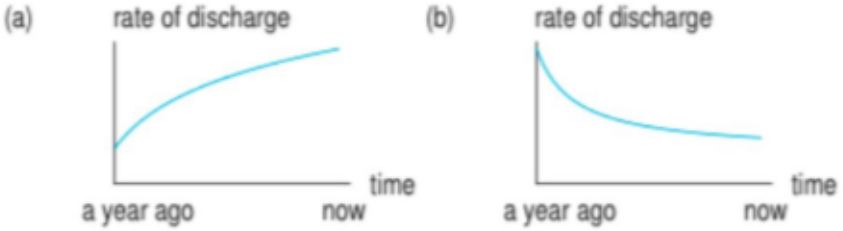


Figure 2.41

(a) Figure 2.41(a)

(b) Figure 2.41(b)

Problem 25 "Winning the war on poverty" has been described cynically as slowing the rate at which people are slipping below the poverty line. Assuming that this is happening:

a. Which describes the graph of the total number of people in poverty against time?

b. If N is the number of people below the poverty line at time t, what are the signs of dN/dt and d^2N/dt^2?

Problem 26 Let P(t) represent the price of a share of stock of a corporation at time t. What does each of the following statements tell us about the signs of the first and second derivatives of P(t)?

a. "The price of the stock is rising faster and faster."

b. "The price of the stock is close to bottoming out."

Problem 27 In economics, *total utility* refers to the total satisfaction from consuming some commodity. According to the economist Samuelson: *As you consume more of the same good, the total (psychological) utility increases. However, …with successive new units of the good, your total utility will grow at a slower and slower rate because of a fundamental tendency for your psychological ability to appreciate more of the good to become less keen.*

a. Which describes the graph of total utility as a function of the number of units consumed?

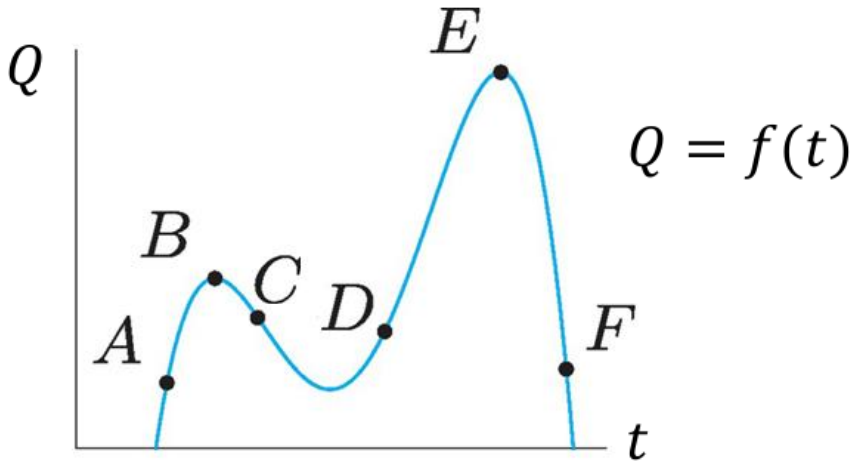
b. In terms of derivatives, what is Samuelson saying?

2.4.4 Test Your Understanding

Test Your Understanding

TYU 6 · P1 The figure below shows the evolution of some quantity, Q, as a function of time, t. Decide whether each of the following statements, comparing the points A through F, is True or False.

1. The rate of change of Q is positive only at Points A and D.
2. The rate of change of Q is negative only at Point C.
3. The most positive rate of change of Q is found at Point D.
4. The highest rate of change of Q is found at Point E.
5. The rate of change of Q at Point F is greater than the rate of change at Point C.
6. The graph of Q is concave up at Points A, B, E and F.
7. The graph of Q is concave down at Points C and D.
8. The rate of change of Q is increasing between Points A and C.
9. The rate of change of Q is decreasing between Points E and F.



Answers · Section 2.4

2.4.1 Concavity and the Sign of f″ from a Graph

Ex 1 **(a) $f'' > 0$ everywhere (concave up everywhere)** — f is concave up everywhere → $f'' > 0$ everywhere.

(b) $g'' < 0$ everywhere (concave down everywhere) — g is concave down everywhere → $g'' < 0$ everywhere.

(c) $h'' < 0$ for $x < 0$ and $h'' > 0$ for $x > 0$ — h is concave down for $x < 0$ → $h'' < 0$ for $x < 0$. h is concave up for $x > 0$ → $h'' > 0$ for $x > 0$.

Ex 2 **Point A** **f −, f′ 0, f″ +** — A: below the x-axis ($f < 0$), at a minimum ($f' = 0$), concave up ($f'' > 0$).

Point B **f +, f′ 0, f″ −** — B: above the x-axis ($f > 0$), at a maximum ($f' = 0$), concave down ($f'' < 0$).

Point C **f +, f′ −, f″ −** — C: above the x-axis ($f > 0$), decreasing ($f' < 0$), concave down ($f'' < 0$).

Point D **f −, f′ +, f″ +** — D: below the x-axis ($f < 0$), increasing ($f' > 0$), concave up ($f'' > 0$).

Problem 1 **a. Negative** — At $x = 2$ the graph is below the x-axis → $f(2)$ is negative.

b. Negative — At $x = 2$ the graph is decreasing (the minimum is a little to the right of 2) → $f'(2)$ is negative.

c. Positive — At $x = 2$ the graph is concave up (bending upward) → $f''(2)$ is positive.

Problem 2 **B** — $dy/dx > 0$ → the graph is increasing. $d^2y/dx^2 > 0$ → the graph is concave up. A is on the decreasing part (before the minimum). B is just after the minimum: the graph is increasing and bending upward. C, D and E are on the part that is concave down (near the maximum). Answer: B.

Problem 3 **f′ positive, f″ positive** — f is increasing and concave up (bending upward). So f′ is positive and f″ is positive.

Problem 4 **f′ zero, f″ zero** — f is a horizontal line: it is constant, so its slope is 0 everywhere and the slope never changes. So f′ is zero and f″ is zero.

Problem 5 **f′ negative, f″ zero** — f is a straight line going down: the slope is negative and constant (it never changes). So f′ is negative and f″ is zero.

Problem 6 **f′ negative, f″ positive** — f is decreasing and concave up (bending upward). So f′ is negative and f″ is positive.

Problem 7 **f′ positive, f″ negative** — f is increasing and concave down (bending downward). So f′ is positive and f″ is negative.

Problem 8 **f′ negative, f″ negative** — f is decreasing and concave down (bending downward). So f′ is negative and f″ is negative.

Problem 16

a. **$g' > 0$ for $0 < t < 0.4$ and $1.7 < t < 3.4$; $g' < 0$ for $0.4 < t < 1.7$ and $3.4 < t < 4$** — From the graph: g has a maximum at $t \approx 0.4$, a minimum at $t \approx 1.7$, and a maximum at $t \approx 3.4$. $g' > 0$ (increasing) for $0 < t < 0.4$ and $1.7 < t < 3.4$; $g' < 0$ (decreasing) for $0.4 < t < 1.7$ and $3.4 < t < 4$.

b. **$g'' < 0$ for $0 < t < 1.1$ and $2.8 < t < 4$; $g'' > 0$ for $1.1 < t < 2.8$** — g changes concavity (inflection points) at $t \approx 1.1$ and $t \approx 2.8$. $g'' < 0$ (concave down) for $0 < t < 1.1$ and $2.8 < t < 4$; $g'' > 0$ (concave up) for $1.1 < t < 2.8$.

Problem 17

a. **$f' > 0$ for $t < -0.5$; $f' < 0$ for $t > -0.5$** — From the graph: f has a maximum at $t \approx -0.5$ and then decreases, leveling off near $f = 1$. $f' > 0$ (increasing) for $t < -0.5$; $f' < 0$ (decreasing) for $t > -0.5$.

b. **$f'' < 0$ for $t < 0.5$; $f'' > 0$ for $t > 0.5$** — f changes concavity at $t \approx 0.5$. $f'' < 0$ (concave down) for $t < 0.5$; $f'' > 0$ (concave up) for $t > 0.5$.

Problem 19

Point A **$f -, f' 0, f'' +$** — A: below the x-axis ($f < 0$), at a minimum ($f' = 0$), concave up ($f'' > 0$).

Point B **$f +, f' 0, f'' -$** — B: above the x-axis ($f > 0$), at a maximum ($f' = 0$), concave down ($f'' < 0$).

Point C **$f +, f' -, f'' -$** — C: above the x-axis ($f > 0$), decreasing ($f' < 0$), concave down ($f'' < 0$).

Point D **$f -, f' +, f'' +$** — D: below the x-axis ($f < 0$), increasing ($f' > 0$), concave up ($f'' > 0$).

Problem 28

a. **(IV)** — Velocity = slope of the position graph. Constant velocity → straight line → (IV).

b. **(III)** — Greatest initial velocity → steepest at $t = 0$ → (III).

c. **(II)** — Average velocity = $(x(5) - x(0))/5$. The largest change in position is in (II) (it goes from below the axis to far above it).

d. **(I)** — Zero average velocity → $x(5) = x(0)$ → (I) (it starts and ends at the same height).

e. **(IV)** — Zero acceleration → velocity does not change → straight line → (IV).

f. **(II)** — Positive acceleration throughout → concave up everywhere → (II).

Second 2024 Q1

a. $f'(s) = dn/ds$ — $f'(s) = dn/ds$.

b. Species of algae per (snails per m²) — Units: species of algae per (snails per m²).

c. At 150 snails per m², increasing the density by 1 snail per m² decreases the number of species by about 0.05 — $f'(150) = -0.05$: when the snail density is 150 snails per m², increasing the density by 1 snail per m² decreases the number of species of algae by about 0.05 (or: 100 more snails per m² → about 5 fewer species).

d. $0 < s < 125$: for densities below about 125 snails per m², more snails means more species of algae — $f'(s) > 0$ where the graph is increasing: $0 < s < 125$ (the maximum is at $s \approx 125$). For densities below about 125 snails per m², more snails means more species of algae.

e. About (60, 6) and (190, 6) — The graph changes concavity at the points (≈ 60, 6) and (≈ 190, 6). These are the inflection points.

f. Concave up for $0 < s < 60$ and $190 < s < 240$; concave down for $60 < s < 190$ — Concave up for $0 < s < 60$ and $190 < s < 240$; concave down for $60 < s < 190$.

First 2024 Q7

a. $(-\infty, 0)$ — The graph of h is an upside-down parabola with its maximum at $x = 0$. h is increasing for $x < 0 \rightarrow h'(x) > 0$ on $(-\infty, 0)$.

b. $(0, \infty)$ — h is decreasing for $x > 0 \rightarrow h'(x) < 0$ on $(0, \infty)$.

c. Nowhere: h' is decreasing everywhere — The graph is concave down everywhere, so h' is decreasing everywhere → h' is never increasing (no interval).

d. $(-\infty, \infty)$ — The graph is concave down everywhere → $h''(x) < 0$ on $(-\infty, \infty)$.

Final 2024 Q1

a. Increasing on $0 < t < 2$ and $4 < t < 5$; decreasing on $2 < t < 4$ — C is increasing where $f'(t) > 0$: $0 < t < 2$ and $4 < t < 5$. C is decreasing where $f'(t) < 0$: $2 < t < 4$.

b. Local maximum at $t = 2$; local minimum at $t = 4$ — At $t = 2$, f' changes from positive to negative → C has a local maximum at $t = 2$. At $t = 4$, f' changes from negative to positive → C has a local minimum at $t = 4$.

c. Concave up on $0 < t < 0.75$ and $3 < t < 4.5$; concave down on $0.75 < t < 3$ and $4.5 < t < 5$ — C is concave up where f' is increasing: $0 < t < 0.75$ and $3 < t < 4.5$. C is concave down where f' is decreasing: $0.75 < t < 3$ and $4.5 < t < 5$.

Midterm Q4

a. Increasing on (1, 7); decreasing on $(0, 1) \cup (7, 8)$ — The given graph is $f'(x)$, not $f(x)$. f is increasing where $f'(x) > 0$ and decreasing where $f'(x) < 0$: f is increasing on (1, 7) and decreasing on $(0, 1) \cup (7, 8)$. (At $x = 4$, $f'(4) = 0$ but f' does not change sign, so f keeps increasing.)

b. Local minimum at $x = 1$; local maximum at $x = 7$; $x = 4$ is neither — $x = 1$ is a local minimum: $f'(1) = 0$ and f' changes from negative to positive. $x = 7$ is a local maximum: $f'(7) = 0$ and f' changes from positive to negative. $x = 4$ is neither: $f'(4) = 0$ but f' stays positive on both sides.

c. $x = 2$, $x = 4$, $x = 6$ — Inflection points of f are where f'' changes sign, i.e. where f' changes from increasing to decreasing or from decreasing to increasing (the peaks and troughs of the graph of f'): $x = 2$, $x = 4$, $x = 6$.

2.4.2 The Second Derivative from Tables and Tangent Lines

Problem 10

a. Negative: the values of w are decreasing — Differences between consecutive values ($\Delta t = 10$): $-4.4, -2.1, -0.7, -0.2$. The values of w are decreasing $\rightarrow$ the derivative is negative.

b. Positive: the differences $-4.4, -2.1, -0.7, -0.2$ are increasing, so w' is increasing — The differences $-4.4, -2.1, -0.7, -0.2$ are increasing (the function decreases more and more slowly), so w' is increasing $\rightarrow$ the second derivative is positive.

Problem 11

a. Positive: the values of s are increasing — Differences ($\Delta t = 1$): 2, 3, 3, 11, 24. The values of s are increasing $\rightarrow$ the derivative is positive.

b. Positive: the differences 2, 3, 3, 11, 24 are increasing, so s' is increasing — The differences 2, 3, 3, 11, 24 never decrease and get much larger (the function increases faster and faster), so s' is increasing $\rightarrow$ the second derivative is positive.

Problem 14

a. First derivative negative; second derivative negative — Differences ($\Delta t = 2$): $-5, -8, -15, -24, -42$. The values of f are decreasing $\rightarrow$ the first derivative is negative. The differences are getting more and more negative (f decreases faster and faster), so f' is decreasing $\rightarrow$ the second derivative is negative.

b. $f'(2) \approx -4$ and $f'(8) \approx -21$ — $f'(2) \approx (f(4) - f(2))/(4 - 2) = (137 - 145)/2 = -4$. $f'(8) \approx (f(10) - f(8))/(10 - 8) = (56 - 98)/2 = -21$.

Problem 15

a. $f'(t)$ positive and $f''(t)$ positive — Differences for 1940–1980 ($\Delta t = 10$): 12.8, 21.4, 27.5, 32.4. C is increasing $\rightarrow$ $f'(t)$ is positive. The differences are increasing (C grows faster and faster) $\rightarrow$ $f''(t)$ is positive.

b. ≈ 3.24 million cars per year: in 1975, the number of passenger cars was increasing at about 3.24 million cars per year — Using the interval 1970–1980, which contains 1975: $f'(1975) \approx (f(1980) - f(1970))/(1980 - 1970) = (121.6 - 89.2)/10 = 3.24$ million cars per year. In 1975, the number of passenger cars in the US was increasing at a rate of about 3.24 million cars per year.

Problem 22

Passes through (2, 5), going up with slope 1/2, concave up (above its tangent line) — The graph passes through (2, 5). The tangent line at $x = 2$ has slope 1/2 (the graph is going up slowly at $x = 2$). $f''(2) > 0$ $\rightarrow$ the graph is concave up at $x = 2$, so it lies above its tangent line near $x = 2$.

Problem 23

a. Impossible — The tangent line at $x = 5$ gives $f(7) \approx f(5) + f'(5)(7 - 5) = 20 + 2(2) = 24$. Since $f'' < 0$, the graph is concave down, so it lies below its tangent line: $f(7) < 24$. So 26 is impossible.

b. Impossible — $f(7)$ must be less than 24 (the graph lies below its tangent line, since f' decreases after $x = 5$). So 24 is impossible.

c. Possible — $22 < 24$, so 22 is possible.

First 2024 Q8

a. **≈ 60** — $f(12) \approx f(10) + f'(10)(12 - 10) = 100 + (-20)(2) = 60$.

b. **Overestimate: $f''(10) < 0$, so the graph is concave down and lies below its tangent line** — $f''(10) < 0 \rightarrow$ the graph is concave down near $x = 10$, so the graph lies below its tangent line. The linear approximation uses the tangent line, so 60 is an overestimate of the true value of $f(12)$.

2.4.3 Interpreting the Second Derivative

Ex 3

i. **$d^2P/dt^2 > 0$ before t^* and $d^2P/dt^2 < 0$ after t^*** — Initially, the population is increasing, and at an increasing rate. So, initially dP/dt is increasing and $d^2P/dt^2 > 0$. At t^*, the rate at which the population is increasing is a maximum; the population is growing fastest then. Beyond t^*, the rate at which the population is growing is decreasing, so $d^2P/dt^2 < 0$.

ii. **t^*: the population is growing fastest; L: the carrying capacity (the largest population the environment can support)** — At t^*, the graph changes from concave up to concave down and $d^2P/dt^2 = 0$: the population is growing fastest. The quantity L represents the limiting value of the population that is approached as t tends to infinity. L is called the carrying capacity of the environment and represents the maximum population that the environment can support.

Problem 9

a. **Increasing, concave up** — $f' > 0 \rightarrow$ increasing; $f'' > 0 \rightarrow$ concave up.

b. **Increasing, concave down** — $f' > 0 \rightarrow$ increasing; $f'' < 0 \rightarrow$ concave down.

c. **Decreasing, concave up** — $f' < 0 \rightarrow$ decreasing; $f'' > 0 \rightarrow$ concave up.

d. **Decreasing, concave down** — $f' < 0 \rightarrow$ decreasing; $f'' < 0 \rightarrow$ concave down.

Problem 12

Always decreasing, concave up in some places and concave down in others — $f' < 0$ everywhere $\rightarrow$ the graph is always decreasing. f'' changes sign $\rightarrow$ the graph is concave up in some places and concave down in others (for example, concave up on the left, concave down on the right).

Problem 13

Positive: sales are rising faster and faster, so the campaign is working; negative: sales are rising more and more slowly, so the campaign is losing effect — Sales are increasing, so the first derivative is positive. A positive second derivative means the rate of increase of sales is itself increasing: sales are going up faster and faster, which suggests the campaign is working well. A negative second derivative means sales are still increasing, but more and more slowly, which suggests the campaign is not very effective (its effect is wearing off).

Problem 18

Always increasing; concave down for $x < 2$ and concave up for $x > 2$ (inflection point at $x = 2$) — $f' > 0 \rightarrow$ f is always increasing. $f'' < 0$ for $x < 2 \rightarrow$ concave down; $f'' > 0$ for $x > 2 \rightarrow$ concave up. So the graph changes concavity at $x = 2$ (inflection point).

Problem 20

The temperature rose in the last minute, but less than it rose in the minute before — $f' > 0 \rightarrow$ the temperature is rising. $f'' < 0 \rightarrow$ it is rising more and more slowly. So in the last minute the temperature rose, but by less than it rose in the minute before. So the correct statement is: the temperature rose in the last minute, but less than it rose in the minute before.

Problem 21 **The temperature rose all day** — The lowest value of $f'(t)$ is 2°C/hour, so $f'(t) \geq 2 > 0$ all day. So the temperature was increasing all day. "It fell in the morning" and "it was lowest at noon" are false, because the temperature never fell. "22°C at 1 pm" is false: $f' > 2$ after noon, so at 1 pm the temperature was more than 22°C. So the temperature rose all day.

Problem 24 **(a) The rate is increasing, but more slowly. EPA: the pollution is getting worse. Industry: the rate is leveling off and will soon stop increasing** — The rate of discharge is increasing (first derivative positive), but it is increasing more slowly (second derivative negative). EPA: the rate at which pollutants are being discharged keeps getting larger — the pollution is getting worse. Industry: the rate of discharge is increasing more and more slowly (concave down); it is leveling off and will soon stop increasing.

(b) The rate is decreasing, but more slowly. EPA: it is leveling off and not going to zero. Industry: the rate is decreasing — they are cleaning up — The rate of discharge is decreasing (first derivative negative), but it is decreasing more slowly (second derivative positive). EPA: the rate of discharge is decreasing more and more slowly and is leveling off, not going to zero — the industry is still dumping pollutants. Industry: the rate of discharge is decreasing — they are cleaning up and reducing the pollution.

Problem 25 **a. Increasing, but more and more slowly (concave down)** — The number of people in poverty keeps increasing, but more and more slowly (the graph is concave down).

b. $dN/dt > 0$ and $d^2N/dt^2 < 0$ — $dN/dt > 0$, because people are still slipping below the poverty line (N is increasing). $d^2N/dt^2 < 0$, because the rate at which N is increasing is getting smaller (the graph is concave down).

Problem 26 **a. $P'(t) > 0$ and $P''(t) > 0$** — The price is rising $\rightarrow$ $P'(t) > 0$. It is rising faster and faster $\rightarrow$ $P''(t) > 0$.

b. $P'(t) < 0$ and $P''(t) > 0$ — The price is still falling $\rightarrow$ $P'(t) < 0$. It is close to reaching its lowest point, so it is falling more and more slowly (the graph is concave up) $\rightarrow$ $P''(t) > 0$.

Problem 27 **a. Increasing, but more and more slowly (concave down)** — Total utility increases, but more and more slowly (concave down).

b. $dU/dx > 0$ and $d^2U/dx^2 < 0$ — If $U = f(x)$ is the total utility of x units, Samuelson is saying that the first derivative is positive, $dU/dx > 0$ (utility increases), and the second derivative is negative, $d^2U/dx^2 < 0$ (it increases at a slower and slower rate).

2.4.4 Test Your Understanding

TYU 6 · P1 From the graph: A and D are on rising parts ($Q' > 0$); B and E are maximums ($Q' = 0$); C and F are on falling parts ($Q' < 0$), and F is much steeper than C.

1. True — B, E: $Q' = 0$; C, F: $Q' < 0$.

2. False — $Q' < 0$ at F too.

3. False — The graph is steeper at A than at D.

4. False — E is a maximum $\rightarrow$ $Q'(E) = 0$.

5. False — F is much steeper going down $\rightarrow$ $Q'(F)$ is more negative.

6. False — B and E are maximums $\rightarrow$ concave down.

7. False — D (just after the minimum) is concave up.

8. False — Q' goes from + (A) to 0 (B) to − (C) $\rightarrow$ decreasing.

9. True — Q' goes from 0 (E) to very negative (F).